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1. MODEM document summary

1.1 Introduction

This document is an extraction from the MODEM model and should be used as a dictionary of MODEM. Descriptive reports can be found in:

- MODAF_M3_and_IDEAS_integration_version_2.00.pdf
  This report deals with the mapping between MODEM and the MODAF M3 meta-model.
- MODAF_M3_and_IDEAS_integration_examplification_version_2.00.pdf
  This report contains modelling examples both of the IDEAS foundation directly as well as models created by the use of MODEM directly, i.e. not filtered through a tool implementation. This is required for verification purposes in order to ensure that the MODEM meta-model can be used directly to create enterprise architecture models. It should be emphasised that this it is not intended that the MODEM model should be used in this manner by modellers. In order for tool manufacturers to implement MODEM properly, the examples shown here are a requirement.

1.1.1 MODEM benefits

- MODEM provides a truly EA tool agnostic representation of MODAF.
- This allows both general EA tools as well as UML based tools to work with a common basis, something that in time will increase the number of different tools that can interwork.
- The semantic created by MODEM underpinning improves the MODAF concepts in a number of areas.
  - Common patterns have also been identified as a result of the MODEM work, something that leads, in a number of areas, to a clearer understanding of the model as well as similarity between different aspects of the model since the patterns are reused throughout the model.
- MODEM is grounded in real-world semantics and provides proper handling of individuals, something that MODAF never did. This also implies that the meta-model has a formalism required to ensure that data can be maintained in a semantically consistent and coherent manner. MODEM can therefore be used to deal with the grave semantic issues that plague the proprietary meta-model that the generic EA tools use (1) as well as the ones identified for the UML based tools (2).
- MODAF M3 was based on UML and this was, at the time, a good decision. UML is a common standard and has had an enormous amount of work invested in it. MODEM has been created by harvesting all of the good points of MODAF and UML and winnowing out all the less good parts by means of the BORO methodology that underpins the work effort to create MODEM.
- The origins of UML are primarily technical and devoted to the design of software systems and this is one of the reasons that UML use brings with it a lot of baggage.
- The UML baggage which in many cases distorts the MODAF meta-model is therefore removed in MODEM and a more enterprise architecture approach has been achieved.
- MODEM development has had the primary objective of covering MODAF, i.e. it is by and large backward compatible to MODAF architectures developed based on the M3 meta-model.

- MODEM, it is felt, answers the need of NATO to have a NAF model without UML dependencies.
- MODEM provides a vehicle for discussions and development of a common enterprise architecture framework for defence (and even outside defence) since it will be based on the same concepts as DoDAF 2.

1.1.2 Document structure

This document is structured in the following manner:

- For each viewpoint (AV, StV, OV, SOV, SV. TV and AcV) the M3 meta-model view MODEM counterparts are contained. In some viewpoints there are some additional view representations that were not contained in MODAF M3 but this is rare.
- The above viewpoint descriptions are followed by by a list of all of the elements owned by a particular viewpoint. It should be noted that there will aften be more elements in this list than the ones that are shown in the above viewpoint views.
- The elements list is followed by a set of figures that show relevant features of the MODEM model in greater detail.
- At the end of the document, the same structure is repeated for the IDEAS foundation, additions made to the Foundation to accommodate MODEM and also the patterns defined in order to bridge the gap between the IDEAD foundation and MODAF.

2. MODEM

2.1 Modelling Notation

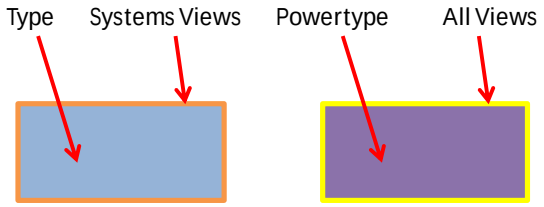
The UML model for MODEM uses a slightly enhance colour scheme from the usual IDEAS approach. The fill colours of each element remain the same:

- Sky-blue = Type
- Orange = IndividualType
- Lilac = Powertype
- Light-green = TupleType
- Dark-grey = Individual
- Yellow = NamingScheme

For MODEM, in addition to, the border colour is used to indicate the MODAF viewpoint, using the standard MODAF colours:

- Yellow = All Views
- Green = Strategic Views
- Purple = Service-Oriented Views
- Blue = Operational Views
- Orange = Systems Views
- Pink = Acquisition Views
- Grey = Technology & Standards Views

Examples are:



In addition, placeable types (Tuples, TupleTypes, TupleTypeTypes, etc.) are displayed with thin borders whilst all other elements have thick borders.

As noted later in this document, some additions to the IDEAS Foundation were necessary. Where these additions are used, they are shown with a red border.

2.2 All Views

2.2.1 AV-1: Overview and summary information

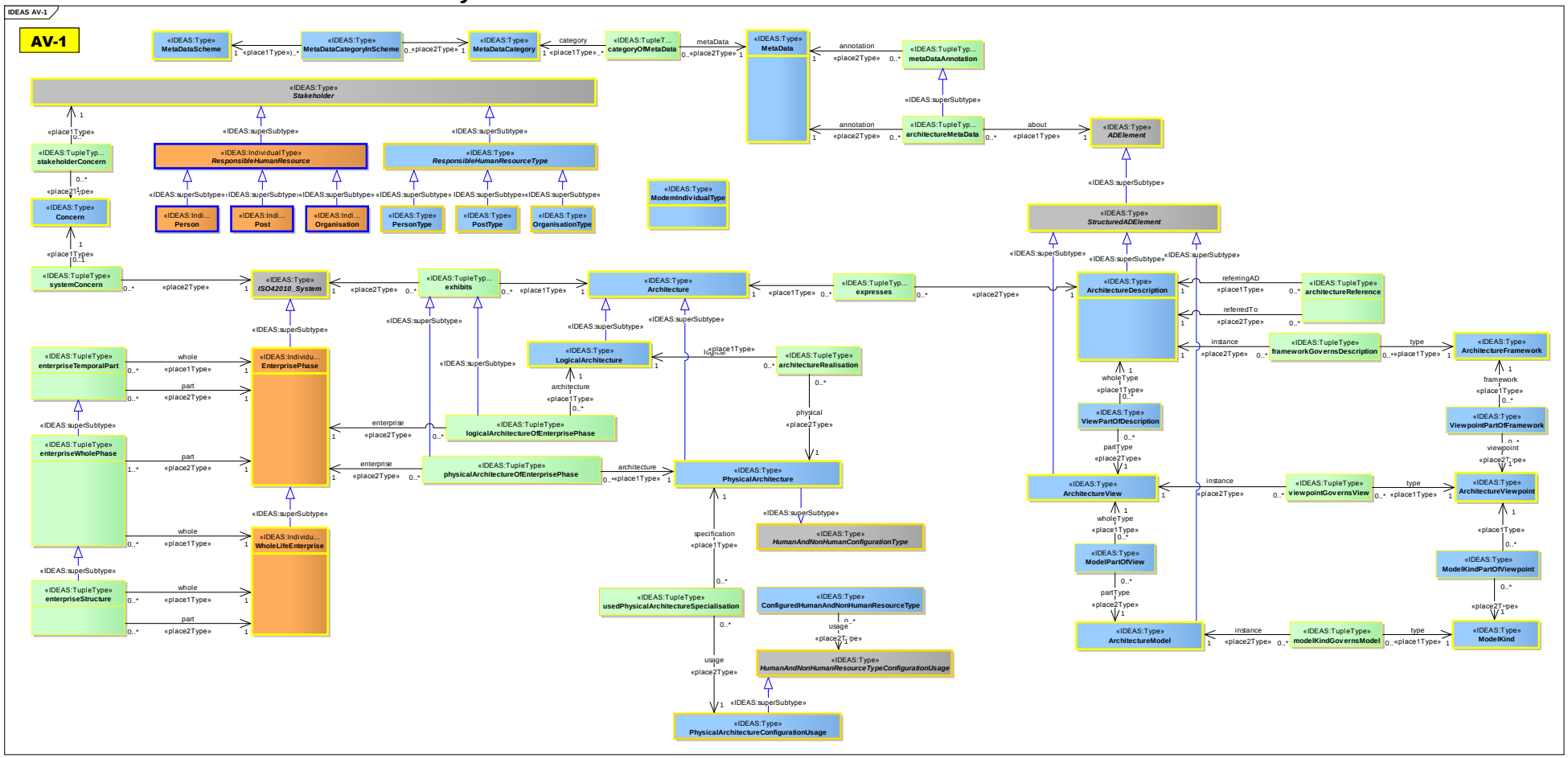


Figure 1 : AV-1

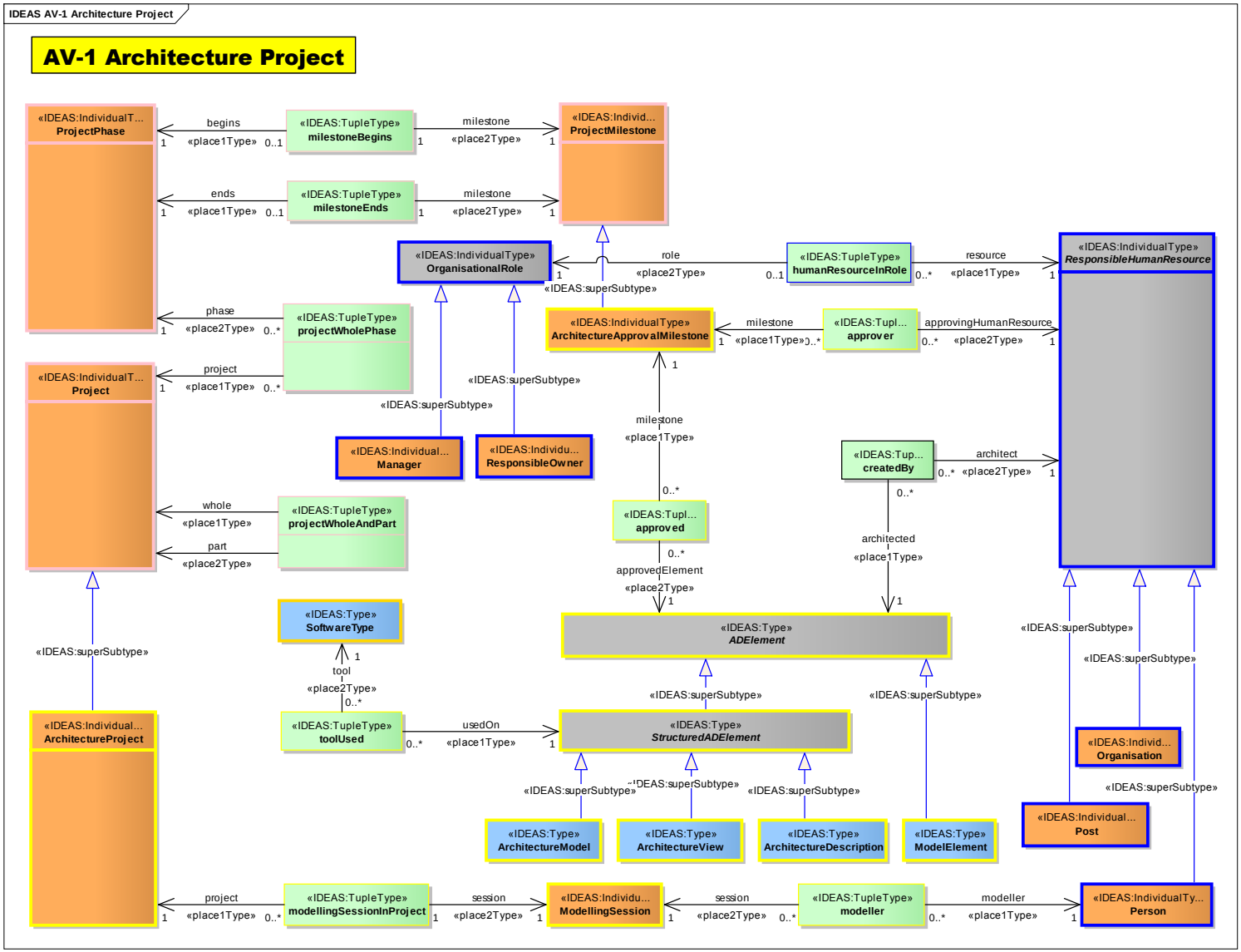


Figure 2 : AV-1 Architecture Project

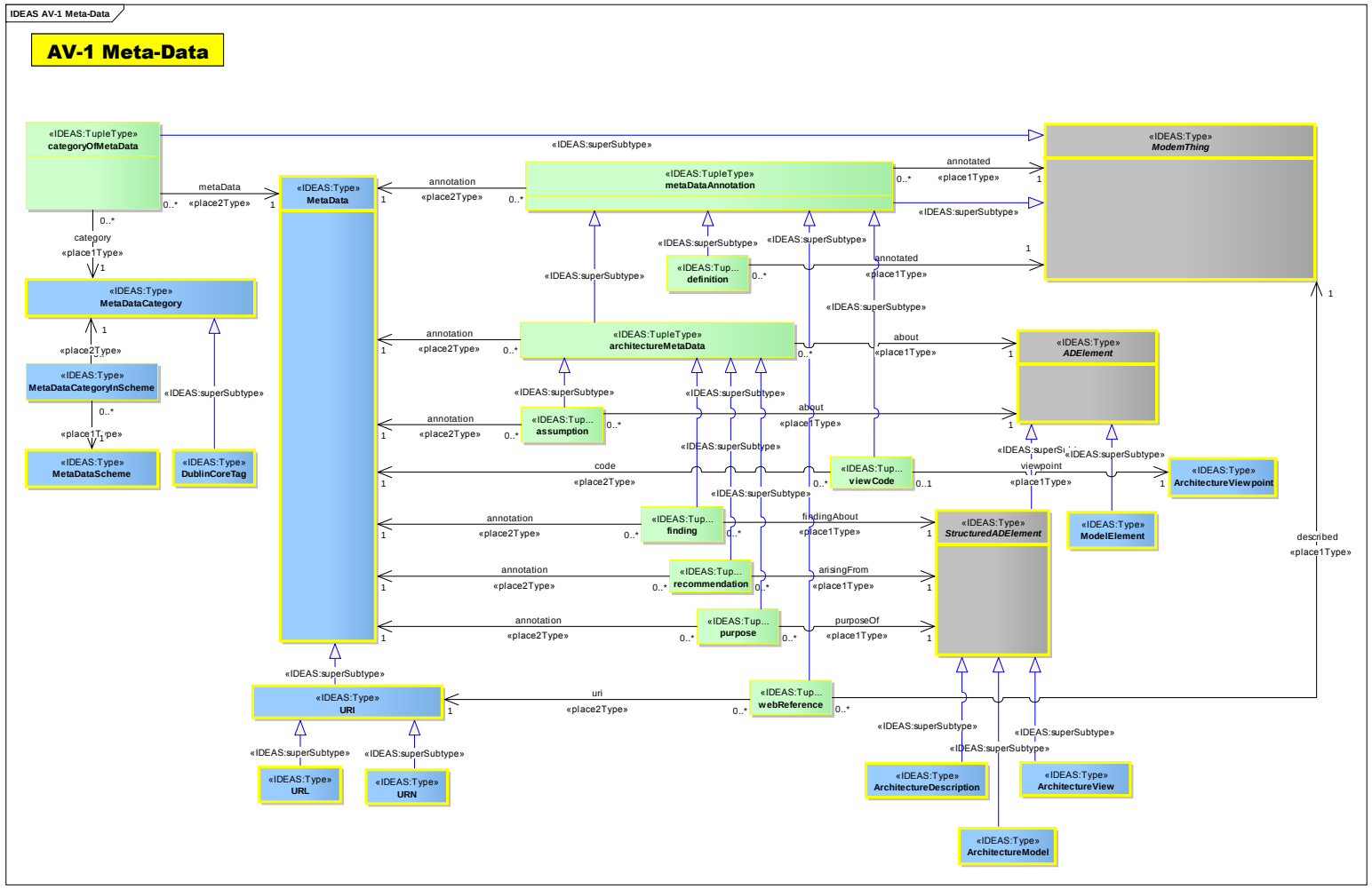


Figure 3 : AV-1 Meta-Data

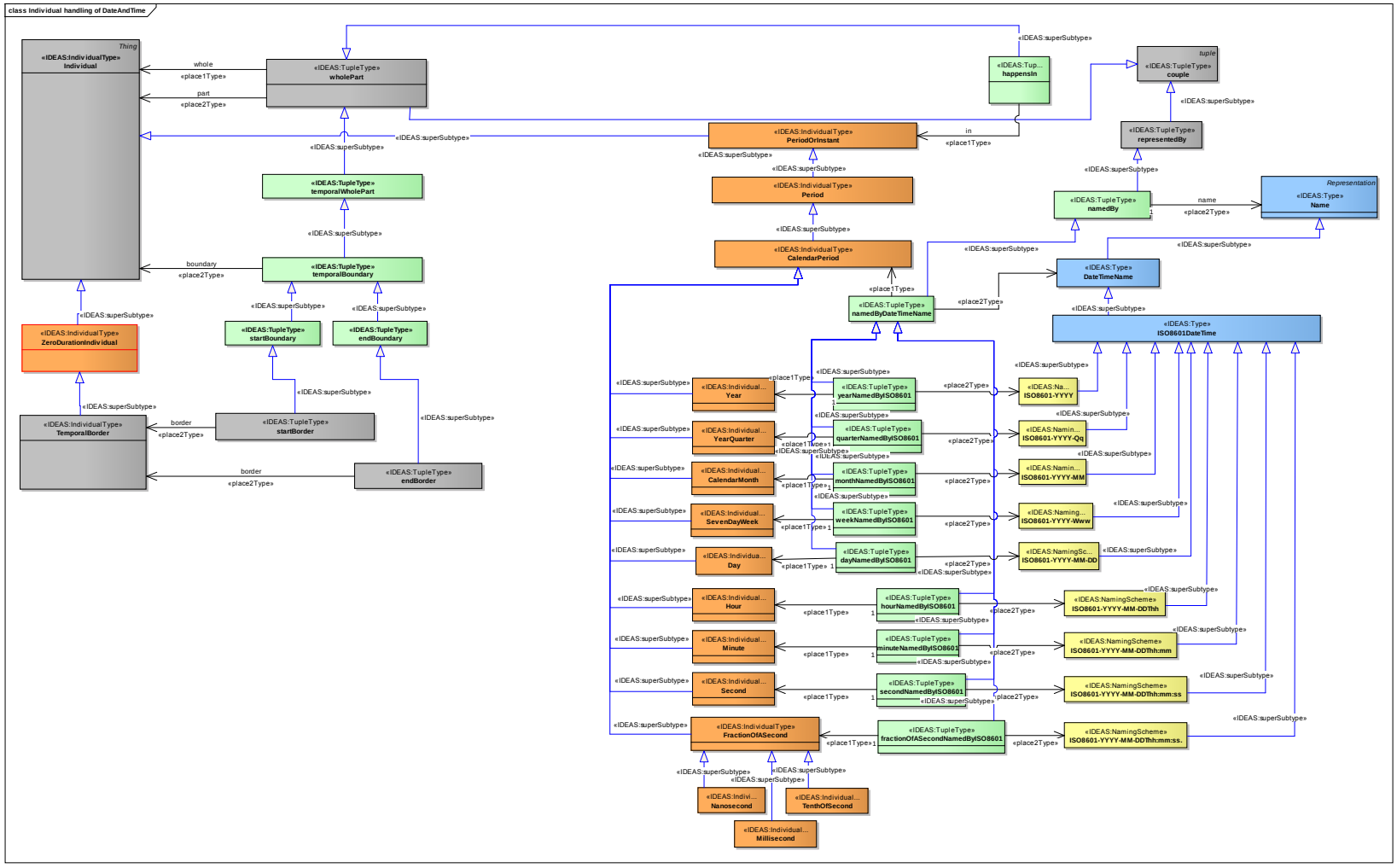


Figure 4 : Individual handling of DateAndTime

.

2.2.2 AV-2: Integrated dictionary

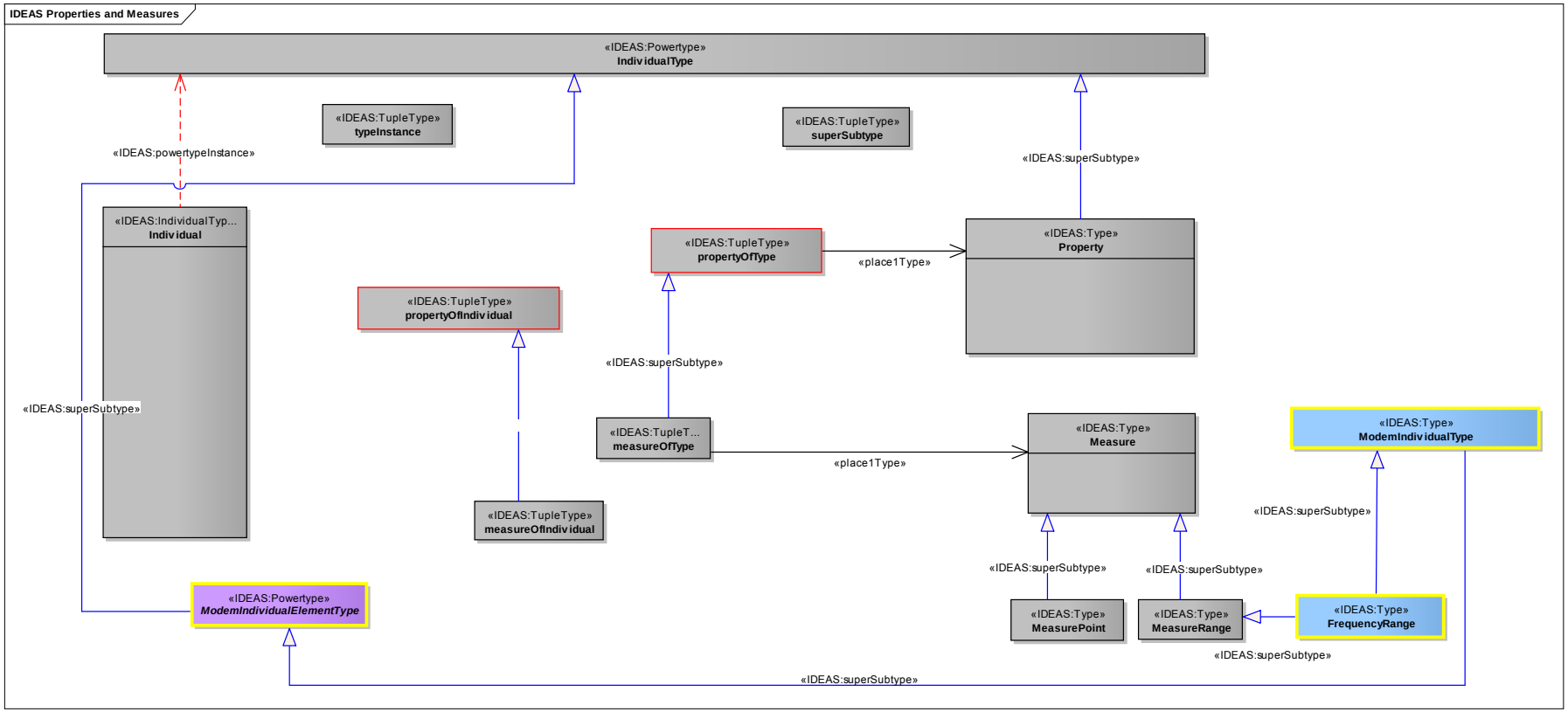


Figure 5 : Properties and Measures

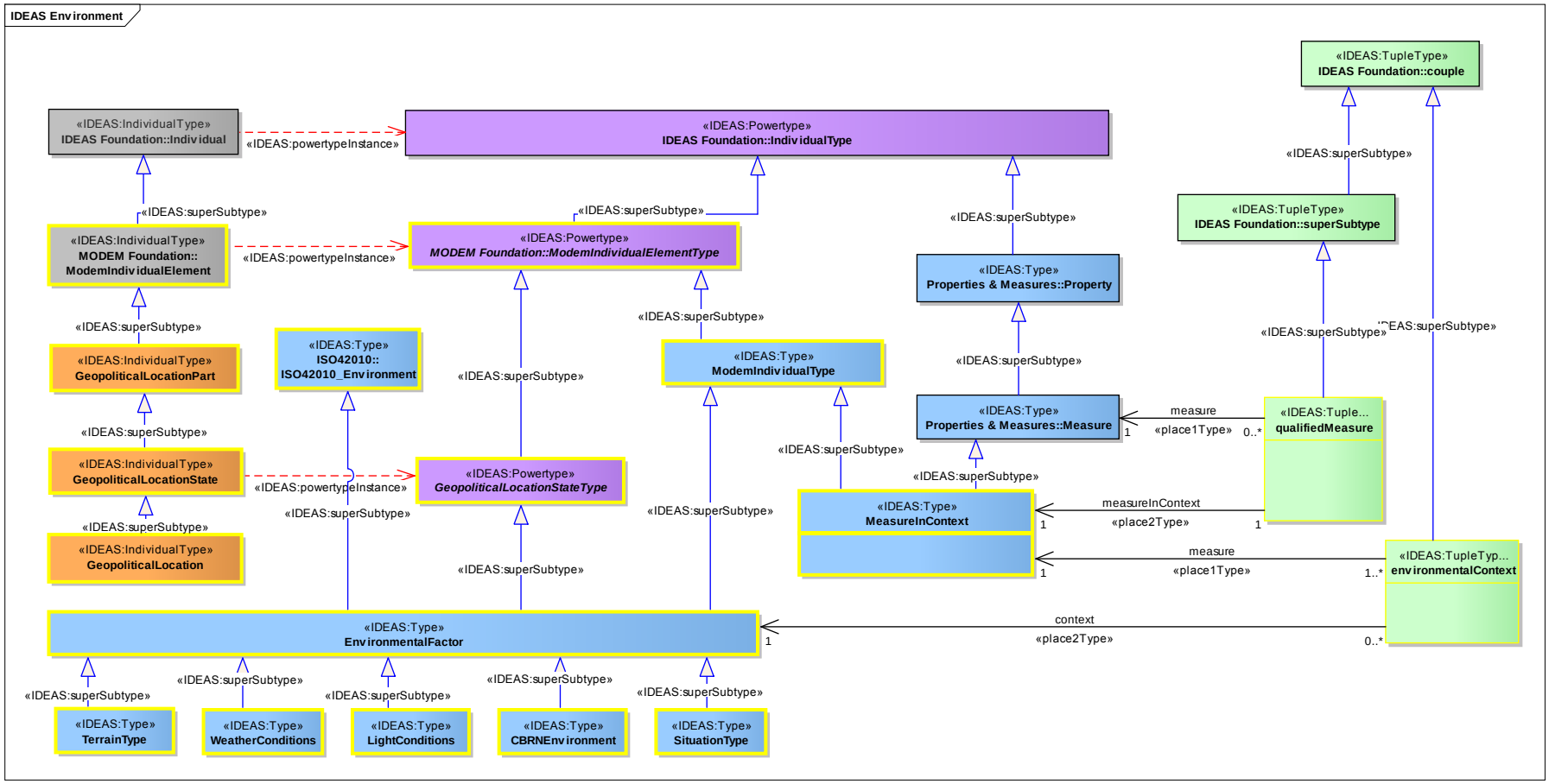


Figure 6 : Environment

2.2.3 All Views elements list

MODEM All Views
After «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» After - BeforeAfterType *Generalization (element - is a subtype of):*«IDEAS:superSubtype» After - StartsAfter *Association (source - target):*«place2Type» After - TriggerItem *Association (source - target):*«place1Type» After - TriggerItem *Attributes:* - A BeforeAfterType where one TriggerItem starts after another has ended. Note: the TriggerItem that happens after may happen at any point in time after the one that comes before it (i.e. there may be an interval of time between them).
AgentCapableOfResponsibilityOrAgentCapableOfResponsibilityType «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AgentCapableOfResponsibilityOrAgentCapableOfResponsibilityType - Type *Attributes:* - The union of AgentCapableOfResponsibility and AgentCapableOfResponsibilityType.
ArchitectureApprovalMilestone «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArchitectureApprovalMilestone - ProjectMilestone *Attributes:* - A ProjectMilestone where an ADElement is approved by a ResponsibleHumanResource. Note: this replaces the dateCompleted tag on ArchitecturalDescription in M3.
ArchitectureProject «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArchitectureProject - Project *Attributes:* - A Project that delivers an ArchitectureDescription.
ArtefactPowertype «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ArtefactPowertype - NonHumanResourcePowertype *Attributes:* - The powertype of Artefact.

CBRNEnvironment «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CBRNEnvironment - EnvironmentalFactor
Attributes:
-
An EnvironmentalFactor that defines the type of chemical, biological, radiological and nuclear environment in which an Enterprise may operate.

Constraint «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Constraint - ModemIndividualType
Attributes:
-
A ModemIndividualType that is the collection of all the objects subject to a particular constraint.

Delay «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Delay – TriggerItem
Attributes:
-
A TriggerItem that is a pause between Processes, Events, etc.

DublinCoreTag «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DublinCoreTag - MetaDataCategory
Attributes:
-
A MetaDataCategory that is a DublinCore tag.

EnduringTaskPart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnduringTaskPart - UndertakingPart
Attributes:
-
An UndertakingPart where the whole is an EnduringTask.

EnterprisePart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnterprisePart - AgentPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnterprisePart - UndertakingPart
Attributes:
-
A part of a WholeLifeEnterprise

EnterprisePhase «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»

FrequencyRange «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FrequencyRange - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FrequencyRange - MeasureRange
Attributes:
-
A MeasureRange that specifies maximum and minimum frequencies, measured in Hertz as real numbers.

GeoName «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeoName - StringName
Attributes:
-
A MeasureRange that specifies maximum and minimum frequencies, measured in Hertz as real numbers.

GeopoliticalLocation «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
GeopoliticalLocation - Location
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeopoliticalLocation - IntentionallyConstructedIndividual
Dependency (element - is instance of):«IDEAS:powertypeInstance»
GeopoliticalLocation - GeopoliticalLocationType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeopoliticalLocation - GeopoliticalLocationState

Attributes:
-
A Location and a GeoPoliticalArea.

GeopoliticalLocationPart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeopoliticalLocationPart - ModemIndividualElement
Attributes:
-
A ModemIndividualElement that is a part of a GeopoliticalLocation**.**

GeopoliticalLocationState «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
GeopoliticalLocationState - GeopoliticalLocationStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeopoliticalLocationState - GeopoliticalLocationPart
Attributes:
-
A GeopoliticalLocationPart that is a temporal state of a GeopoliticalLocation - i.e. all of the location for a period of time.

GeopoliticalLocationStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeopoliticalLocationStateType - ModemIndividualElementType
Attributes:
-
The powertype of GeopoliticalLocationState.

GeopoliticalLocationType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
GeopoliticalLocationType - GeopoliticalLocationStateType
Attributes:
-
The powertype of GeopoliticalLocation.

ISO6709Representation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ISO6709Representation - GeoName
Attributes:
-
A GeoName expressed using the ISO6709:2008 standard notaton
Examples: "-90+000+2800CRSWGS_84/" "+48.8577+002.295/"

ImmediatelyAfter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ImmediatelyAfter - ImmediateBeforeAfterType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ImmediatelyAfter - After
Association (source - target):«place2Type»
ImmediatelyAfter - TriggerItem
Association (source - target):«place1Type»
ImmediatelyAfter - TriggerItem
Attributes:
-
An After where the subsequent TriggerItem starts immediately as the preceeding TriggerItem ends.

InformationInstance «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationInstance - Sign
Dependency (element - is instance of): «IDEAS:powertypeInstance»
InformationInstance - InformationInstanceType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationInstance - ModemIndividualElement
Attributes:
-
A Sign that is an individual item of information (e.g. an utterance, an individual instance of a paper or electronic document).

InformationInstanceType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationInstanceType - SignType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationInstanceType - ModemIndividualElementType
Attributes:
-
The powertype of InformationInstance.

ItemInScenario «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ItemInScenario – TypicalWholePart
Association (source - target): «place1Type»
ItemInScenario - Scenario
Attributes:
-
A TypicalWholePart where the whole is a Scenario.

LightConditions «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LightConditions - EnvironmentalFactor
Attributes:
-
An EnvironmentmentalFactor that defines the types of light (e.g. broad daylight, dusk, moonlit, etc.) in which an Enterprise may operate.

Location «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Location - ModemIndividualElement
Attributes:
-
A location anywhere on the earth. The means of describing the location is a string (locationDescription). The information contained in that string is governed by the taxonomy reference - e.g. if the Location is a "GPS reference", the string will contain the GPS coordinates.
Note: was called "ActualLocation" in M3 v1.2

LogicalArchitecture «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalArchitecture - Architecture
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalArchitecture - NodeParent
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalArchitecture - EnteprisePhaseType
Attributes:
-
A NodeParent whose parts are either Nodes, KnownResources or LogicalDomains.

MeasureInContext «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MeasureInContext - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MeasureInContext - Measure
Attributes:
-
A ModemIndividualType that brings together EnvironmentalFactors with a Measure in order to qualify the measure. Examples: 40mph in desert, 1km range in cloudy conditions.

MetaData «IDEAS:Type»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
MetaData - MetaDataType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MetaData - StringRepresentation
Attributes:
-
A StringRepresentation that can be applied to any element in the architecture. Note: wherever possible, standard Meta-Data types should be used - e.g. conforming to Dublin Core. Note for MOD Users: The MOD Meta Data Standard categories shall be used.

MetaDataCategory «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
MetaDataCategory - MetaDataType
Attributes:
-
A MetaDataType that defines the category of a MetaData element
example: http://purl.org/dc/terms/abstract

MetaDataCategoryInScheme «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MetaDataCategoryInScheme - WholePartTypeType
Association (source - target):«place2Type»
MetaDataCategoryInScheme - MetaDataCategory
Association (source - target):«place1Type»
MetaDataCategoryInScheme - MetaDataScheme
Attributes:
-
A WholePartTypeType that asserts a MetaDataCategory belongs to a MetaDataScheme.

MetaDataScheme «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
MetaDataScheme - RepresentationScheme

0 Tf1 6.30 2.83 1q65.76 1 BDC q61C 8.04 Tf1 0 0 1 134.18 137.42 Tm0 g0 G[( )] TJ2tionen-UTfoul28 440.620137.42 Tm0 g0 G[()-6(e)11(me)13( )] T.57 146.69 Tm0 g0 g0 G44 r.

MetaDataType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
MetaDataType - RepresentationType
Attributes:
-
The powertype of MetaData.

ModellingSession «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ModellingSession - AgentPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ModellingSession - ProjectPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ModellingSession - ParticipationExtent
Attributes:
-
A ProjectPart where ArchitectureDescriptions are worked on.

ModemIndividualType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ModemIndividualType - ModemIndividualElementOrModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModemIndividualType - ModafIndividualElementType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModemIndividualType - ModemThing
Attributes:
-
The parent (supertype) of all MODEM elements that are types of Individuals
e.g. tank, computer, etc.

ModemTemporalWholePartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModemTemporalWholePartType - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModemTemporalWholePartType - TemporalWholePartType
Association (source - target): «place2Type»
ModemTemporalWholePartType - ModemIndividualElementType
Association (source - target): «place1Type»
ModemTemporalWholePartType - ModemIndividualElementType
Attributes:
-
The powertype of modemTemporalWholePart.

ModemWholePartType

The powertype of ProjectPhase
ProjectPowertype «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ProjectPowertype - ProjectPhaseType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ProjectPowertype - UndertakingType *Attributes:* - The powertype of Project
ProjectThreadPowertype «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ProjectThreadPowertype - ProjectPartType *Attributes:* - The powertype of ProjectThread.
ResourcePackageType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ResourcePackageType - IndividualResourcePowertype *Attributes:* - The powertype of ResourcePackage.
Scenario «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Scenario - ModemIndividualType *Attributes:* - A ModemIndividualType whose typical parts are other ModemIndividualTypes that are organised into a typical temporal sequence.
ServiceDelivery «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ServiceDelivery - ServiceDeliveryState *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ServiceDelivery - ServiceDeliveryType *Attributes:* - An Undertaking that is a realization of a ServiceSpecification - i.e. the delivery of Service according to that specification. Note a ServiceDelivery is an Individual whose extent is the fusion of all the processes, people, systems that go into delivering a service. Example: the ongoing building management service provided to MOD by Amey for Main Building.
ServiceDeliveryPart «IDEAS:IndividualType» *Connectors:* *GGeneralization (element - is a subtype of):*«IDEAS:superSubtype» ServiceDeliveryPart - ModemIndividualElement *Dependency (element - is instance of):*«IDEAS:powertypeInstance» ServiceDeliveryPart - ServiceDeliveryPartType *Attributes:*

-
A ModemIndividualType that is part of a ServiceDelivery.

ServiceDeliveryPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryPartType - ModemIndividualElementType
Attributes:
-
The powertype of ServiceDeliveryPart.

ServiceDeliveryState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryState - ServiceDeliveryPart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ServiceDeliveryState - ServiceDeliveryStateType
Attributes:
-
A ServiceDeliveryPart that is a temporal part of a ServiceDelivery - i.e. all of the ServiceDelivery for a period of time.

ServiceDeliveryStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryStateType - ServiceDeliveryPartType
Attributes:
-
The powertype of ServiceDeliveryState.

ServiceDeliveryType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryType - ServiceDeliveryStateType
Attributes:
-
The powertype of ServiceDelivery.

ServiceDeliveryWholeAndPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceDeliveryWholeAndPartType - ServiceDeliveryWholeStateType
Association (source - target):«place2Type»
ServiceDeliveryWholeAndPartType - ServiceDeliveryType
Attributes:
-
The Powertype of serviceDeliveryWholeAndPart.

ServiceDeliveryWholeFacadeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryWholeFacadeType - ServiceDeliveryWholePartType
Association (source - target): «place2Type»
ServiceDeliveryWholeFacadeType - ServiceFacadeType
Attributes:
-
The powertype of serviceDeliveryWholeFacade.

ServiceDeliveryWholePartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryWholePartType - ModemWholePartType
Association (source - target): «place2Type»
ServiceDeliveryWholePartType - ServiceDeliveryPartType
Association (source - target): «place1Type»
ServiceDeliveryWholePartType - ServiceDeliveryType
Attributes:
-
The powertype of serviceDeliveryWholePart.

ServiceDeliveryWholeStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryWholeStateType - ModemTemporalWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceDeliveryWholeStateType - ServiceDeliveryWholePartType
Association (source - target): «place2Type»
ServiceDeliveryWholeStateType - ServiceDeliveryStateType
Attributes:
-
The powertype of serviceDeliveryWholeState.

ServiceFacade «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFacade - ServiceDeliveryPart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ServiceFacade - ServiceFacadeType
Attributes:
-
A ServiceDeliveryPart that is the extent of all the parts of a ServiceDelivery that are concerned with communication with consumers. A ServiceFacade may be specified by a ServiceInterface.

ServiceFacadeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFacadeType - ServiceDeliveryPartType
Attributes:
-
The powertype of ServiceFacade.

SituationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SituationType - EnvironmentalFactor
Attributes:
-
An EnvironmentalFactor used to describe the types and levels of threat under which an Enterprise may operate. Examples: Corrosive, Fire, Smoke, Peaceful, Under Fire, Under Heavy Fire, etc.

StartsAfter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StartsAfter - WeakTemporalOrderingType
Association (source - target):«place2Type»
StartsAfter - TriggerItem
Association (source - target):«place1Type»
StartsAfter - TriggerItem
Attributes:
-
A WeakTemporalOrderingType that asserts one TriggerItem starts before another. Note: there is constraint on when either TriggerItem ends - hence if A starts before B, it is possible that B ends before A and indeed that A ends before B.

StartsImmediatelyAfter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StartsImmediatelyAfter - StartsAfter
Association (source - target):«place2Type»
StartsImmediatelyAfter - TriggerItem
Association (source - target):«place1Type»
StartsImmediatelyAfter - TriggerItem
Attributes:
-
A StartsAfter where the subsequent TriggerItem starts immediately after the preceeding TriggerItem.

StateMachine «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateMachine - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateMachine - StateMachineViews
Attributes:
-
A StateMachineViews used to model typical states and transitions for ModemIndividualElementTypes.

StateMachineRegion «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateMachineRegion - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateMachineRegion - StateMachineRegions
Attributes:
-
A StateMachineRegions which is part of a StateMachine.

StateSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateSpecification - TriggerItem
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateSpecification - OwnedStateSets
Attributes:
-
An OwnedStateSets used in a MODEM state machine.

StateTransition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateTransition - StateSetTransitions
Association (source - target):«place2Type»
StateTransition - StateSpecification
Association (source - target):«place1Type»
StateTransition - StateSpecification
Attributes:
-
A StateSuccessionType indicating there is a possible transition between StateSpecifications.

StatusOfThreadType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StatusOfThreadType - ModemTemporalWholePartType
Attributes:
-
The powertype of statusOfThread.

TerrainType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
TerrainType - EnvironmentalFactor
Attributes:
-
An EnvironmentalFactor that defines the type of ground conditions that an Enterprise may operate in.
Note: TerrainType is a subtype of GeopoliticalLocationStateType as the terrain may change over time (e.g. muddy, frozen ground, deep snow, etc.)

ThreadStatusType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ThreadStatusType - ProjectPartType
Attributes:
-
The powertype of ThreadStatus.

TriggerItem «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
TriggerItem - ModemIndividualType
Attributes:
-

A ModafIndividualType that can be the cause or effect of a Trigger.
TypicalTemporalWholePart «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» TypicalTemporalWholePart - ModemTemporalWholePartType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» TypicalTemporalWholePart - TypicalWholePart *Association (source - target):* «place2Type» TypicalTemporalWholePart - ModemIndividualType *Association (source - target):* «place1Type» TypicalTemporalWholePart - ModemIndividualType *Attributes:* - A TypicalWholePart where the instances of the partType are temporal parts of instances of the wholeType.
TypicalWholePart «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» TypicalWholePart - ModemWholePartType *Association (source - target):* «place2Type» TypicalWholePart - ModemIndividualType *Association (source - target):* «place1Type» TypicalWholePart - ModemIndividualType *Attributes:* - A ModafWholePartType where types of whole and part are ModafIndividalTypeElements Note : this is used in AV-2 to model typical whole-part relationships between ModafIndividualTypeElements
URI «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» URI - MetaData *Attributes:* - A MetaData that is a uniform resource identifier.
URL «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» URL - URI *Attributes:* - A URI that is a uniform resource location.
URN «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» URN - URI *Attributes:* - A URI that is a uniform resource name.

Undertaking «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Undertaking - Process
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Undertaking - UndertakingState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Undertaking - UndertakingType
Attributes:
-
A Process that is intended to deliver something.

UndertakingPart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingPart - ModemIndividualElement
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingPart - ProcessPart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
UndertakingPart - UndertakingPartType
Attributes:
-
A ModemIndividualElement that is part of an Undertaking.

UndertakingPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingPartType - ProcessPartType
Attributes:
-
The powertype of UndertakingPart.

UndertakingState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingState - UndertakingPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingState - ProcessState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
UndertakingState - UndertakingStateType
Attributes:
-
An UndertakingPart that is a temporal part of an Undertaking.

UndertakingStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingStateType - UndertakingPartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
UndertakingStateType - ProcessStateType
Attributes:
-

The powertype of UndertakingState.
UndertakingType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» UndertakingType - ProcessType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» UndertakingType - UndertakingStateType *Attributes:* - The powertype of Undertaking.
UndertakingWholeAndPartType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» UndertakingWholeAndPartType - UndertakingWholeStateType *Association (source - target):* «place2Type» UndertakingWholeAndPartType - UndertakingType *Attributes:* - The powertype of undertakingWholeAndPart.
UndertakingWholePartType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» UndertakingWholePartType - ModemWholePartType *Association (source - target):* «place2Type» UndertakingWholePartType - UndertakingPartType *Association (source - target):* «place1Type» UndertakingWholePartType - UndertakingType *Attributes:* - The powertype of undertakingWholePart.
UndertakingWholeStateType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» UndertakingWholeStateType - UndertakingWholePartType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» UndertakingWholeStateType - ModemTemporalWholePartType *Association (source - target):* «place2Type» UndertakingWholeStateType - UndertakingStateType *Attributes:* - The powertype of undertakingWholeState.
WeatherConditions «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» WeatherConditions - EnvironmentalFactor *Attributes:* - An EnvironmentalFactor that defines the type of weather in which an Enterprise may operate.

WholeLifeEnterprise «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
WholeLifeEnterprise - EnterprisePhase
Generalization (element - is a subtype of): «IDEAS:superSubtype»
WholeLifeEnterprise - Undertaking
Generalization (element - is a subtype of): «IDEAS:superSubtype»
WholeLifeEnterprise - Agent
Attributes:
-
An EnterprisePhase that represents the whole existance of an enterprise.

appliedStateMachine «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
appliedStateMachine - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
appliedStateMachine - stateMachineViewStateTypeOwners
Association (source - target):«place1Type»
appliedStateMachine - ModemIndividualElementType
Association (source - target):«place2Type»
appliedStateMachine - StateMachine
Attributes:
-
A stateMachineViewStateTypeOwners that relates a ModemIndividualElemetType to its state machine.

approved «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
approved - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
approved - couple
Association (source - target):«place2Type»
approved - ADElement
Association (source - target):«place1Type»
approved - ArchitectureApprovalMilestone
Attributes:
-
A couple that relates an ArchitectureApprovalMilestone to the ADElement that is approved.

approver «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
approver - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
approver - ModemThing
Association (source - target):«place2Type»
approver - ResponsibleHumanResource
Association (source - target):«place1Type»
approver - ArchitectureApprovalMilestone
Attributes:
-

A couple that relates an ArchitectureApprovalMilestone to the ResponsibleHumanResource that approved it.

architectureMetaData «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
architectureMetaData - metaDataAnnotation
Association (source - target):«place1Type»
architectureMetaData - ADElement
Association (source - target):«place2Type»
architectureMetaData - MetaData
Attributes:
-
A metaDataAnnotation that relates a MetaData element to the ArchitectureDescription it annotates.

architectureRealisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
architectureRealisation - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»
architectureRealisation - LogicalArchitecture
Association (source - target): «place2Type»
architectureRealisation - PhysicalArchiutecture
Attributes:
-
A modemIndividualTypeSpecialisation that asserts that a PhysicalArchitecture is a realisation of a LogicalArchitecture

architectureReference «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
architectureReference - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
architectureReference - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
architectureReference - correspondence
Association (source - target):«place2Type»
architectureReference - ArchitectureDescription
Association (source - target):«place1Type»
architectureReference - ArchitectureDescription
Attributes:
-
A couple that relates an ArchitectureDescription to another ArchitectureDescription it refers to.

assumption «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
assumption - architectureMetaData
Association (source - target):«place1Type»
assumption - ADElement
Association (source - target):«place2Type»
assumption - MetaData
Attributes:
-

A describedBy that states an assumption about an ADElement. Note: Any given ADElement may have zero to many assumptions.

categoryOfMetaData «IDEAS:TupleType»
Connectors:
Association (source - target):«place2Type»
categoryOfMetaData - MetaData
Generalization (element - is a subtype of):«IDEAS:superSubtype»
categoryOfMetaData - typeInstance
Generalization (element - is a subtype of):«IDEAS:superSubtype»
categoryOfMetaData - ModemThing
Association (source - target):«place1Type»
categoryOfMetaData - MetaDataCategory
Attributes:
-
A typeInstance that relates a MetaData element to its category.

constraintOnIndividual «IDEAS:TupleType»
Connectors:
Association (source - target):«place2Type»
constraintOnIndividual - ModemIndividualElement
Association (source - target):«place1Type»
constraintOnIndividual - Constraint
Generalization (element - is a subtype of):«IDEAS:superSubtype»
constraintOnIndividual - modemIndividualTypeInstance
Attributes:
-
A couple that asserts a constraint placed upon a ModemThing related to a ModemThing.

constraintOnType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
constraintOnType - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
constraintOnType - ModemIndividualType
Association (source - target):«place1Type»
constraintOnType - Constraint
Attributes:
-
A superSubtype that asserts all the instances of the subType object are subject to the constraint.

createdBy «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
createdBy - couple
Association (source - target):«place2Type»
createdBy - ResponsibleHumanResource
Association (source - target):«place1Type»
createdBy - ADElement
Attributes:
-
A couple that asserts a ResponsibleHumanResource is the creator of an ADElement. Note: this covers the creatingOrganisation and architect tags that were applied to ArchitectureDescription in M3.

definition «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
definition - metaDataAnnotation
Association (source - target):«place1Type»
definition - ModemThing
Attributes:
-
A metaDataAnnotation that provides the definition for a ModemThing.

delayRange «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
delayRange - Delay
Generalization (element - is a subtype of): «IDEAS:superSubtype»
delayRange - measureOfType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
delayRange - ModemThing
Association (source - target): «place1Type»
delayRange - TimeRange
Attributes:
-
A measureOfType that relates a LogicalDelay to the TimeRange in which it falls.

delayTime «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
delayTime - ModemThing
Association (source - target): «place2Type»
delayRange - Delay
Generalization (element - is a subtype of): «IDEAS:superSubtype»
delayRange - measureOfType
Association (source - target): «place1Type»
delayRange - Time
Attributes:
-
A measureOfType that relates a LogicalDelay to its Time.

designReleasedAtMilestone «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
designReleasedAtMilestone - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
designReleasedAtMilestone - ModemThing
Association (source - target):«place1Type»
designReleasedAtMilestone - ResourceType
Association (source - target):«place2Type»
designReleasedAtMilestone - ProjectMilestone
Attributes:
-
A couple that indicates a ResourceType is released as a design at a ProjectMilestone.

designWithdrawnAtMilestone «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
designWithdrawnAtMilestone - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
designWithdrawnAtMilestone - ModemThing
Association (source - target):«place1Type»
designWithdrawnAtMilestone - ResourceType
Association (source - target):«place2Type»
designWithdrawnAtMilestone - ProjectMilestone
Attributes:
-
A couple that indicates a ResourceType was withdrawn as a design at a ProjectMilestone.

enduringTaskWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enduringTaskWholePart - undertakingWholePart
Association (source - target): «place2Type»
enduringTaskWholePart - EnduringTaskPart
Association (source - target): «place1Type»
enduringTaskWholePart - EnduringTask
Attributes:
-
An undertakingWholePart where the whole is an EnduringTask.

enterpriseStructure «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseStructure - agentWholeAndPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseStructure - undertakingWholeAndPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseStructure - enterpriseWholePhase
Association (source - target): «place2Type»
enterpriseStructure - WholeLifeEnterprise
Association (source - target): «place1Type»
enterpriseStructure - WholeLifeEnterprise
Attributes:
-
A wholePart that asserts that one EnterprisePhase is a spatial part of another.

enterpriseTemporalPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseTemporalPart - undertakingWholeState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseTemporalPart - agentWholeState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseTemporalPart - enterpriseWholePart
Association (source - target): «place2Type»
enterpriseTemporalPart - EnterprisePhase

Association (source - target): «place1Type»
enterpriseTemporalPart - EnterprisePhase
Attributes:
-
An enterpriseStructure and a temporalWholePart that asserts that one EnterprisePhase is a temporal part of another (i.e. it is a phase of the other).

enterpriseWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseWholePart - agentWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseWholePart - undertakingWholePart
Association (source - target): «place2Type»
enterpriseWholePart - EnterprisePart
Association (source - target): «place1Type»
enterpriseWholePart - WholeLifeEnterprise
Attributes:
-
An agentWholePart where the whole is a WholeLifeEnterprise and the part is an EnterprisePart.

enterpriseWholePhase «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseWholePhase - enterpriseTemporalPart
Association (source - target): «place2Type»
enterpriseWholePhase - EnterprisePhase
Association (source - target): «place1Type»
enterpriseWholePhase - WholeLifeEnterprise
Attributes:
-
An enterpriseTemporalPart where whole is a WholeLifeEnterprise.

environmentalContext «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
environmentalContext - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
environmentalContext - ModemThing
Association (source - target):«place2Type»
environmentalContext - EnvironmentalFactor
Association (source - target):«place1Type»
environmentalContext - MeasureInContext
Attributes:
-
A couple that relates a MeasureInContext to an EnvironmentalFactor in order to qualify the measure.

finding «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
finding - architectureMetaData
Association (source - target):«place2Type»
finding - MetaData

Association (source - target):«place1Type»
finding - StructuredADElement
Attributes:
-
A describedBy that describes a finding about an ArchitectureDescription. Note: Any given ADElement may have zero to many findings.

individualFacade «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
individualFacade - IndividualFacadeType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualFacade - modemWholePart
Attributes:
-
A modemWholePart where the part is an outer part of another ModemIndividualElement.

locatedAt «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
locatedAt - modemWholePart
Association (source - target): «place1Type»
locatedAt - ModemIndividualElement
Association (source - target): «place2Type»
locatedAt - PointLocation
Attributes:
-
A modafWholePart relating an MODAIndividualElement to a PointLocation that is within the extent of the MODAIndividualElement.

locatedIn «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
locatedIn - modemWholePart
Association (source - target): «place2Type»
locatedIn - ModemIndividualElement
Association (source - target): «place1Type»
locatedIn - GeopoliticalLocation
Attributes:
-
A modafWholePart that relates a ModafIndividualElement to the GeopoliticalLocation it is in.

locationNamedBy «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
locationNamedBy - namedBy
Association (source - target):«place1Type»
locationNamedBy - Location
Association (source - target):«place2Type»
locationNamedBy - GeoName
Attributes:
-
A namedBy that identifies a Location.

logicalArchitectureOfEnterprisePhase «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
logicalArchitectureOfEnterprisePhase - exhibits
Generalization (element - is a subtype of): «IDEAS:superSubtype»
logicalArchitectureOfEnterprisePhase - modemIndividualTypeInstance
Association (source - target): «place1Type»
logicalArchitectureOfEnterprisePhase - LogicalArchitecture
Association (source - target): «place2Type»
logicalArchitectureOfEnterprisePhase - EnterprisePhase
Attributes:
-
Relates an EnterprisePhase to a LogicalArchitecture that specifies its (logical) structure and behaviour.

metaDataAnnotation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
metaDataAnnotation - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
metaDataAnnotation - representedBy
Association (source - target): «place1Type»
metaDataAnnotation - ModemThing
Association (source - target): «place2Type»
metaDataAnnotation - MetaData
Attributes:
-
A representedBy that relates a MetaData element to the ModafThing it describes.

modeller «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
modeller - modemWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
modeller - agentParticipation
Association (source - target):«place1Type»
modeller - Person
Association (source - target):«place2Type»
modeller - ModellingSession
Attributes:
-
An agentParticipation where a Person conducts a ModellingSession.

modellingSessionInProject «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
modellingSessionInProject - processWholeRoleExtentPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
modellingSessionInProject - projectWholePart
Association (source - target):«place1Type»
modellingSessionInProject - ArchitectureProject
Association (source - target):«place2Type»
modellingSessionInProject - ModellingSession

Attributes:
-
A projectWholePart relating a ModellingSession to the ArchitectureProject it is part of.

modemIndividualTypeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemIndividualTypeInstance - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemIndividualTypeInstance - typeInstance
Association (source - target): «place2Type»
modemIndividualTypeInstance - ModemIndividualElement
Association (source - target): «place1Type»
modemIndividualTypeInstance - ModemIndividualType
Attributes:
-
A typeInstance used to assert that a ModemIndividualElement is an instance of a ModemIndividualType.

modemIndividualTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemIndividualTypeSpecialisation - superSubtype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemIndividualTypeSpecialisation - ModemThing
Association (source - target): «place2Type»
modemIndividualTypeSpecialisation - ModemIndividualType
Association (source - target): «place1Type»
modemIndividualTypeSpecialisation - ModemIndividualType
Attributes:
-
A superSubtype that expresses a specialisation relationship between ModemIndividualTypeElements.
Note: This relationship is used to build specialisation hierarchies of ModemIndividualTypeElements in an AV-2.

modemTemporalWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemTemporalWholePart - modemWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemTemporalWholePart - temporalWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
modemTemporalWholePart - ModemTemporalWholePartType
Association (source - target): «place2Type»
modemTemporalWholePart - ModemIndividualElement
Association (source - target): «place1Type»
modemTemporalWholePart - ModemIndividualElement
Attributes:
-
A temporalWholePart relationship between two ModemIndividualElements.

modemWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemWholePart - wholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
modemWholePart - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modemWholePart - ModemThing
Association (source - target): «place2Type»
modemWholePart - ModemIndividualElement
Association (source - target): «place1Type»
modemWholePart - ModemIndividualElement
Attributes:
-
A wholePart relationship between ModemIndividualElements.

organisationWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
organisationWholePart - undertakingWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
organisationWholePart - responsibleHumanResourceWholePart
Association (source - target): «place2Type»
organisationWholePart - OrganisationPart
Association (source - target): «place1Type»
organisationWholePart - Organisation
Attributes:
-
A modafWholePart where the whole is an Organisation and the part is an OrganisationPart (i.e. a Post or Organisation).

physicalArchitectureOfEnterprisePhase «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
physicalArchitectureOfEnterprisePhase - modemIndividualTypeInstance
Generalization (element - is a subtype of): «IDEAS:superSubtype»
physicalArchitectureOfEnterprisePhase - exhibits
Association (source - target): «place1Type»
physicalArchitectureOfEnterprisePhase - PhysicalArchitecture
Association (source - target): «place2Type»
physicalArchitectureOfEnterprisePhase - EnterprisePhase
Attributes:
-
Relates an EnterprisePhase to a ResourceType that specifies its structure and behaviour.

projectTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
projectTypeSpecialisation - modemIndividualTypeSpecialisation
Association (source - target): «place2Type»
projectTypeSpecialisation - ProjectType
Association (source - target): «place1Type»
projectTypeSpecialisation - ProjectType

Attributes:
-
A modafIndividualTypeSpecialisation that asserts one ProjectType (subtype) is a special type of another (supertype).

purpose «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
purpose - architectureMetaData
Association (source - target):«place2Type»
purpose - MetaData
Association (source - target):«place1Type»
purpose - StructuredADElement
Attributes:
-
An architectureMetaData that describes the purpose of a StructuredADElement.

qualifiedMeasure «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
qualifiedMeasure - superSubtype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
qualifiedMeasure - ModemThing
Association (source - target):«place1Type»
qualifiedMeasure - Measure
Association (source - target):«place2Type»
qualifiedMeasure - MeasureInContext
Attributes:
-
A superSubtype that relates a MeasureInContext to the measure it qualifies.

recommendation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
recommendation - architectureMetaData
Association (source - target):«place2Type»
recommendation - MetaData
Association (source - target):«place1Type»
recommendation - StructuredADElement
Attributes:
-
An architectureMetaData that expresses a recommendation arising from a StructuredADElement.

regionOfStateMachine «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
regionOfStateMachine - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
regionOfStateMachine - stateMachineViewTypesRegionInstances
Association (source - target):«place2Type»
regionOfStateMachine - StateMachineRegion
Association (source - target):«place1Type»
regionOfStateMachine - StateMachine

Attributes:
-
A stateMachineViewTypesRegionInstances which relates a StateMachineRegion to a StateMachine.

serviceDeliveryWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
serviceDeliveryWholeAndPart - serviceDeliveryWholeState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
serviceDeliveryWholeAndPart - ServiceDeliveryWholeAndPart
Association (source - target): «place2Type»
serviceDeliveryWholeAndPart - ServiceDelivery
Attributes:
-
A ServiceDeleiveryWholeState where both the whole and part are ServiceDeliveries.

serviceDeliveryWholeFacade «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
serviceDeliveryWholeFacade - serviceDeliveryWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
serviceDeliveryWholeFacade - ServiceDeliveryWholeFacadeType
Association (source - target): «place2Type»
serviceDeliveryWholeFacade - ServiceFacade
Attributes:
-
A serviceDeliveryWholePart where the part is a ServiceFacade.

serviceDeliveryWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
serviceDeliveryWholePart - modemWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
serviceDeliveryWholePart - ServiceDeliveryWholePartType
Association (source - target): «place2Type»
serviceDeliveryWholePart - ServiceDeliveryPart
Association (source - target): «place1Type»
serviceDeliveryWholePart - ServiceDelivery
Attributes:
-
A modemWholePart where the whole is a ServiceDelivery and the part is a ServiceDeliveryPart.

serviceDeliveryWholeState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
serviceDeliveryWholeState - modemTemporalWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
serviceDeliveryWholeState - serviceDeliveryWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
serviceDeliveryWholeState - ServiceDeliveryWholeStateType
Association (source - target): «place2Type»
serviceDeliveryWholeState - ServiceDeliveryState

Association (source - target): «place2Type»
undertakingWholeAndPart - Undertaking
Attributes:
-
An UndertakingWholeState where both the whole and part are Undertakings.

undertakingWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
undertakingWholePart - modemWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
undertakingWholePart - processWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
undertakingWholePart - UndertakingWholePartType
Association (source - target): «place2Type»
undertakingWholePart - UndertakingPart
Association (source - target): «place1Type»
undertakingWholePart - Undertaking
Attributes:
-
A modemWholePart where an UndertakingPart is part of an Undertaking.

undertakingWholeState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
undertakingWholeState - undertakingWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
undertakingWholeState - modemTemporalWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
undertakingWholeState - processWholeState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
undertakingWholeState - UndertakingWholeStateType
Association (source - target): «place2Type»
undertakingWholeState - UndertakingState
Attributes:
-
An undertakingWholePart where the part is a temporal part of an Undertaking.

viewCode «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
viewCode - metaDataAnnotation
Association (source - target):«place2Type»
viewCode - MetaData
Association (source - target):«place1Type»
viewCode - ArchitectureViewpoint
Attributes:
-
A metaDataAnnotation that uses MetaData to represent the short code that identifies an ArchitectureViewpoint. Note that viewCode and viewDescription from M3 are handled using the core IDEAS description and naming patterns.

webReference «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
webReference - metaDataAnnotation
Association (source - target):«place1Type»
webReference - ModemThing
Association (source - target):«place2Type»
webReference - URI
Attributes:
-
A metaDataAnnotation that asserts URI contains information about a ModemThing.

All Views Foundation
ModemIndividualElement «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ModemIndividualElement - ModemThing *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ModemIndividualElement - ModemIndividualElementOrModemIndividualType *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ModemIndividualElement - ModafIndividualElementType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ModemIndividualElement - Individual *Attributes:* - An Individual that can feature in a MODEM architecture.
ModafIndividualElementType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ModafIndividualElementType - IndividualType *Attributes:* - The powertype of ModafIndividualElement. Note - this is simply used to specify the set-theretic logic at the top of MODA. It should never be used in an architecture.
ModemThing «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ModemThing - Thing *Attributes:* - Any Thing that can feature in a MODEM Architecture. Note: things that appear in the MODEM metamodel will not necessarily be instances, unless they appear in an architecture.

All Views ISO42010
ADElement «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ADElement - Representation *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ADElement - ISO42010_Thing *Attributes:* - From ISO42010: An AD element is any construct in an architecture description. AD elements are the most primitive constructs discussed in this International Standard. Every stakeholder, concern, architecture viewpoint, architecture view, model kind, architecture model, architecture decision and rationale (see 4.2.7) is considered an AD element. When viewpoints and model kinds are defined and their models are populated, additional AD elements are introduced.
Architecture «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» Architecture - ISO42010_Thing *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Architecture - IndividualType *Attributes:* - Fundamental concepts or properties of a system in its environment embodied in its elements, relationships, and in the principles of its design and evolution.
ArchitectureDescription «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ArchitectureDescription - StructuredADElement *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ArchitectureDescription - ArchitectureDescriptionType *Attributes:* - A work product used to express an architecture.
ArchitectureDescriptionType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArchitectureDescriptionType - ISO42010_Thing *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ArchitectureDescriptionType - RepresentationType *Attributes:* - The powertype of ArchitectureDescription.
ArchitectureFramework «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ArchitectureFramework - ArchitectureDescriptionType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ArchitectureFramework - RepresentationScheme *Attributes:* -

From ISO42010:
Uses of architecture frameworks include, but are not limited to: creating architecture descriptions; developing architecture modelling tools and architecting methods; and establishing processes to facilitate communication, commitments and interoperation across multiple projects and/or organizations.
NOTE 1 Architecture frameworks frequently encompass both provisions for architecture description and additional architecting practices.
EXAMPLES The following are architecture frameworks in the terms of this International Standard: Zachman's information systems architecture framework [44], UK Ministry of Defence Architecture ramework [27], The Open Group's Architecture Framework (TOGAF) [41], Kruchten's "4+1" view model [23], Siemens' 4 views method [10], Reference Model for Open Distributed Processing (RM-ODP), [ISO/IEC 10746] and Generalized Enterprise Reference Architecture (GERA) [ISO 15704].

ArchitectureModel «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ArchitectureModel - ArchitectureModelType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureModel - StructuredRepresentation
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureModel - StructuredADElement
Attributes:
-
There is no no specific definition provided in ISO42010.

ArchitectureModelType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ArchitectureModelType - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureModelType - RepresentationType
Attributes:
-
The powertype of ArchitectureModel.

ArchitectureView «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ArchitectureView - ArchitectureViewType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureView - StructuredADElement
Attributes:
-
work product expressing the architecture of a system from the perspective of specific system concerns

ArchitectureViewType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureModelType - RepresentationType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ArchitectureViewType - ISO42010_Thing
Attributes:
-
The powertype of ArchitectureView.

ArchitectureViewpoint «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureViewpoint - ArchitectureViewType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ArchitectureViewpoint - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArchitectureViewpoint - RepresentationScheme
Attributes:
-
Work product establishing the conventions for the construction, interpretation and use of architecture views to frame specific system concerns

Concern «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Concern - ISO42010_Thing
Attributes:
-
From ISO42010:
A concern could be held by one or more stakeholders. Concerns arise throughout the life cycle from system needs and requirements, from design choices and from implementation and operating considerations. A concern could be manifest in many forms, such as in relation to one or more stakeholder needs, goals, expectations, responsibilities, requirements, design constraints, assumptions, dependencies, quality attributes, architecture decisions, risks or other issues pertaining to the system.
EXAMPLES The following are concerns in the terms of this International Standard: functionality, feasibility, usage, system purposes, system features, system properties, known limitations, structure, behavior, performance, resource utilization, reliability, security, information assurance, complexity, evolvability, openness, concurrency, autonomy, cost, schedule, quality of service, flexibility, agility, modifiability, modularity, control, inter-process communication, deadlock, state change, subsystem integration, data accessibility, privacy, compliance to regulation, assurance, business goals and strategies, customer experience, maintainability, affordability and disposability. The distribution transparencies described in the Reference Model of Open Distributed Processing [ISO/IEC 10746-1] are concerns in the terms of this International Standard. Software properties as described in SQUARE [ISO/IEC 25010:2011, 4.2] name concerns in the terms of this International Standard.

CorrespondenceRule «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
CorrespondenceRule - ISO42010_Thing
Dependency (element - is instance of): «IDEAS:typeInstance»
CorrespondenceRule - PlaceableType
Attributes:
-
From ISO42010:
Correspondence rules are used to enforce relations within an architecture description (or between architecture descriptions).

ISO42010_Environment «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ISO42010_Environment - ISO42010_Thing
Attributes:
--
Context determining the setting and circumstances of all influences upon a system

ISO42010_System «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ISO42010_System - ISO42010_Thing
Attributes:

-
The term system is used in this International Standard to refer to entities whose architectures are of interest. The term is intended to encompass, but is not limited to, entities within the following domains:
- systems as described in [ISO/IEC 15288]: “systems that are man-made and may be configured with one or more of the following: hardware, software, data, humans, processes (e.g., processes for providing service to users), procedures (e.g. operator instructions), facilities, materials and naturally occurring entities”;
- software products and services as described in [ISO/IEC 12207];
- software-intensive systems as described in [IEEE Std 1471TM:2000]: “any system where software contributes essential influences to the design, construction, deployment, and evolution of the system as a whole” to encompass “individual applications, systems in the traditional sense, subsystems, systems of systems, product lines, product families, whole enterprises, and other aggregations of interest”.
This International Standard takes no position on what constitutes a system within those domains—or elsewhere. The nature of systems is not defined by this International Standard.

ISO42010_Thing «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ISO42010_Thing - Thing
Attributes:
-
A Thing that is described by the standard ISO 421010.

ModelKind «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModelKind - ArchitectureModelType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModelKind - RepresentationScheme
Attributes:
-
Conventions for a type of modelling.
NOTE Examples of model kinds include: data flow diagrams, class diagrams, Petri nets, balance sheets, organization charts and state transition models.

ModelKindPartOfViewpoint «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModelKindPartOfViewpoint - ModelPartOfViewType
Association (source - target): «place1Type»
ModelKindPartOfViewpoint - ArchitectureViewpoint
Association (source - target): «place2Type»
ModelKindPartOfViewpoint - ModelKind
Attributes:
-
A ModelPartOfViewType where a ModelKind is a typical part of an ArchitectureViewpoint.

ModelPartOfView «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ModelPartOfView - ISO42010_Thing
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ModelPartOfView - ModelPartOfViewType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModelPartOfView - WholePartType
Association (source - target): «place2Type»
ModelPartOfView - ArchitectureModel
Association (source - target): «place1Type»
ModelPartOfView - ArchitectureView

Attributes:
-
A WholePartType where an ArchitectureModel is part of an ArchitectureView.

ModelPartOfViewType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ModelPartOfViewType - WholePartTypeType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ModelPartOfViewType - ISO42010_Thing
Attributes:
-
The powertype of ModelPartOfView

Stakeholder «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Stakeholder - AgentCapableOfResponsibilityOrAgentCapableOfResponsibilityType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Stakeholder - ISO42010_Thing
Attributes:
-
An individual, team, organization, or classes thereof, having an interest in a system.

StructuredADElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StructuredADElement - ADElement
Attributes:
-
An ADElement that has other ADElements as part of it. Note: this is not in ISO42010, but is required if the model is to be useful.

ViewPartOfDescription «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ViewPartOfDescription - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ViewPartOfDescription - WholePartType
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ViewPartOfDescription - ViewPartOfDescriptionType
Association (source - target): «place2Type»
ViewPartOfDescription - ArchitectureView
Association (source - target): «place1Type»
ViewPartOfDescription - ArchitectureDescription
Attributes:
-
A WholePartType where an ArchitectureView is part of an ArchitectureDescription.

ViewPartOfDescriptionType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ViewPartOfDescriptionType - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»

ViewPartOfDescriptionType - WholePartTypeType
Attributes:
-
The powertype of ViewPartOfDescription.

ViewpointPartOfFramework «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ViewpointPartOfFramework - ViewPartOfDescriptionType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ViewpointPartOfFramework - WholePartTypeType
Association (source - target): «place1Type»
ViewpointPartOfFramework - ArchitectureFramework
Association (source - target): «place2Type»
ViewpointPartOfFramework - ArchitectureViewpoint
Attributes:
-
A WholePartTypeType that asserts an ArchitectureViewpoint is part of an ArchitectureFramework.

correspondence «IDEAS:Type»
Connectors:
Association (source - target): «place1Type»
correspondence - ADElement
Dependency (element - is instance of): «IDEAS:typeInstance»
correspondence - PlaceableType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
correspondence - ISO42010_Thing
Association (source - target): «place2Type»
correspondence - ADElement
Attributes:
-
From ISO42010:
A correspondence defines a relation between AD elements. Correspondences are used to express architecture relations of interest within an architecture description (or between architecture descriptions). Correspondences can be governed by correspondence rules. Correspondence rules are used to enforce relations within an architecture description (or between architecture descriptions).

exhibits «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
exhibits - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exhibits - couple
Association (source - target): «place2Type»
exhibits - ISO42010_System
Association (source - target): «place1Type»
exhibits - Architecture
Attributes:
-
A couple that asserts an ISO42010_System has an Architecture.

expresses «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
expresses - Architecture
Association (source - target): «place2Type»
expresses - ArchitectureDescription
Generalization (element - is a subtype of):«IDEAS:superSubtype»
expresses - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
expresses - representedBy
Attributes:
-
A representedBy that asserts and an ArchitectureDescription represents an Architecture.

frameworkGovernsDescription «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
frameworkGovernsDescription - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
frameworkGovernsDescription - typeInstance
Association (source - target): «place2Type»
frameworkGovernsDescription - ArchitectureDescription
Association (source - target): «place1Type»
frameworkGovernsDescription - ArchitectureFramework
Attributes:
-
A typeInstance relating an ArchitectureDescription to the ArchitectureFramework it conforms to.

modelKindGovernsModel «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modelKindGovernsModel - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
modelKindGovernsModel - typeInstance
Association (source - target): «place2Type»
modelKindGovernsModel - ArchitectureModel
Association (source - target): «place1Type»
modelKindGovernsModel - ModelKind
Attributes:
-
A typeInstance where an ArchitecturalModel conforms to a ModelKind.

ruleGovernsCorrespondence «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ruleGovernsCorrespondence - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ruleGovernsCorrespondence - typeInstance
Association (source - target): «place2Type»
ruleGovernsCorrespondence - correspondence
Association (source - target): «place1Type»
ruleGovernsCorrespondence - CorrespondenceRule

Attributes: - A typeInstance relating a correspondence to the CorrespondenceRule that governs it.
situatedIn «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» situatedIn - ISO42010_Thing *Generalization (element - is a subtype of):*«IDEAS:superSubtype» situatedIn - couple *Association (source - target):*«place1Type» situatedIn - ISO42010_Environment *Association (source - target):*«place2Type» situatedIn - ISO42010_System *Attributes:* - A couple that relates an ISO42010_Environment toa ISO42010_system. Note: This is probably a subtype of the union of wholePart and WholePartType.
stakeholderConcern «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» stakeholderConcern - couple *Association (source - target):* «place2Type» stakeholderConcern - Concern *Generalization (element - is a subtype of):*«IDEAS:superSubtype» stakeholderConcern - ISO42010_Thing *Association (source - target):* «place1Type» stakeholderConcern - Stakeholder *Attributes:* - A couple that relates a Concern to a stakeholder that have the Concern. Note: a concern may be held by more than one Stakeholder, hence there maybe multiple stakeholderConcerns.
systemConcern «IDEAS:TupleType» *Connectors:* *Association (source - target):* «place1Type» systemConcern - Concern *Association (source - target):* «place2Type» systemConcern - ISO42010_System *Generalization (element - is a subtype of):*«IDEAS:superSubtype» systemConcern - ISO42010_Thing *Generalization (element - is a subtype of):* «IDEAS:superSubtype» systemConcern - couple *Attributes:* - A couple that relates an ISO42010_System to a Concern that is held against the system. NOTE A concern pertains to any influence on a system in its environment including: developmental, technological, business, operational, organizational, political, economic, legal, regulatory, ecological and social influences.

viewpointGovernsView «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
viewpointGovernsView - ISO42010_Thing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
viewpointGovernsView - typeInstance
Association (source - target): «place2Type»
viewpointGovernsView - ArchitectureView
Association (source - target): «place1Type»
viewpointGovernsView - ArchitectureViewpoint
Attributes:
-
A typeInstance where an ArchitectureView conforms to an ArchitectureViewpoint.

All Views Representation
ElementInModel «IDEAS:Type» *Connectors:* *Association (source - target):* «place1Type» ElementInModel - ArchitectureModel *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ElementInModel - RepresentationInStructure *Association (source - target):* «place2Type» ElementInModel - ModelElement *Attributes:* - A WholePartType relating a RepresentationElement to the ArchitecuralModel it is shown in.
ModelElement «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ModelElement - ADElement *Attributes:* - A graphical element in an ArchitectureModel.
ModemIndividualElementOrModemIndividualType «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ModemIndividualElementOrModemIndividualType - ModemThing *Attributes:* - A ModemThing that collects all ModemIndividualElements and all ModemIndividualType elements.
SvgCanvas «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SvgCanvas - ArchitectureModel *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SvgCanvas - SvgRepresentation *Attributes:* -

An ArchitectureModel which uses the Scalable Vector Graphics (1.1) standard to encode the model canvas. The exemplar attribute is used to store the XML SVG code that defines the canvas - e.g.
<svg:svg width="4cm" height="8cm" version="1.1">.
</svg:svg>

SvgElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SvgElement - SvgRepresentation
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SvgElement - ModelElement
Attributes:
-
A ModelElement that is encoded using the Scaleable Vector Graphics (1.1) standard. An SvgElement can be any graphical element that is allowed by SVG v1.1 - e.g.
The 'rect' element
The 'circle' element
The 'ellipse' element
The 'line' element
The 'polyline' element
The 'polygon' element
If the element is a group of other elements, then the SvgElementGroup should be used.

SvgElementGroup «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SvgElementGroup - SvgElement
Attributes:
-
An SvgElement that groups together other SvgElements. This corresponds to the <g> element in the Scaleable Vector Graphics Standard (v1.1).

SvgElementInGroup «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SvgElementInGroup - RepresentationInStructure
Association (source - target): «place2Type»
SvgElementInGroup - SvgElement
Association (source - target): «place1Type»
SvgElementInGroup - SvgElementGroup
Attributes:
-
A RepresentationStructure.

SvgElementOnCanvas «IDEAS:Type»
Connectors:
Association (source - target): «place1Type»
SvgElementOnCanvas - SvgCanvas
Association (source - target): «place2Type»
SvgElementOnCanvas - SvgElement
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SvgElementOnCanvas -ElementInModel
Attributes:
-
An ElementInModel relating an SvgElement to the SvgCanvas on which it is displayed.

SvgRepresentation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SvgRepresentation - Representation
Attributes:
exemplar
A Representation where the exemplar text for the representation is and encoding in SVG XML that represents a graphical symbol.

representedModemThing «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
representedModemThing - ModemIndividualElementOrModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
representedModemThing - representedBy
Association (source - target): «place2Type»
representedModemThing - ModelElement
Attributes:
-
A representedBy relating a ModemIndividualElementOrModemIndividualType to the ModelRepresentation that depicts it.

2.2.4 All Views additional diagrams

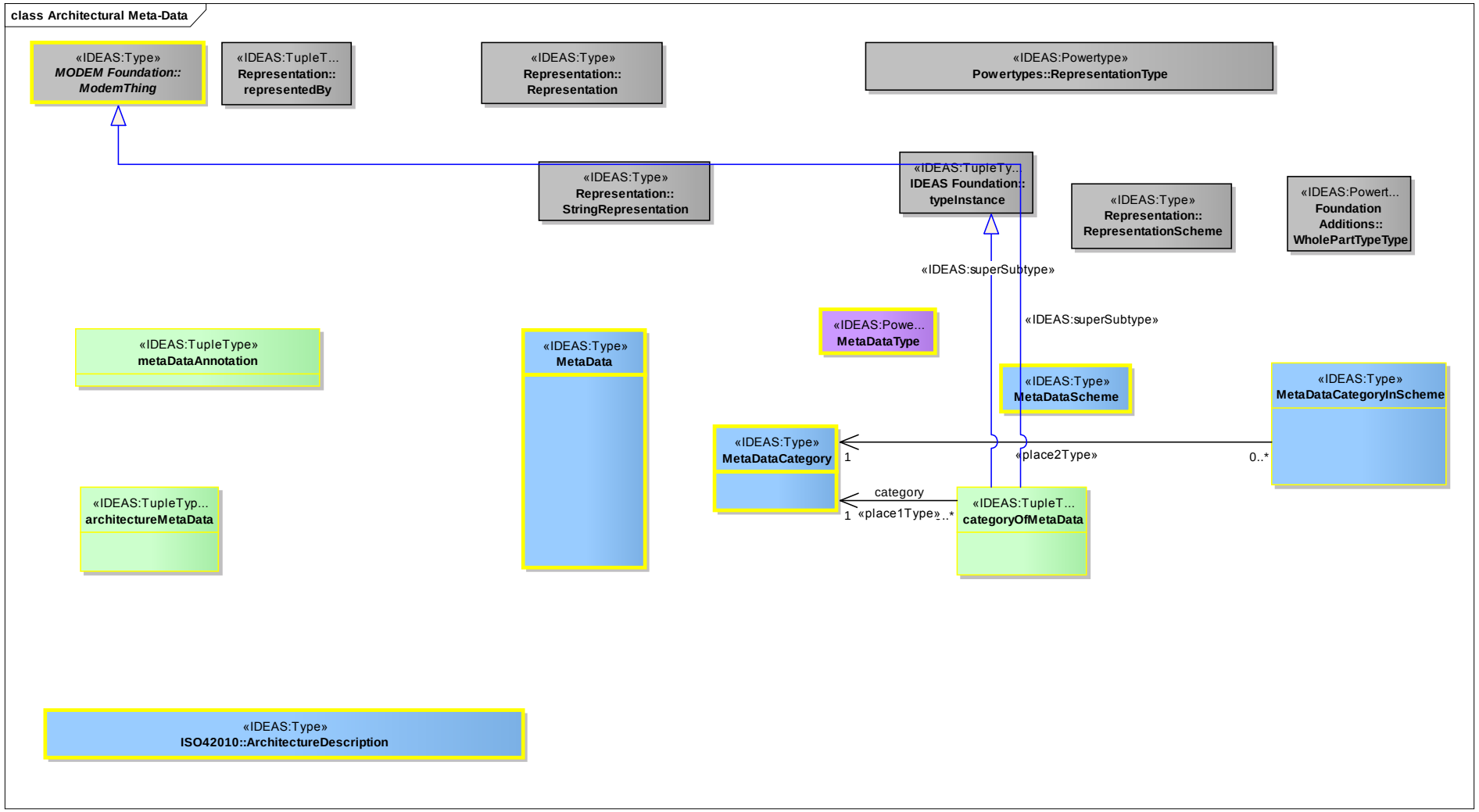


Figure 7 : Architectural Meta-Data

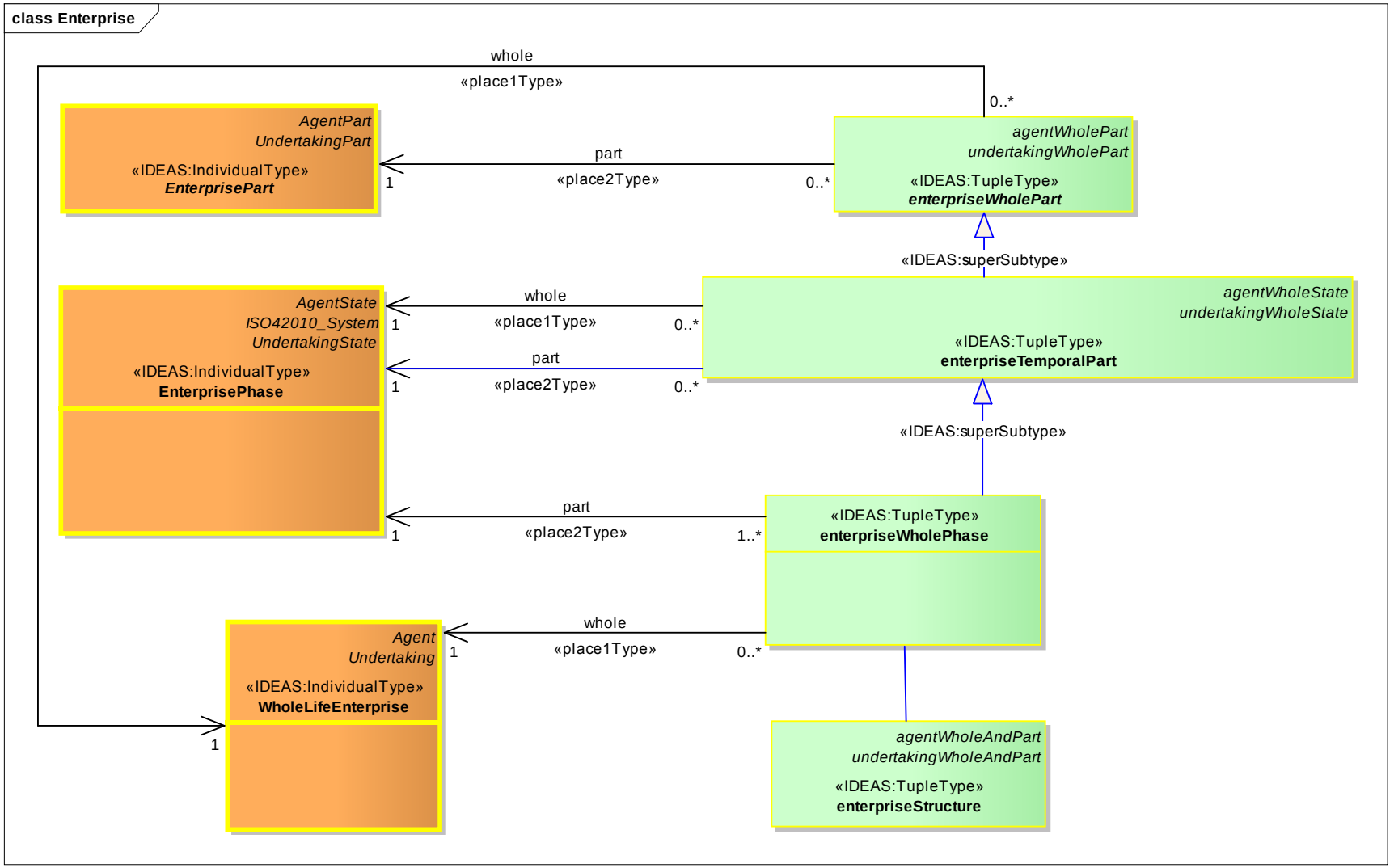


Figure 8 : Enterprise

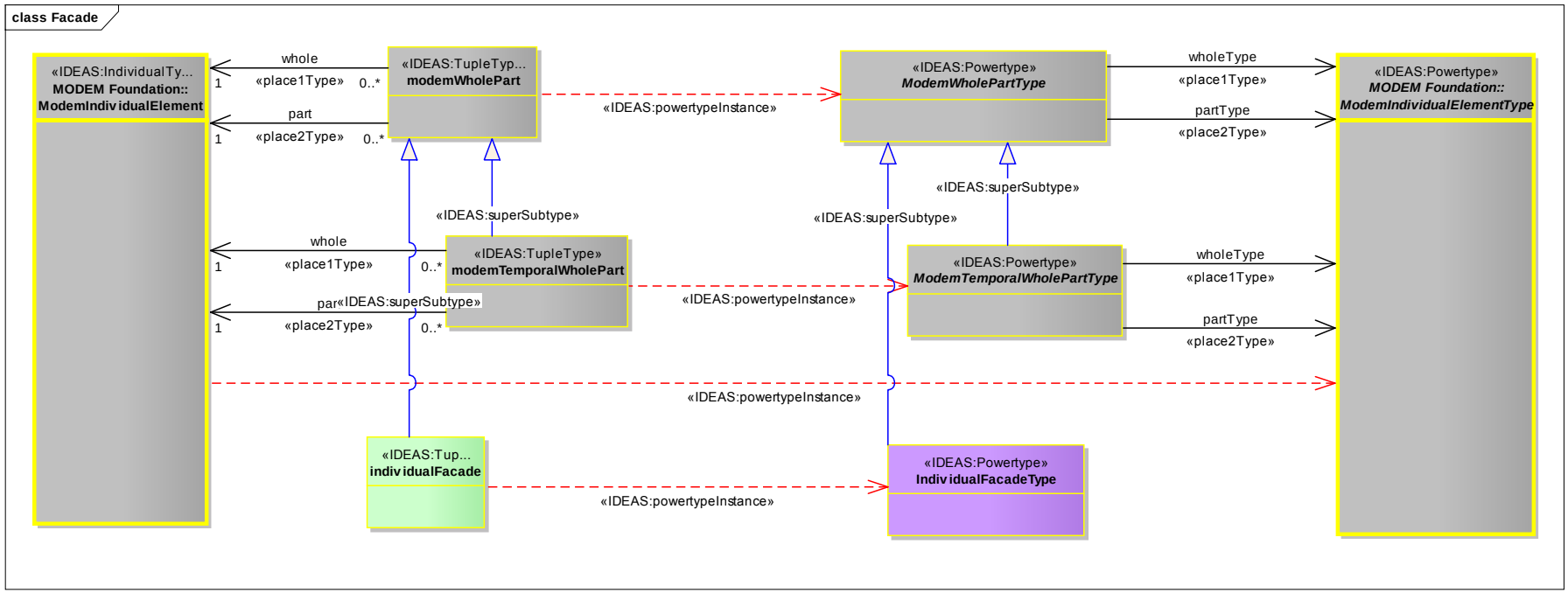


Figure 9 : Facade

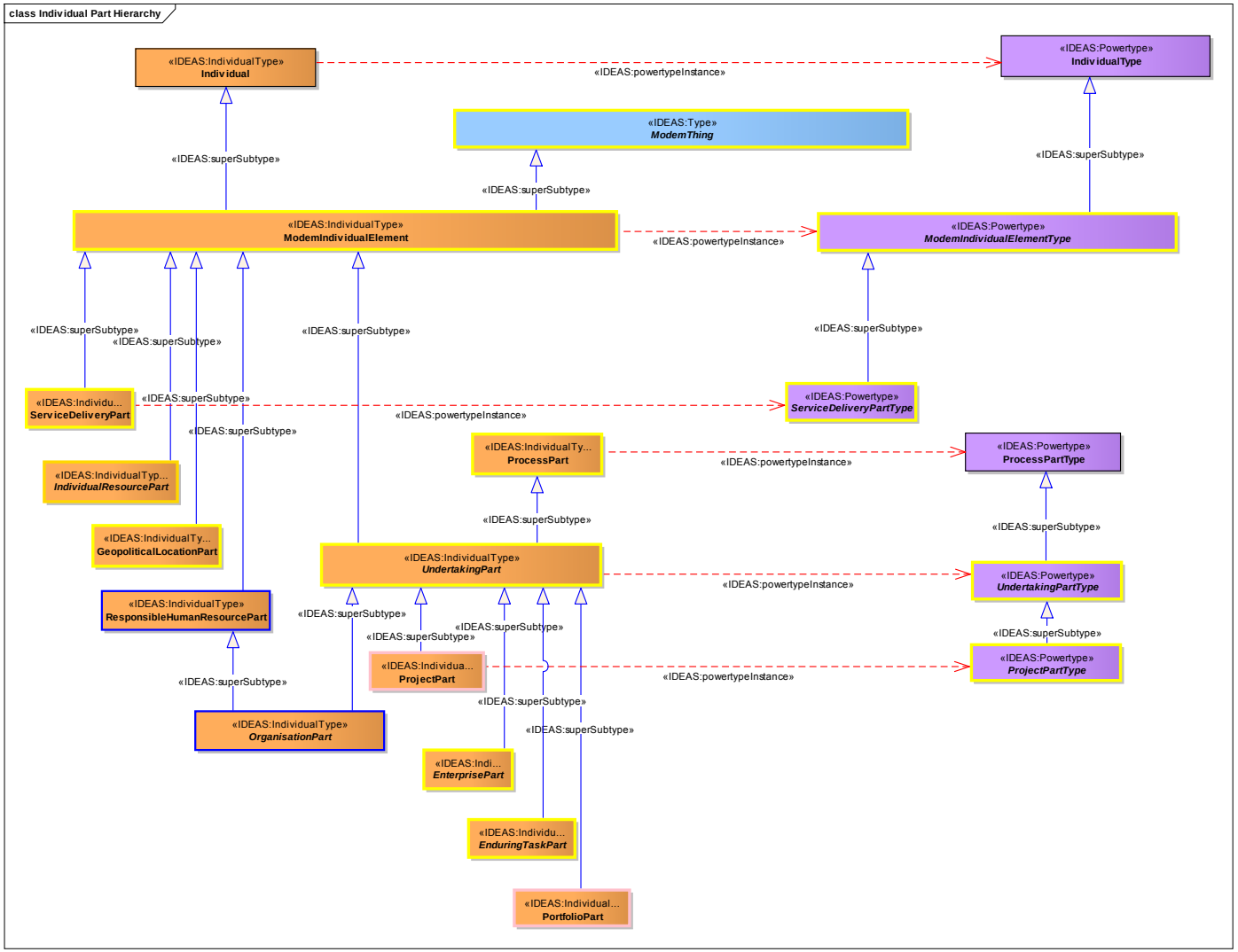


Figure 10 : Individual Part Hierarchy

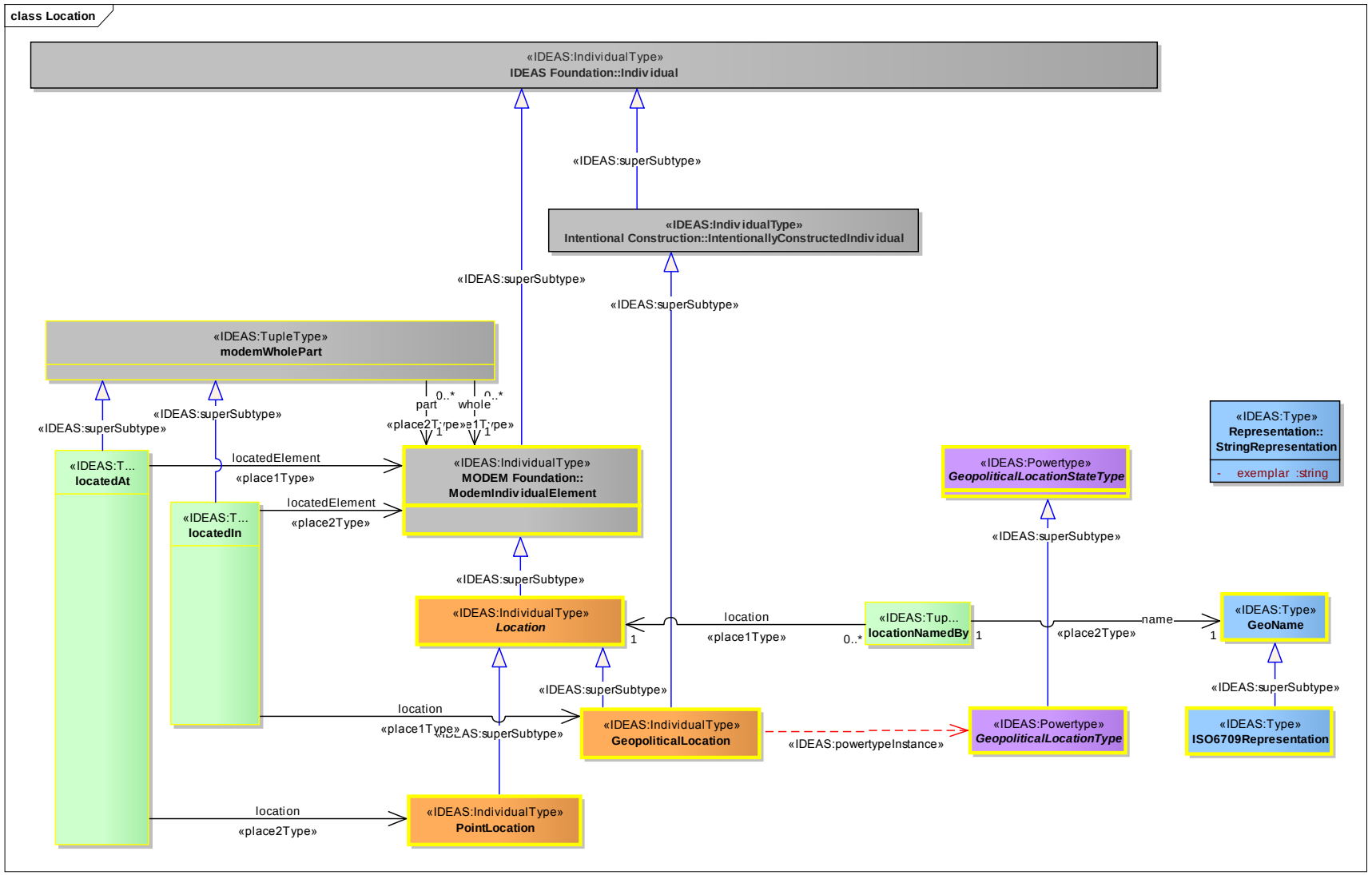


Figure 11 : Location

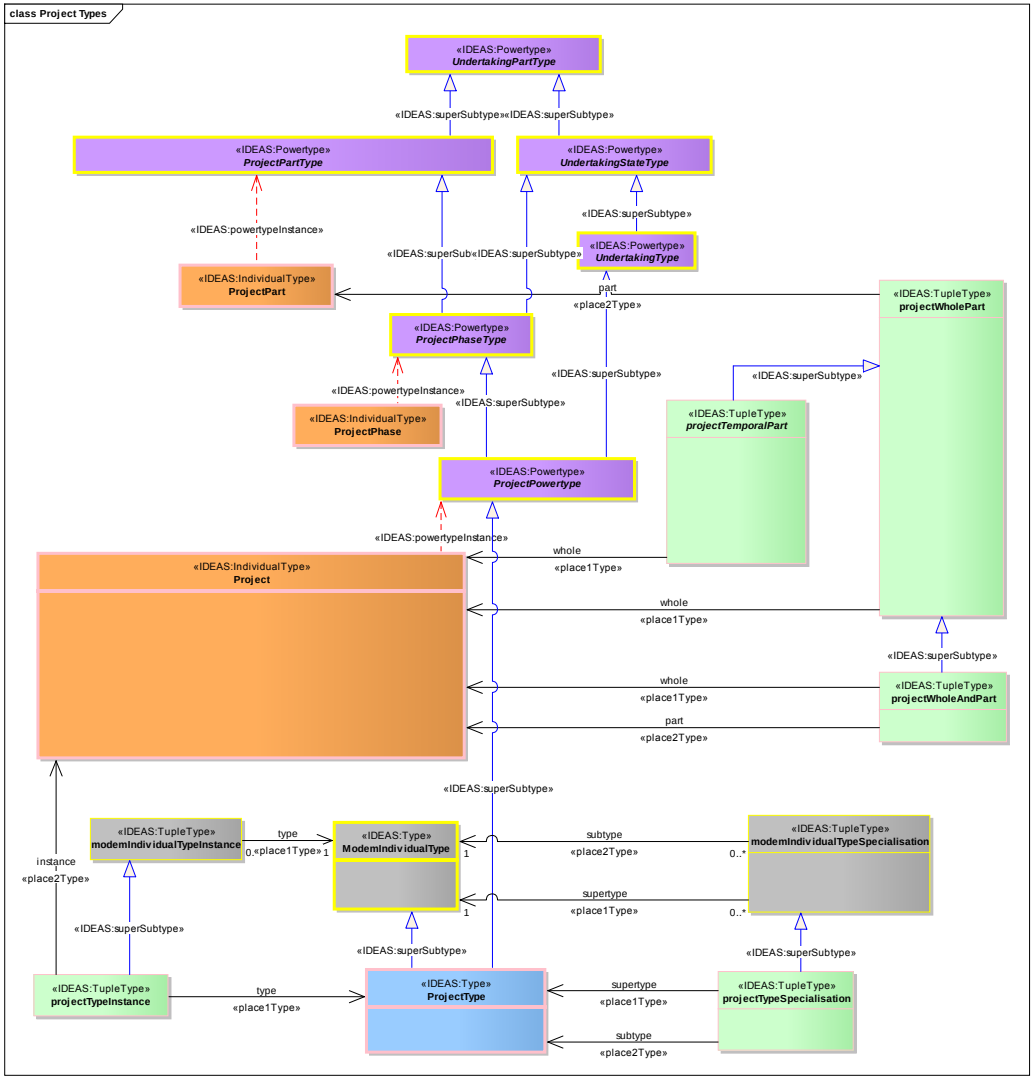


Figure 12 : Project Types

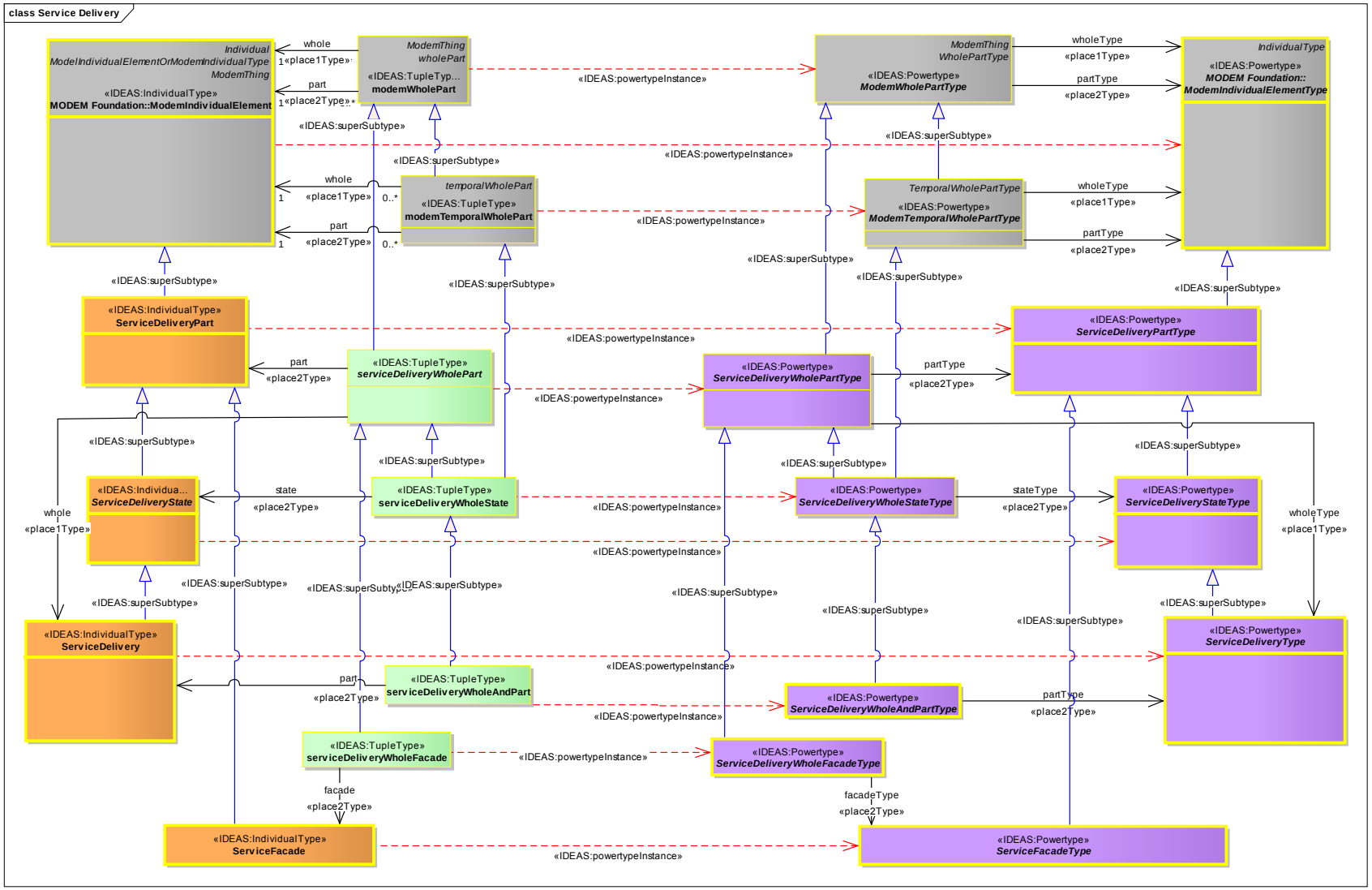


Figure 13 : Service Delivery

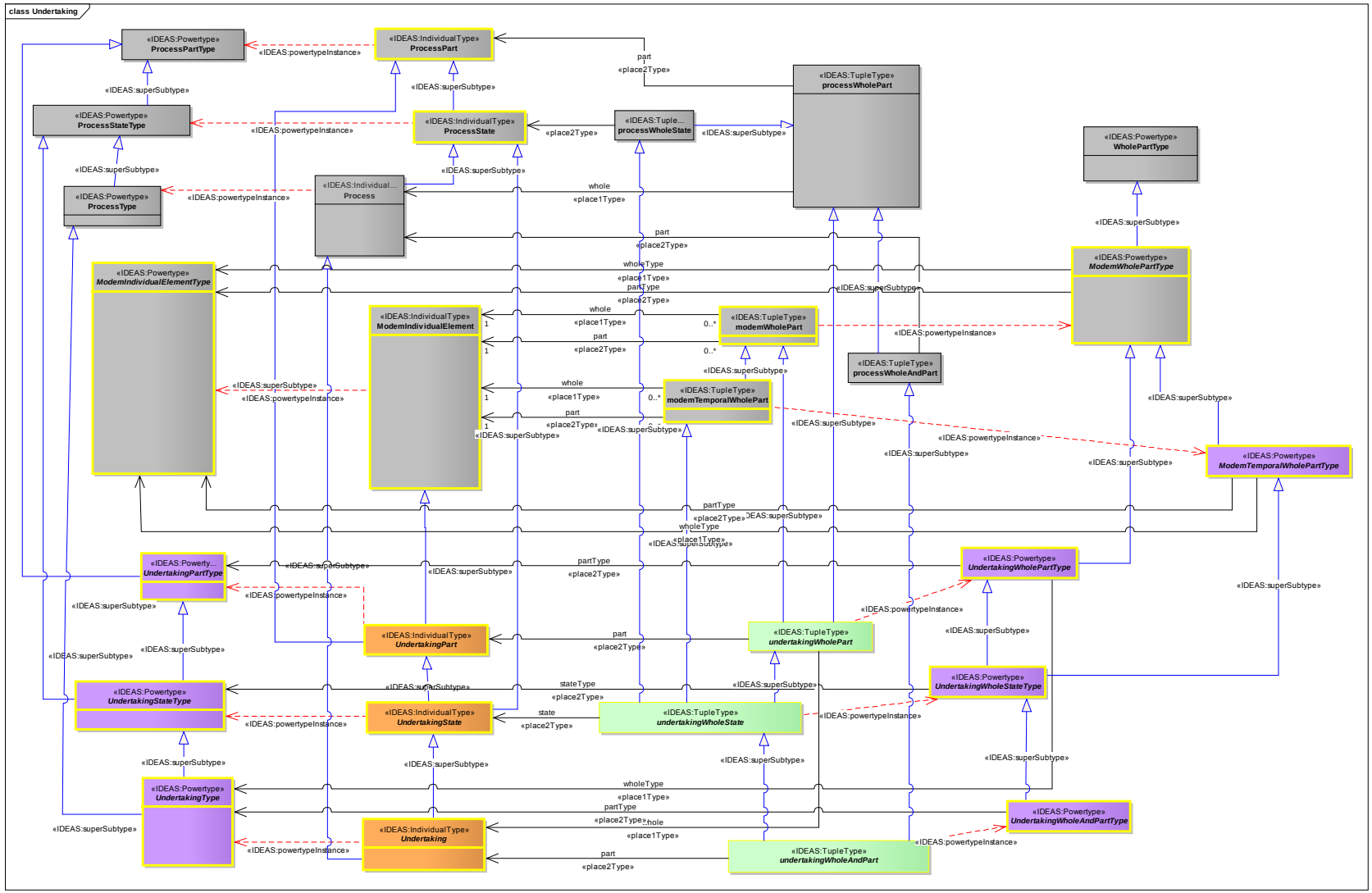


Figure 14 : Undertaking

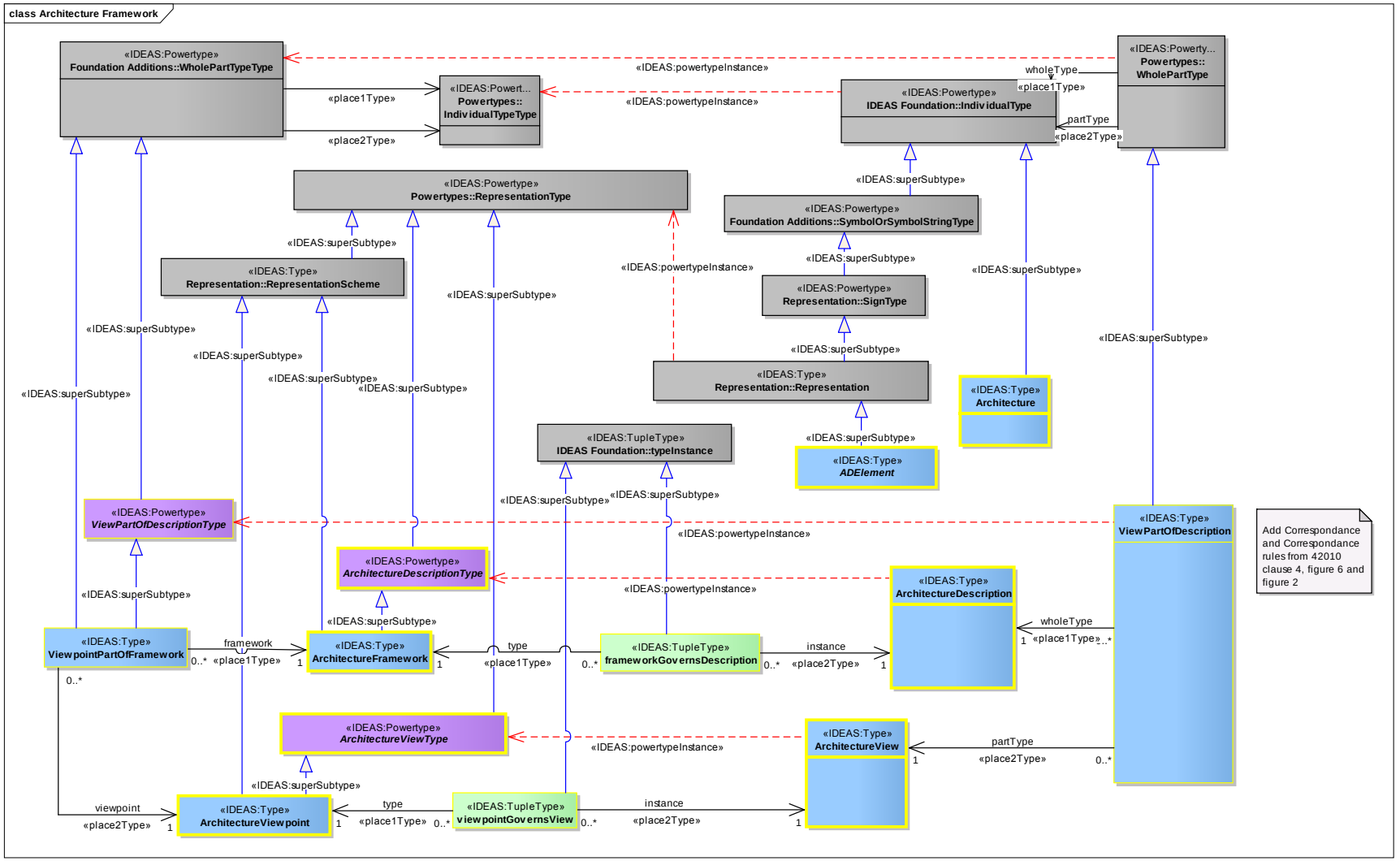


Figure 15 : Architecture Framework

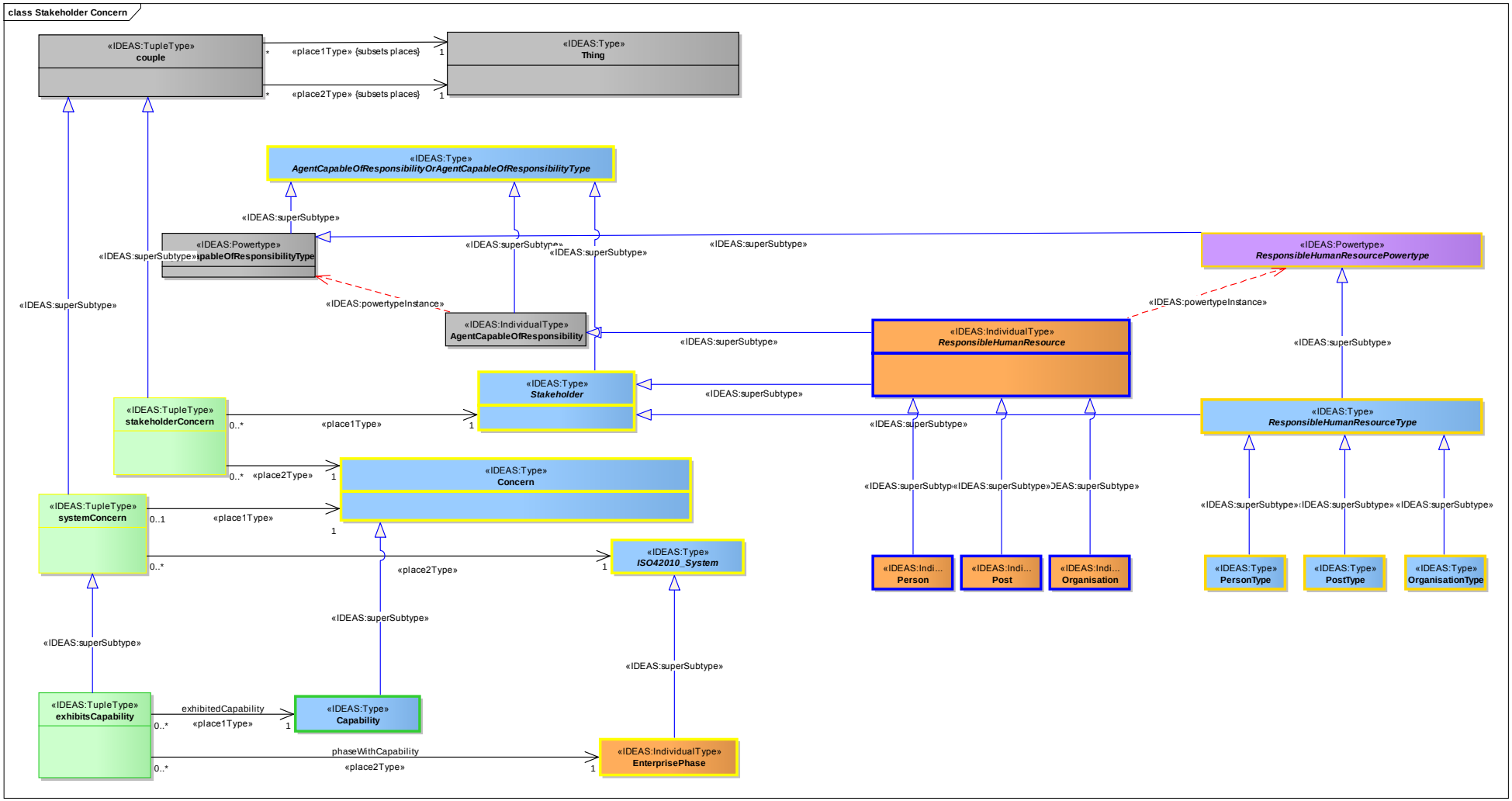


Figure 16 : Stakeholder Concern

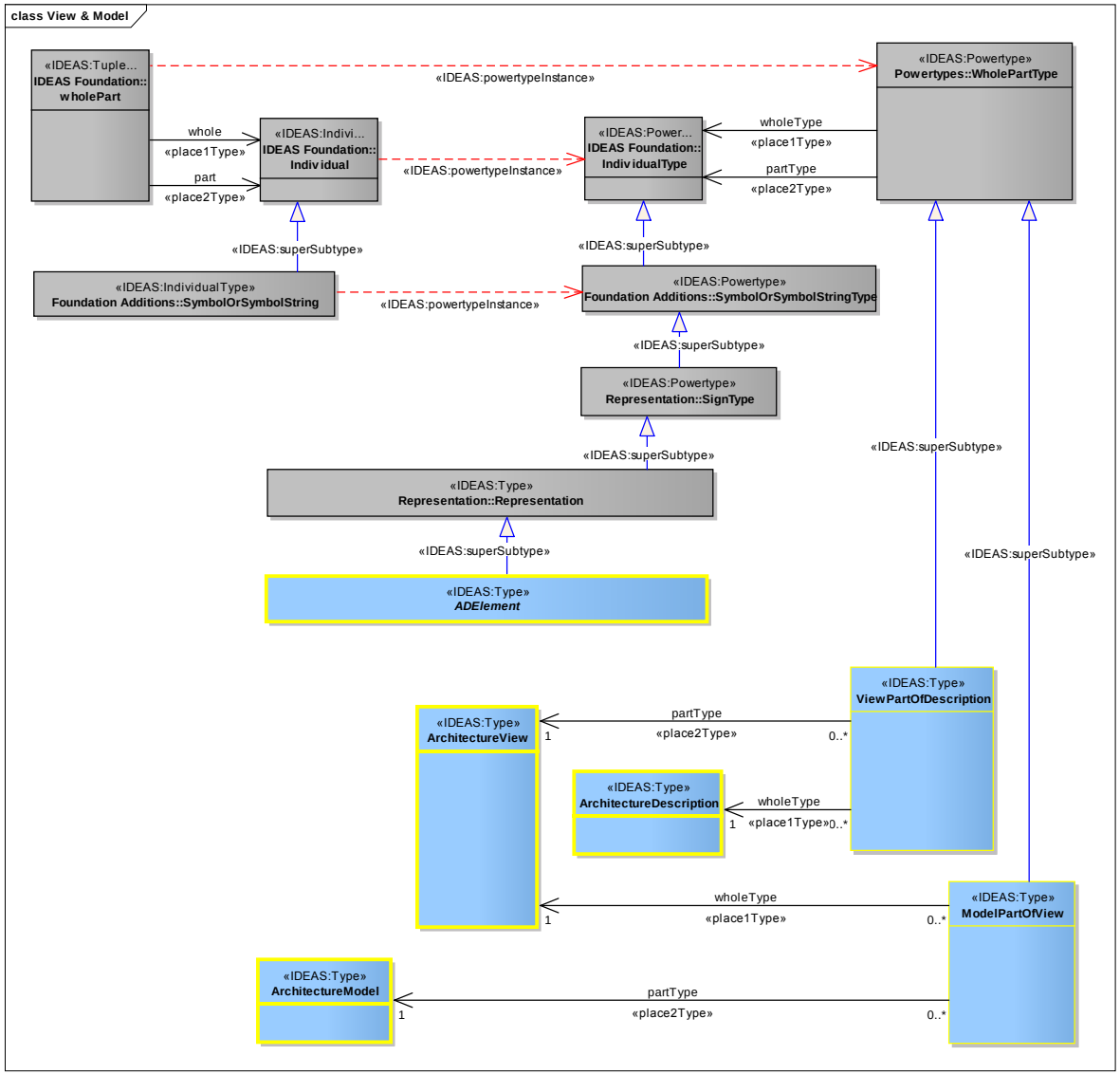


Figure 17 : View and model

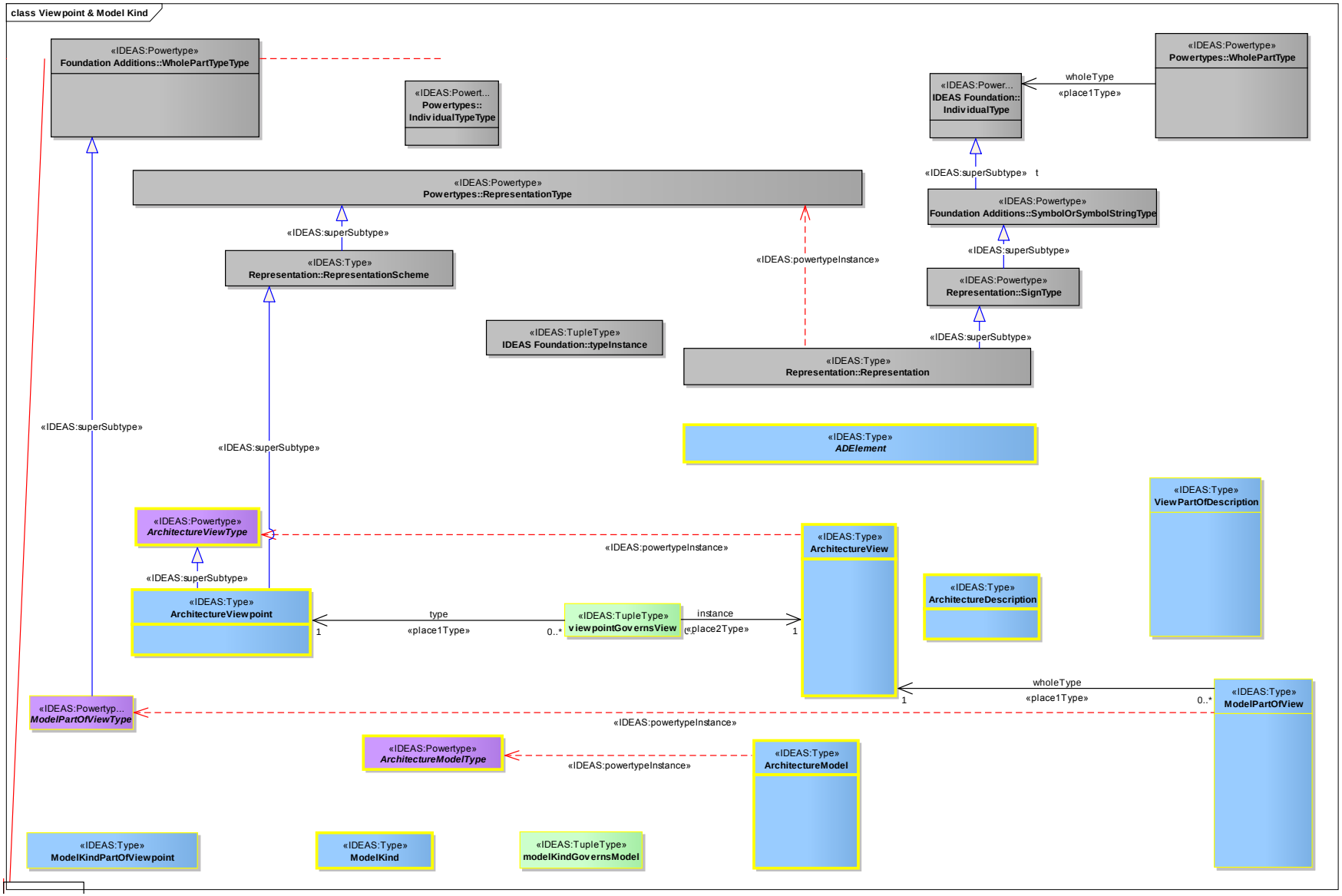


Figure 18 : Viewpoint & Model kind

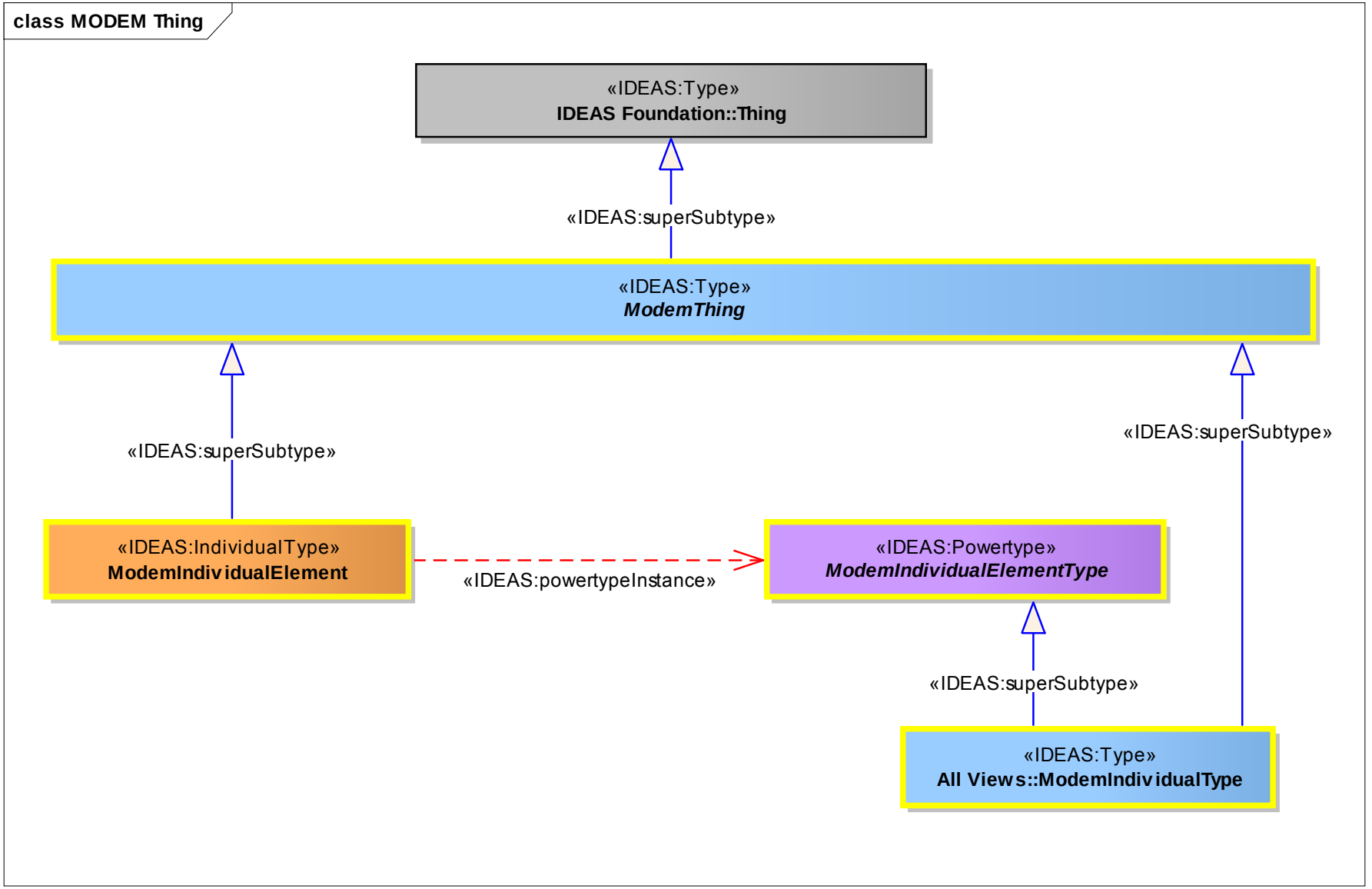


Figure 19 : Modem Thing

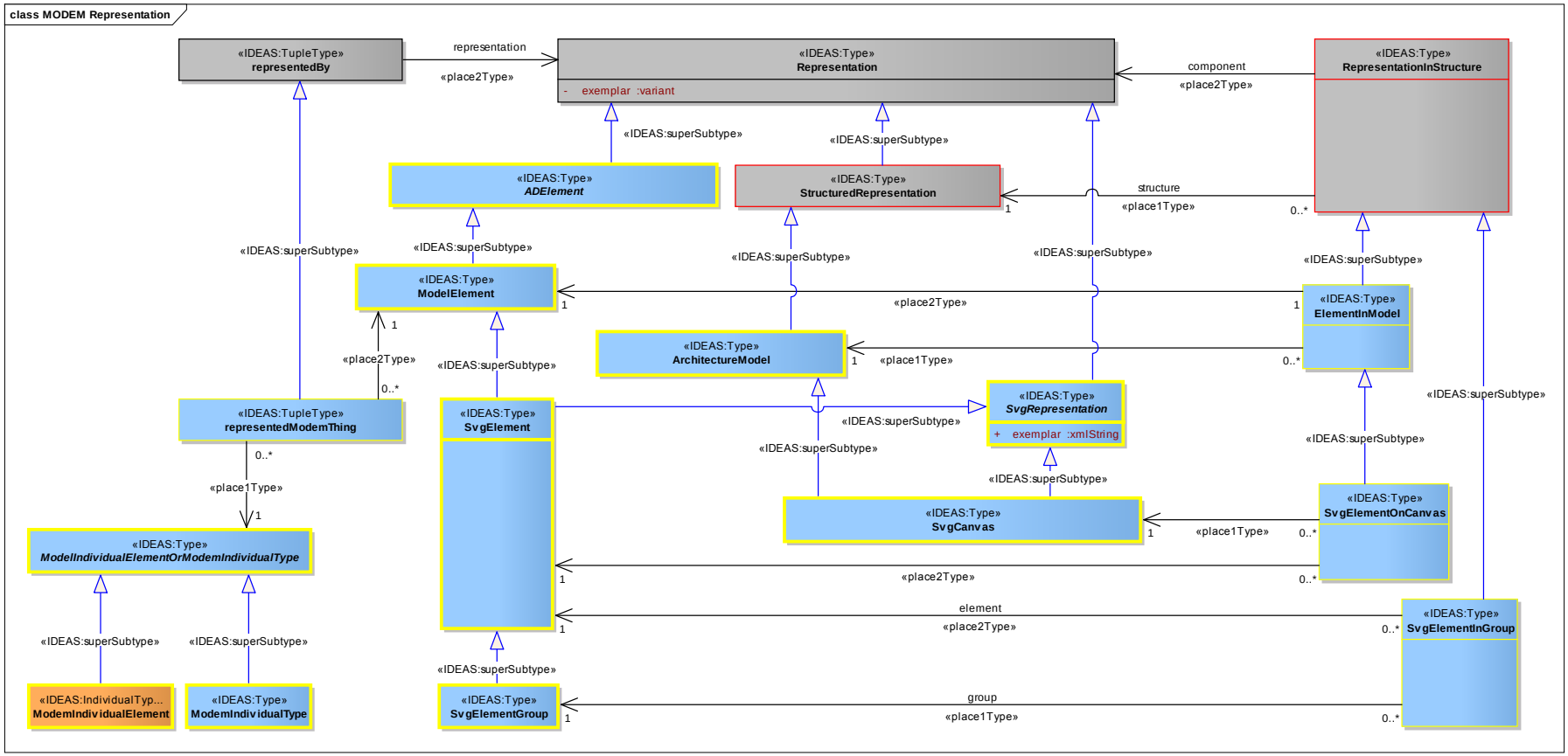


Figure 20 : Modem representation

2.3 Strategic views

2.3.1 StV-1: Enterprise vision

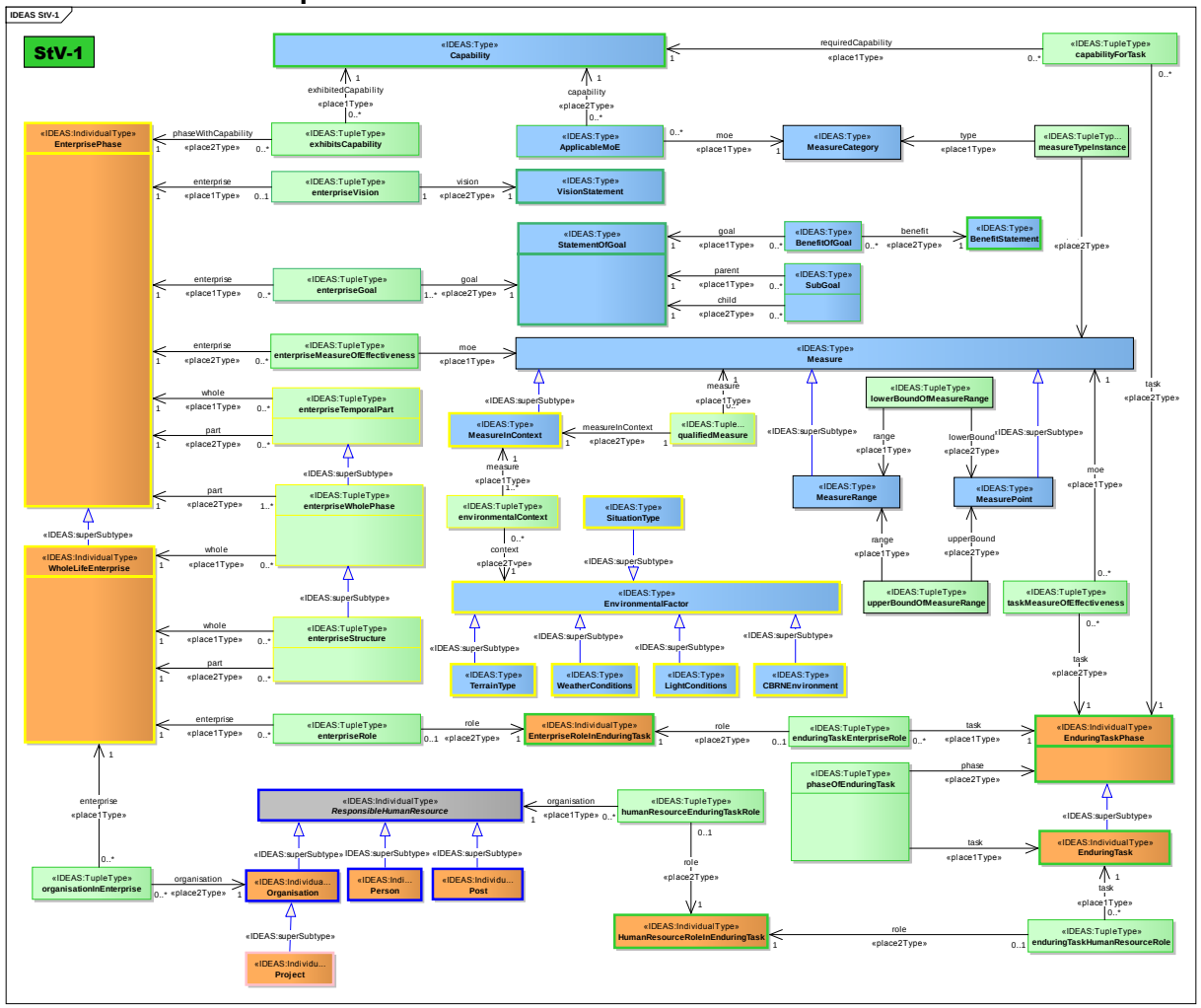


Figure 21 : StV-1

2.3.2 StV-2: Capability taxonomy

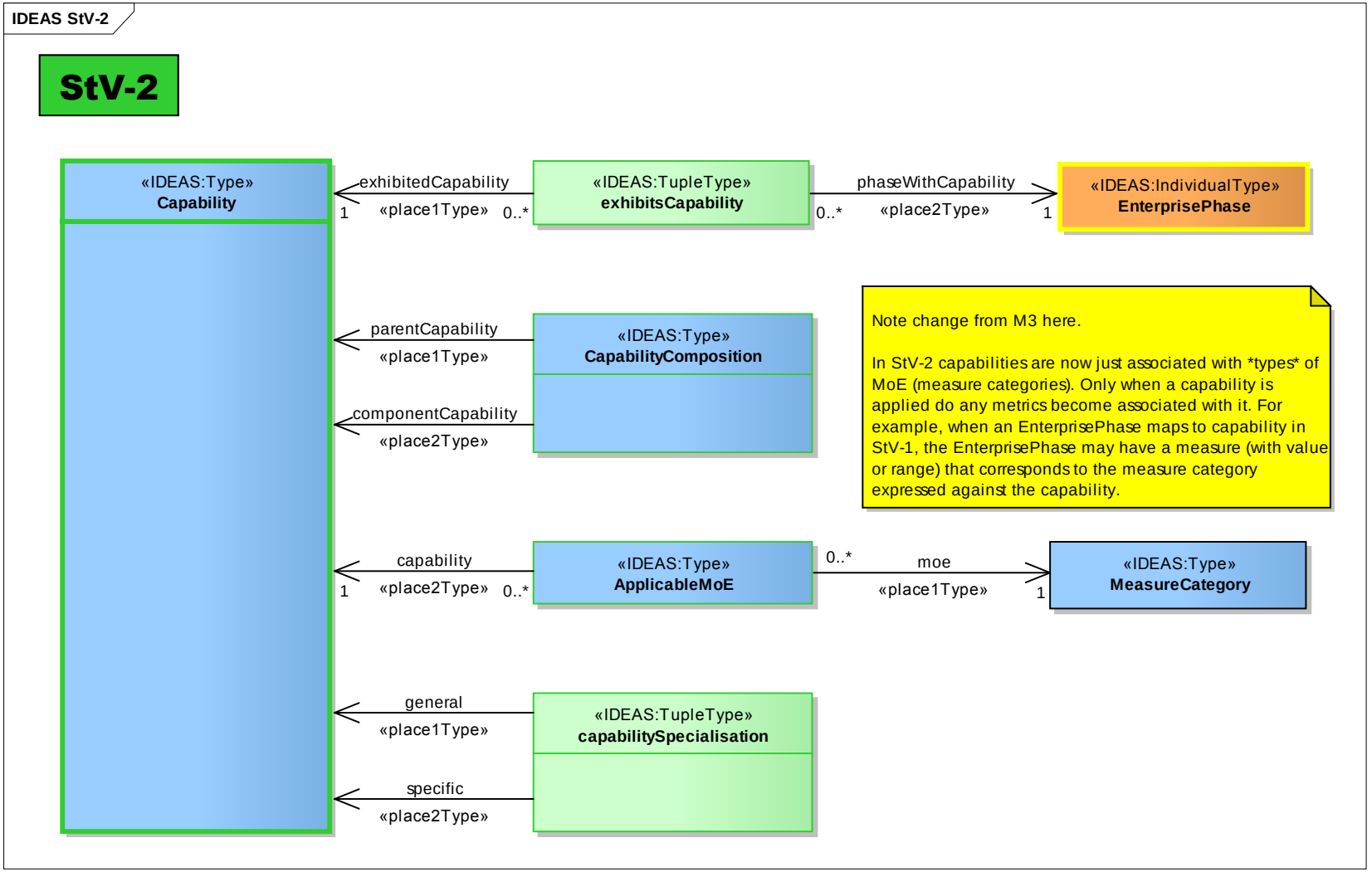


Figure 22 : StV-2

2.3.3 StV-3: Capability phasing

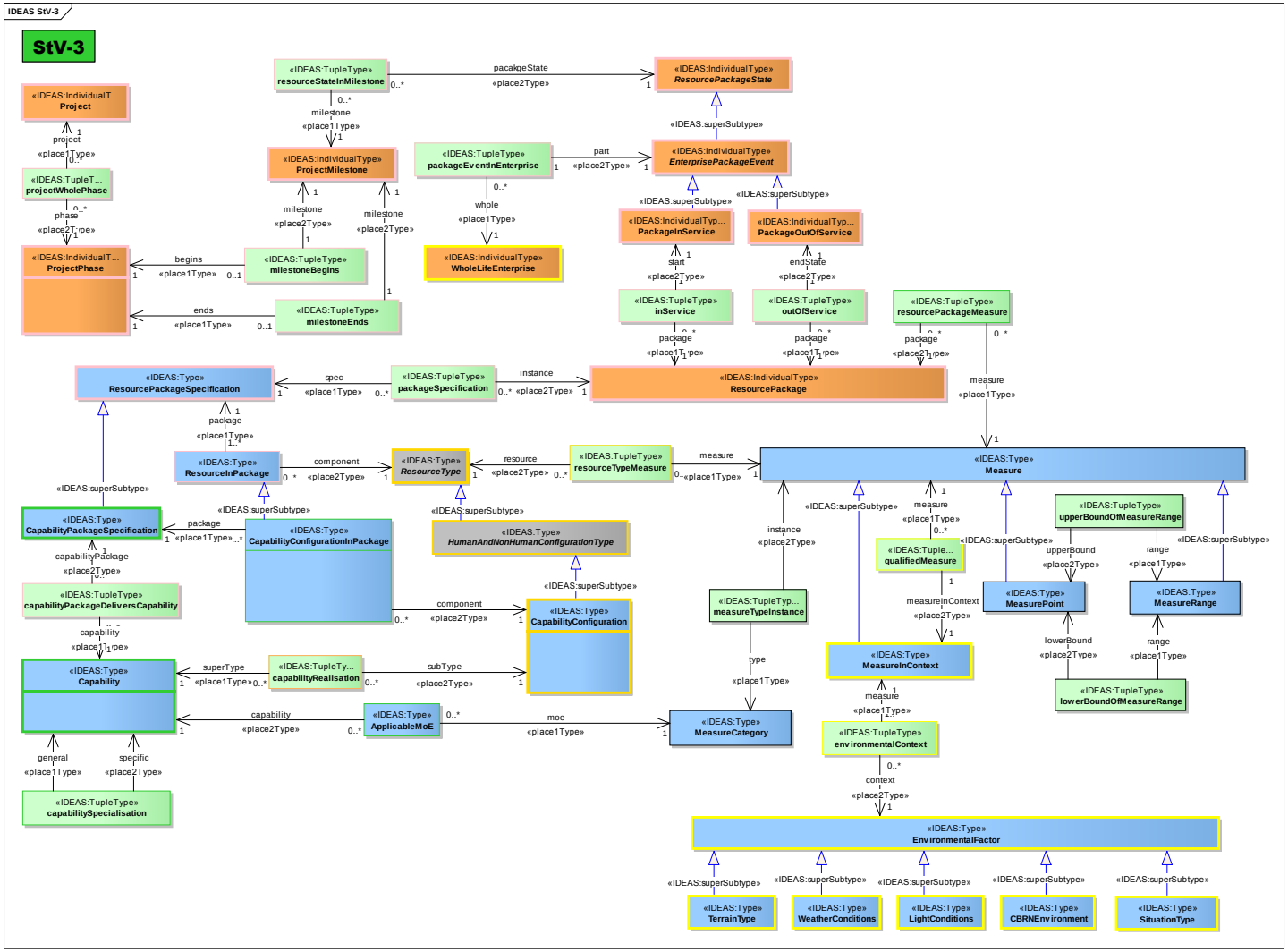


Figure 23 : StV-3

2.3.4 StV-4: Capability dependencies

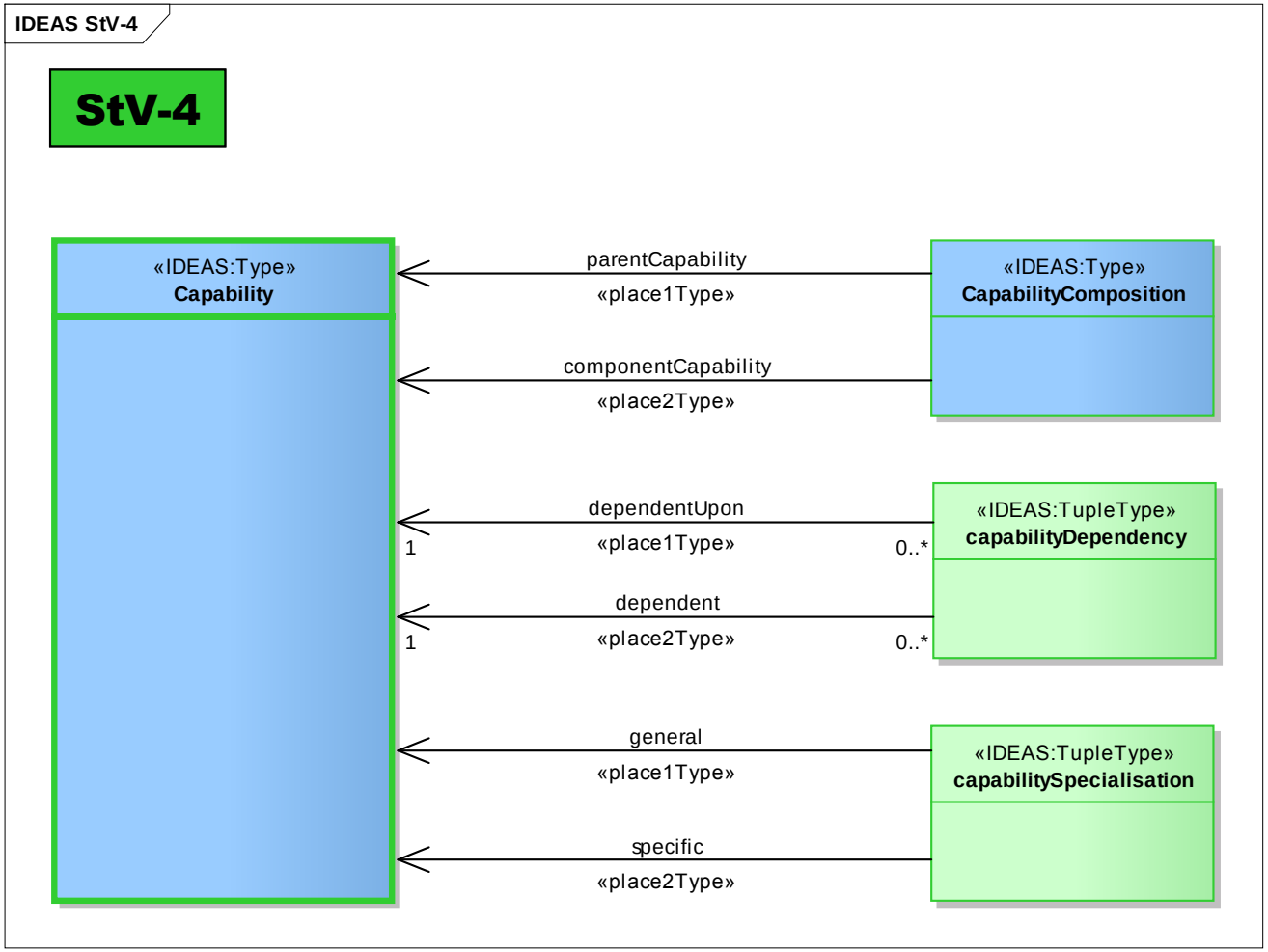


Figure 24 : StV-4

2.3.5 StV-5: Capability to organisational deployment mapping

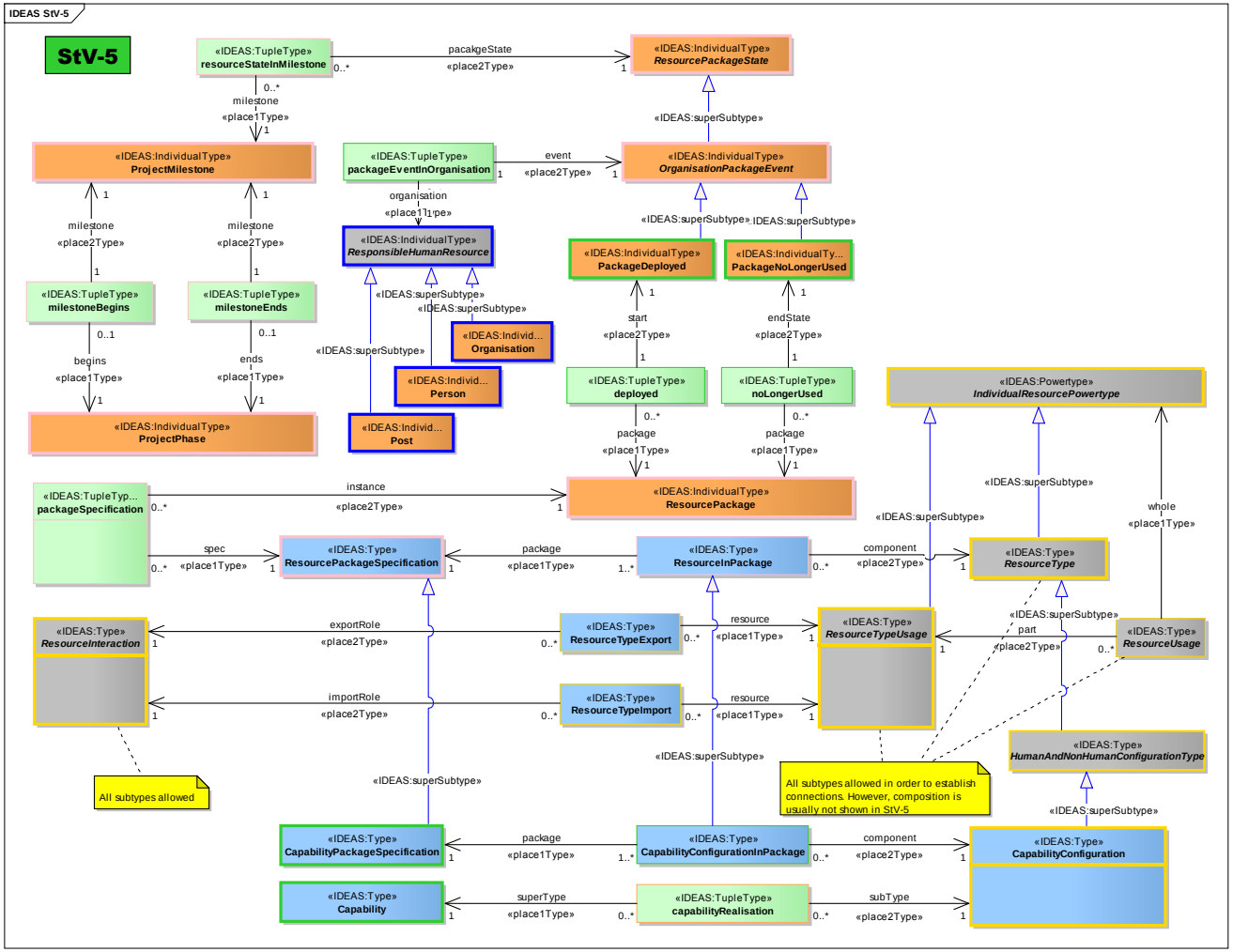


Figure 25 : StV-5

2.3.6 StV-6: Operational activity to capability mapping

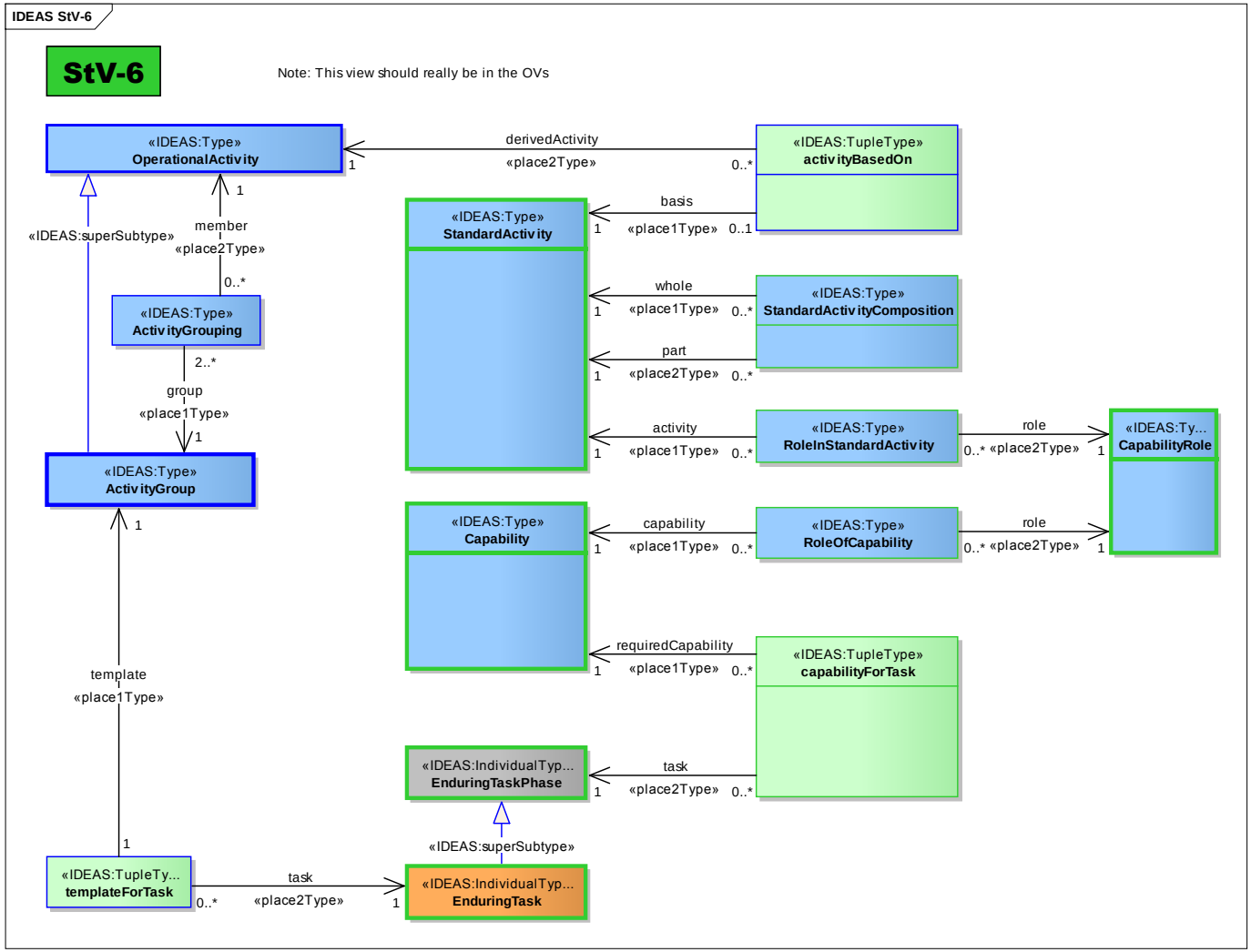


Figure 26 : StV-6

2.3.7 Strategic Views elements list

Strategic Views

ApplicableMoE «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ApplicableMoE - ApplicableMeasureCategory
Association (source - target):«place2Type»
ApplicableMoE - Capability
Association (source - target):«place1Type»
ApplicableMoE - MeasureCategory
Attributes:
-
An ApplicableMeasureCategory where the categories are MeasureOfEffectivenessCategories and the things being measured are Capabilities.

BenefitOfGoal «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
BenefitOfGoal - RepresentationInStructure
Association (source - target):«place2Type»
BenefitOfGoal - BenefitStatement
Association (source - target):«place1Type»
BenefitOfGoal - StatementOfGoal
Attributes:
-
A RepresentationInStructure where a BenefitStatement is part of a StatementOfGoal.

BenefitStatement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
BenefitStatement - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
BenefitStatement - StringDescription
Generalization (element - is a subtype of):«IDEAS:superSubtype»
BenefitStatement - StructuredRepresentation
Attributes:
-
A StringDescription that is part of a StatementOfGoal which describes a benefit realised by achieving the goal.

Capability «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Capability - Concern
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Capability - DispositionalProperty
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Capability - StrategicIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Capability - AgentType
Attributes:
-
A DispositionalProperty that is the set of all things that are capable of achieving a particular outcome.

CapabilityComposition «IDEAS:Type»
Connectors:

Generalization (element - is a subtype of): «IDEAS:superSubtype»
CapabilityComposition - TypicalWholePart
Association (source - target): «place2Type»
CapabilityComposition - Capability
Association (source - target): «place1Type»
CapabilityComposition - Capability
Attributes:
-
A WholePartType that asserts one Capability is part of another.

CapabilityConfigurationInPackage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CapabilityConfigurationInPackage - ResourceInPackage
Association (source - target):«place2Type»
CapabilityConfigurationInPackage - CapabilityConfiguration
Association (source - target):«place1Type»
CapabilityConfigurationInPackage - CapabilityPackageSpecification
Attributes:
-
A ResourceInPackage where the ResourceType is a CapabilityConfiguration and the package is a ResourcePackageSpecification.

CapabilityPackageSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CapabilityPackageSpecification - ResourcePackageSpecification
Attributes:
-
A ResourcePackageSpecification that contains at least one CapabilityConfiguration.

CapabilityRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
CapabilityRole - ParticipationExtentType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CapabilityRole - StrategicIndividualType
Attributes:
-
A ParticipationExtentType which is the extent of a Capability's participation in a StandardActivity.

EnduringTask «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnduringTask - Undertaking
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnduringTask - EnduringTaskPhase
Attributes:
-
An Undertaking recognised by an enterprise as being essential to achieving its goals - i.e. a strategic specification of what the enterprise does.

EnduringTaskPhase «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnduringTaskPhase - UndertakingState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnduringTaskPhase - EnduringTaskPart
Attributes:
-
A UndertakingState that is a temporal part of an EnduringTask.

EnterpriseRoleInEnduringTask «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnterpriseRoleInEnduringTask - ParticipationExtent
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnterpriseRoleInEnduringTask - EnterprisePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EnterpriseRoleInEnduringTask - EnduringTaskPart
Attributes:
-
A ParticipationExtent whose extent is the participation of an EnterprisePhase (or WholeLifeEnterprise) in an EnduringTask.

HumanResourceRoleInEnduringTask «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanResourceRoleInEnduringTask - ParticipationExtent
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanResourceRoleInEnduringTask - OrganisationPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanResourceRoleInEnduringTask - EnduringTaskPart
Attributes:
-
A ParticipationExtent where the participant is a ResponsibleHumanResource and the Process is an EnduringTask.

PackageDeployed «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PackageDeployed - OrganisationPackageEvent
Attributes:
-
An OrganisationPackageEvent which marks the delivery of a particular ResourcePackage to an Organisation.

PackageNoLongerUsed «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PackageNoLongerUsed - OrganisationPackageEvent

Attributes:
-
A OrganisationPackageEvent marking the point when a ResourcePacakge is no longer used by an Organisation.

RoleInStandardActivity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInStandardActivity - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInStandardActivity - ProcessWholeRoleExtentPartType
Association (source - target): «place1Type»
RoleInStandardActivity - StandardActivity
Association (source - target): «place2Type»
RoleInStandardActivity - CapabilityRole
Attributes:
-
A ProcessWholeRoleExtentPartType that relates a StandardActivity to a CapabilityRole.

RoleOfCapability «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleOfCapability - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleOfCapability - AgentParticipationType
Association (source - target): «place1Type»
RoleOfCapability - Capability
Association (source - target): «place2Type»
RoleOfCapability - CapabilityRole
Attributes:
-
An AgentParticipationType that relates a Capability to its role in a StandardOperationalActivity.

StandardActivity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StandardActivity - ProcessType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StandardActivity - StrategicIndividualType
Attributes:
-
An ProcessType that is a standard procedure (e.g. doctrinal tasks). Note: This is equivalent to what some defence organisations call JETLs. Note: was called "StandardOperationalActivity" in M3.

StandardActivityComposition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StandardActivityComposition - TypicalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StandardActivityComposition - ProcessWholeAndPartType
Association (source - target):«place2Type»
StandardActivityComposition - StandardActivity
Association (source - target):«place1Type»
StandardActivityComposition - StandardActivity
Attributes:
-
A TypicalWholePart that asserts one StandardActivity is part of another.

StatementOfGoal «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StatementOfGoal - StructuredRepresentation
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StatementOfGoal - StringDescription
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StatementOfGoal - ModemIndividualType
Attributes:
-
A StringDescription that is a specific, required objective for an EnterprisePhase.

StrategicIndividualType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StrategicIndividualType - ModemIndividualType
Attributes:
-
A ModemIndividualType which is used in Strategic modelling - i.e. a type of individual that may have relevance across more than one architecture.

SubGoal «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubGoal - RepresentationInStructure
Association (source - target):«place2Type»
SubGoal - StatementOfGoal
Association (source - target):«place1Type»
SubGoal - StatementOfGoal
Attributes:
-
A RepresentationInStructure where one StatementOfGoal is part of another.

VisionStatement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
VisionStatement - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
VisionStatement - StringDescription
Attributes:
-
A StringDescription that is a short paragraph outlining the vision for a given phase of an enterprise.

capabilityDependency «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilityDependency - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilityDependency - couple
Association (source - target):«place2Type»
capabilityDependency - Capability
Association (source - target):«place1Type»
capabilityDependency - Capability

Attributes:
-
A couple that relates a (dependent) Capability to a Capability it is dependent upon.

capabilityForTask «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilityForTask - propertyOfIndividual
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilityForTask - modemIndividualTypeInstance
Association (source - target):«place1Type»
capabilityForTask - Capability
Association (source - target):«place2Type»
capabilityForTask - EnduringTaskPhase
Attributes:
-
A propertyOfIndividual that asserts a Capability is required in order for an Enterprise to conduct a phase of an EnduringTask.

capabilitySpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilitySpecialisation - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
capabilitySpecialisation - Capability
Association (source - target):«place2Type»
capabilitySpecialisation - Capability
Attributes:
-
A superSubtype that relates one Capability (supertype) to a more specialised Capability (subtype).

deployed «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
deployed - modemTemporalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
deployed - individualResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
deployed - startBorder
Association (source - target):«place1Type»
deployed - ResourcePackage
Association (source - target):«place1Type»
deployed - PackageDeployed
Attributes:
-
A startBorder that indicates that an PackageDeployed marks the introduction into an Organisation of a ResourcePackage.

enduringTaskEnterpriseRole «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enduringTaskEnterpriseRole - processWholeRoleExtentPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enduringTaskEnterpriseRole - enduringTaskWholePart
Association (source - target): «place2Type»
enduringTaskEnterpriseRole - EnterpriseRoleInEnduringTask
Association (source - target): «place1Type»
enduringTaskEnterpriseRole - EnduringTaskPhase
Attributes:
-
A processWholeRoleExtentPart which relates an EnduringTaskPhase to an EnterpriseRoleInEnduringTask.

enduringTaskHumanResourceRole «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enduringTaskHumanResourceRole - enduringTaskWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enduringTaskHumanResourceRole - processWholeRoleExtentPart
Association (source - target): «place2Type»
enduringTaskHumanResourceRole -HumanResourceRoleInEnduringTask
Association (source - target): «place1Type»
enduringTaskHumanResourceRole - EnduringTask
Attributes:
-
A processWholeRoleExtentPart where the Process is an EnduringTask and the involved Individual is a ResponsibleHumanResource.

enterpriseGoal «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enterpriseGoal - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enterpriseGoal - describedBy
Association (source - target):«place1Type»
enterpriseGoal - EnterprisePhase
Association (source - target):«place2Type»
enterpriseGoal - StatementOfGoal
Attributes:
-
A describedBy that relates a StatementOfGoal to the EnterprisePhase it describes.

enterpriseMeasureOfEffectiveness «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enterpriseMeasureOfEffectiveness - measureOfIndividual
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enterpriseMeasureOfEffectiveness - modemIndividualTypeInstance
Association (source - target):«place2Type»
enterpriseMeasureOfEffectiveness - EnterprisePhase
Association (source - target):«place1Type»
enterpriseMeasureOfEffectiveness - Measure

Attributes:
-
A measureOfIndividual where the Individual is an EnterprisePhase and the measure is a MeasureOfEffectiveness.

enterpriseRole «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseRole - agentParticipation
Generalization (element - is a subtype of): «IDEAS:superSubtype»
enterpriseRole - enterpriseWholePart
Association (source - target): «place1Type»
enterpriseRole - WholeLifeEnterprise
Association (source - target): «place2Type»
enterpriseRole - EnterpriseRoleInEnduringTask
Attributes:
-
An agentParticipation where the agent is a WholeLifeEnterprise and the participation is an EnterpriseRoleInEnduringTask. An enterpriseRole relates a WholeLifeEnterprise to a role it performs in an EnduringTask.

enterpriseVision «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enterpriseVision - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enterpriseVision - describedBy
Association (source - target):«place1Type»
enterpriseVision - EnterprisePhase
Association (source - target):«place2Type»
enterpriseVision - VisionStatement
Attributes:
-
A describedBy that relates a VisionStatement to the EnteprisePhase it describes.

exhibitsCapability «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exhibitsCapability - propertyOfIndividual
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exhibitsCapability - modemIndividualTypeInstance
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exhibitsCapability - systemConcern
Association (source - target): «place1Type»
exhibitsCapability - Capability
Association (source - target): «place2Type»
exhibitsCapability - EnterprisePhase
Attributes:
-
A propertyOfIndividual that relates an EnteprisePhase to a Capability that it exhibits.
Note: replaces "exhibits" tagged value in M3.

humanResourceEnduringTaskRole «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
humanResourceEnduringTaskRole - agentParticipation
Generalization (element - is a subtype of):«IDEAS:superSubtype»
humanResourceEnduringTaskRole - organisationWholePart
Association (source - target):«place1Type»
humanResourceEnduringTaskRole - ResponsibleHumanResource
Association (source - target):«place2Type»
humanResourceEnduringTaskRole - HumanResourceRoleInEnduringTask
Attributes:
-
An agentParticipation where the Agent is a ResponsibleHumanResource and the Process is an EnduringTask.

noLongerUsed «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
noLongerUsed - endBorder
Generalization (element - is a subtype of):«IDEAS:superSubtype»
noLongerUsed - individualResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
noLongerUsed - modemTemporalWholePart
Association (source - target):«place2Type»
noLongerUsed - PackageNoLongerUsed
Association (source - target):«place1Type»
noLongerUsed - ResourcePackage

Attributes:
-
An endBorder that indicates that an PackageNoLongerUsed marks the retirement from an Organisation of a ResourcePackage.

organisationInEnterprise «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
organisationInEnterprise - enterpriseWholePart
Association (source - target):«place1Type»
organisationInEnterprise - WholeLifeEnterprise
Association (source - target):«place2Type»
organisationInEnterprise - Organisation
Attributes:
-
An enterpriseWholePart that asserts an Organisation is part of a WholeLifeEnterprise. Note: this includes the limit case where the Organisation *is* the Enterprise, and cases where the Organisation is a Project.

packageEventInOrganisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
packageEventInOrganisation - organisationWholePart
Association (source - target):«place2Type»
packageEventInOrganisation - OrganisationPackageEvent
Association (source - target):«place1Type»
packageEventInOrganisation - ResponsibleHumanResource

Attributes:

-

An organisationWholePart where an OrganisationPackageEvent is part of an Organisation - e.g. the package is rolled-out into the organisation.

phaseOfEnduringTask

2.3.8 Strategic Views additional diagrams.

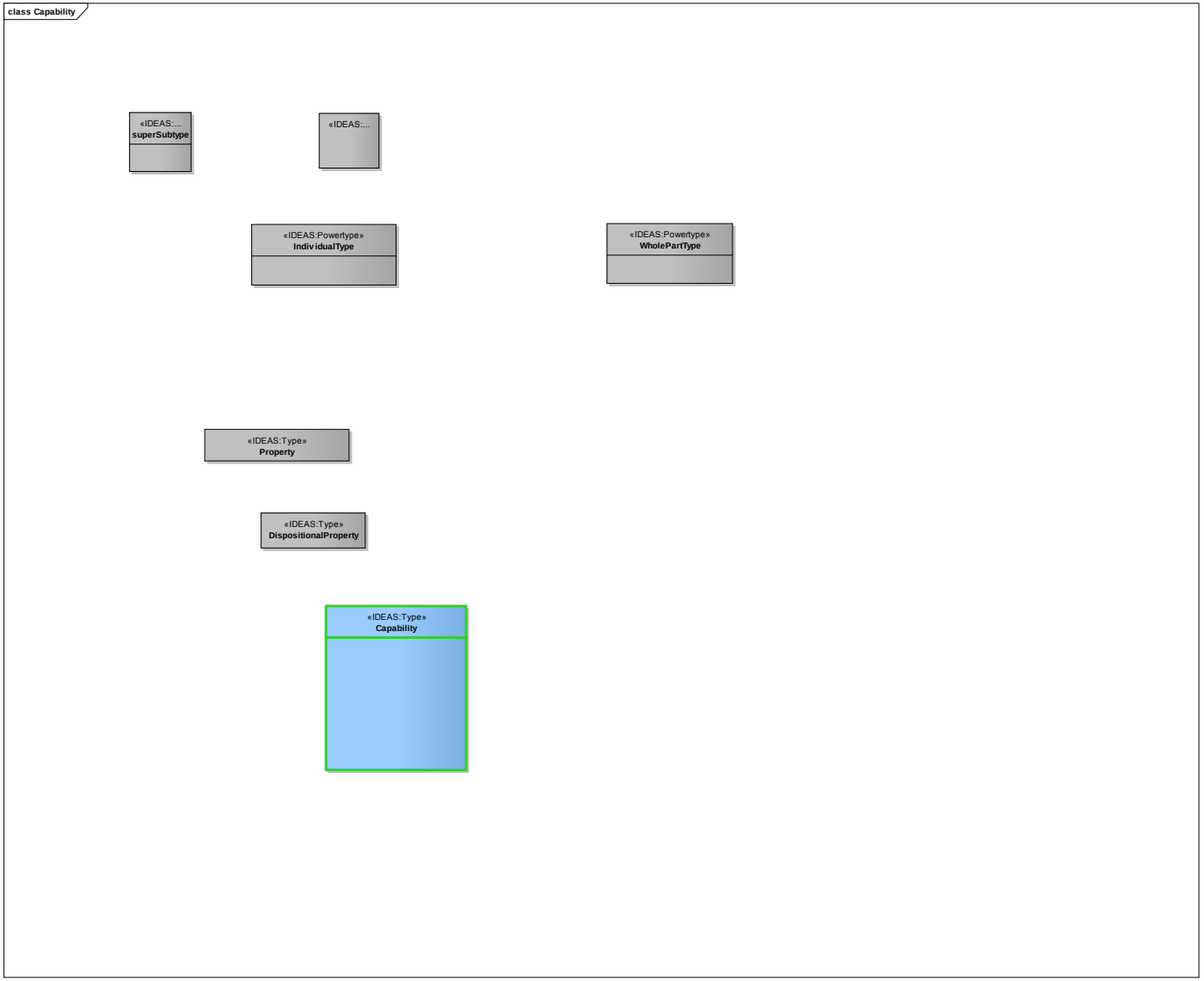


Figure 27 : Capability

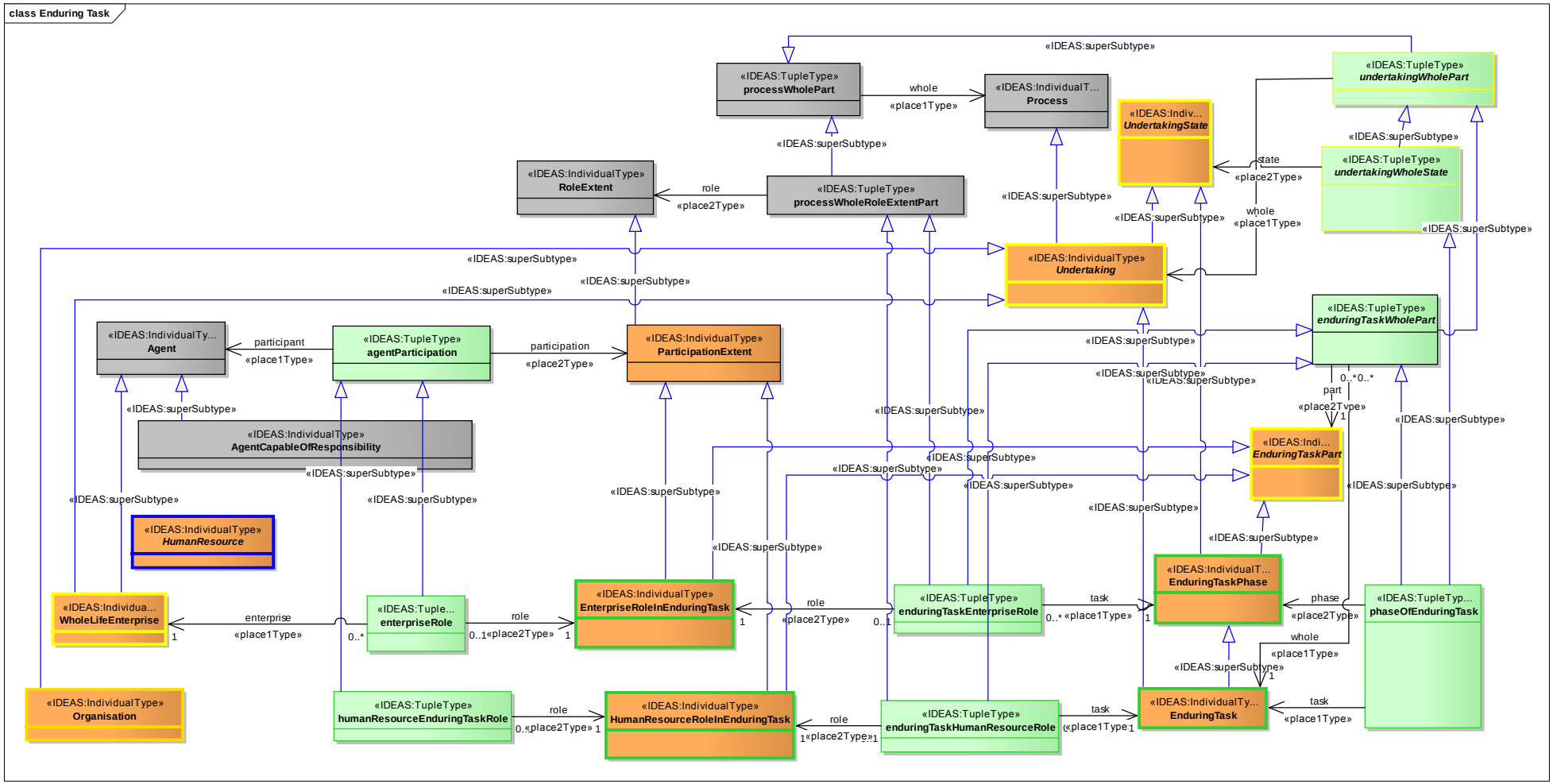


Figure 28 : Enduring Task

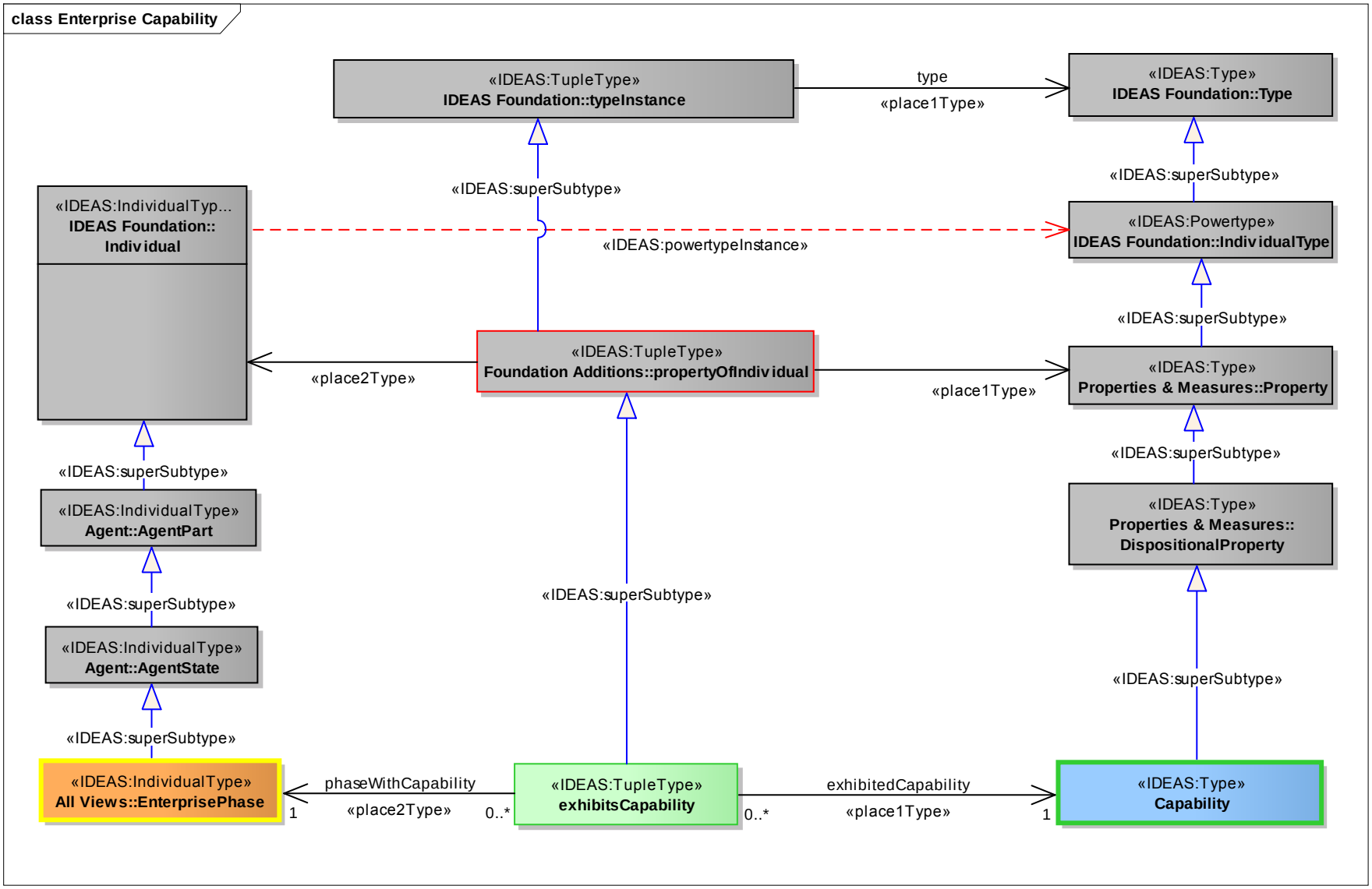


Figure 29 : Enterprise Capability

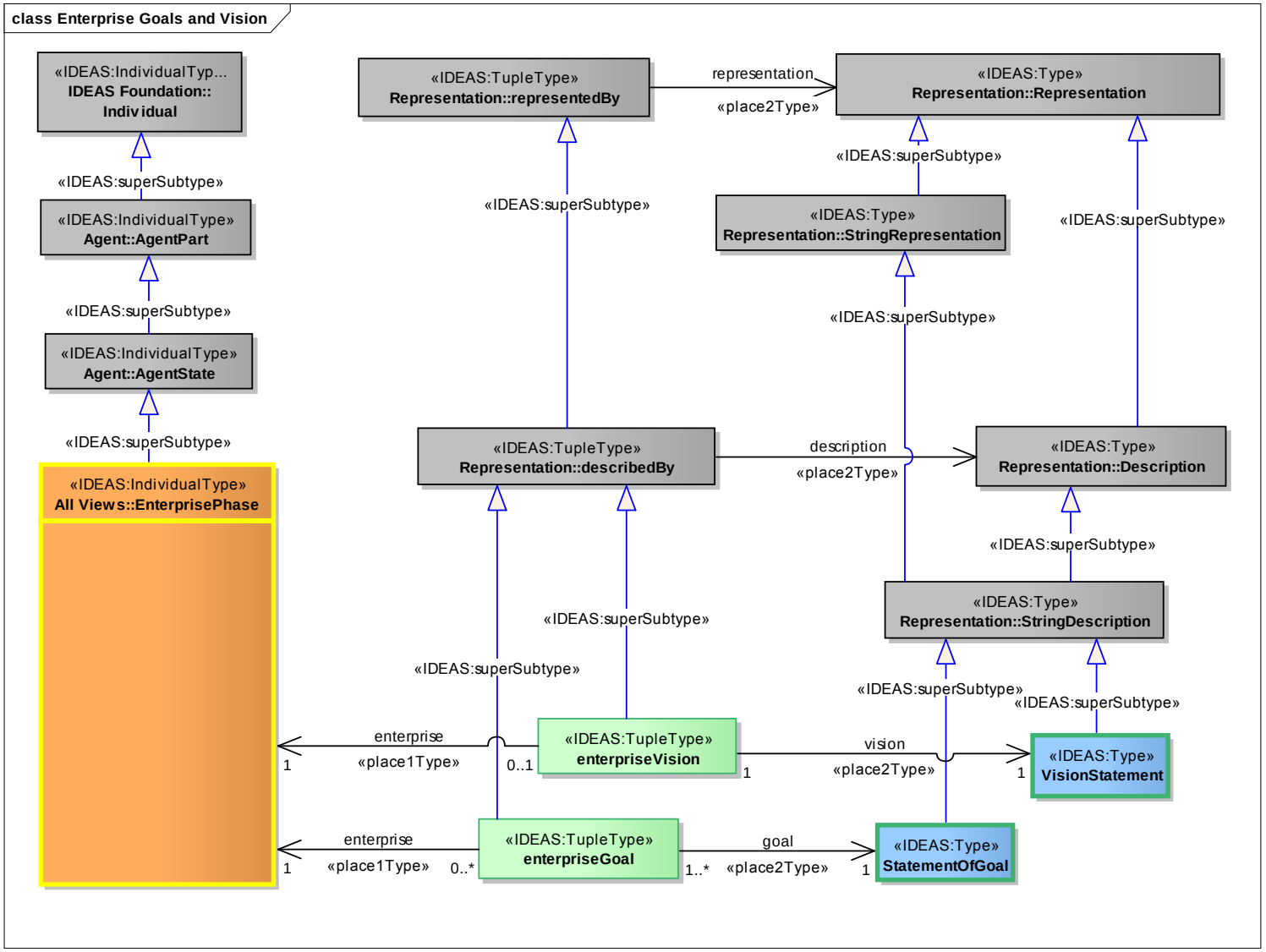


Figure 30 : Enterprise Goals and Vision

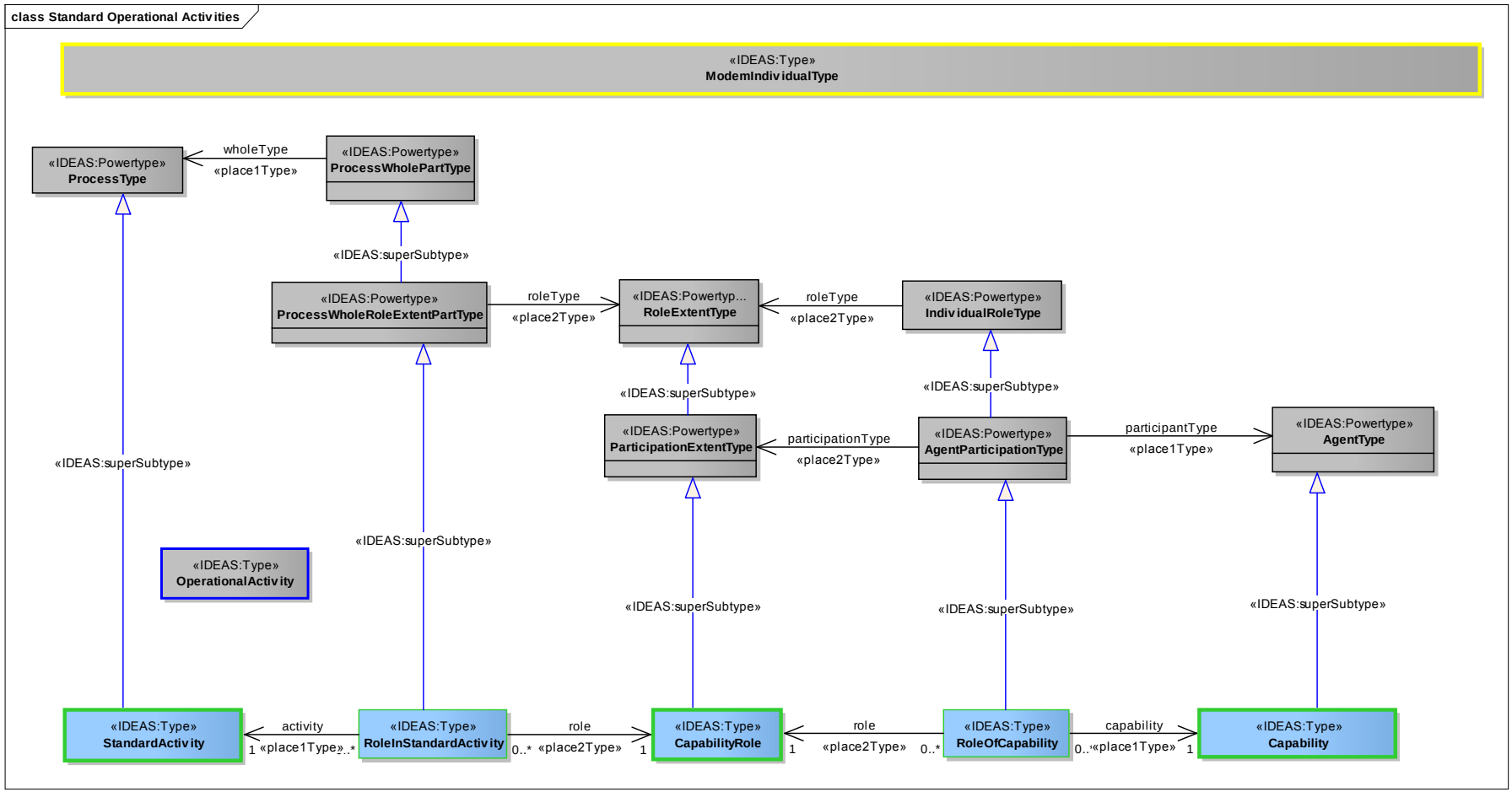


Figure 31 : Standard Operational Activities

2.4 Operational views

2.4.1 OV-1: High level operational concept graphic (a, b, c)

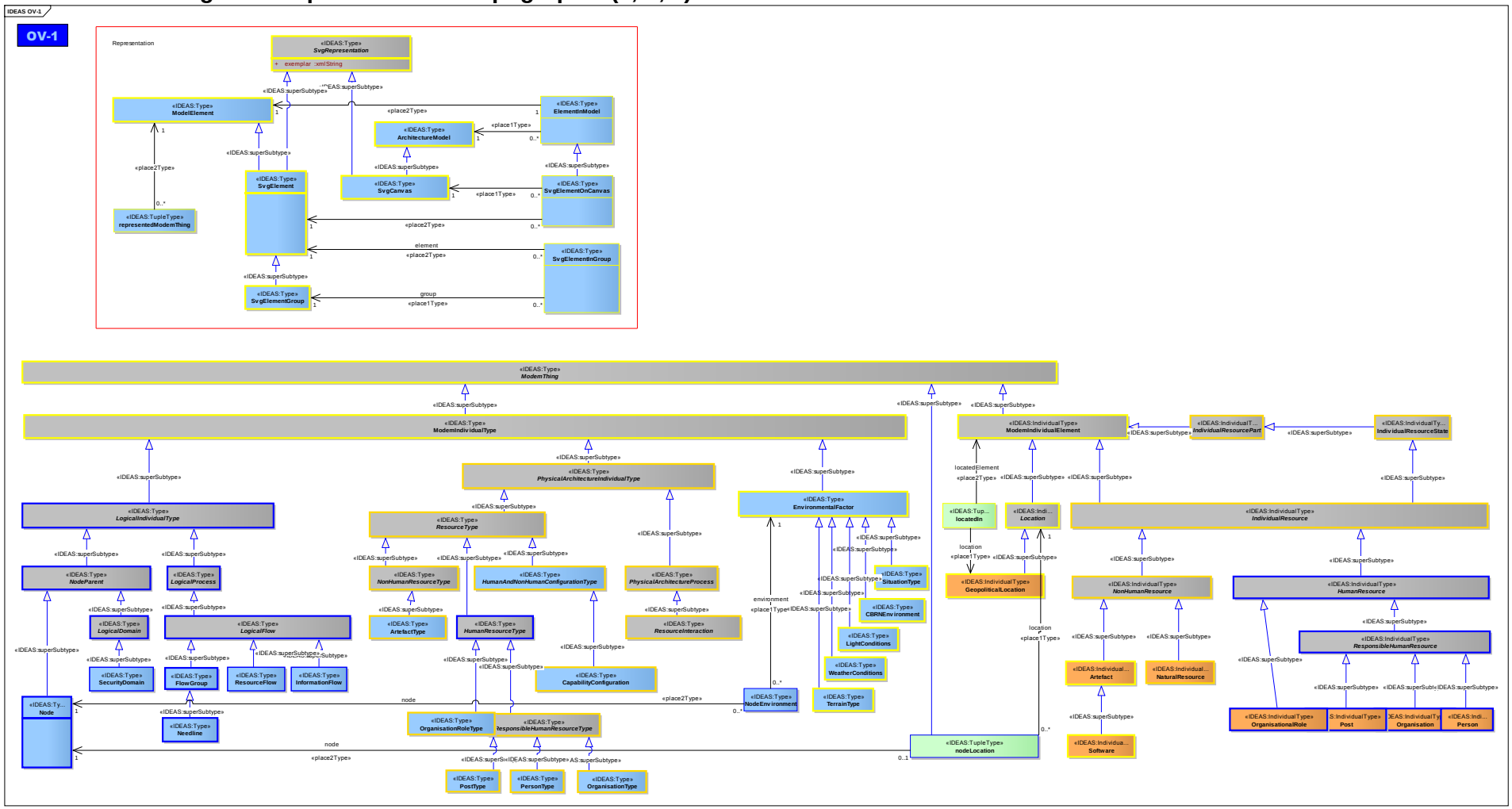


Figure 32 : OV-1

2.4.2 OV-2: Operational node relationship description

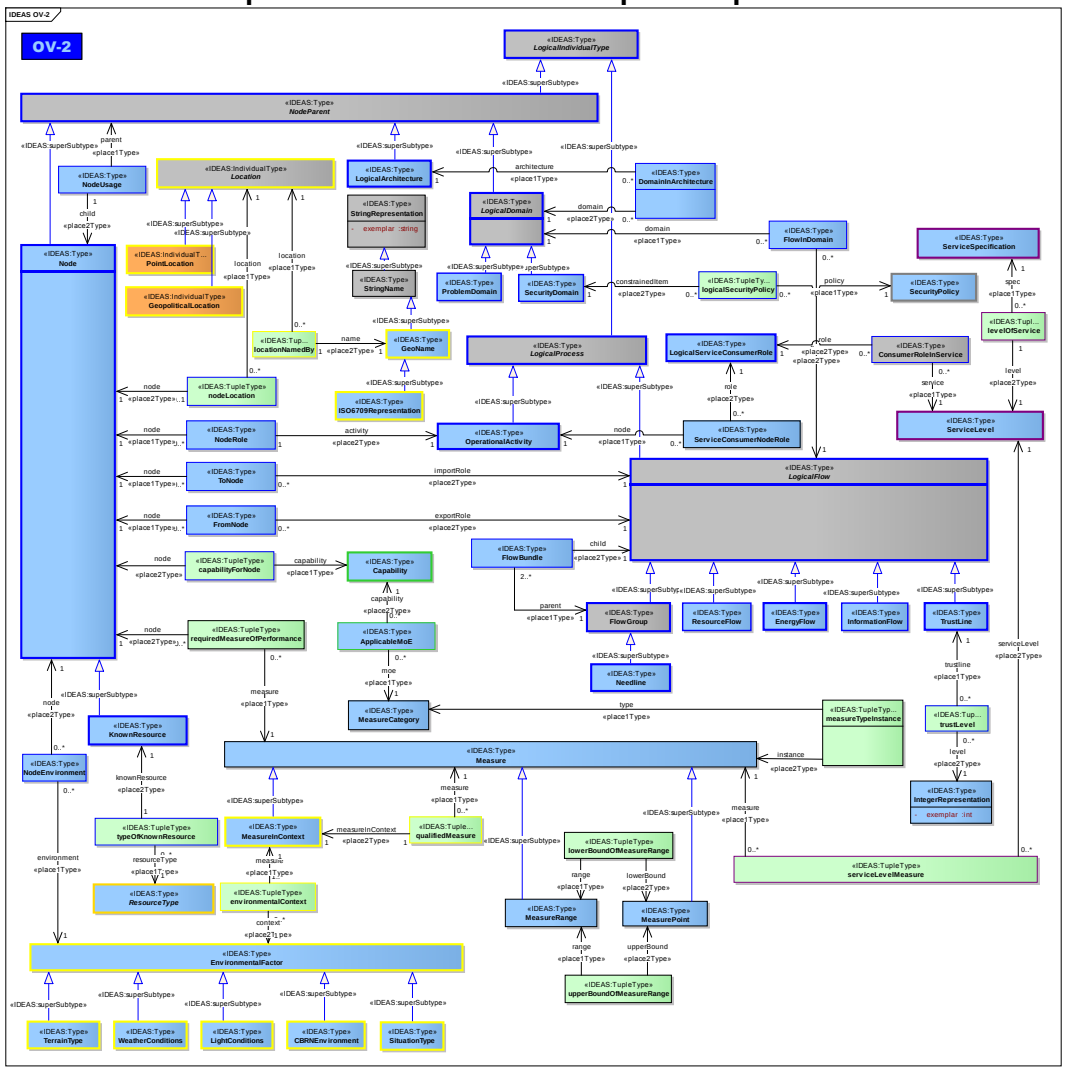


Figure 33 : OV-2

2.4.3 OV-3: Operational information exchange matrix

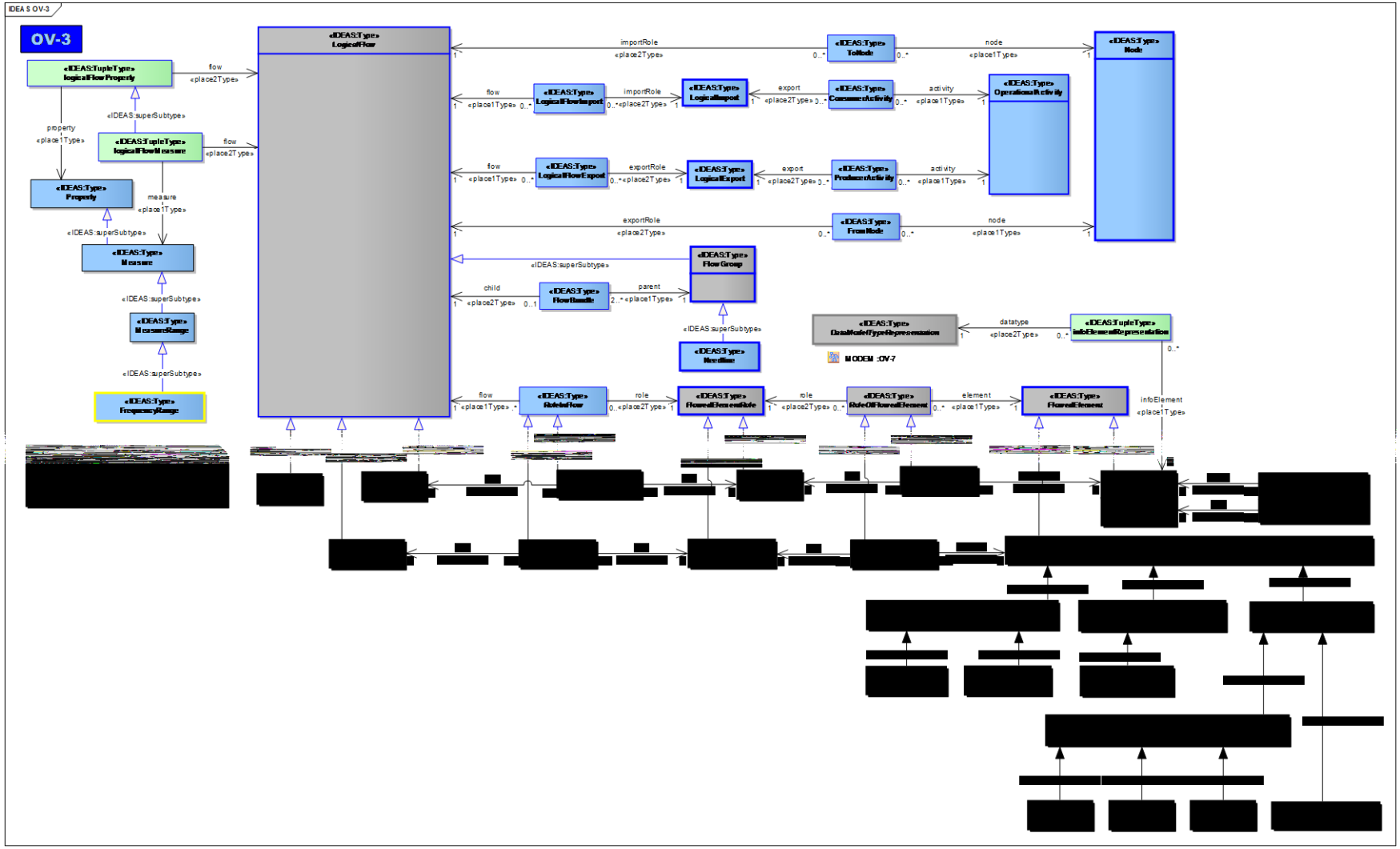


Figure 34 : OV-3

2.4.4 OV-4: Organisational relationships chart

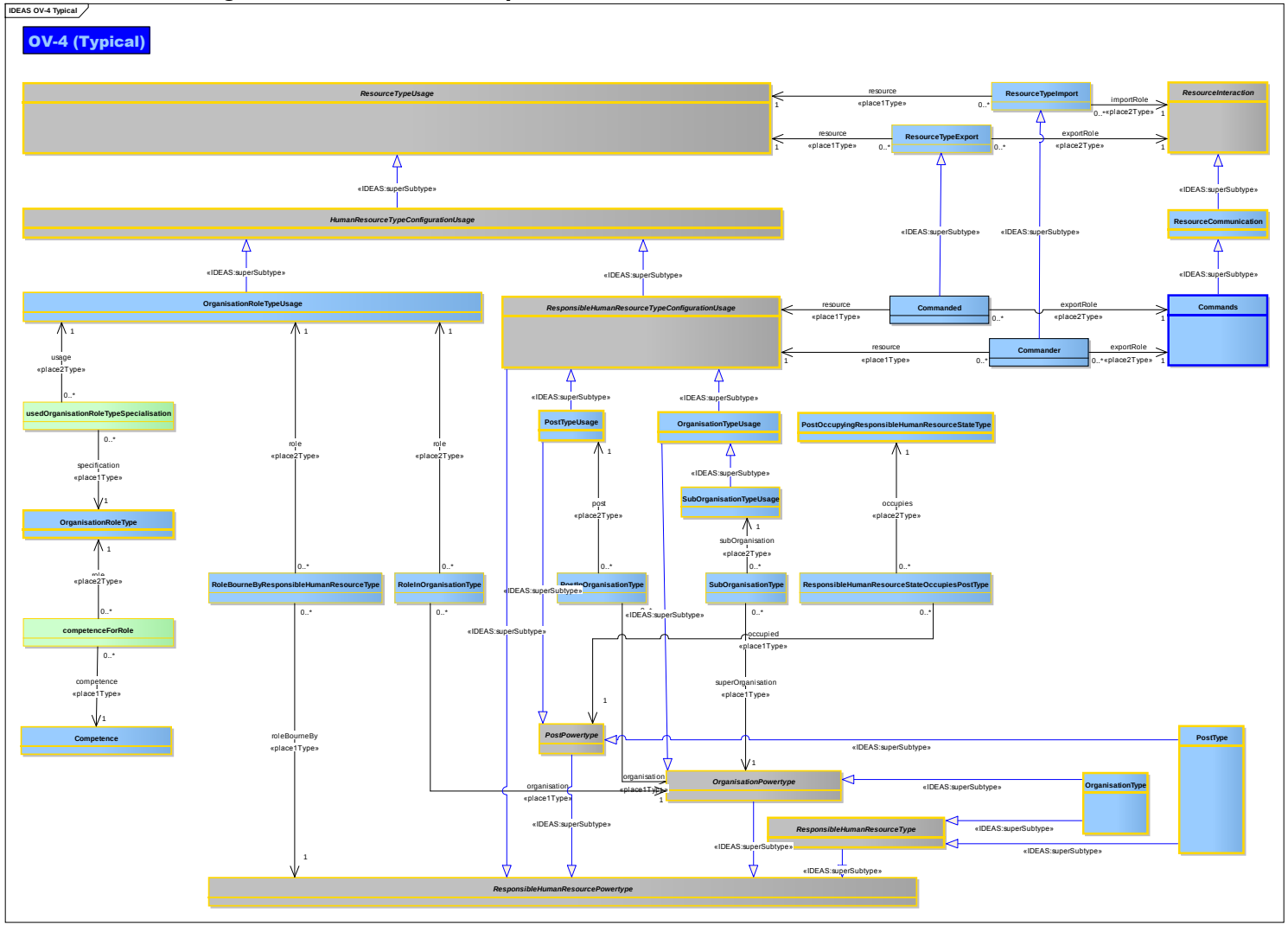


Figure 35 : OV-4 Typical

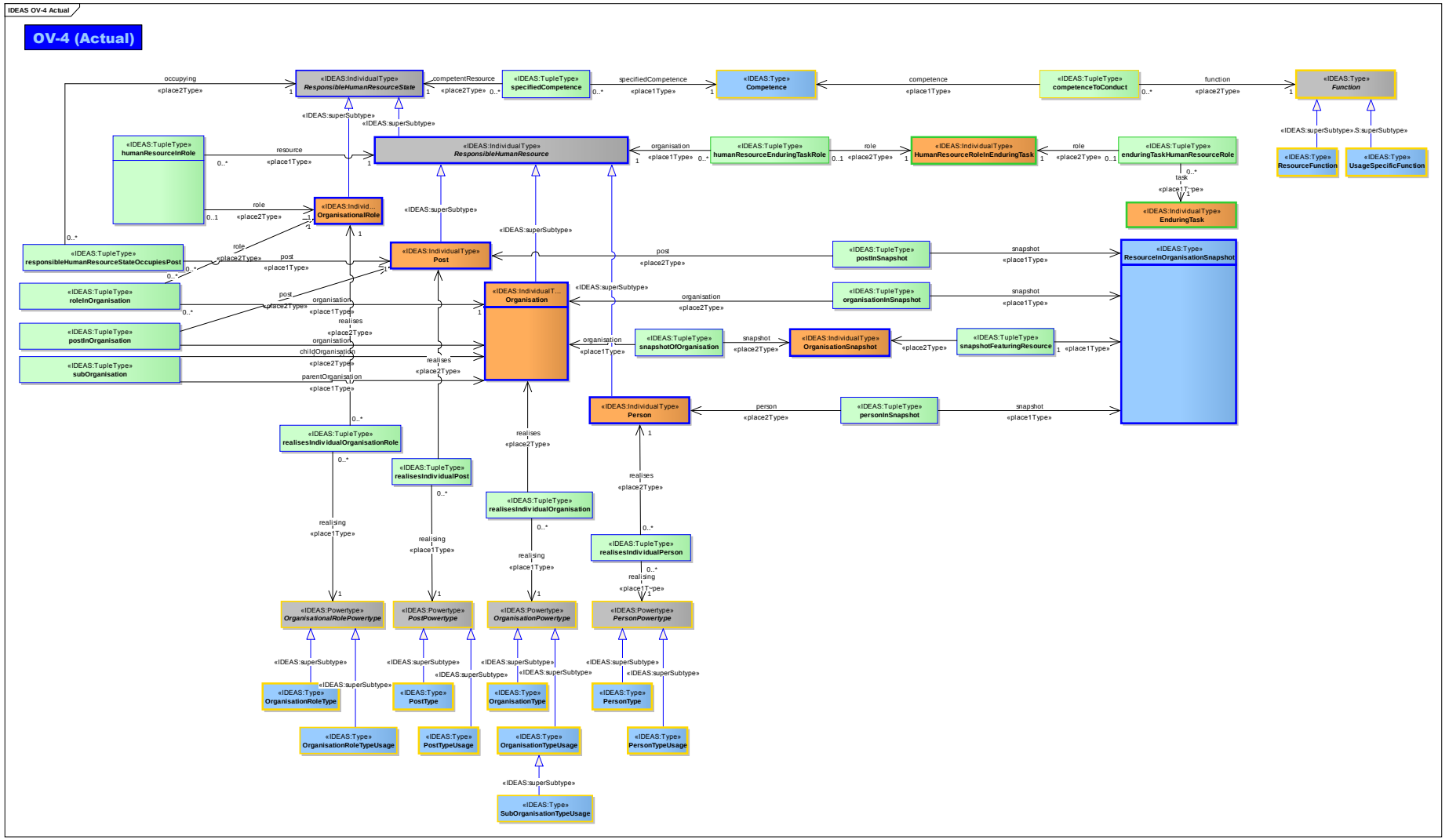


Figure 36 : OV-4 Actual

2.4.5 OV-5: Operational activity model

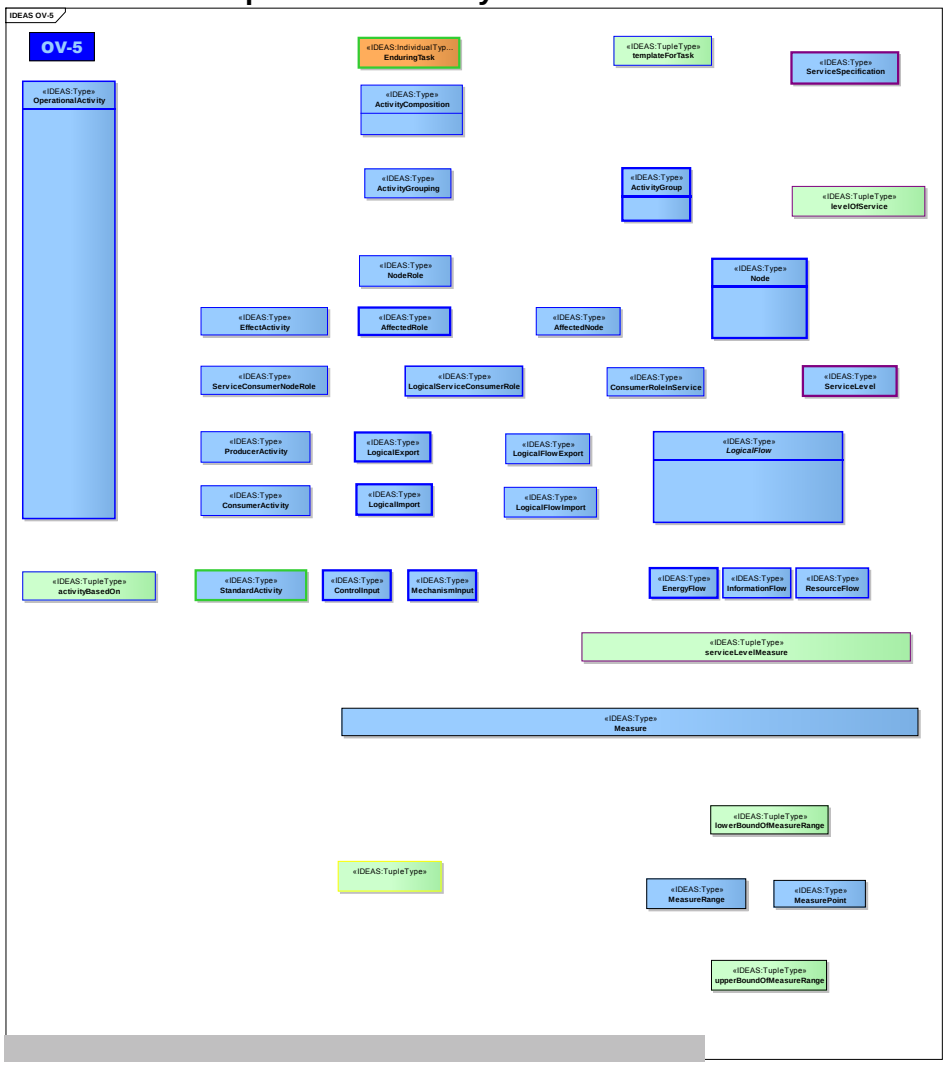


Figure 37 : OV-5

2.4.6 OV-6: Operational rules, state descriptions and event-trace description

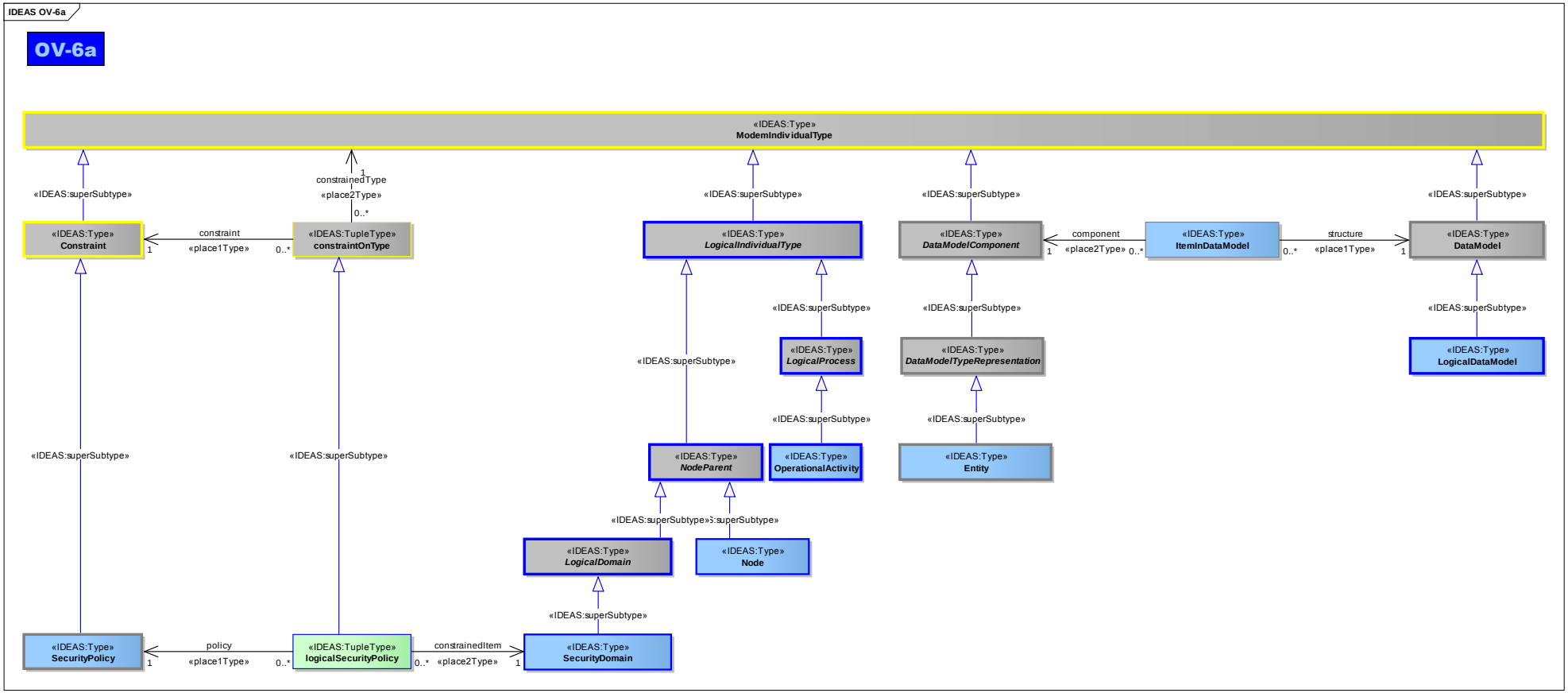


Figure 38 : OV-6a

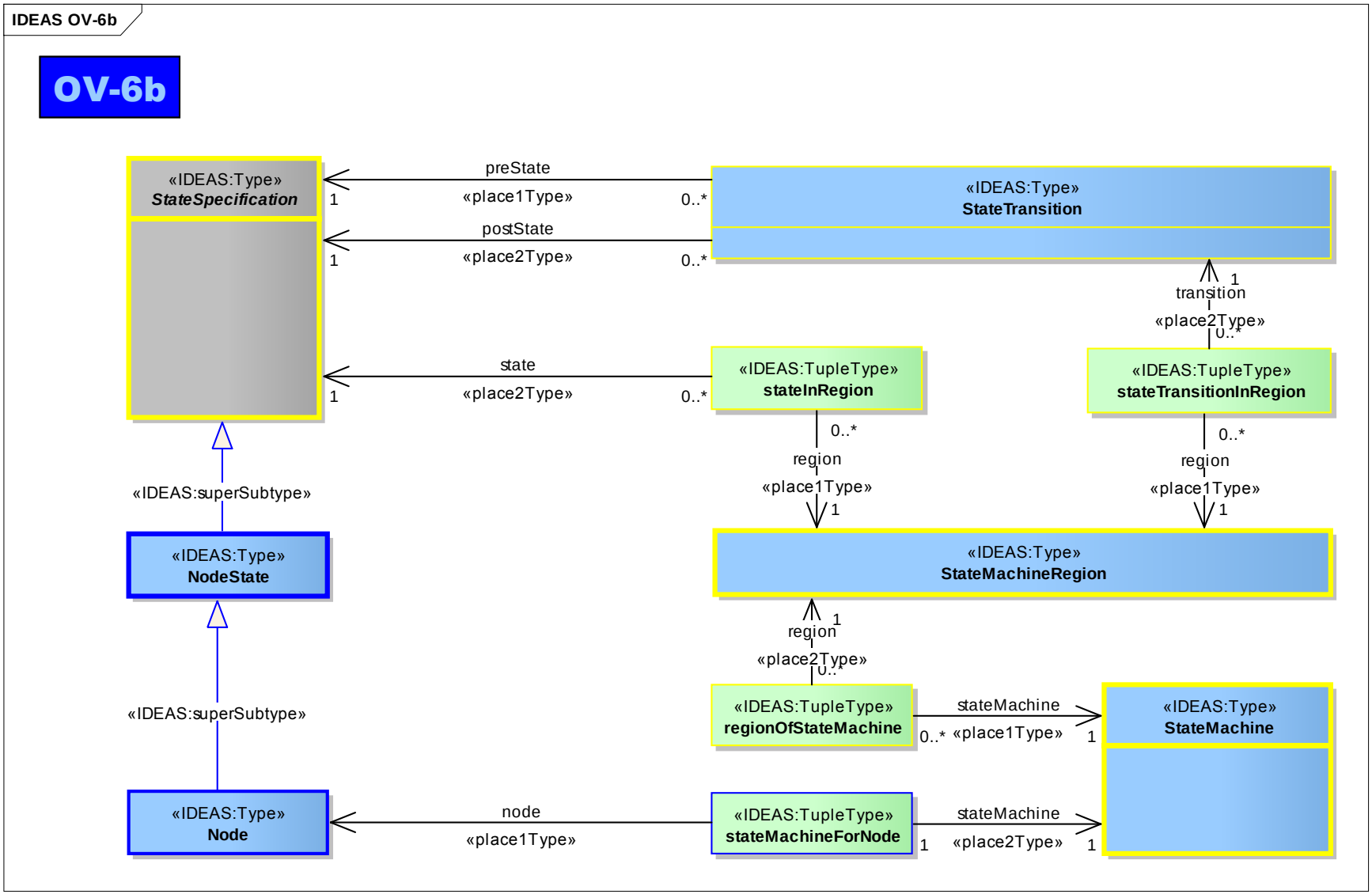


Figure 39 : OV-6b

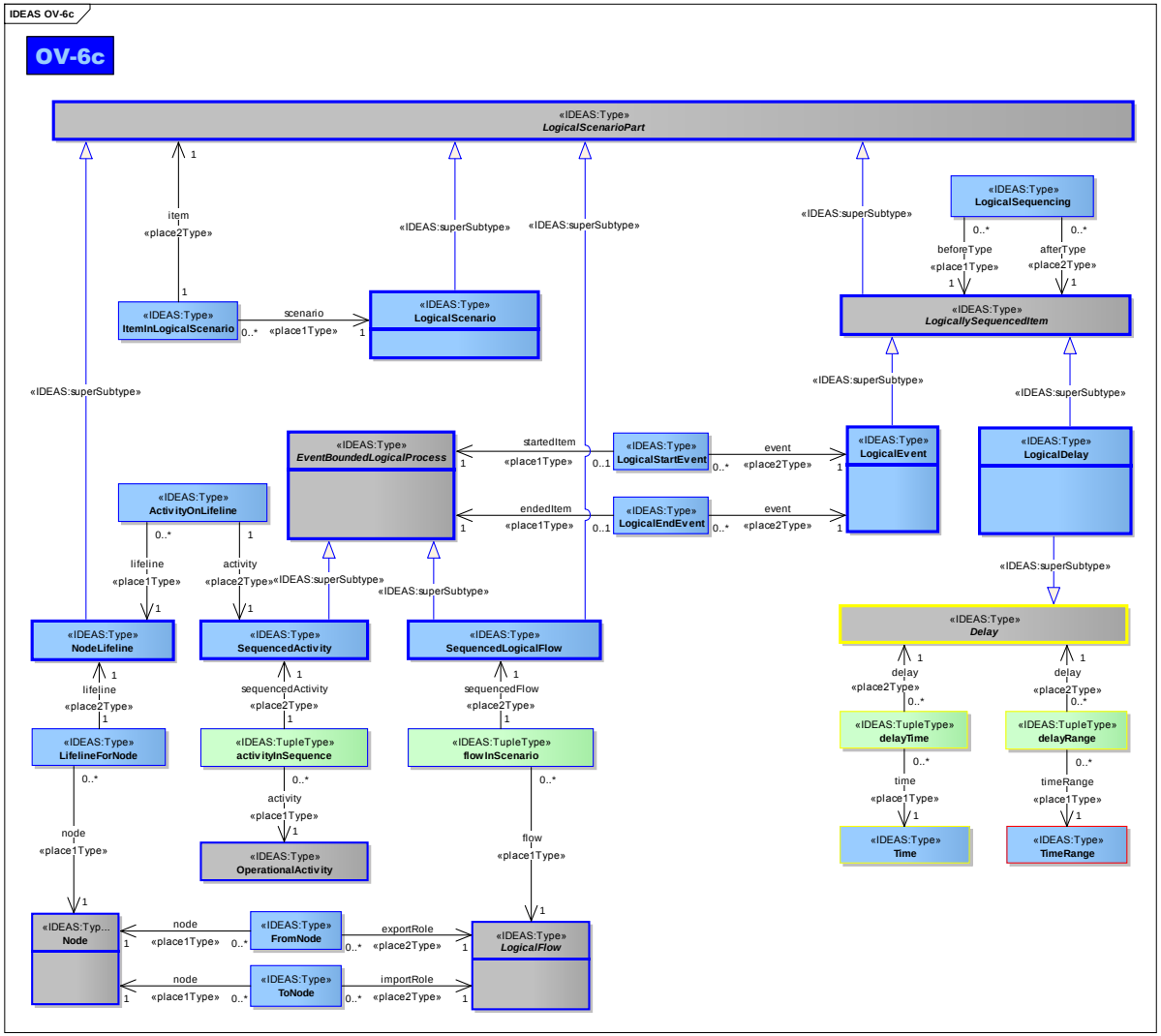


Figure 40 : OV-6c

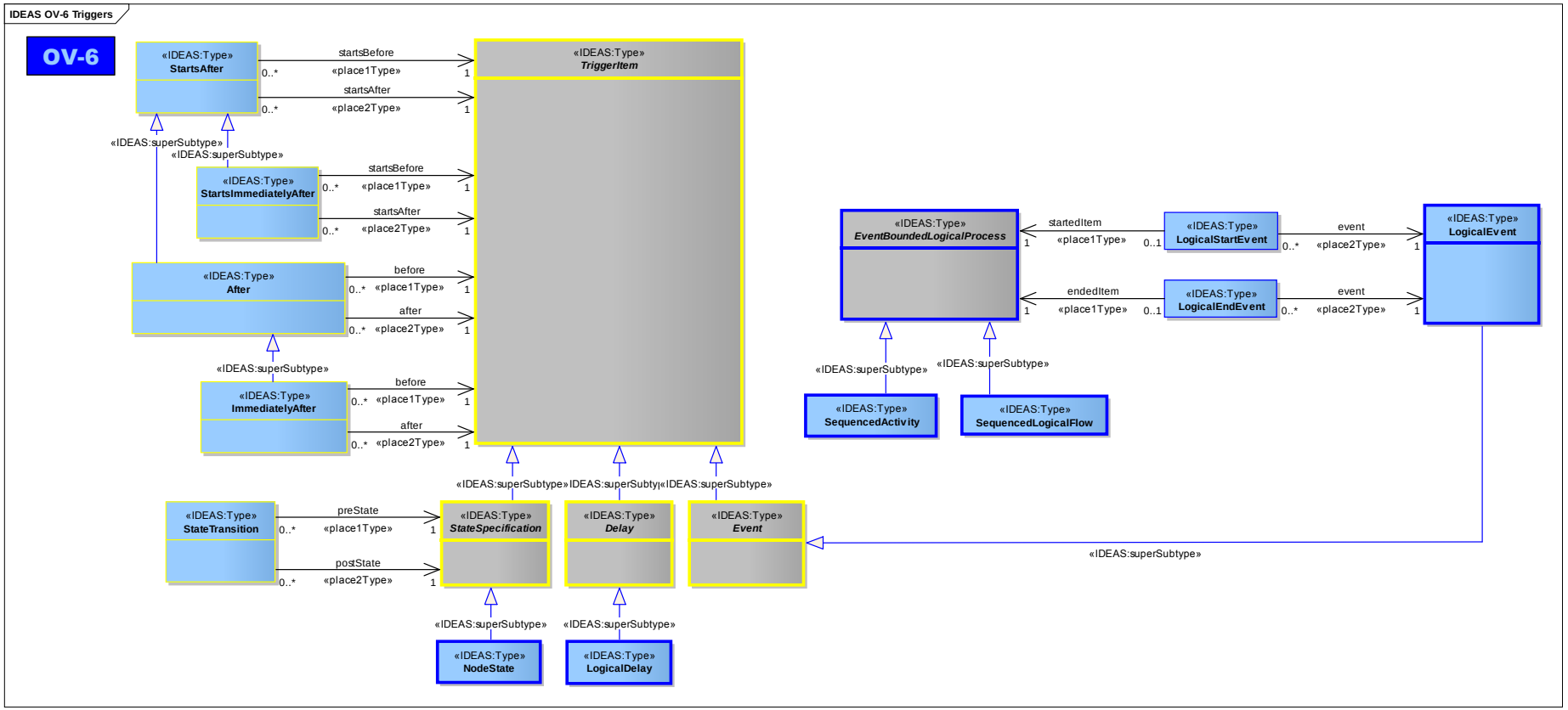


Figure 41 : OV-6 Triggers

2.4.7 OV-7: Information model

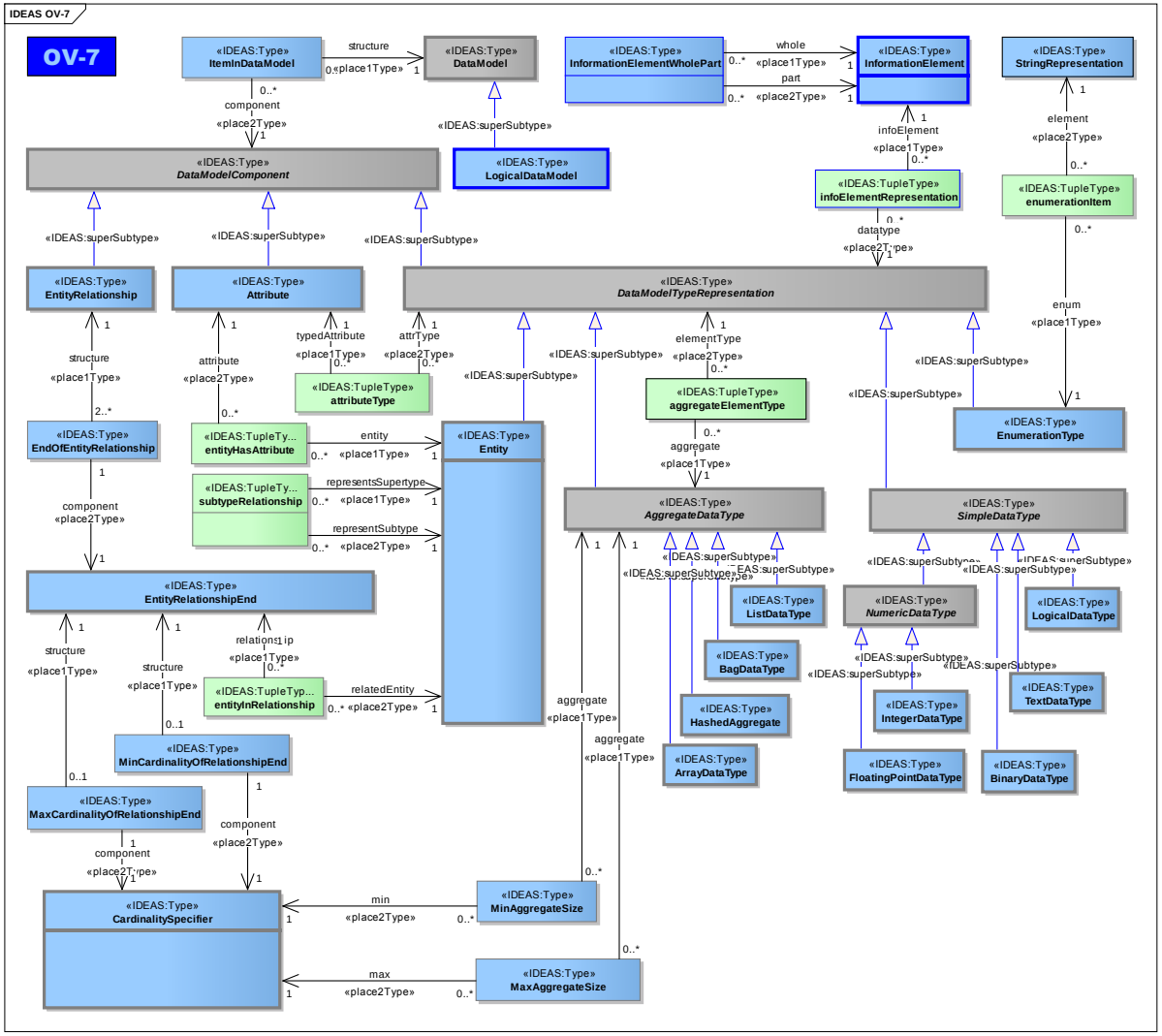


Figure 42 : OV-7

2.4.8 Operational Views elements list

Operational Views
Commands «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» Commands - ResourceCommunication *Attributes:* - A ResourceCommunication where one ResponsibleHumanResourceTypeConfigurationUsage commands another.
HumanResource «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» HumanResource - HumanResourceState *Generalization (element - is a subtype of):*«IDEAS:superSubtype» HumanResource - IndividualResource *Dependency (element - is instance of):*«IDEAS:powertypeInstance» HumanResource - HumanResourcePowertype *Attributes:* - An IndividualResource that is composed entirely of human resources. Note: was called "OrganisationalResource" in M3.
Manager «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» Manager - OrganisationalRole *Attributes:* - An OrganisationalRole where the Person's role in the Organisation is as a Manager. Example: when the Organisation is a Project, the role would be as project manager.
NodeRole «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» NodeRole - RoleInLogicalProcess *Generalization (element - is a subtype of):*«IDEAS:superSubtype» NodeRole - ParticipationExtentType *Dependency (element - is instance of):* *Association (source - target):*«place2Type» NodeRole - OperationalActivity *Association (source - target):*«place1Type» NodeRole - Node *Attributes:* - A RoleInLogicalProcess which is the extent of a Node's participation in an OperationalActivity. Note: An OperationalActivity can only be conducted by one Node.
OperationalActivity «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» OperationalActivity - LogicalProcess *Attributes:* - A ProcessType that is a type of logical process, specified independently of how the process is carried out. Note: an OperationalActivity may only be carried out by a logical Node.

OrganisationalRole «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationalRole - HumanResource
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationalRole - OrganisationPart
Dependency (element - is instance of):«IDEAS:powertypeInstance»
OrganisationalRole - OrganisationalRolePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationalRole - ConstructedHumanResource
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationalRole - ResponsibleHumanResourceState
Attributes:
-
A ConstructedHumanResource that is the state of the ResponsibleHumanResource (that part of its life) where it has the role in an Organisation. Where a role carries the authority to undertake a function - though the human resource given the role has the responsibility.

Person «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
Person - PersonPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Person - ResponsibleHumanResource
Attributes:
-
An individual human being.

ResourceInOrganisationSnapshot «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInOrganisationSnapshot - Doubleton
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInOrganisationSnapshot - SetOfOverlappingIndividuals
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInOrganisationSnapshot - ModemIndividualType
Attributes:
-
A SetOfOverlappingIndividuals whose members are an OrganisationSnapshot and

ResponsibleHumanResource «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResource - ResponsibleHumanResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResource - AgentCapableOfResponsibility
Dependency (element - is instance of):«IDEAS:powertypeInstance»
ResponsibleHumanResource - ResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResource - Stakeholder
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResource - HumanResource
Attributes:

-
A Person, Post or Organisation. These can be held responsible for their actions, hence are responsible human resources.

ResponsibleHumanResourceState «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
ResponsibleHumanResourceState - ResponsibleHumanResourceStatePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceState - AgentCapableOfResponsibilityState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceState - ResponsibleHumanResourcePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceState - IndividualResourceState
Attributes:
-
A temporal stage of a ResponsibleHumanResource.

ResponsibleOwner «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleOwner - OrganisationalRole
Attributes:
-
An OrganisationProjectRole where the ResponsibleHumanResource is the responsible for the Organisation - e.g. a project owner.

ServiceConsumerNodeRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceConsumerNodeRole - ProcessWholeRoleExtentPartType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceConsumerNodeRole - ModemWholePartType
Association (source - target):«place1Type»
ServiceConsumerNodeRole - OperationalActivity
Association (source - target):«place2Type»
ServiceConsumerNodeRole - LogicalServiceConsumerRole
Attributes:
-
A ProcessWholeRoleExtentType that relates an OperationalActivity to the role of a ServiceSpecification that supports it.

humanResourceInRole «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
humanResourceInRole - RoleBourneByResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
humanResourceInRole - individualResourceUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
humanResourceInRole - stateOfResponsibleHumanResourceState
Association (source - target):«place1Type»
humanResourceInRole - ResponsibleHumanResource
Association (source - target):«place2Type»
humanResourceInRole - OrganisationalRole
Attributes:

-
A responsibleHumanResourceState relationship between the OrganisationRole and the ResponsibleHumanResource that bears the responsibility. Note: the OrganisationRole cannot be passed on. Instead, a new instance of the role is created.

personInSnapshot «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
personInSnapshot - resourceInSnapshot
Association (source - target):«place2Type»
personInSnapshot - Person
Association (source - target):«place1Type»
personInSnapshot - ResourceInOrganisationSnapshot
Attributes:
-
A resourceInSnapshot where the resource is a Person.

postInOrganisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
postInOrganisation - organisationWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
postInOrganisation - agentCapableOfResponsibilityWholeAndPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
postInOrganisation - individualResourceUsage
Dependency (element - is instance of):«IDEAS:powertypeInstance»
postInOrganisation - PostInOrganisationPowertype
Association (source - target):«place1Type»
postInOrganisation - Organisation
Association (source - target):«place2Type»
postInOrganisation - Post
Attributes:
-
An organisationWholePart that asserts a Post is part of an Organisation.

responsibleHumanResourceStateOccupiesPost «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
responsibleHumanResourceStateOccupiesPost - stateOfResponsibleHumanResourceState
Dependency (element - is instance of):«IDEAS:powertypeInstance»
responsibleHumanResourceStateOccupiesPost - ResponsibleHumanResourceStateOccupiesPostPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
responsibleHumanResourceStateOccupiesPost - individualResourceStateUsage
Association (source - target):«place2Type»
responsibleHumanResourceStateOccupiesPost - ResponsibleHumanResourceState
Association (source - target):«place1Type»
responsibleHumanResourceStateOccupiesPost - Post
Attributes:
-
A ResponsibleHumanResourceWholeState relationship between a Post and state of the ResponsibleHumanResource that is occupying the Post. Note: Any type of ResponsibleHumanResource's state can occupy a post.

roleInOrganisation «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
roleInOrganisation - RoleInOrganisationPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
roleInOrganisation - individualResourceUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
roleInOrganisation - responsibleHumanResourceWholePart
Association (source - target):«place1Type»
roleInOrganisation - Organisation
Association (source - target):«place2Type»
roleInOrganisation - OrganisationalRole
Attributes:
-
A ResponsibleHumanResourceWholePart relationship between an Organisation and one of its OrganisationalRoles.

subOrganisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
subOrganisation - individualResourceUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
subOrganisation - organisationWholePart
Dependency (element - is instance of):«IDEAS:powertypeInstance»
subOrganisation - SubOrganisationPowertype
Association (source - target):«place2Type»
subOrganisation - Organisation
Association (source - target):«place1Type»
subOrganisation - Organisation
Attributes:
-
An organisationWholePart that asserts one Organisation is a component of another Organisation. Note: The childOrganisation is a component of the parentOrganisation, so this relation is not intended to model situations where a subsidiary Organisation moves from one parent to another.

ActivityComposition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ActivityComposition - ProcessWholeAndPartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ActivityComposition - TypicalWholePart
Association (source - target): «place2Type»
ActivityComposition - OperationalActivity
Association (source - target): «place1Type»
ActivityComposition - OperationalActivity
Attributes:
-
A TypicalWholePart that relates a parent (whole) OperationalActivity to its child (part).

ActivityGroup «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ActivityGroup - OperationalActivity
Attributes:

-
An OperationalActivity that is entirely composed of other OperationalActivities.

ActivityGrouping «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ActivityGrouping - ActivityComposition
Association (source - target):«place1Type»
ActivityGrouping - ActivityGroup
Association (source - target):«place2Type»
ActivityGrouping - OperationalActivity
Attributes:
-
An ActivityComposition where the parent Activity is an ActivityGroup.

ActivityOnLifeline «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ActivityOnLifeline - TypicalWholePart
Association (source - target): «place2Type»
ActivityOnLifeline - SequencedActivity
Association (source - target): «place1Type»
ActivityOnLifeline - NodeLifeline
Attributes:
-
A TypicalWholePart where a SequencedActivity is part of a NodeLifeline. Note: a given SequencedActivity may appear on one and only one NodeLifeline.

AffectedNode «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
AffectedNode - ModemWholePartType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
AffectedNode - TypicalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
AffectedNode - IndividualRoleType
Association (source - target):«place1Type»
AffectedNode - Node
Association (source - target):«place2Type»
AffectedNode - AffectedRole
Attributes:
-
An IndividualRoleType where the role extent is an AffectedRole and the whole is a Node.

AffectedRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
AffectedRole - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
AffectedRole - RoleExtentType
Attributes:
-
A RoleExtentType that corresponds to the part of a Node affected by an Activity that acts upon it. Note: by "part", this includes temporal parts, so all of the Node may be affected for a period of time. This was

previously <<ActsUpon>> in M3.

Commanded «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Commanded - ResourceTypeExport
Association (source - target):«place2Type»
Commanded - Commands
Association (source - target):«place1Type»
Commanded - ResponsibleHumanResourceTypeConfigurationUsage
Attributes:
-
A ResourceTypeExport that asserts the ResponsibleHumanResourceTypeConfigurationUsage commanded participation in command.

Commander «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Commander - ResourceTypeImport
Association (source - target):«place2Type»
Commander - Commands
Association (source - target):«place1Type»
Commander - ResponsibleHumanResourceTypeConfigurationUsage
Attributes:
-
A ResourceTypeExport that asserts the ResponsibleHumanResourceTypeConfigurationUsage participation as the commander in the command.

ConstructedHumanResource «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConstructedHumanResource - HumanResource
Attributes:
-
A HumanResource that is intentionally constructed. An OrganisationRole, Post, or Organisation.

ConsumerActivity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConsumerActivity - IndividualExchangeRoleType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConsumerActivity - ModemWholePartType
Association (source - target): «place1Type»
ConsumerActivity - OperationalActivity
Association (source - target): «place2Type»
ConsumerActivity - LogicalImport
Attributes:
-
An IndividualExchangeRoleType where an OperationalActivity is the consumer of a LogicalFlow.

ConsumerRoleInService «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConsumerRoleInService - AgentParticipationType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConsumerRoleInService - ModemWholePartType
Association (source - target): «place2Type»
ConsumerRoleInService - LogicalServiceConsumerRole
Association (source - target): «place1Type»
ConsumerRoleInService - ServiceLevel
Attributes:
-
A AgentParticipationType that relates a ServiceSpecification to its role in supporting an OperationalActivity.

ControlInput «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ControlInput - LogicalImport
Attributes:
-
A LogicalImport where the imported LogicalFlow controls the OperationalActivity. Note: this exists to provide compatibility with IDEF0.

DomainInArchitecture «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DomainInArchitecture - ModemWholePartType
Association (source - target):«place1Type»
DomainInArchitecture - LogicalArchitecture
Association (source - target):«place2Type»
DomainInArchitecture - LogicalDomain
Attributes:
-
A ModemWholePartType that asserts a LogicalDomain is part of a LogicalArchitecture.

EffectActivity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EffectActivity - TypicalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EffectActivity - ModemWholePartType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EffectActivity - ProcessWholeRoleExtentPartType
Association (source - target):«place1Type»
EffectActivity - OperationalActivity
Association (source - target):«place2Type»
EffectActivity - AffectedRole
Attributes:
-
A ProcessWholeRoleExtentPartType where the ProcessType is an OperationalActivity.

EnergyFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EnergyFlow - LogicalFlow
Attributes:
-
A LogicalFlow where energy is transferred from one Node to another.

EventBoundedLogicalProcess «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EventBoundedLogicalProcess - LogicalProcess
Attributes:
-
A LogicalProcess that can have LogicalEvents marking its start and end points.

FlowBundle «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FlowBundle - TypicalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FlowBundle - ExchangeWholeAndPartType
Association (source - target):«place2Type»
FlowBundle - LogicalFlow
Association (source - target):«place1Type»
FlowBundle - FlowGroup
Attributes:
-
A TypicalWholePart where the whole is a FlowGroup and the part is a LogicalFlow.

FlowGroup «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FlowGroup - LogicalFlow
Attributes:
-
A LogicalFlow that is composed of other LogicalFlows.

FlowInDomain «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FlowInDomain - ModemWholePartType
Association (source - target):«place1Type»
FlowInDomain - LogicalDomain
Association (source - target):«place2Type»
FlowInDomain - LogicalFlow
Attributes:
-
A ModemWholePartType that asserts a LogicalFlow lies within an LogicalDomain.

FlowedElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FlowedElement - ModemIndividualType
Attributes:
-
A ModafIndividualType that can be flowed along a LogicalFlow.

FlowedElementRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FlowedElementRole - ModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FlowedElementRole - ExchangedItemRoleType
Attributes:
-
An ExchangedItemRoleType where a FlowedElement is exchanged along a LogicalFlow.

FromNode «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FromNode - RoleInLogicalProcess
Association (source - target):«place2Type»
FromNode - LogicalFlow
Association (source - target):«place1Type»
FromNode - Node
Attributes:
-
An RoleInLogicalProcess where a LogicalFlow flows from a Node.

InformationElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationElement - InformationInstanceType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationElement - Information
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InformationElement - FlowedElement
Attributes:
-
An InformationInstanceType that flows between OperationalActivities and Nodes. The structure of an InformationElement may be defined using a LogicalDataModel.

InformationElementWholePart «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InformationElementWholePart - TypicalWholePart
Association (source - target): «place2Type»
InformationElementWholePart - InformationElement
Association (source - target): «place1Type»
InformationElementWholePart - InformationElement
Attributes:
-

A TypicalWholePart where one InformationElement is a part of another.
InformationFlow «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» InformationFlow - LogicalFlow *Attributes:* - A LogicalFlow where the FlowedElement is information.
InformationRole «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» InformationRole - FlowedElementRole *Attributes:* - A FlowedElementRole where information is flowed.
ItemInLogicalScenario «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ItemInLogicalScenario - ItemInScenario *Association (source - target):* «place2Type» ItemInLogicalScenario - LogicalScenarioPart *Association (source - target):* «place1Type» ItemInLogicalScenario - LogicalScenario *Attributes:* - An ItemInScenario where the item (part) is a LogicalScenarioItem and the scenario (whole) is a LogicalScenario.
KnownResource «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» KnownResource - Node *Attributes:* - A ResourceType that plays a part in a LogicalArchitecture. Note: An OV-2 is meant to show logical interactions between nodes. However, sometimes it is known that a connection runs to/from a particular type of resource.
LifelineForNode «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» LifelineForNode - StateOfNode *Association (source - target):* «place2Type» LifelineForNode - NodeLifeline *Association (source - target):* «place1Type» LifelineForNode - Node *Attributes:* - A StateOfNode that asserts that a NodeLifeLine is a typical temporal part of a Node.

LogicalDataModel «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalDataModel - DataModel
Attributes:
-
A DataModel that is a specification of business information requirements as a formal data structure, where relationships and classes (entities) are used to specify the logic which underpins the information.

LogicalDelay «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalDelay - LogicallySequencedItem
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalDelay - Delay
Attributes:
-
A LogicalSequencedItem that is part of a LogicalScenario that has a specified temporal extent, but an unspecified spatial extent.

LogicalDomain «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalDomain - NodeParent
Attributes:
-
A NodeParent that is a collection of Nodes that share some common feature.

LogicalEndEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalEndEvent - ModemTemporalWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalEndEvent - EndBorderType
Association (source - target): «place1Type»
LogicalEndEvent - EventBoundedLogicalProcess
Association (source - target): «place2Type»
LogicalEndEvent - LogicalEvent
Attributes:
-
An EndtBorderType that relates a LogicallySequencedItem to the LogicalEvent that marks its end
Note: there may be no more than one LogicalEndEvent for a given LogicallySequencedItem.

LogicalEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalEvent - Event
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalEvent - LogicallySequencedItem
Attributes:
-
An Event that marks the beginning or end of a LogicalActivity.

LogicalExport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalExport - SendType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalExport - LogicalProcess
Attributes:
-
A SendType where a LogicalFlow exports from a Node or OperationalActivity. Note: this is the equivalent of OpActivityOutputPin in M3.

LogicalFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalFlow - LogicalProcess
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalFlow - ExchangeType
Attributes:
-
An ExchangeType that flows between OperationalActivities and/or Nodes.

LogicalFlowExport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalFlowExport - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalFlowExport - SendInExchangeType
Association (source - target): «place2Type»
LogicalFlowExport - LogicalExport
Association (source - target): «place1Type»
LogicalFlowExport - LogicalFlow
Attributes:
-
A SendInExchangeType where a LogicalFlow exports from a Node or OperationalActivity.

LogicalFlowImport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalFlowImport - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalFlowImport - ReceiveInExchangeType
Association (source - target): «place1Type»
LogicalFlowImport - LogicalFlow
Association (source - target): «place2Type»
LogicalFlowImport - LogicalImport
Attributes:
-
A ReceiveInExchangeType where a LogicalFlow imports to a Node or OperationalActivity.

LogicalImport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalImport - ReceiveType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalImport - LogicalProcess
Attributes:
-
A ReceiveType where a LogicalFlow imports to a Node or OperationalActivity. Note: this is the equivalent of OpActivityInputPin in M3.

LogicalIndividualType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalIndividualType - ModemIndividualType
Attributes:
-
A ModemIndividualType that is specified independently of any implemenation mechanism (i.e. without specifying the ResourceType).

LogicalProcess «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalProcess - ProcessType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalProcess - LogicalIndividualType
Attributes:
-
A ProcessType used to specify functionality without being specific about the type of Resource that provides the functionality.

LogicalScenario «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalScenario - LogicalScenarioPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalScenario - Scenario
Attributes:
-
A Scenario that does not specify particular ResourceTypes - i.e. one that consists of Nodes and LogicalProcesses.

LogicalScenarioPart «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalScenarioPart - LogicalIndividualType
Attributes:
-
A LogicalIndividualType that is part of a LogicalScenario - note this can include other LogicalScenarios.

LogicalSequencing «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalSequencing - ImmediateBeforeAfterType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalSequencing - ModemThing
Association (source - target): «place2Type»

LogicalSequencing - LogicallySequencedItem
Association (source - target): «place1Type»
LogicalSequencing - LogicallySequencedItem
Attributes:
-
An ImmediateBeforeAfterType that asserts one LogicallySequencedItem occurs immediately after the other.

LogicalServiceConsumerRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalServiceConsumerRole - ModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalServiceConsumerRole - ParticipationExtentType
Attributes:
-
A ParticipationExtentType which is the extent of an OperationalActivity's participation in as the consumer of a ServiceSpecification.

LogicalStartEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalStartEvent - ModemTemporalWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicalStartEvent - StartBorderType
Association (source - target): «place2Type»
LogicalStartEvent - LogicalEvent
Association (source - target): «place1Type»
LogicalStartEvent - EventBoundedLogicalProcess
Attributes:
-
A StartBorderType that relates an EventBoundedLogicalProcess to the LogicalEvent that marks its start. Note: there may be no more than one LogicalStartEvent for a given EventBoundedLogicalProcess.

LogicallySequencedItem «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LogicallySequencedItem - LogicalScenarioPart
Attributes:
-
A LogicalScenarioPart which may be temporally ordered using LogicalSequencing.

MechanismInput «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MechanismInput - LogicalImport
Attributes:
-
A LogicalImport where the imported LogicalFlow provides a mechanism for conducting the OperationalActivity. Note: this exists to provide compatibility with IDEF0.

Needline «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Needline - FlowGroup
Attributes:
-

A FlowGroup that is a bundle of LogicalFlows between Nodes.
Node «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Node - NodeState *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Node - BodyType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Node - NodeParent *Attributes:* - A NodeState that is used in context of a NodeParent.
NodeEnvironment «IDEAS:Type» *Connectors:* *Association (source - target):*«place1Type» NodeEnvironment - EnvironmentalFactor *Association (source - target):*«place2Type» NodeEnvironment - Node *Generalization (element - is a subtype of):*«IDEAS:superSubtype» NodeEnvironment - TypicalWholePart *Attributes:* - A TypicalWholePart that indicates an of EnvironmentalFactor of the environment in which the Node will operate.
NodeLifeline «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» NodeLifeline - NodeState *Generalization (element - is a subtype of):* «IDEAS:superSubtype» NodeLifeline - LogicalScenarioPart *Attributes:* - A NodeState whose extent is defined by a LogicalScenario.
NodeParent «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» NodeParent - AgentType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» NodeParent - LogicalIndividualType *Attributes:* - A LogicalIndividualType that is any type of thing that has parts that are Nodes.
NodeState «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» NodeState - LogicalIndividualType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» NodeState - StateSpecification *Attributes:*

-
A LogicalIndividualType that is a type of state that a Node can be in. This includes the limit case of the whole-life state type - i.e. the Node itself.

NodeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NodeUsage - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NodeUsage - AgentWholeAndPartType
Association (source - target): «place2Type»
NodeUsage - Node
Association (source - target): «place1Type»
NodeUsage - NodeParent
Attributes:
-
An AgentWholeAndPartType where a NodeParent has a Node as a part.

Organisation «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Organisation - OrganisationState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Organisation - OrganisationPowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Organisation - ConstructedHumanResource
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Organisation - Undertaking
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Organisation - ResponsibleHumanResource
Attributes:
-
A ConstructedHumanResource which is an Organisation.

OrganisationPart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationPart - ResponsibleHumanResourcePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationPart - UndertakingPart
Attributes:
-
A ResponsibleHumanResourcePart that is part of an Organisation.

OrganisationSnapshot «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationSnapshot - OrganisationPart
Attributes:
-
An OrganisationPart that is a timeslice of an Organisation or part of an Organisation (i.e. and incomplete snapshot) at a particular point in time. Note: most org charts represent a OrganisationSnapshot.

OrganisationState «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
OrganisationState - OrganisationStatePowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationState - OrganisationPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationState - ResponsibleHumanResourceState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationState - UndertakingState
Attributes:
-
A ResponsibleHumanResourceState which is a temporal part of an Organisation.

Post «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Post - ResponsibleHumanResource
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Post - OrganisationPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Post - ConstructedHumanResource
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Post - PostPowertype
Attributes:
-
A ConstructedHumanResource that is a position in an Organisation that may be filled wholly or partly by a ResponsibleHumanResource; in other words, by an Organisation, Person or Post. As the position is in the Organisation, it is a part of the Organisation.

ProblemDomain «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProblemDomain - LogicalDomain
Attributes:
-
A LogicalDomain that contains (has parts that are) those Nodes which may be realised by physical resources specified in SV-1. There may be more than one alternative solution for a given ProblemDomain specified as a set of SV suites. There may be only one ProblemDomain in a LogicalArchitecture.

ProducerActivity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProducerActivity - IndividualExchangeRoleType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProducerActivity - ModemWholePartType
Association (source - target): «place1Type»
ProducerActivity - OperationalActivity
Association (source - target): «place2Type»
ProducerActivity - LogicalExport
Attributes:
-
An IndividualExchangeRoleType where the involved ProcessType is an OperationalActivity that is the producer of a LogicalFlow.

ResourceFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceFlow - LogicalFlow
Attributes:
-
A LogicalFlow where the flowed element is a ResourceType.

ResourceFlowRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceFlowRole - FlowedElementRole
Attributes:
-
A FlowedElementRole where a ResourceType is flowed.

ResponsibleHumanResourcePart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourcePart - ModemIndividualElement
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourcePart - AgentCapableOfResponsibilityPart
Attributes:
-
An AgentCapableOfResponsibilityPart that is a part of a ResponsibleHumanResource.

RoleInFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInFlow - TypicalWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInFlow - ExchangedItemRoleInExchangeType
Association (source - target): «place1Type»
RoleInFlow - LogicalFlow
Association (source - target): «place2Type»
RoleInFlow - FlowedElementRole
Attributes:
-
An ExchangedItemRoleInExchangeType where the role in exchange is a LogicalFlow.

RoleInInformationFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInInformationFlow - RoleInFlow
Association (source - target): «place1Type»
RoleInInformationFlow - InformationFlow
Association (source - target): «place2Type»
RoleInInformationFlow - InformationRole
Attributes:
-
A RoleInFlow where Information is being flowed.

RoleInLogicalProcess «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInLogicalProcess - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInLogicalProcess - CapableOfType
Association (source - target):«place1Type»
RoleInLogicalProcess - Node
Association (source - target):«place2Type»
RoleInLogicalProcess - LogicalProcess
Attributes:
-
A CapableOfType that asserts that a Node conducts a LogicalProcess.

RoleInResourceFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInResourceFlow - RoleInFlow
Association (source - target): «place1Type»
RoleInResourceFlow - ResourceFlow
Association (source - target): «place2Type»
RoleInResourceFlow - ResourceFlowRole
Attributes:
-
A RoleInFlow where a ResourceType is being flowed.

RoleOfFlowedElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleOfFlowedElement - IndividualRoleAsExchangedItemType
Association (source - target): «place1Type»
RoleOfFlowedElement - FlowedElement
Association (source - target): «place2Type»
RoleOfFlowedElement - FlowedElementRole
Attributes:
-
An IndividualRoleAsExchangedItemType where a FlowedElement is exchanged.

RoleOfInformation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleOfInformation - RoleOfFlowedElement
Association (source - target): «place1Type»
RoleOfInformation - InformationElement
Association (source - target): «place2Type»
RoleOfInformation - InformationRole
Attributes:
-
A RoleOfFlowedElement where the flowed element is Information.

RoleOfResourceInFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleOfResourceInFlow - RoleOfFlowedElement
Association (source - target): «place1Type»
RoleOfResourceInFlow - ResourceType
Association (source - target): «place2Type»
RoleOfResourceInFlow - ResourceFlowRole
Attributes:
-
A RoleOfFlowedElement where the flowed element is ResourceType.

SecurityDomain «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SecurityDomain - LogicalDomain
Attributes:
-
A LogicalDomain whose parts all share a common SecurityPolicy.

SequencedActivity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SequencedActivity - EventBoundedLogicalProcess
Attributes:
-
A LogicalProcess that is the typical useage of an OperationalActivity in a NodeLifeLine.

SequencedLogicalFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SequencedLogicalFlow - EventBoundedLogicalProcess
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SequencedLogicalFlow - LogicalScenarioPart
Attributes:
-
A LogicalProcess that is the typical useage of a LogicalFlow between two NodeLifeLines.

StateOfNode «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StateOfNode - TypicalTemporalWholePart
Association (source - target): «place2Type»
StateOfNode - NodeState
Association (source - target): «place1Type»
StateOfNode - Node
Attributes:
-
A TypicalTemporalWholePart that relates a Node to a NodeState.

ToNode «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ToNode - RoleInLogicalProcess
Association (source - target):«place2Type»
ToNode - LogicalFlow
Association (source - target):«place1Type»
ToNode - Node
Attributes:
-
An RoleInLogicalProcess where a LogicalFlow flows to a Node.

TrustLine «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
TrustLine - LogicalFlow
Attributes:
-
A LogicalFlow that asserts that the trusting Party (either a Node or a KnownResource) trusts the trustedParty to a given level (indicated by the level attribute). Note: No unit of measure is associated with the level - security architects must define their own scale of trust levels for a given architecture or set of architectures.

activityBasedOn «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
activityBasedOn - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
activityBasedOn - StandardActivity
Association (source - target):«place2Type»
activityBasedOn - OperationalActivity
Attributes:
-
A modemIndividualTypeSpecialisation that asserts and OperationalActivity is based on a StandardActivity - e.g. a specialist usage of doctrine.

activityInSequence «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
activityInSequence - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»
activityInSequence - OperationalActivity
Association (source - target): «place2Type»
activityInSequence - SequencedActivity
Attributes:
-
A modemIndividualTypeSpecialisation that relates an OperationaActivity to its usage (as a SequencedActivity) on a NodeLifeLine. Note: A SequencedActivity is based on only one OperationaActivity.

capabilityForNode «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
capabilityForNode - BodyTypeSuperSubType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
capabilityForNode - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»

capabilityForNode - Capability
Association (source - target): «place2Type»
capabilityForNode - Node
Attributes:
-
A BodyTypeSuperSubType that asserts that a Node exhibits or is required to exhibit a Capability.

flowInScenario «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
flowInScenario - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»
flowInScenario - LogicalFlow
Association (source - target): «place2Type»
flowInScenario - SequencedLogicalFlow
Attributes:
-
A modemIndividualTypeSpecialisation that relates a LogicalFlow to its usage (as a SequencedLogicalFlow) in a LogicalScenario. Note: A SequencedLogicalFlow is based on only one LogicalFlow.

infoElementRepresentation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
infoElementRepresentation - representedByDataType
Association (source - target):«place2Type»
infoElementRepresentation - DataModelTypeRepresentation
Association (source - target):«place1Type»
infoElementRepresentation - InformationElement
Attributes:
-
A representedByDataType that asserts an InformationElement is represented by a DataModelTypeRepresentation.

logicalFlowMeasure «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
logicalFlowMeasure - measureOfType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
logicalFlowMeasure - logicalFlowProperty
Association (source - target): «place2Type»
logicalFlowMeasure - LogicalFlow
Association (source - target): «place1Type»
logicalFlowMeasure - MeasureableProperty
Attributes:
-
A logicalFlowProperty and a measureOfType - i.e. an assignment of a MeasureableProperty to a LogicalFlow.

logicalFlowProperty «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
logicalFlowProperty - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
logicalFlowProperty - propertyOfType
Association (source - target): «place2Type»

logicalFlowProperty - LogicalFlow
Association (source - target): «place1Type»
logicalFlowProperty - Property
Attributes:
-
A propertyOfType where the Property applies to a LogicalFlow.

logicalSecurityPolicy «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
logicalSecurityPolicy - constraintOnType
Association (source - target):«place2Type»
logicalSecurityPolicy - SecurityDomain
Association (source - target):«place1Type»
logicalSecurityPolicy - SecurityPolicy
Attributes:
-
A constraintOnType that sets the security policy for LogicalIndividualType.

nodeLocation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
nodeLocation - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
nodeLocation - couple
Association (source - target):«place1Type»
nodeLocation - Location
Association (source - target):«place2Type»
nodeLocation - Node
Attributes:
-
A couple used to assert the Location at/ in which a Node resides. Note: given that OV-2 is a logical model, more often than not, the environment rather than the actual location should be specified - i.e. use NodeEnvironment.

organisationInSnapshot «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
organisationInSnapshot - resourceInSnapshot
Association (source - target): «place1Type»
organisationInSnapshot - ResourceInOrganisationSnapshot
Association (source - target): «place2Type»
organisationInSnapshot - Organisation
Attributes:
-
A resourceInSnapshot where the resource is an Organisation.

postInOrganisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
postInOrganisation - organisationWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
postInOrganisation - agentCapableOfResponsibilityWholeAndPart

Generalization (element - is a subtype of):«IDEAS:superSubtype»
postInOrganisation - individualResourceUsage
Dependency (element - is instance of):«IDEAS:powertypeInstance»
postInOrganisation - PostInOrganisationPowertype
Association (source - target):«place1Type»
postInOrganisation - Organisation
Association (source - target):«place2Type»
postInOrganisation - Post

Attributes:
-
An organisationWholePart that asserts a Post is part of an Organisation.

postInSnapshot «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
postInSnapshot - resourceInSnapshot
Association (source - target): «place1Type»
postInSnapshot - ResourceInOrganisationSnapshot
Association (source - target): «place2Type»
postInSnapshot - Post
Attributes:
-
A resourceInSnapshot where the resource is a Post.

realisesIndividualOrganisation «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
realisesIndividualOrganisation - OrganisationPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
realisesIndividualOrganisation - realisesIndividualResource
Association (source - target):«place2Type»
realisesIndividualOrganisation - Organisation
Attributes:
-
A realisesIndividualResource that asserts a type of organisation is realised by an organisation.

realisesIndividualOrganisationRole «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
realisesIndividualOrganisationRole - OrganisationalRolePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
realisesIndividualOrganisationRole - realisesIndividualResource
Association (source - target):«place2Type»
realisesIndividualOrganisationRole - OrganisationalRole
Attributes:
-
A realisesIndividualResource that asserts a type of organisation role is realised by an organisation role.

realisesIndividualPerson «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
realisesIndividualPerson - PersonPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
realisesIndividualPerson - realisesIndividualResource
Association (source - target):«place2Type»
realisesIndividualPerson - Person
Attributes:
-
A realisesIndividualResource that asserts a type of person is realised by a person.

realisesIndividualPost «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
realisesIndividualPost - PostPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
realisesIndividualPost - realisesIndividualResource
Association (source - target):«place2Type»
realisesIndividualPost - Post
Attributes:
-
A realisesIndividualResource that asserts a type of post is realised by a post.

realisesIndividualResource «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
realisesIndividualResource - modemIndividualTypeInstance
Association (source - target):«place1Type»
realisesIndividualResource - IndividualResourcePowertype
Association (source - target):«place2Type»
realisesIndividualResource - IndividualResource
Attributes:
-
A modemIndividualTypeInstance that asserts a type of Resource is realised by a resource.

requiredMeasureOfPerformance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
requiredMeasureOfPerformance - measureOfType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
requiredMeasureOfPerformance - ModemThing
Association (source - target):«place1Type»
requiredMeasureOfPerformance - Measure
Association (source - target):«place2Type»
requiredMeasureOfPerformance - Node
Attributes:
-
A measureOfType that asserts a Node is required to achieve a level of performance specified by a Measure.

resourceInSnapshot «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
resourceInSnapshot - ResourceInOrganisationSnapshot
Association (source - target): «place2Type»
resourceInSnapshot - HumanResource
Generalization (element - is a subtype of): «IDEAS:superSubtype»
resourceInSnapshot - doubletonTypeInstance
Generalization (element - is a subtype of): «IDEAS:superSubtype»
resourceInSnapshot - overlapTypeIndividualInstance
Generalization (element - is a subtype of): «IDEAS:superSubtype»
resourceInSnapshot - modemIndividualTypeInstance
Attributes:
-
An overlapTypeIndividualInstance where the instance is an OrganisationalResource.

responsibleHumanResourceWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
responsibleHumanResourceWholePart - individualResourceWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
responsibleHumanResourceWholePart - agentCapableOfResponsibilityWholePart
Association (source - target):«place2Type»
responsibleHumanResourceWholePart - ResponsibleHumanResourcePart
Association (source - target):«place1Type»
responsibleHumanResourceWholePart - ResponsibleHumanResource
Attributes:
-
An agentCapableOfResponsibilityWholePart relationship between an OrganisationRole and the Organisation within which it has the role.

snapshotFeaturingResource «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
snapshotFeaturingResource - overlapTypeIndividualInstance
Generalization (element - is a subtype of): «IDEAS:superSubtype»
snapshotFeaturingResource - doubletonTypeInstance
Association (source - target): «place1Type»
snapshotFeaturingResource - ResourceInOrganisationSnapshot
Association (source - target): «place2Type»
snapshotFeaturingResource - OrganisationSnapshot
Generalization (element - is a subtype of): «IDEAS:superSubtype»
snapshotFeaturingResource - modemIndividualTypeInstance
Attributes:
-
An overlapTypeIndividualInstance where the instance is an OrganisationSnapshot.

snapshotOfOrganisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
snapshotOfOrganisation - organisationWholePart
Association (source - target): «place1Type»
snapshotOfOrganisation - Organisation
Association (source - target): «place2Type»
snapshotOfOrganisation - OrganisationSnapshot
Attributes:
-
An organisationWholePart where the whole is an Organisation and the part is an OrganisationSnapshot.

specifiedCompetence «IDEAS:TupleType»
Connectors:
Association (source - target):«place2Type»
specifiedCompetence - ResponsibleHumanResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
specifiedCompetence - ModemThing
Association (source - target):«place1Type»
specifiedCompetence - Competence
Generalization (element - is a subtype of):«IDEAS:superSubtype»
specifiedCompetence - couple
Attributes:
-
A modemIndividualTypeSpecialisation that asserts an ResponsibleHumanResourceState is specified to have a Competence. Note: Was called "actualCompetence" in M3.

stateMachineForNode «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateMachineForNode - appliedStateMachine
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateMachineForNode - ModemThing
Association (source - target): «place1Type»
stateMachineForNode - Node
Association (source - target): «place2Type»
stateMachineForNode - StateMachine
Attributes:
-
An appliedStateMachine that relates a Node to its state machine.

stateOfResponsibleHumanResourceState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateOfResponsibleHumanResourceState - agentCapableOfResponsibilityWholeState
Association (source - target):«place2Type»
stateOfResponsibleHumanResourceState - ResponsibleHumanResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateOfResponsibleHumanResourceState - individualResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateOfResponsibleHumanResourceState - responsibleHumanResourceWholePart
Association (source - target):«place1Type»
stateOfResponsibleHumanResourceState - ResponsibleHumanResource

Attributes:
-
An agentCapableOfResponsibilityWholeState relationship between a ResponsibleHumanResource and its ResponsibleHumanResourceState.

templateForTask «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
templateForTask - modemIndividualTypeInstance
Association (source - target):«place2Type»
templateForTask - EnduringTask
Association (source - target):«place1Type»
templateForTask - ActivityGroup
Attributes:
-
A modemIndividualTypeInstance that relates an EnduringTask to an EnduringTaskTemplate that specifies it.

trustLevel «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
trustLevel - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
trustLevel - representedBy
Association (source - target):«place2Type»
trustLevel - IntegerRepresentation
Association (source - target):«place1Type»
trustLevel - TrustLine
Attributes:
-
A representedBy that uses an IntegerRepresentation to specify an arbitrary level of trust between the Nodes connected by a Trustline.

typeOfKnownResource «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
typeOfKnownResource - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
typeOfKnownResource - KnownResource
Association (source - target):«place1Type»
typeOfKnownResource - ResourceType
Attributes:
-
A modemIndividualTypeSpecialisation where a KnownResource is a subtype of a ResourceType.

2.4.9 Operational Views additional diagrams.

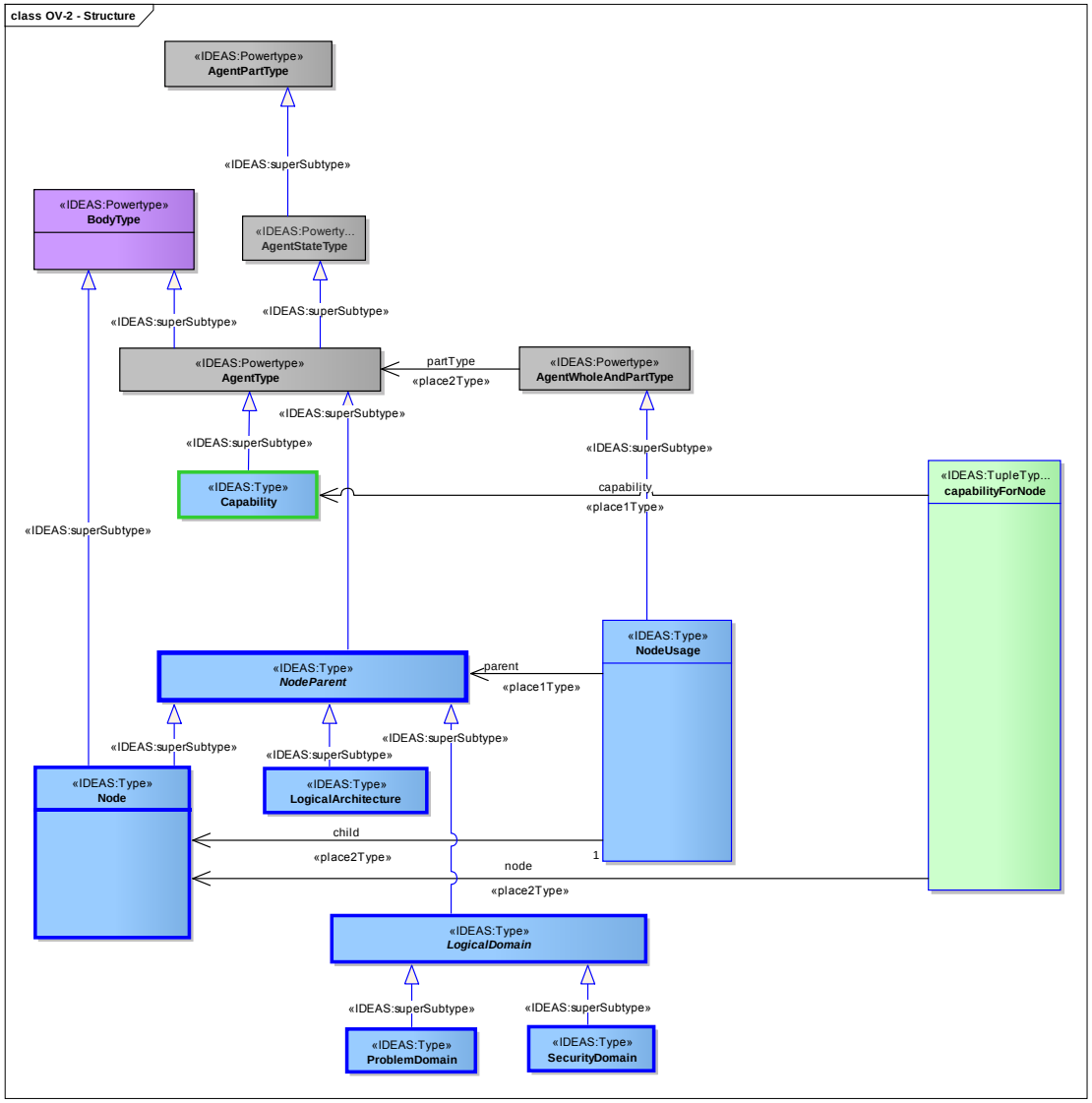


Figure 43 : OV-2 Structure

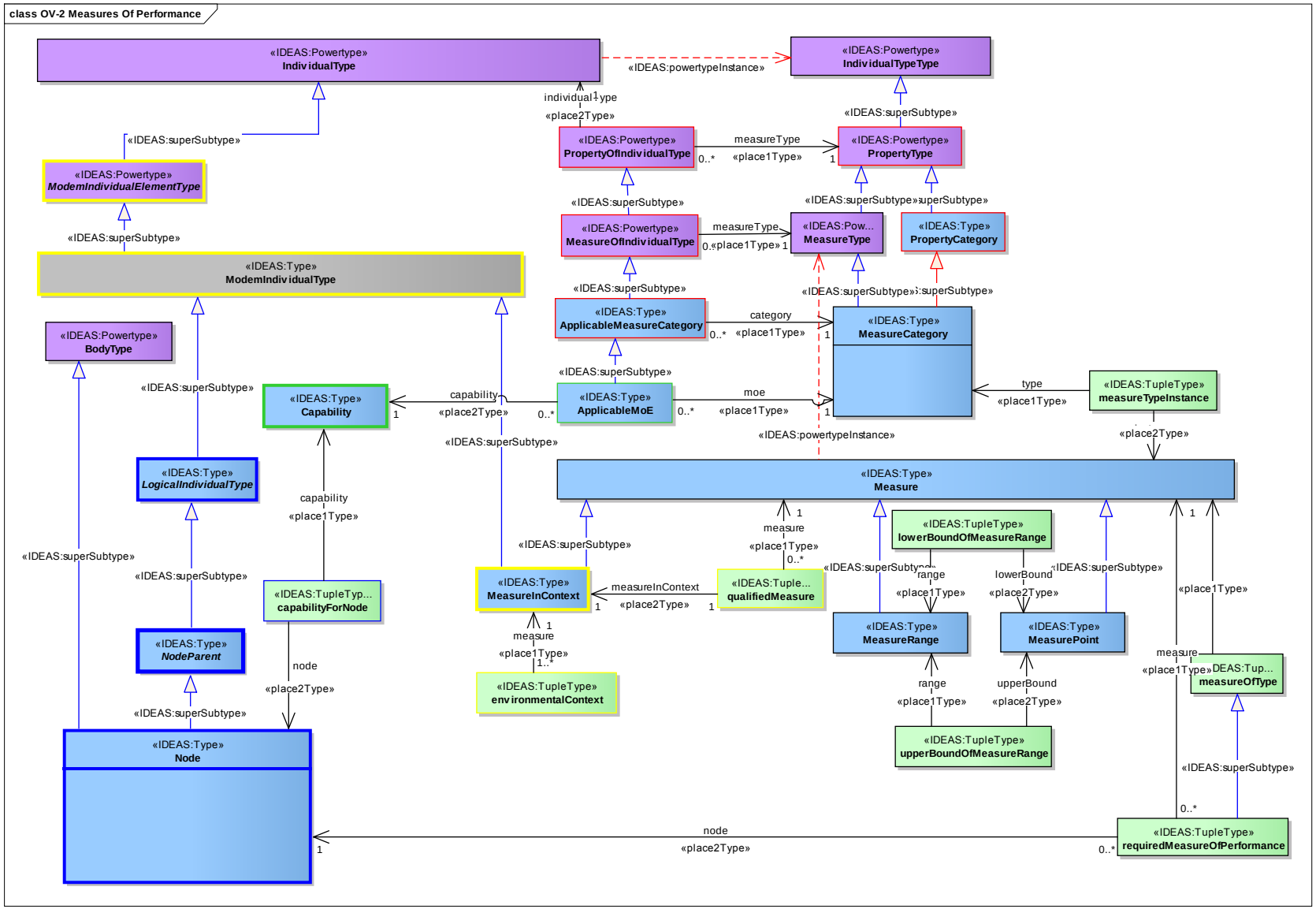


Figure 44 : Measures of performance

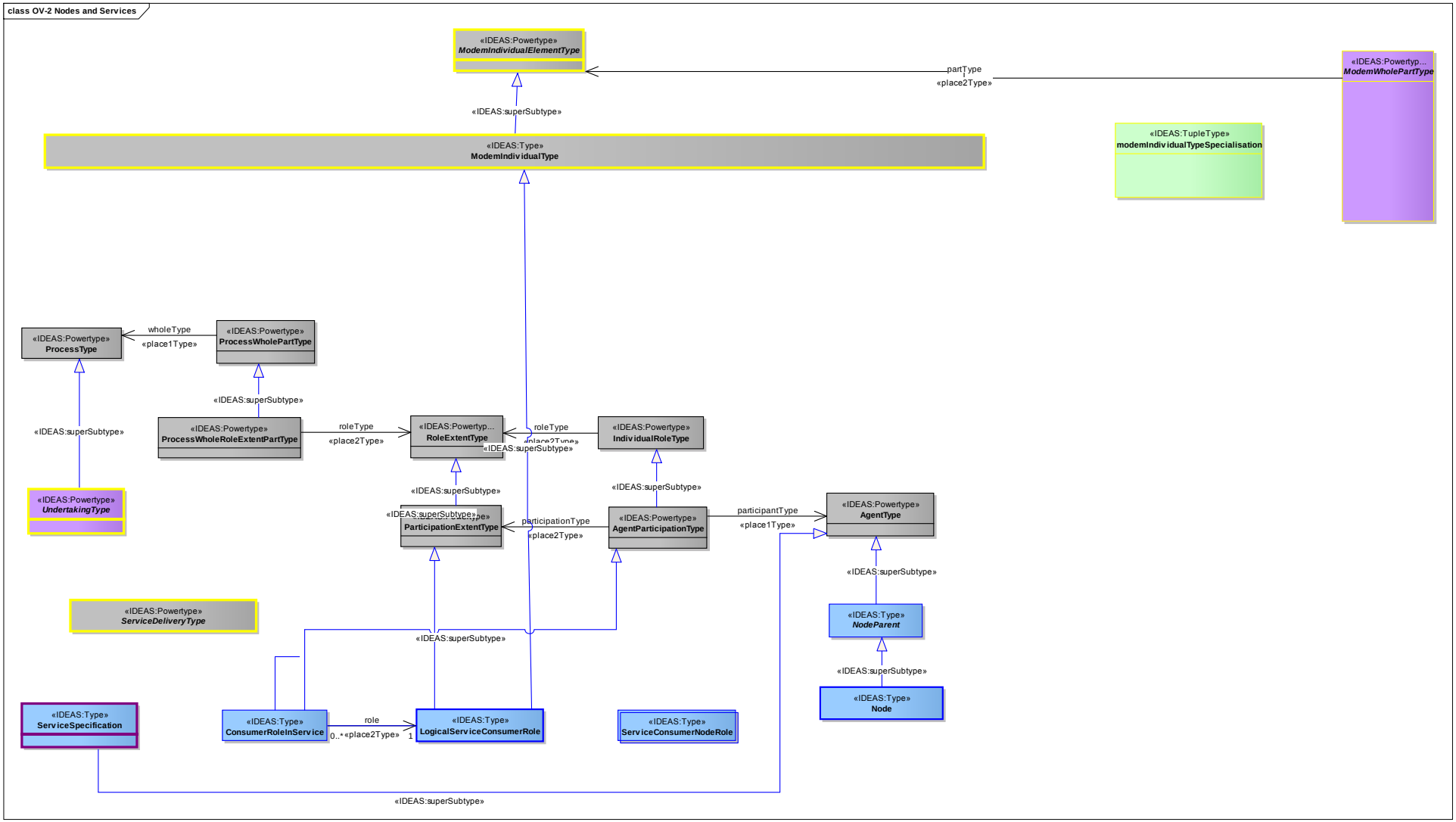


Figure 45 : Nodes and Services

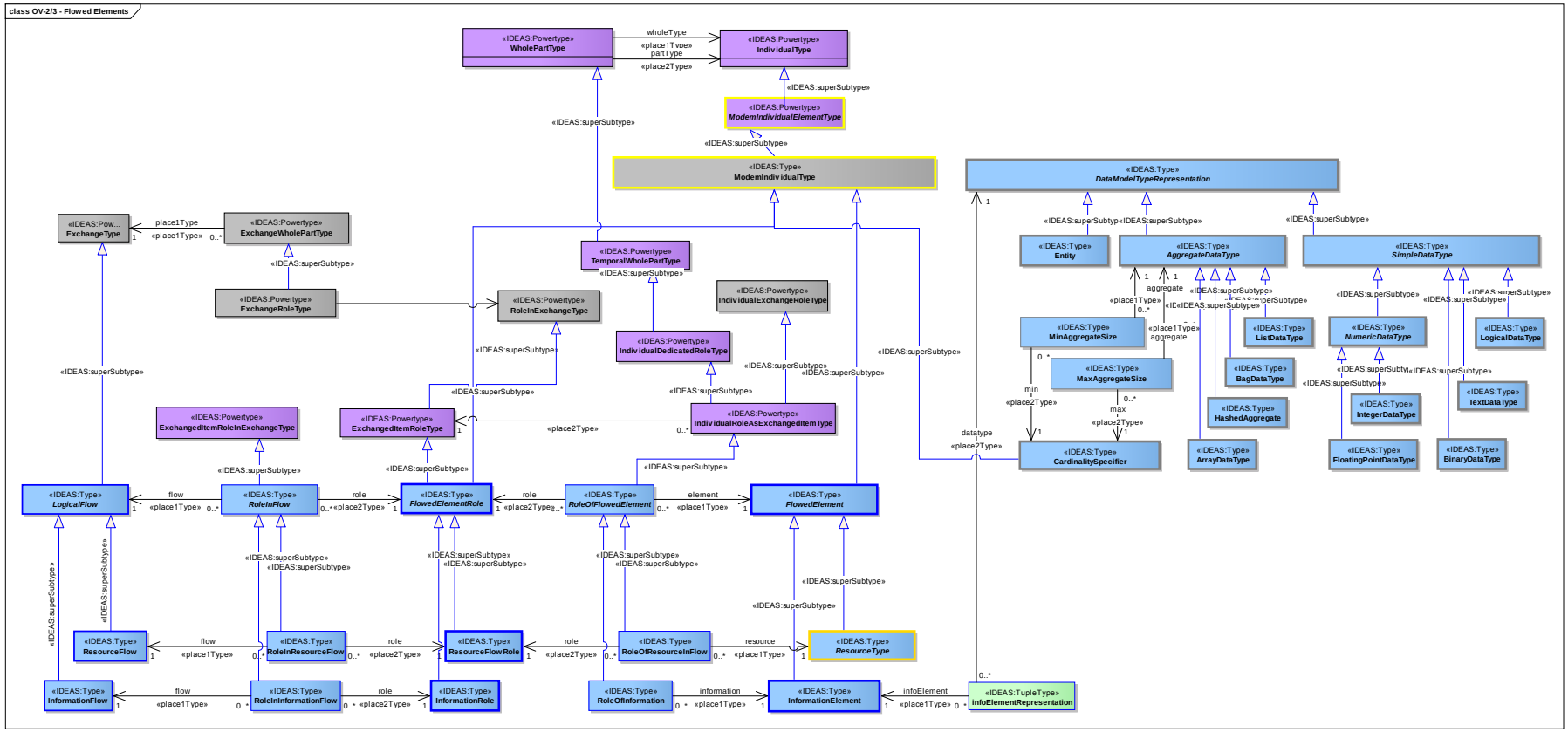


Figure 46 : OV-2/3 – Flowed Elements

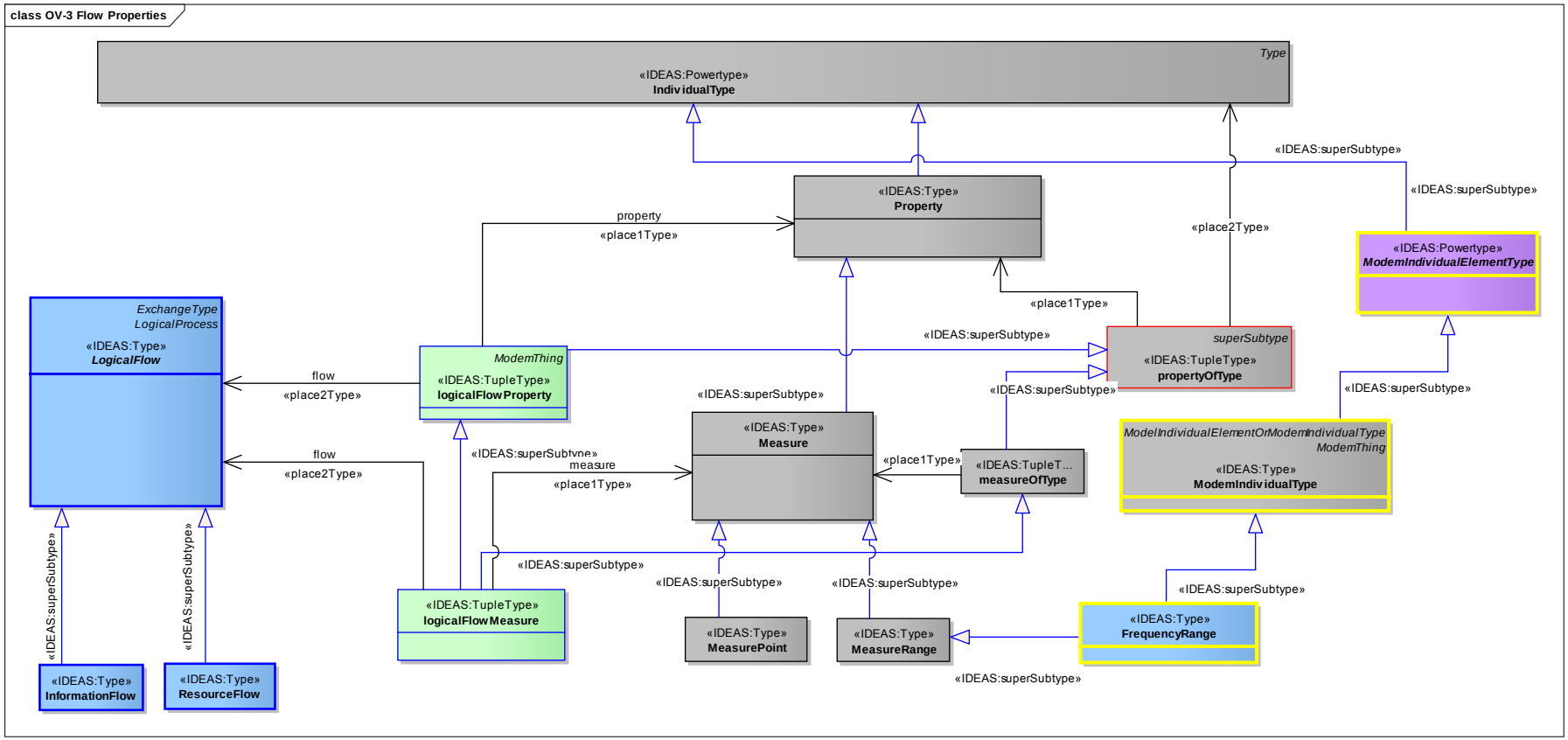


Figure 47 : Flow Properties

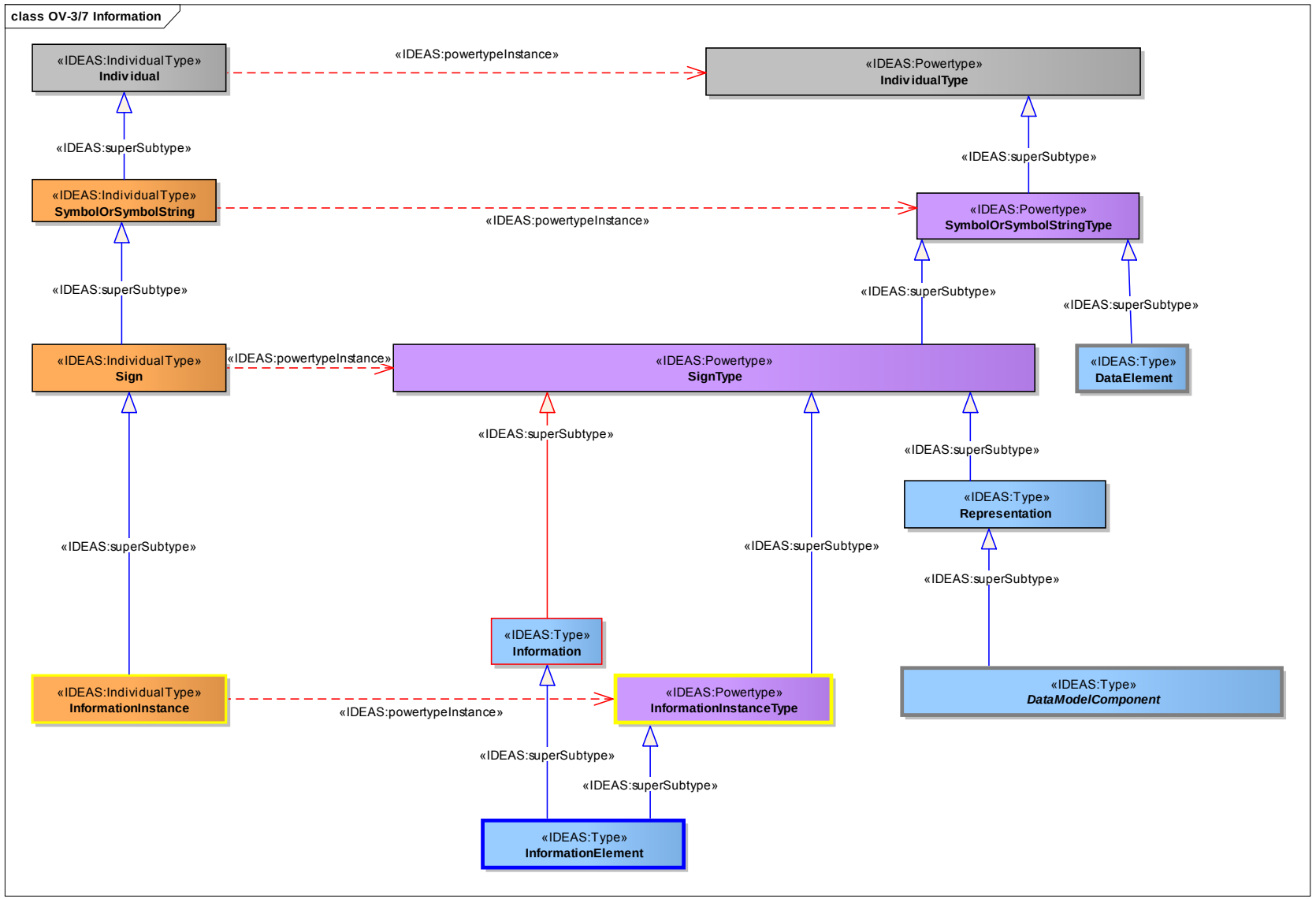


Figure 48 : OV-3/7 Information

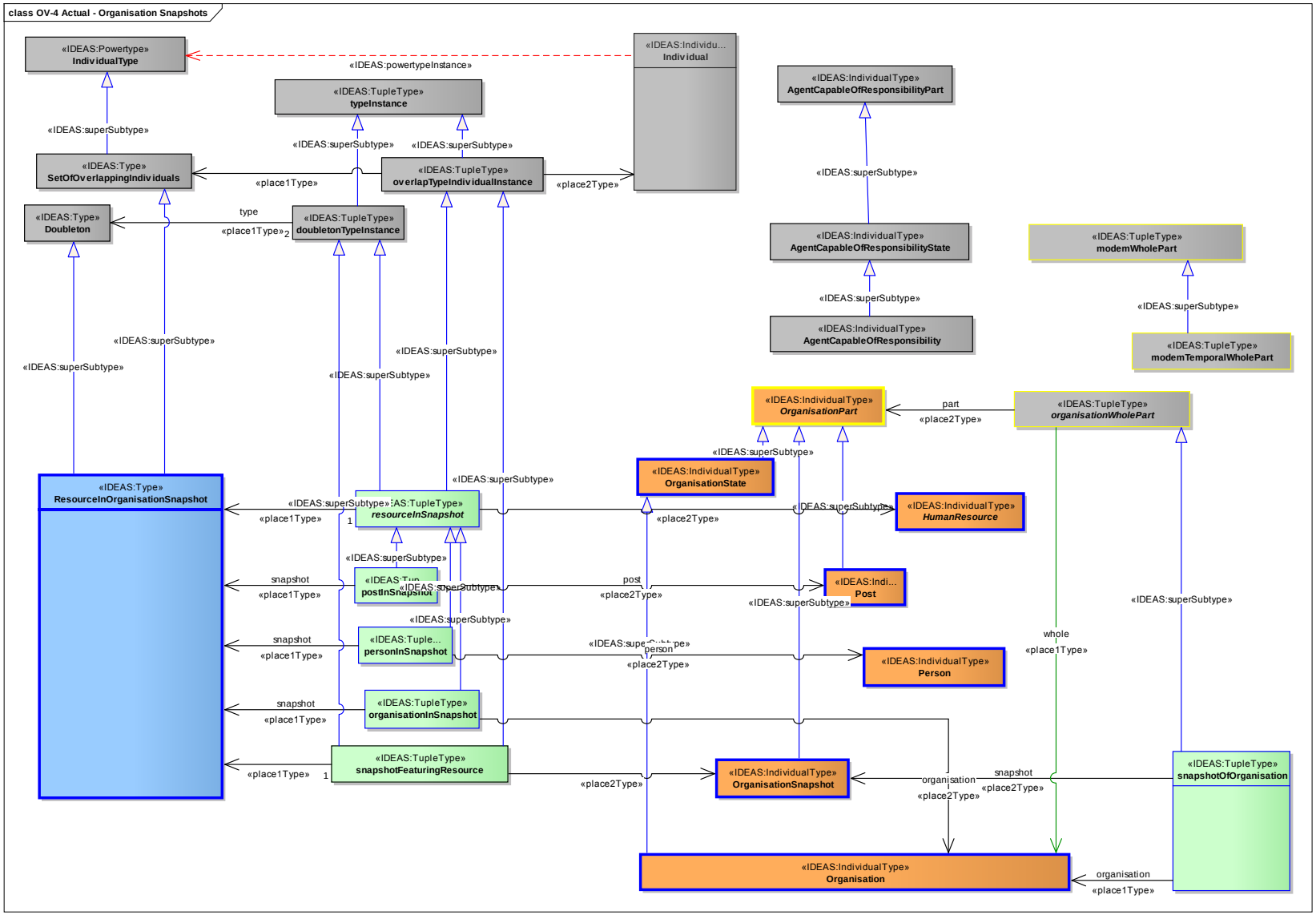


Figure 49 : OV-4 Actual Organisation Snapshots

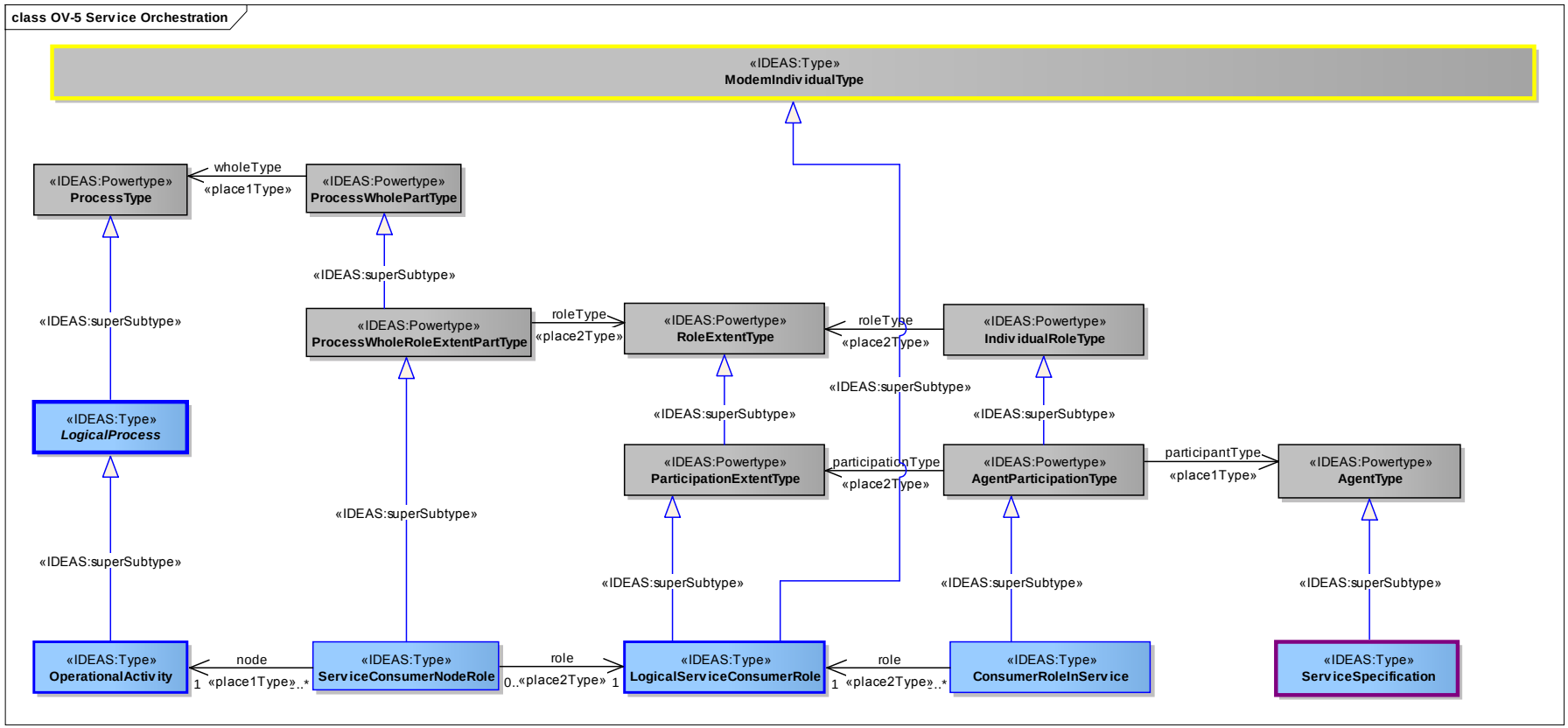


Figure 50 : OV-5 Service Orchestration

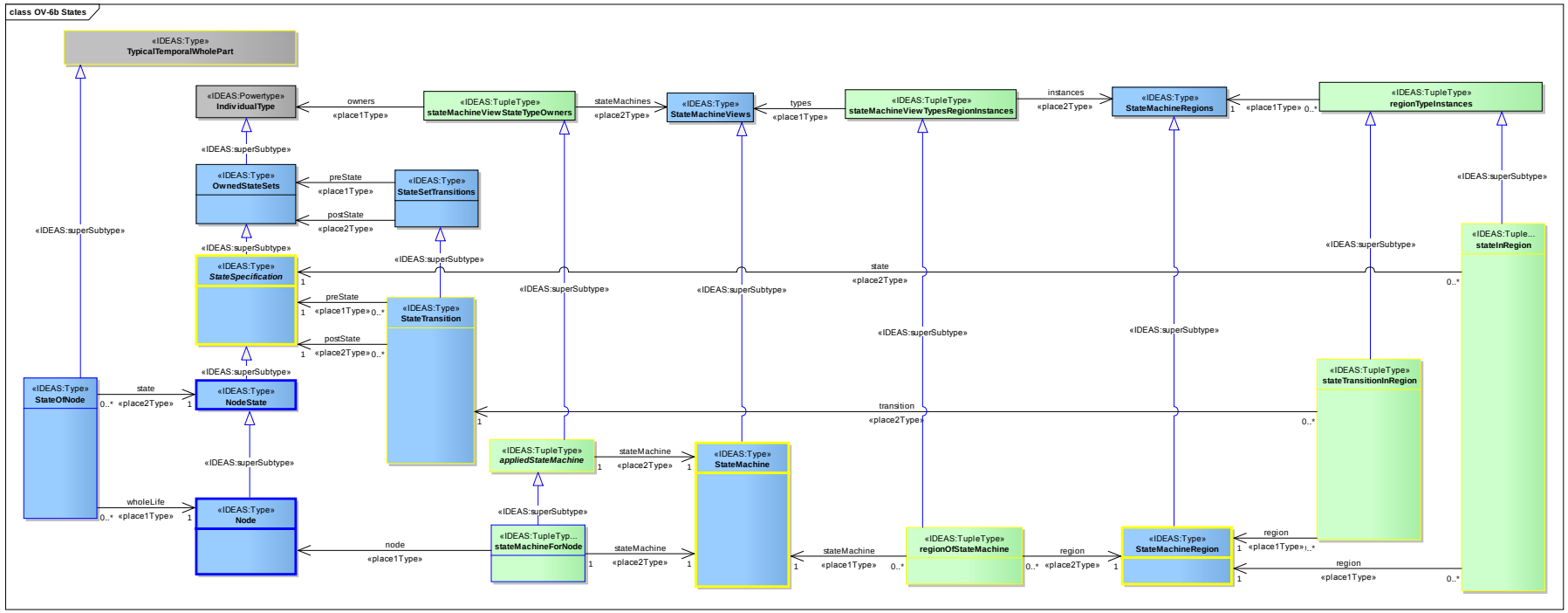


Figure 51 : OV-6b States

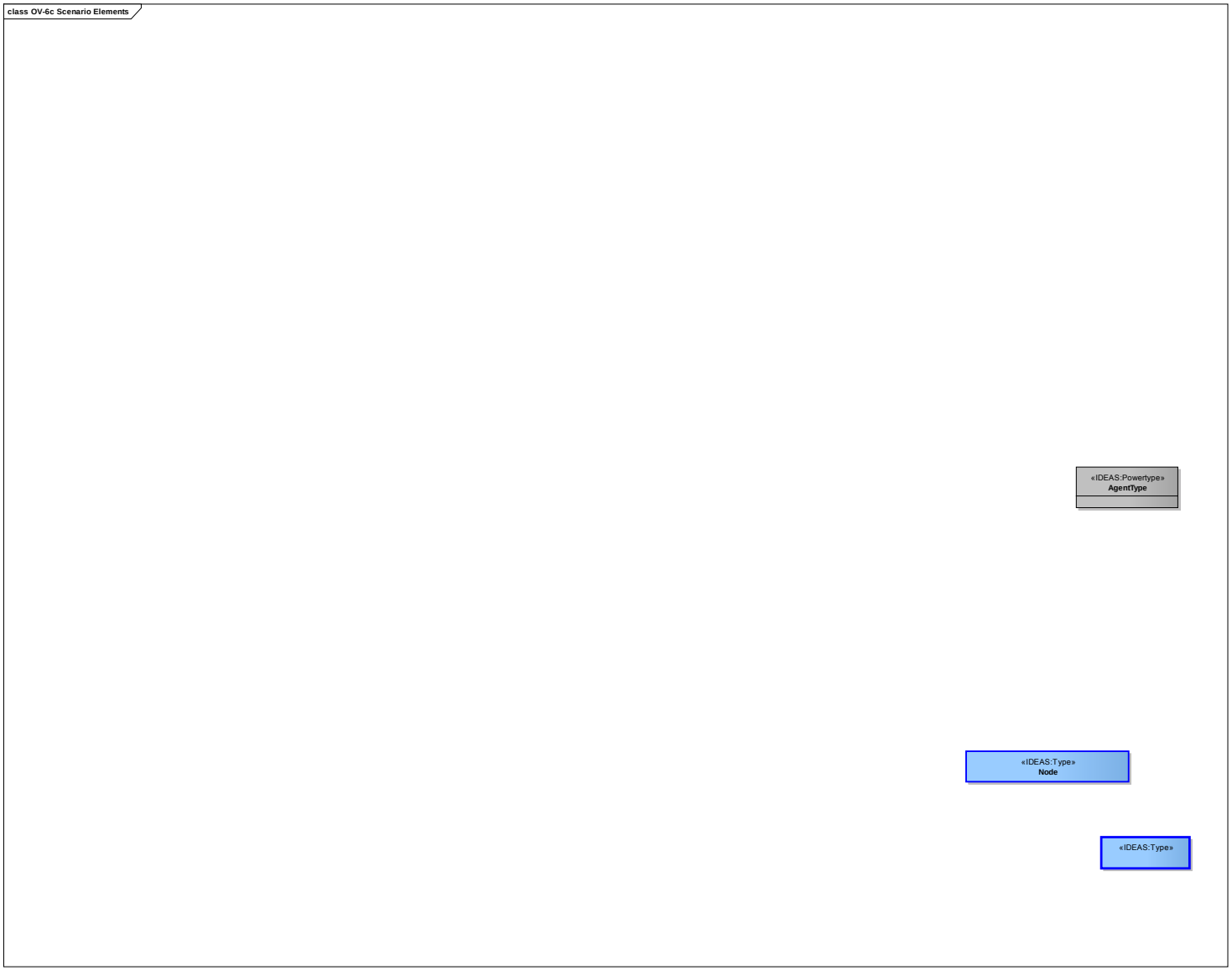


Figure 52 : OV-6c Scenario Elements

2.5 Service views

2.5.1 SOV-1: Service taxonomy

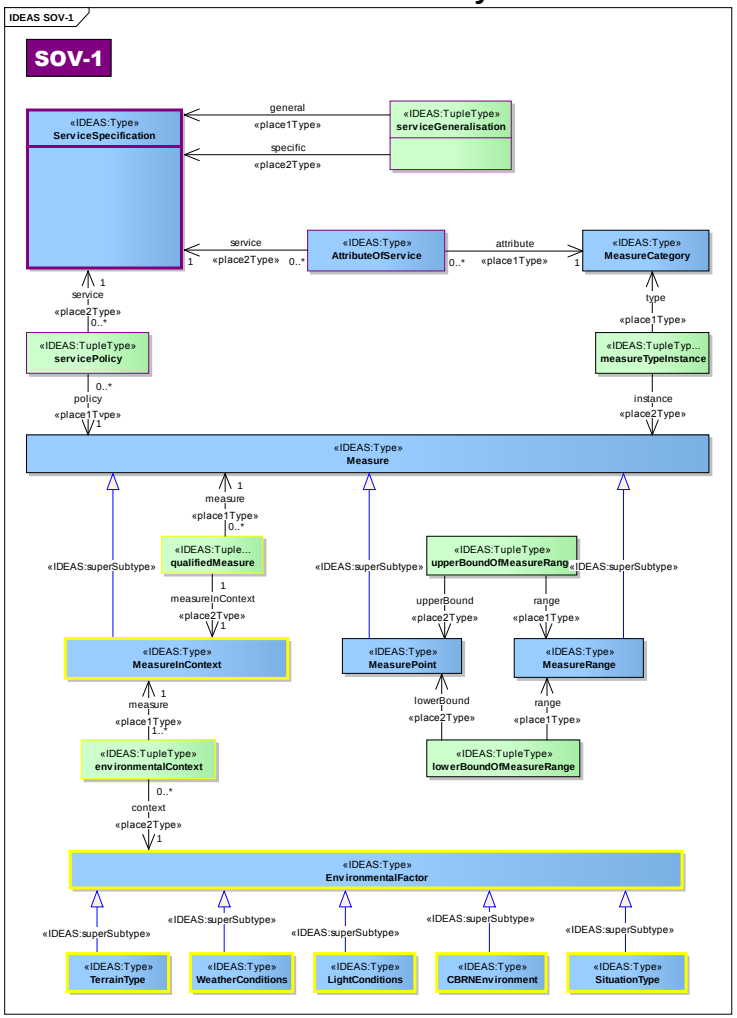


Figure 53 : SOV-1

2.5.2 SOV-2: Service interface specification

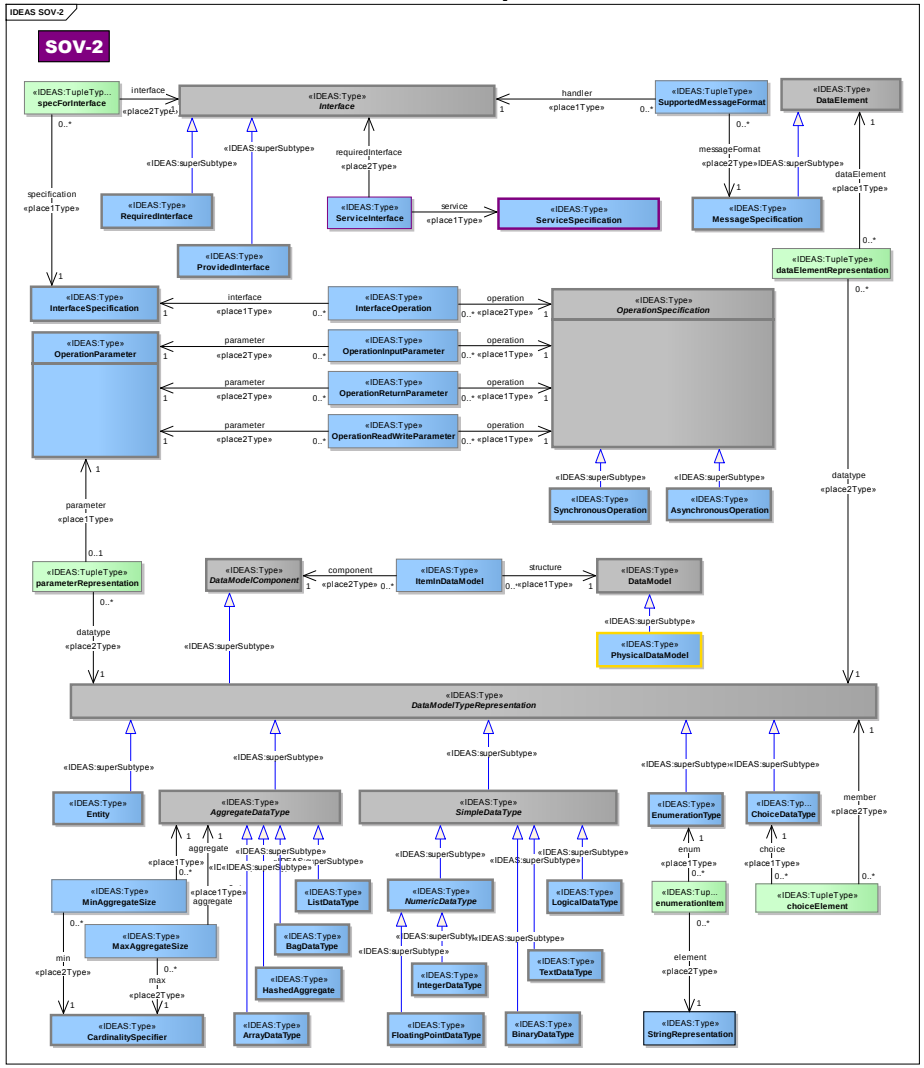


Figure 54 : SOV-2

2.5.3 SOV-3: Capability to service mapping

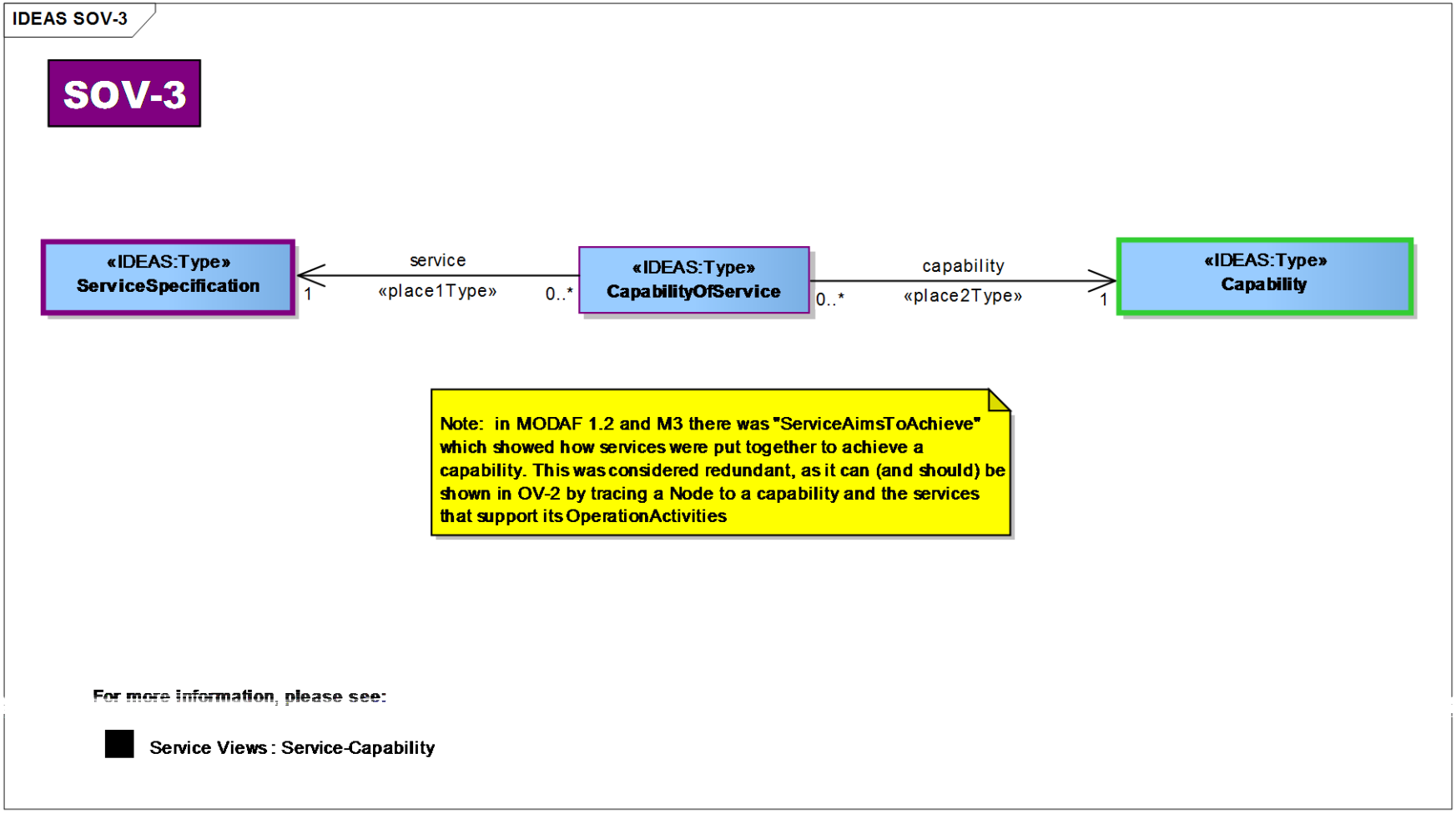


Figure 55 : SOV-3

2.5.4 SOV-4: Service constraints, state model and interaction specification

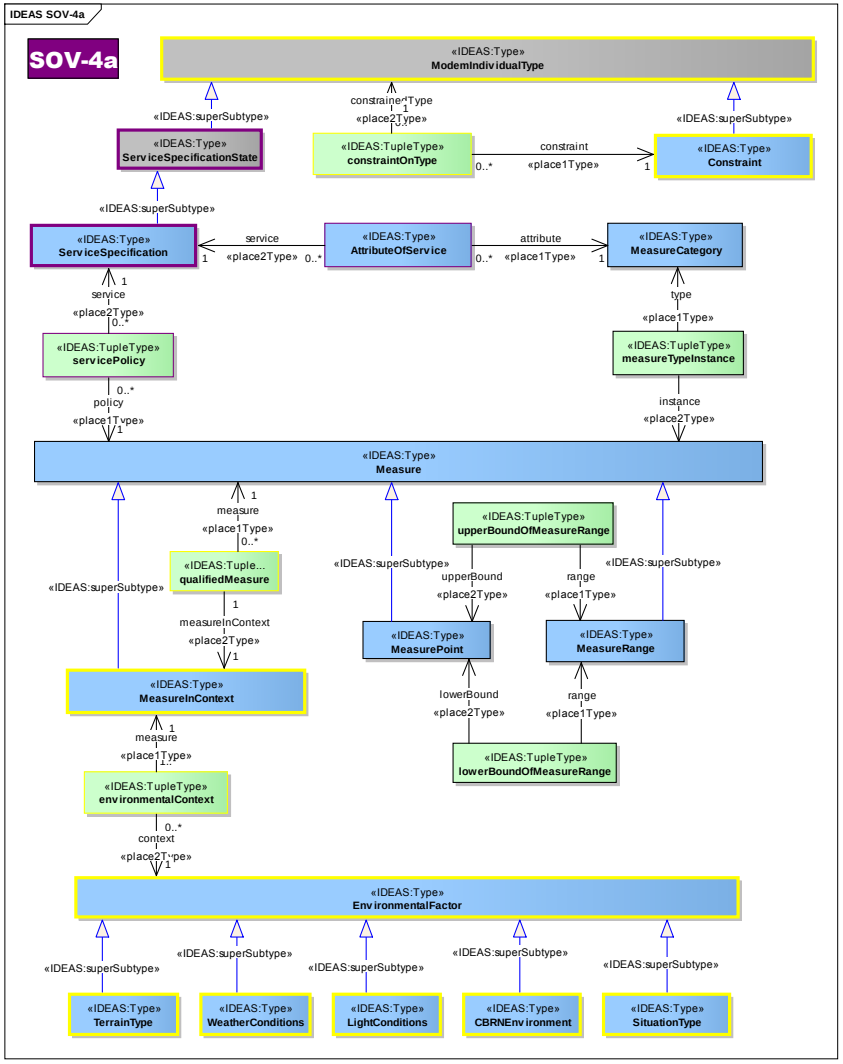


Figure 56 : SOV-4a

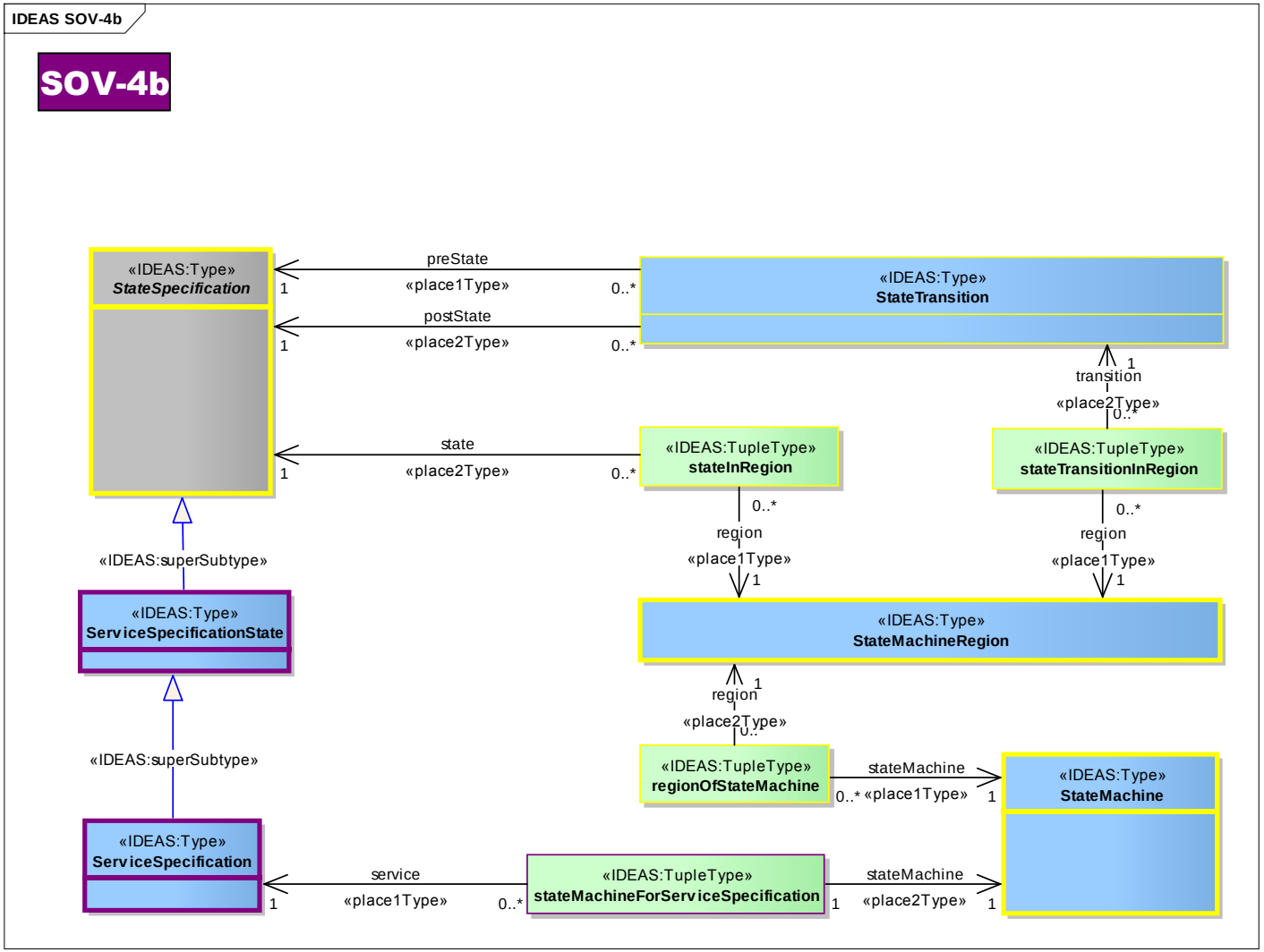


Figure 57 : SOV-4b

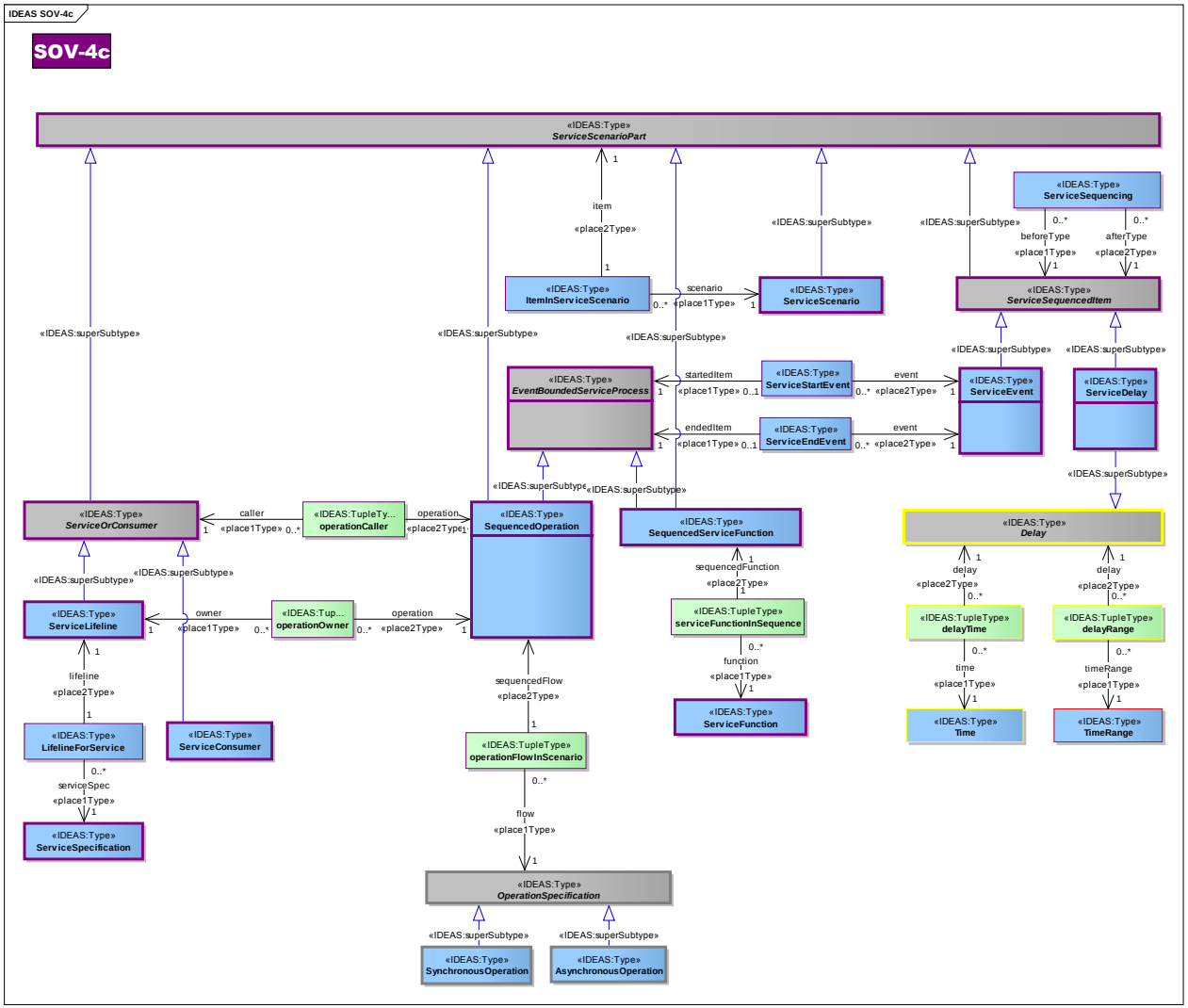


Figure 58 : SOV-4c

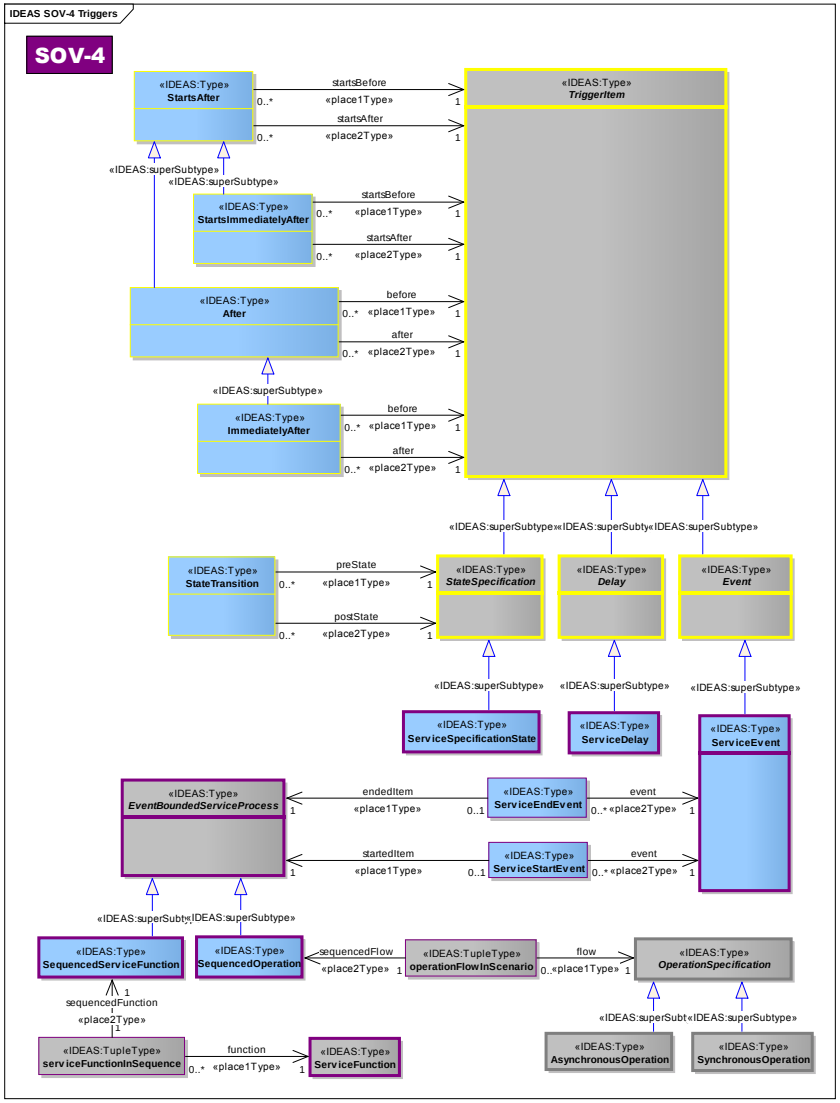


Figure 59 : SOV-4 Triggers

2.5.5 SOV-5: Service functionality

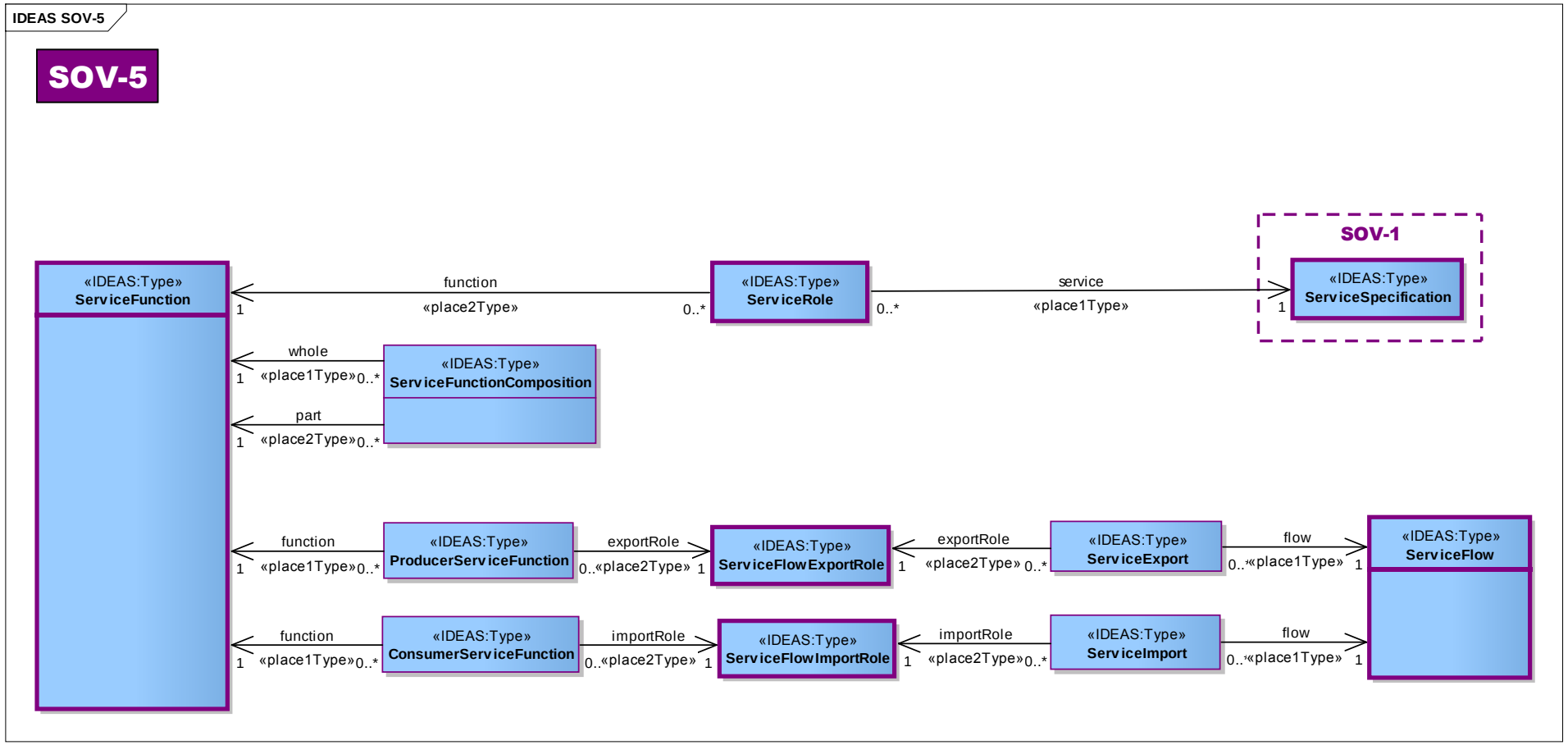


Figure 60 : SOV-5

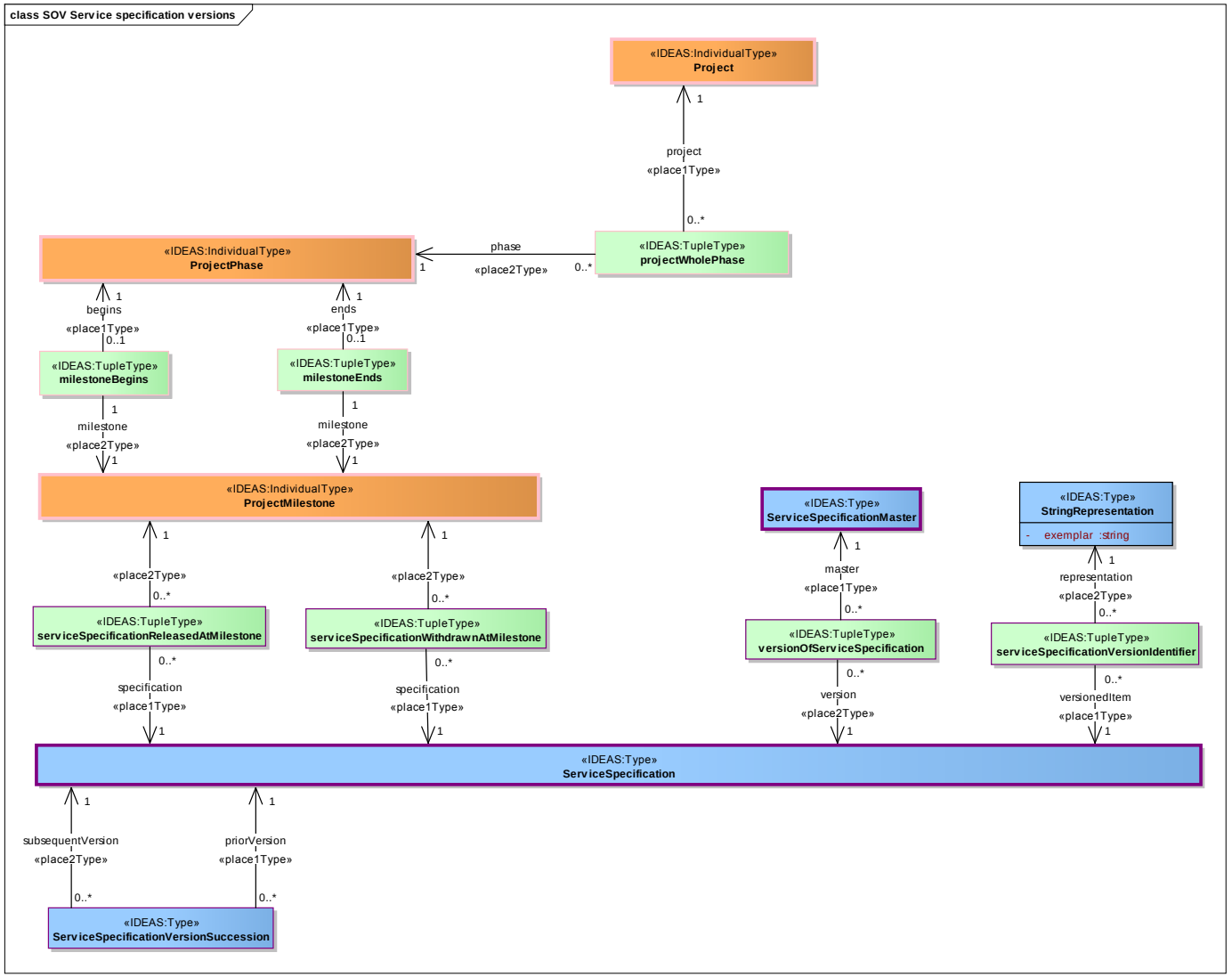


Figure 61 : SOV Service specification versions

2.5.6 Service Views elements list

Service Views

AttributeOfService «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»

EventBoundedServiceProcess «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EventBoundedServiceProcess - ServiceProcess
Attributes:
-
A ServiceProcess that has ServiceEvents marking its beginning and end.

ItemInServiceScenario «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ItemInServiceScenario - ItemInScenario
Association (source - target):«place2Type»
ItemInServiceScenario - ServiceScenarioPart
Association (source - target):«place1Type»
ItemInServiceScenario - ServiceScenario
Attributes:
-
An ItemInScenario where the Scenario is a ServiceScenario.

LifelineForService «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LifelineForService - TypicalTemporalWholePart
Association (source - target):«place2Type»
LifelineForService - ServiceLifeline
Association (source - target):«place1Type»
LifelineForService - ServiceSpecification
Attributes:
-
A TypicalTemporalWholePart that asserts a ServiceLifeLine is a typical temporal part of a ServiceSpecification.

ProducerServiceFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProducerServiceFunction - IndividualExchangeRoleType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProducerServiceFunction - ModemWholePartType
Association (source - target):«place2Type»
ProducerServiceFunction - ServiceFlowExportRole
Association (source - target):«place1Type»
ProducerServiceFunction - ServiceFunction
Attributes:
-
An IndividualExchangeRoleType where the role is a ServiceFlowExportRole and the producer is a ServiceFunction.

SequencedOperation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SequencedOperation - ServiceScenarioPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SequencedOperation - EventBoundedServiceProcess

Attributes:
-
A ServiceScenarioPart that is the typical occurance of an OperationSpecification.

SequencedServiceFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SequencedServiceFunction - ServiceScenarioPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SequencedServiceFunction - ServiceProcess
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SequencedServiceFunction - EventBoundedServiceProcess
Attributes:
-
An EventBoundedServiceProcess whose instances are special cases of ServiceFunctions that take part in ServiceScenarios.

ServiceConsumer «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceConsumer - ServiceOrConsumer
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceConsumer - Node
Attributes:
-
A Node that interacts with one or more services.

ServiceDelay «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceDelay - ServiceSequencedItem
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceDelay - Delay
Attributes:
-
A ServiceSequencedItem that has a specified temporal extent, but an unspecified spatial extent.

ServiceEndEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceEndEvent - EndBorderType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceEndEvent - ModemTemporalWholePartType
Association (source - target):«place1Type»
ServiceEndEvent - EventBoundedServiceProcess
Association (source - target):«place2Type»
ServiceEndEvent - ServiceEvent
Attributes:
-
An EndBorderType that relates a EventBoundedServiceProcess to the ServiceEvent that marks its end. Note: there may be no more than one ServiceEndEvent for a given EventBoundedServiceProcess.

ServiceEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceEvent - ServiceSequencedItem
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceEvent - Event
Attributes:
-
An Event that marks the beginning or end of a EventBoundedServiceProcess.

ServiceExport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceImport - SendInExchangeType
Association (source - target):«place1Type»
ServiceImport - ServiceFlow
Association (source - target):«place2Type»
ServiceImport - ServiceFlowExportRole
Attributes:
-
A ReceiveInExchangeType where the receiver is a ServiceSpecification or ServiceFunction.

ServiceFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceFlow - ServiceProcess
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceFlow - ExchangeType
Attributes:
-
An ExchangeType where two ServiceSpecifications interact.

ServiceFlowExportRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceFlowExportRole - ModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFlowExportRole - SendType
Attributes:
-
A SendType where the sender is a ServiceSpecification or ServiceFunction.

ServiceFlowImportRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFlowImportRole - ReceiveType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceFlowImportRole - ModemIndividualType
Attributes:
-
A RecieveType where the receiver is a ServiceSpecification or ServiceFunction.

ServiceFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFunction - ModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFunction - ServiceProcess
Attributes:
-
A ServiceProcess carried out by a ServiceSpecification

ServiceFunctionComposition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFunctionComposition - TypicalWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ServiceFunctionComposition - ProcessWholeAndPartType
Association (source - target): «place2Type»
ServiceFunctionComposition - ServiceFunction
Association (source - target): «place1Type»
ServiceFunctionComposition - ServiceFunction
Attributes:
-
A TypicalWholePart that relates a parent (whole) ServiceFunction to its child (part) ServiceFunction. Note: was called "ActivityComposition" in M3.

ServiceImport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceImport - ReceiveInExchangeType
Association (source - target):«place1Type»
ServiceImport - ServiceFlow
Association (source - target):«place2Type»
ServiceImport - ServiceFlowImportRole
Attributes:
-
A ReceiveInExchangeType where the receiver is a ServiceSpecification or ServiceFunction.

ServiceInterface «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceInterface - TypicalWholePart
Association (source - target):«place2Type»
ServiceInterface - Interface
Association (source - target):«place1Type»
ServiceInterface - ServiceSpecification
Attributes:
-
A TypicalWholePart that relates a ServiceSpecification to an Interface that it requires or provides.

ServiceLevel «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLevel - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLevel - BodyType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLevel - AgentType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLevel - DispositionalProperty
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLevel - ServiceDeliveryType
Attributes:
-
A ServiceDeliveryType based on a ServiceSpecification that sets a level of service using of Measures that correspond to ServiceAttributes.

ServiceLifeline «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLifeline - ServiceOrConsumer
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceLifeline - ServiceSpecificationState
Attributes:
-
A ServiceSpecificationState whose extent is defined by a ServiceScenario.

ServiceOrConsumer «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceOrConsumer - ServiceScenarioPart
Attributes:
-
A ServiceScenarioPart that is either a ServiceLifeline or a ServiceConsumer.
[ABSTRACT]

ServiceProcess «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceProcess - ProcessType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceProcess - ModemWholePartType
Attributes:
-
A ProcessType conducted by a ServiceSpecification.

ServiceRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceRole - ProcessPartOfBodyType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceRole - TypicalWholePart
Association (source - target):«place2Type»

ServiceRole - ServiceFunction *Association (source - target):*«place1Type» ServiceRole - ServiceSpecification *Attributes:* - A ProcessPartOfBodyType that asserts that a ServiceFunction is part of a ServiceSpecification.
ServiceScenario «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceScenario - ServiceScenarioPart *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceScenario - Scenario *Attributes:* - A Scenario that describes the order of interactions with a ServiceSpecification.
ServiceScenarioPart «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceScenarioPart - ModemIndividualType *Attributes:* - A ModemIndividualType that features in (i.e. is part of) a ServiceScenario.
ServiceSequencedItem «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceSequencedItem - ServiceScenarioPart *Attributes:* - A ServiceScenarioPart that can be sequenced by ServiceSequencing.
ServiceSequencing «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceSequencing - ImmediateBeforeAfterType *Association (source - target):*«place2Type» ServiceSequencing - ServiceSequencedItem *Association (source - target):*«place1Type» ServiceSequencing - ServiceSequencedItem *Attributes:* - An ImmediateBeforeAfterType that asserts one ServiceSequencedItem happens immediately before another.
ServiceSpecification «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceSpecification - ServiceDeliveryType *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceSpecification - BodyType *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ServiceSpecification - AgentType

Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecification - DispositionalProperty
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecification - ServiceSpecificationState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecification - SubjectOfForecast
Attributes:
-
A ServiceDeliveryType that is the specification of a ServiceDelivery. Note: was called "Service" in M3.

ServiceSpecificationComposition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationComposition - TypicalWholePart
Association (source - target):«place1Type»
ServiceSpecificationComposition - ServiceSpecification
Association (source - target):«place2Type»
ServiceSpecificationComposition - ServiceSpecification
Attributes:
-
A TypicalWholePart that states that a ServiceSpecification reuses other ServiceSpecifications, i.e. is specified on top of the ones reused.

ServiceSpecificationMaster «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationMaster - ModemIndividualType
Attributes:
-
A ModemIndividualType that is the master specification from which ServiceSpecifications are versioned.

ServiceSpecificationState «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationState - ServiceDeliveryStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationState - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationState - StateSpecification
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationState - AgentStateType
Attributes:
-
A ServiceDeliveryStateType that is a type of temporal state typical of a ServiceSpecification.

ServiceSpecificationVersionSuccession «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationVersionSuccession - BeforeAfterType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceSpecificationVersionSuccession - ModemThing
Association (source - target):«place1Type»
ServiceSpecificationVersionSuccession - ServiceSpecification

Association (source - target):«place2Type»
ServiceSpecificationVersionSuccession - ServiceSpecification
Attributes:
-
A BeforeAfterType that asserts that one ServiceSpecification succeeds another. Note: both ServiceSpecifications must be versions of the same ServiceSpecificationMaster.

ServiceStartEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceStartEvent - ModemTemporalWholePartType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ServiceStartEvent - StartBorderType
Association (source - target):«place2Type»
ServiceStartEvent - ServiceEvent
Association (source - target):«place1Type»
ServiceStartEvent - EventBoundedServiceProcess
Attributes:
-
A StartBorderType that relates an EventBoundedServicelProcess to the ServiceEvent that marks its start. Note: there may be no more than one ServiceStartEvent for a given ServiceSequencedProcess.

StateOfServiceSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateOfServiceSpecification - AgentWholeStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateOfServiceSpecification - ServiceDeliveryWholeStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateOfServiceSpecification - TypicalTemporalWholePart
Association (source - target):«place1Type»
StateOfServiceSpecification - ServiceSpecification
Association (source - target):«place2Type»
StateOfServiceSpecification - ServiceSpecificationState
Attributes:
-
A ServiceDeliveryWholeStateType that relates a ServiceSpecification to one of its temporal states.

levelOfService «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
levelOfService - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
levelOfService - ServiceLevel
Association (source - target):«place1Type»
levelOfService - ServiceSpecification
Attributes:
-
A modemIndividualTypeSpecialisation where a ServiceLevel is sets levels of service based on a ServiceSpecification.

operationCaller «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
operationCaller - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
operationCaller - couple
Association (source - target):«place1Type»
operationCaller - ServiceOrConsumer
Association (source - target):«place2Type»
operationCaller - SequencedOperation
Attributes:
-
A couple that asserts a ServiceOrConsumer invokes a SequencedOperation on a ServiceLifeline.

operationFlowInScenario «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
operationFlowInScenario - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
operationFlowInScenario - OperationSpecification
Association (source - target):«place2Type»
operationFlowInScenario - SequencedOperation
Attributes:
-
A modemIndividualTypeSpecialisation that relates an OperationSpecification to its usage (as a SequencedOperation) in a ServiceScenario. Note: A SequencedOperation is based on only one OperationSpecification.

operationOwner «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
operationOwner - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
operationOwner - couple
Association (source - target):«place2Type»
operationOwner - SequencedOperation
Association (source - target):«place1Type»
operationOwner - ServiceLifeline
Attributes:
-
A couple where a SequencedOperation is run by a ServiceLifeline.

serviceFunctionInSequence «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceFunctionInSequence - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
serviceFunctionInSequence - ServiceFunction
Association (source - target):«place2Type»
serviceFunctionInSequence - SequencedServiceFunction
Attributes:
-
A modemIndividualTypeSpecialisation that relates a ServiceFunction to a SequencedServiceFunction that is a case of it being used in a ServiceScenario.

serviceGeneralisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceGeneralisation - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
serviceGeneralisation - ServiceSpecification
Association (source - target):«place2Type»
serviceGeneralisation - ServiceSpecification
Attributes:
-
A modemIndividualTypeSpecialisation where one ServiceSpecification is a specialisation of another.

serviceLevelMeasure «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceLevelMeasure - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceLevelMeasure - measureOfType
Association (source - target):«place1Type»
serviceLevelMeasure - Measure
Association (source - target):«place2Type»
serviceLevelMeasure - ServiceLevel
Attributes:
-
A measureOfType that specifies a ServiceLevel.

servicePolicy «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
servicePolicy - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
servicePolicy - measureOfType
Association (source - target):«place1Type»
servicePolicy - Measure
Association (source - target):«place2Type»
servicePolicy - ServiceSpecification
Attributes:
-
A measureOfType where the Measure specifies a policy for a ServiceSpecification. Note: The Measure must correspond to a given MeasureCategory that is an attributeOfService for the ServiceSpecification.

serviceSpecificationReleasedAtMilestone «IDEAS:TupleType»
Connectors:
Association (source - target):«place2Type»
serviceSpecificationReleasedAtMilestone - ProjectMilestone
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceSpecificationReleasedAtMilestone - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceSpecificationReleasedAtMilestone - ModemThing
Association (source - target):«place1Type»
serviceSpecificationReleasedAtMilestone - ServiceSpecification
Attributes:
-

A couple that indicates that a ServiceSpecification is released at a ProjectMilestone.

serviceSpecificationVersionIdentifier «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceSpecificationVersionIdentifier - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceSpecificationVersionIdentifier - representedBy
Association (source - target):«place2Type»
serviceSpecificationVersionIdentifier - StringRepresentation
Association (source - target):«place1Type»
serviceSpecificationVersionIdentifier - ServiceSpecification
Attributes:
-
A representedBy that asserts that a StringRepresentation represents the version identifier of a ServiceSpecification.

serviceSpecificationWithdrawnAtMilestone «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceSpecificationWithdrawnAtMilestone - ModemThing
Association (source - target):«place2Type»
serviceSpecificationWithdrawnAtMilestone - ProjectMilestone
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceSpecificationWithdrawnAtMilestone - couple
Association (source - target):«place1Type»
serviceSpecificationWithdrawnAtMilestone - ServiceSpecification
Attributes:
-
A couple that indicates that a ServiceSpecification is withdrawn at a ProjectMilestone.

stateMachineForServiceSpecification «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateMachineForServiceSpecification - appliedStateMachine
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateMachineForServiceSpecification - ModemThing
Association (source - target):«place1Type»
stateMachineForServiceSpecification - ServiceSpecification
Association (source - target):«place2Type»
stateMachineForServiceSpecification - StateMachine
Attributes:
-
An appliedStateMachine that relates a ServiceSpecification to its state machine.

versionOfServiceSpecification «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
versionOfServiceSpecification - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
versionOfServiceSpecification - ServiceSpecificationMaster
Association (source - target):«place2Type»
versionOfServiceSpecification - ServiceSpecification
Attributes:
-
A modemIndividualTypeSpecialisation that asserts that a ServiceSpecification is a version of a ServiceSpecificationMaster.

2.5.7 Service Views additional diagrams.

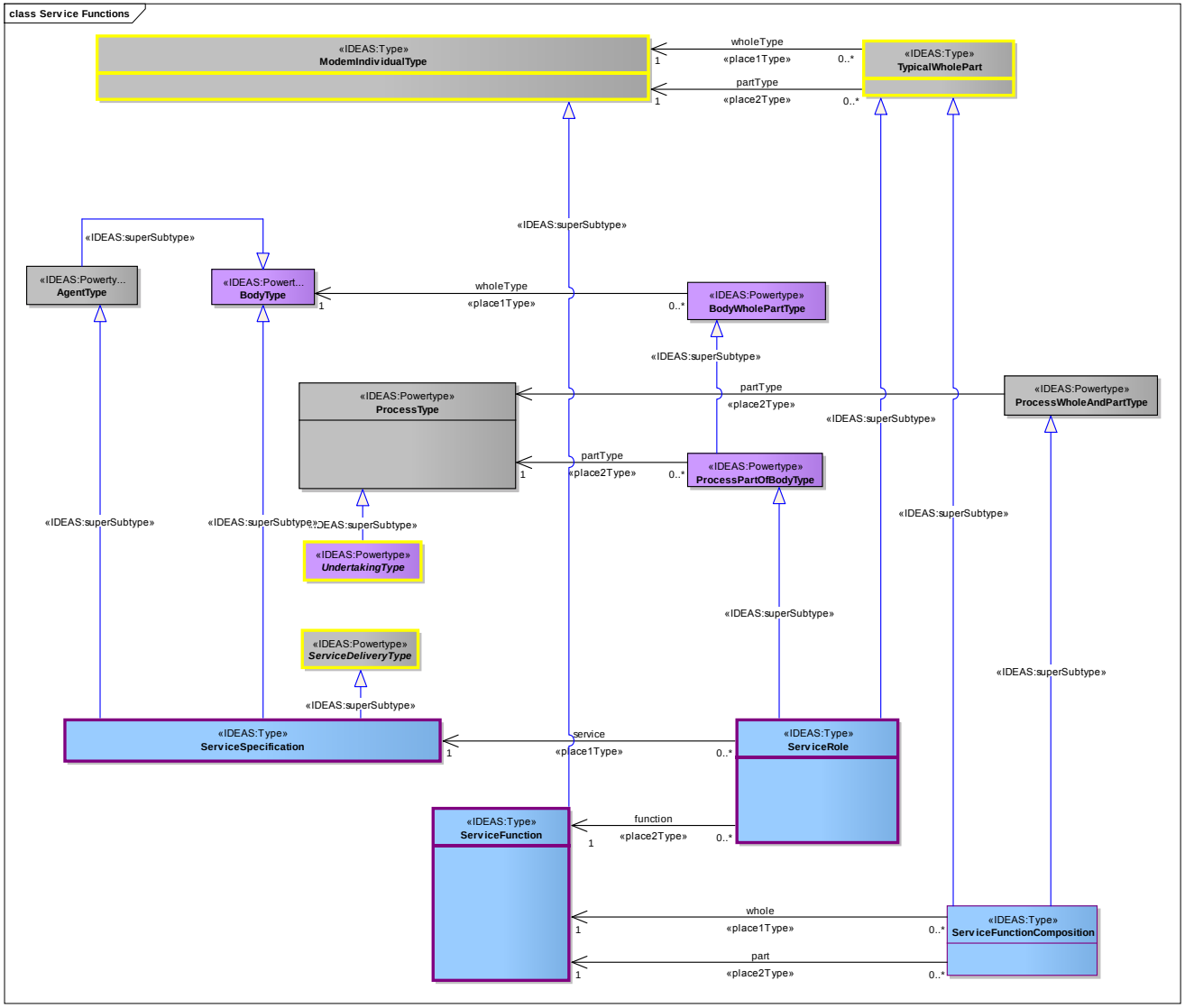


Figure 62 : Service Functions

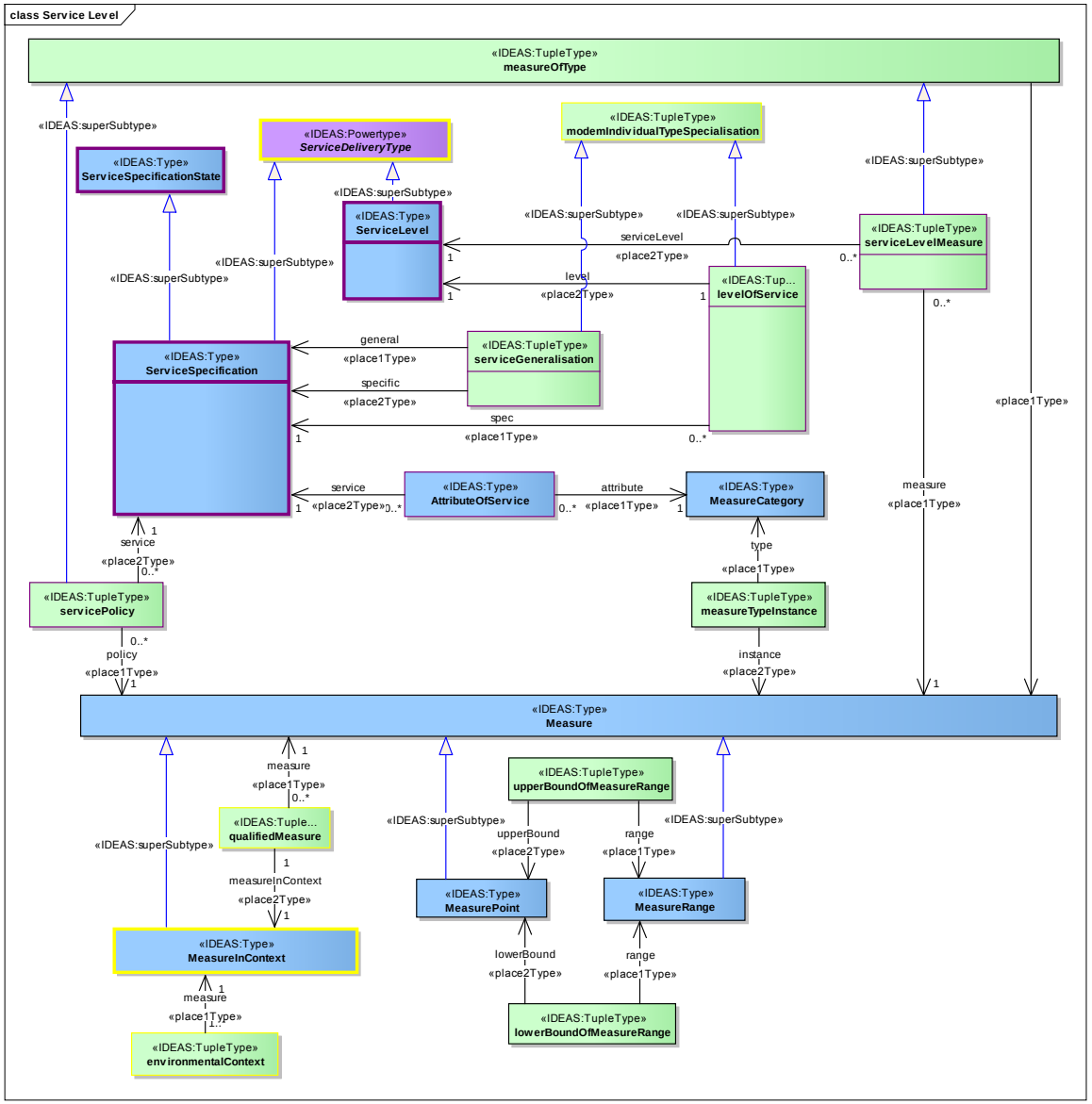


Figure 63 : Service Level

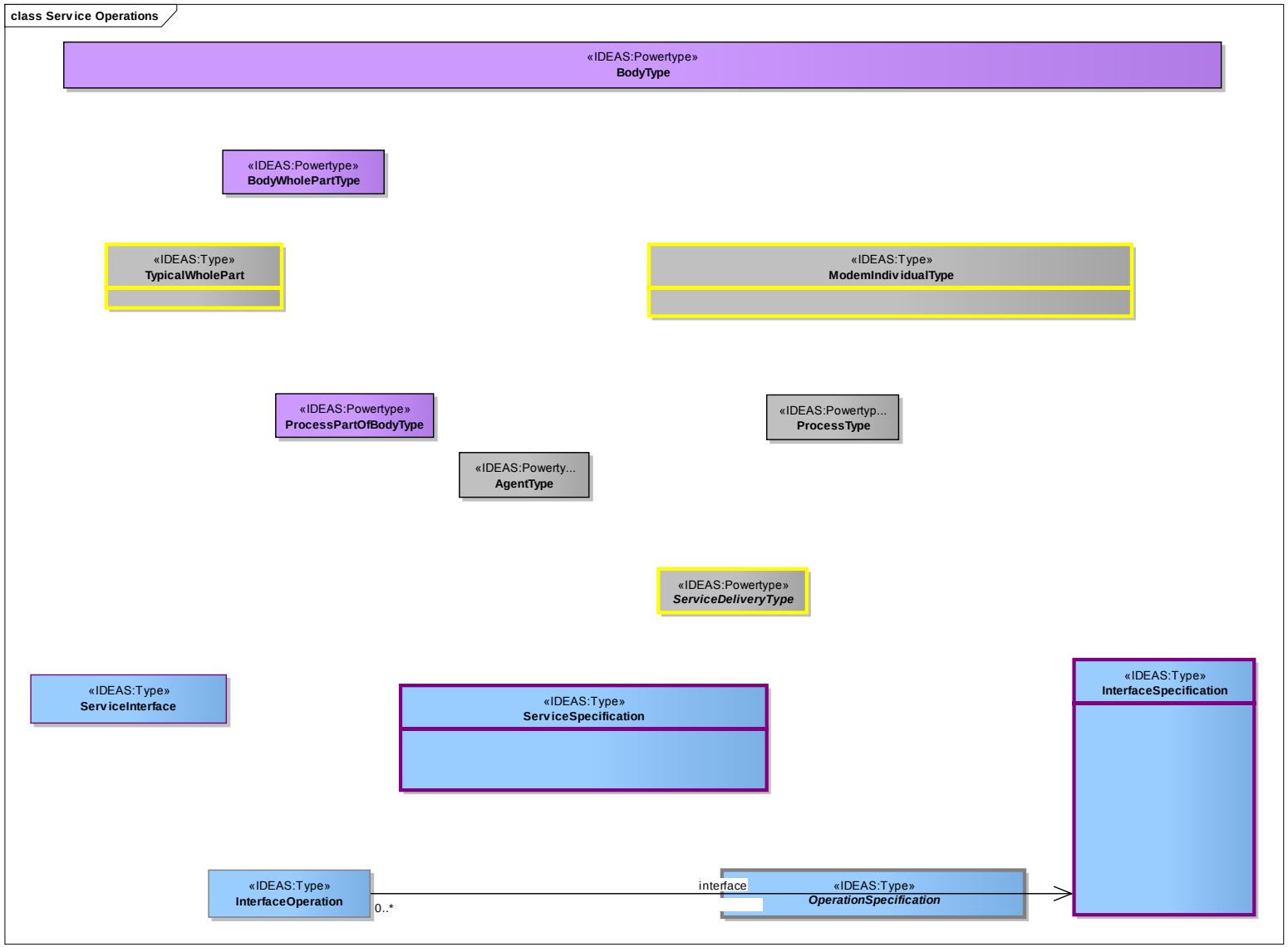


Figure 64 : Service Operations

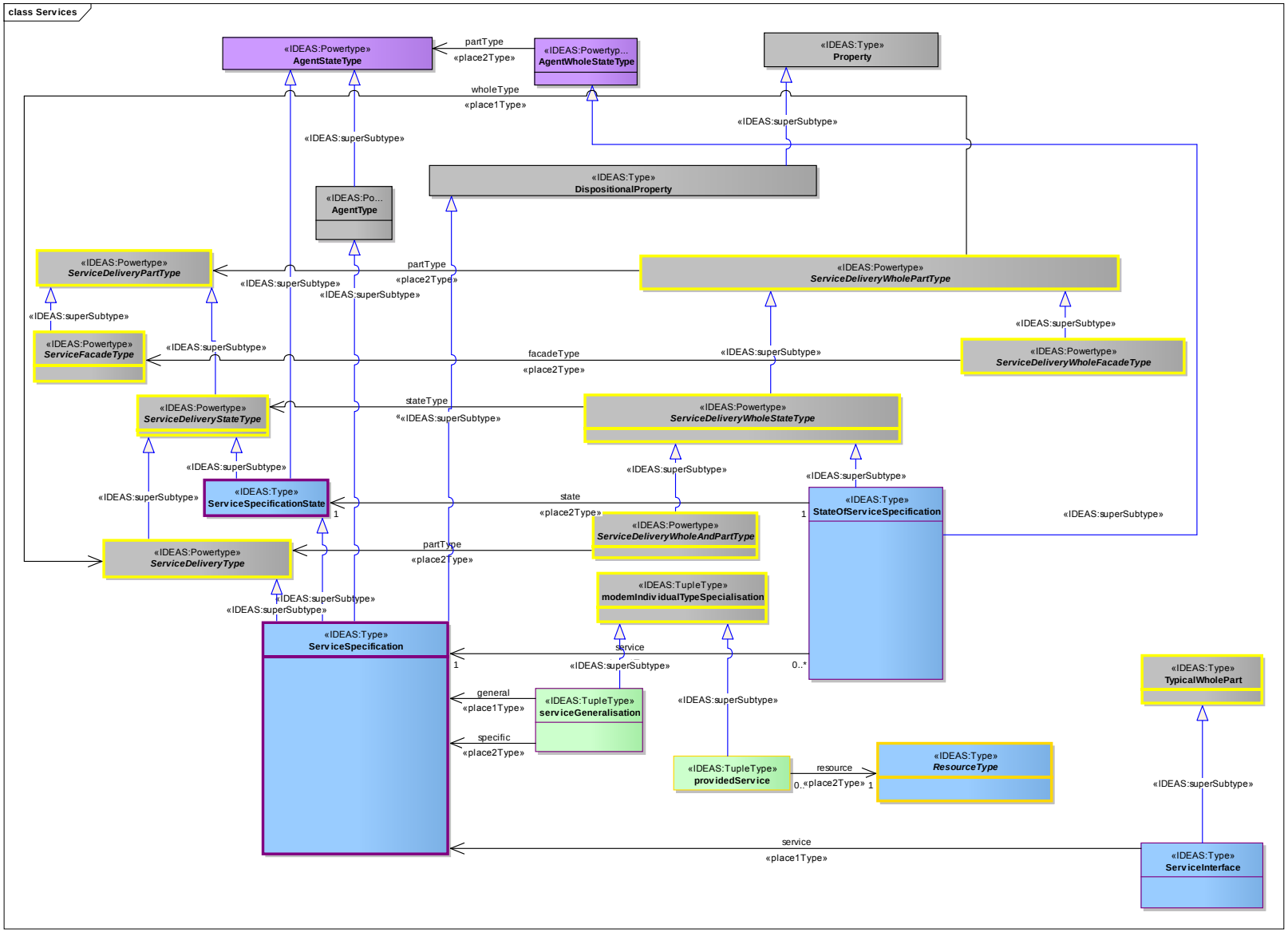


Figure 65 : Services

2.6 System views

2.6.1 SV-1: Resource interaction specification

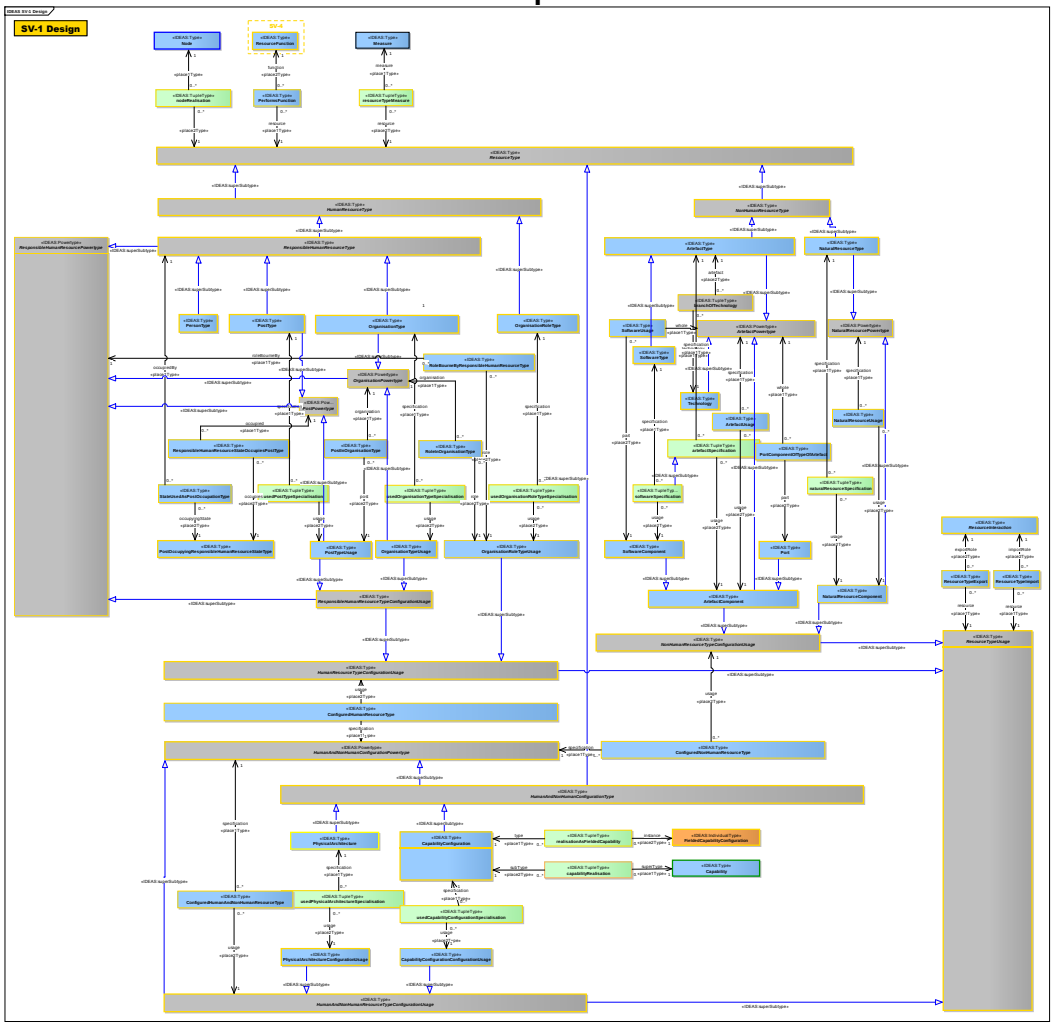


Figure 66 : SV-1 Design

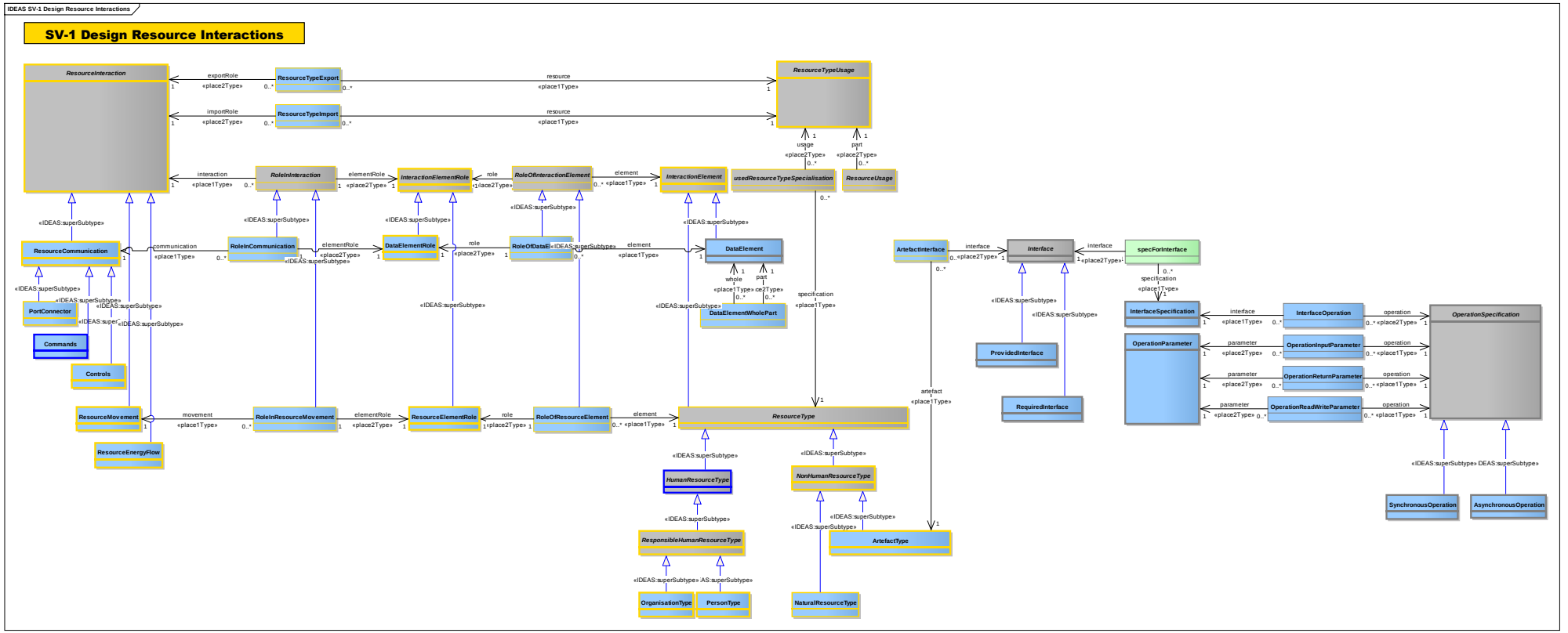


Figure 67 : SV-1 Design Resource Interactions

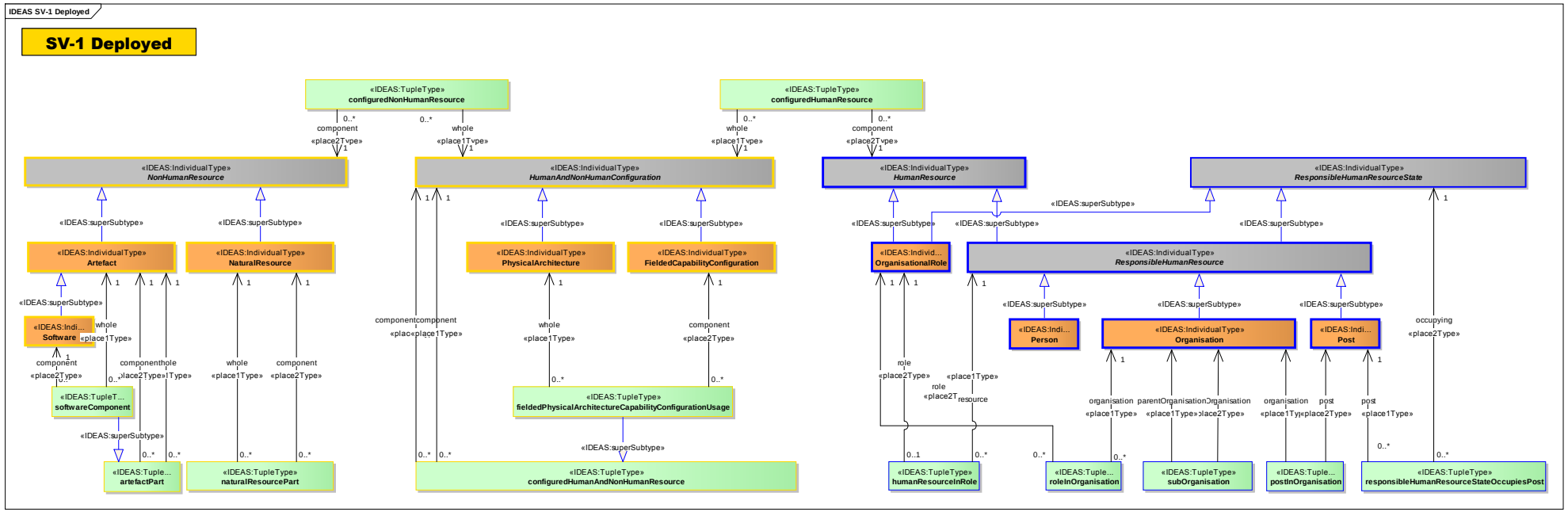


Figure 68 : SV-1 Deployed

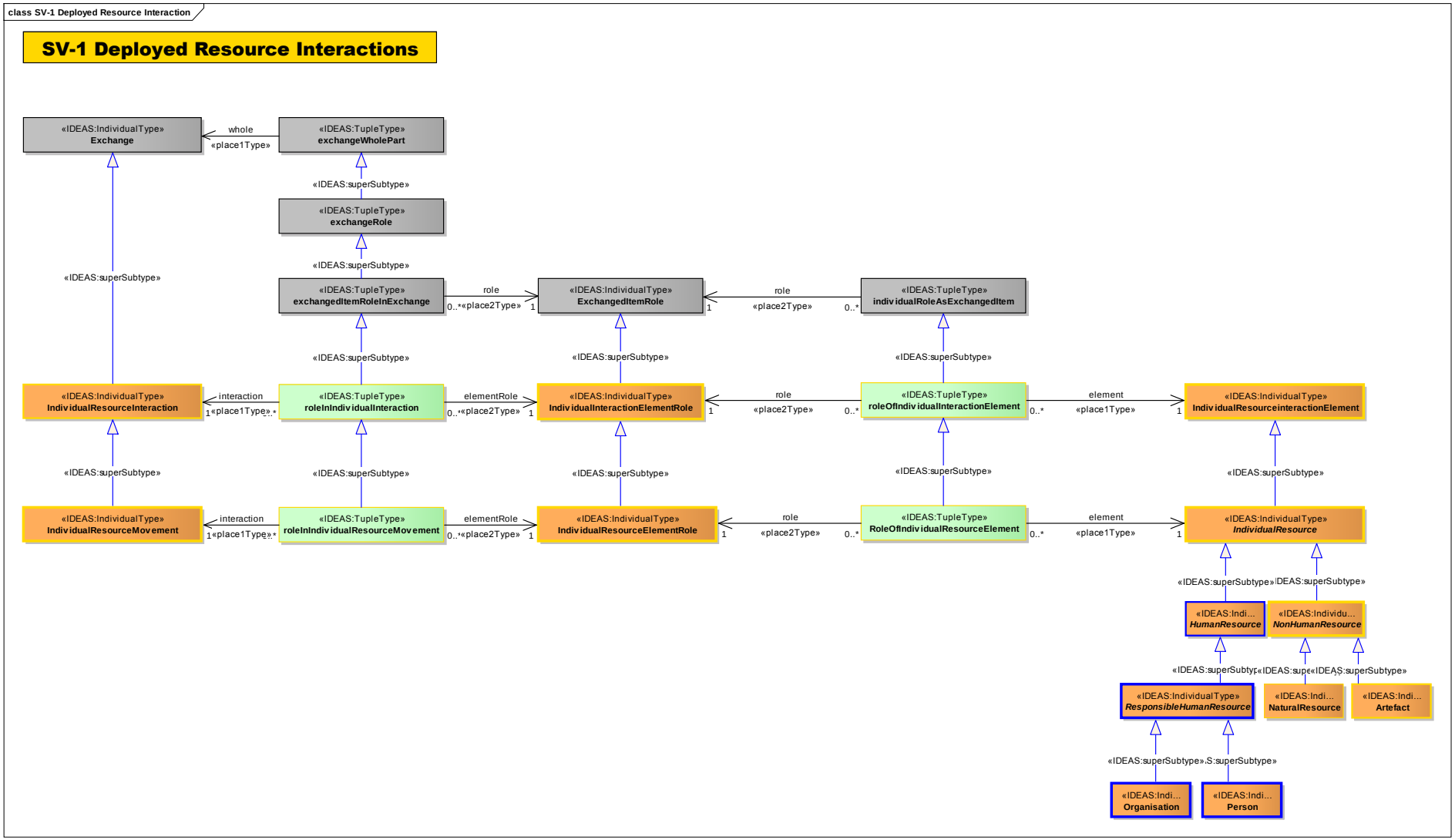


Figure 69 : SV-1 Deployed Resource Interaction

2.6.2 SV-2: System port specification, connectivity description and clusters

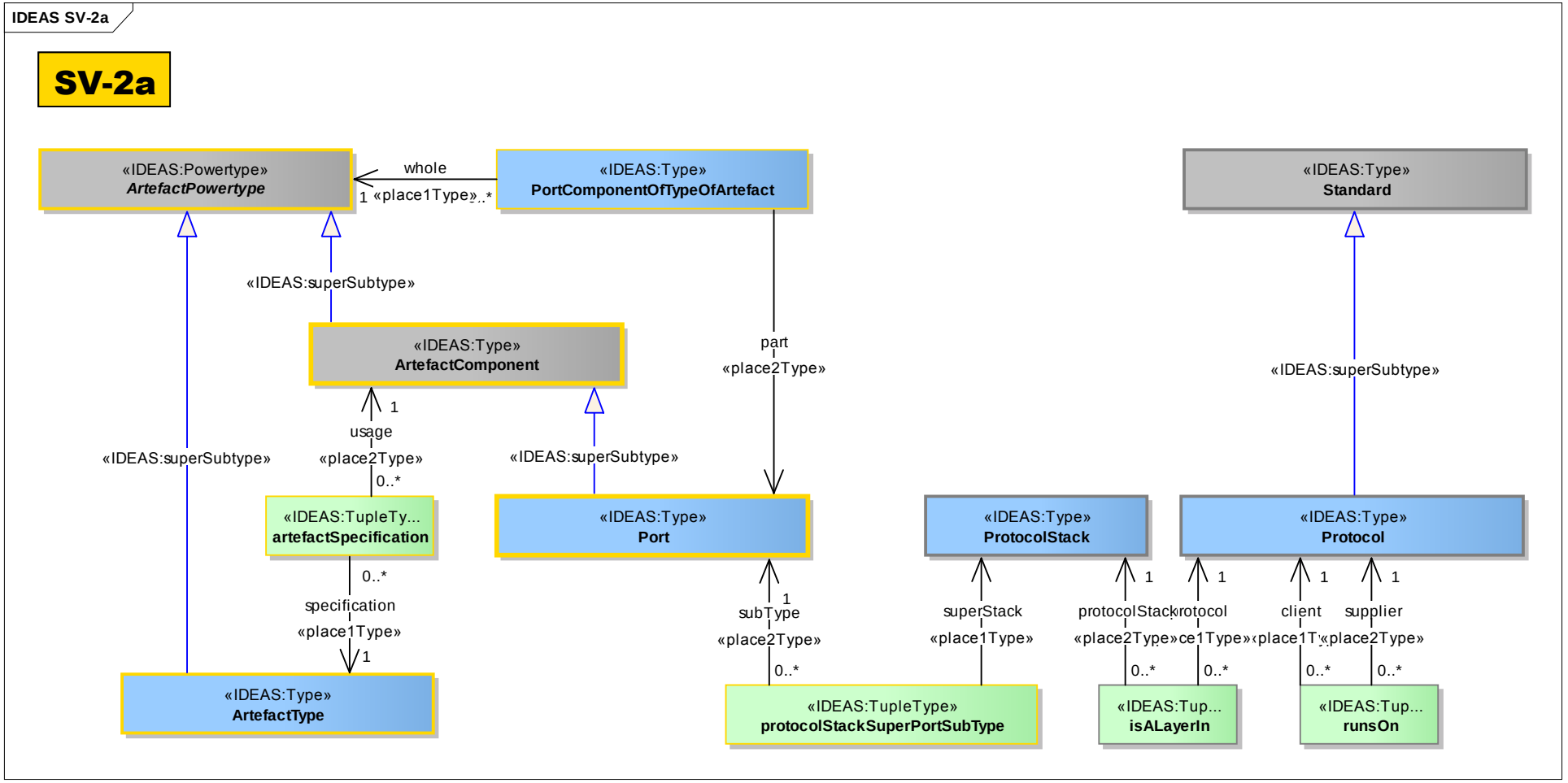


Figure 70 : SV-2a

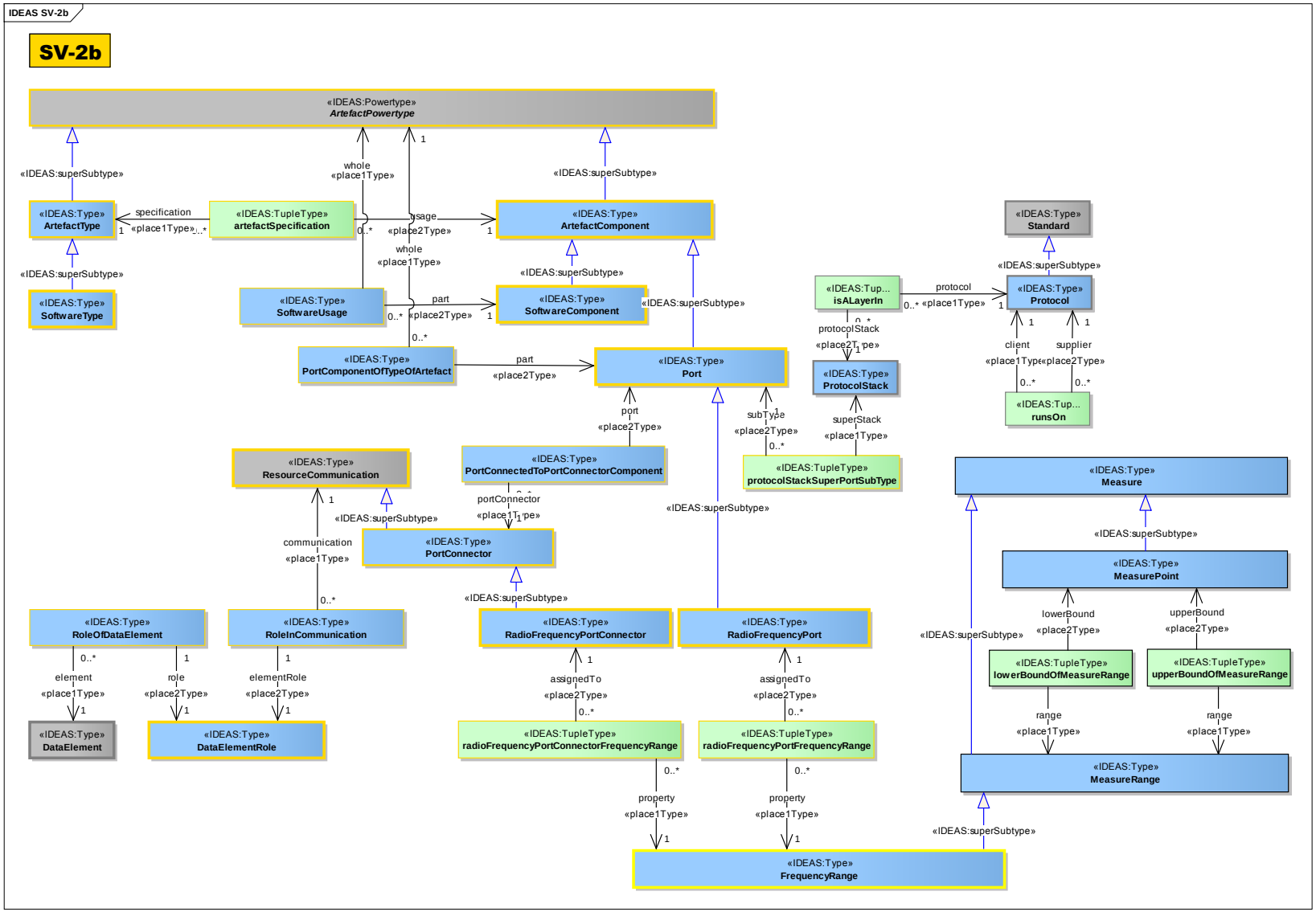


Figure 71 : SV-2b

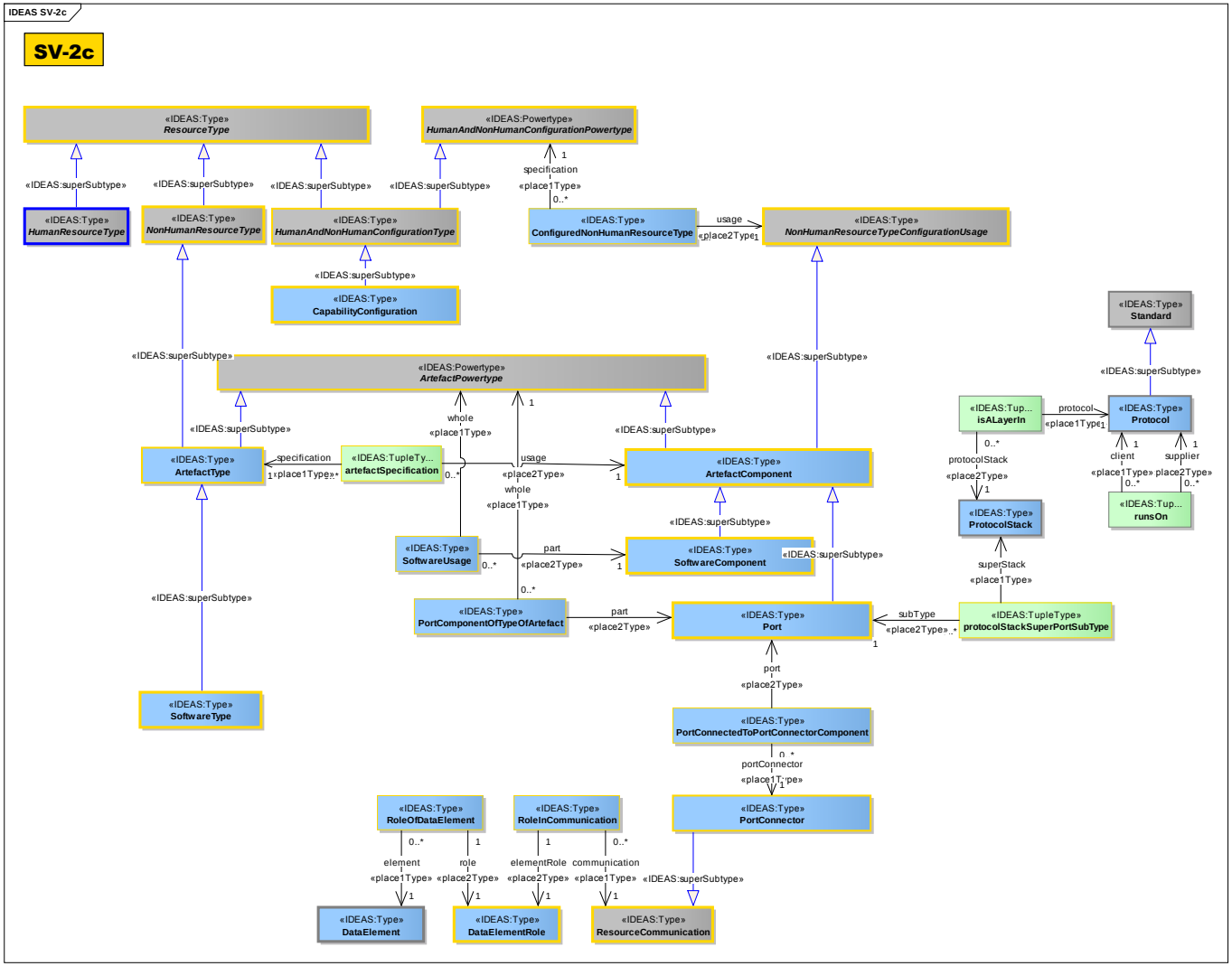


Figure 72 : SV-2c

2.6.3 SV-3: Resource interaction matrix

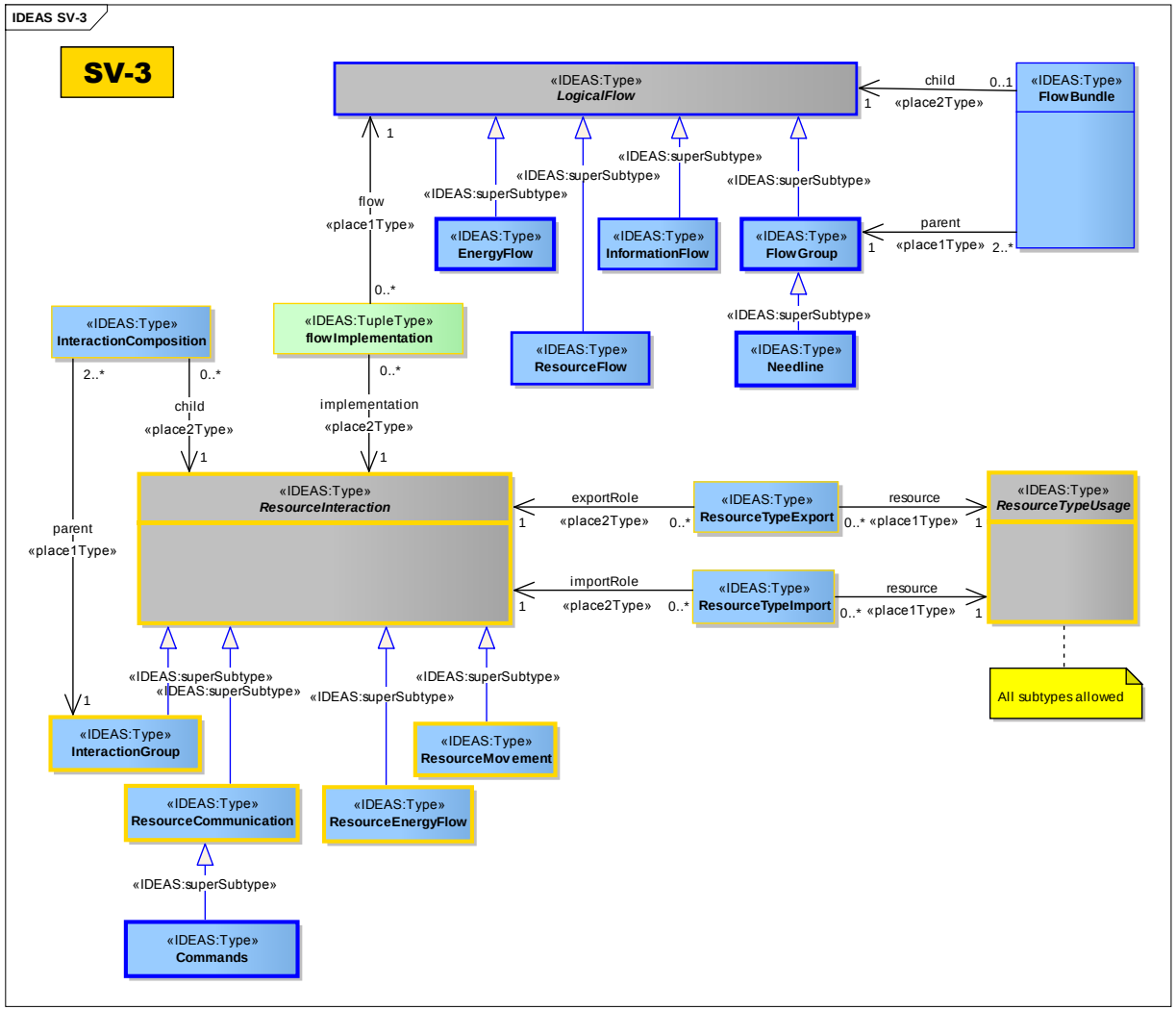


Figure 73 : SV-3

2.6.4 SV-4: Functionality description

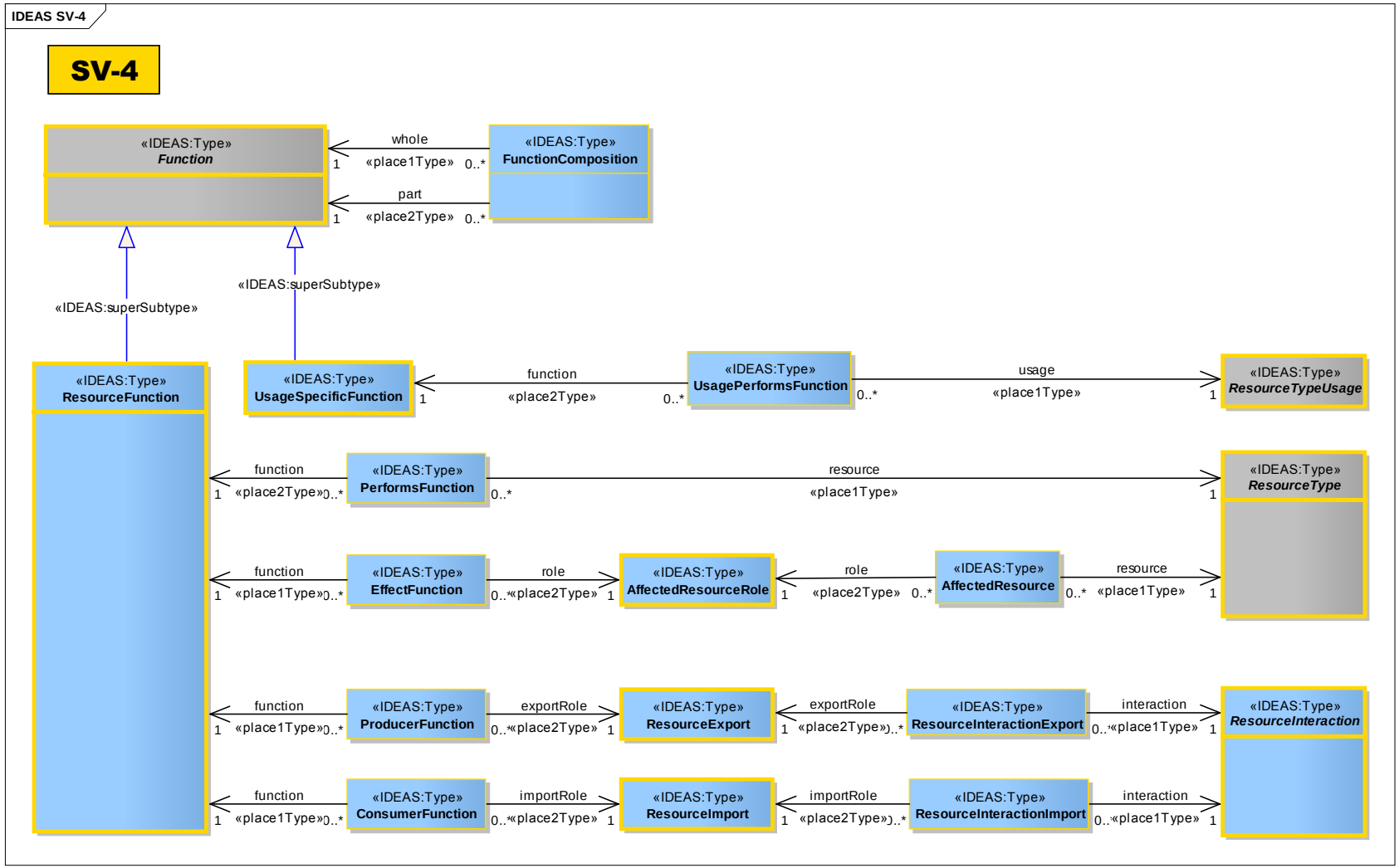


Figure 74 : SV-4

2.6.5 SV-5: Function operational activity/ service function traceability matrix

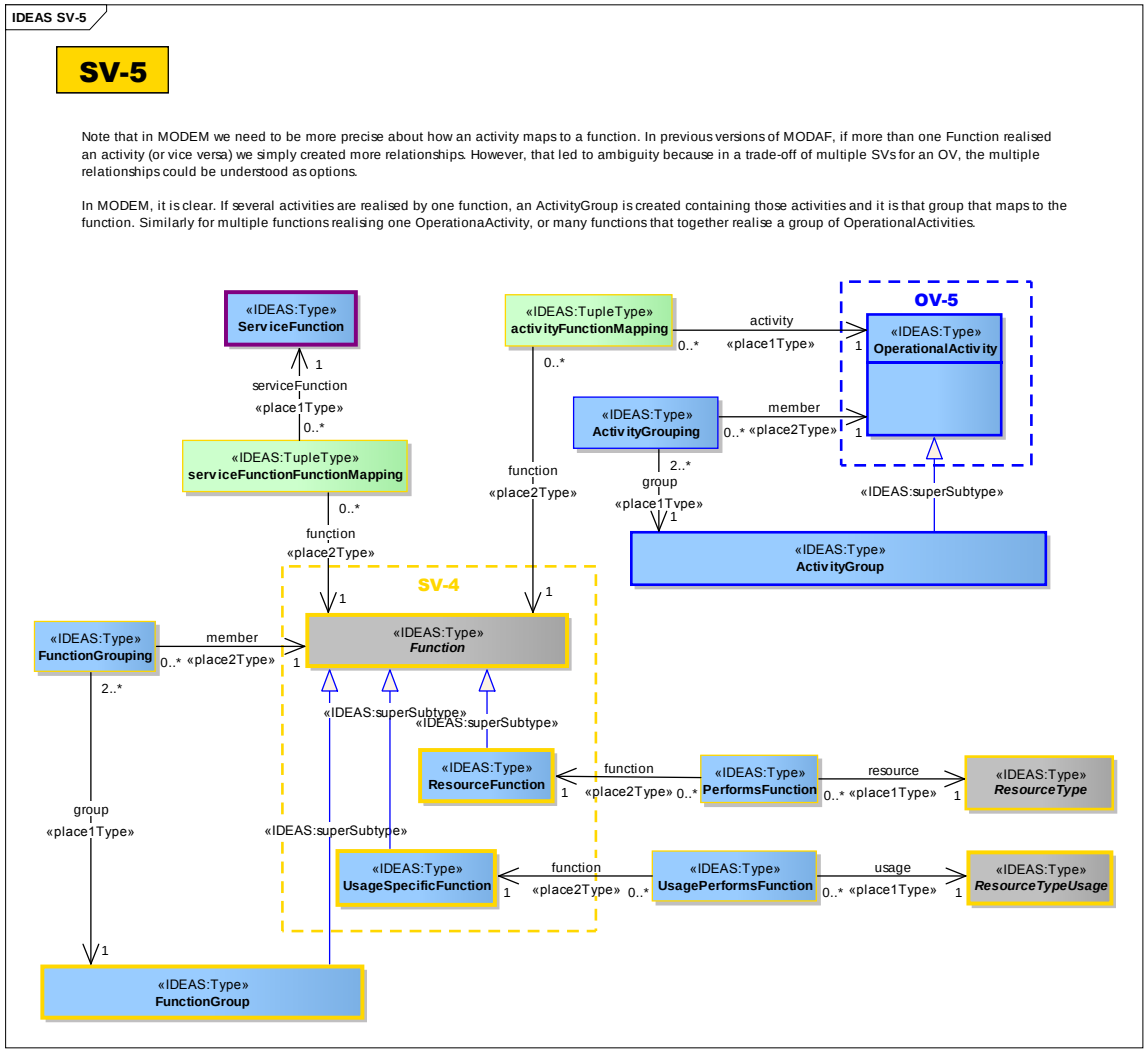


Figure 75 : SV-5

2.6.6 SV-6: Systems data exchange matrix

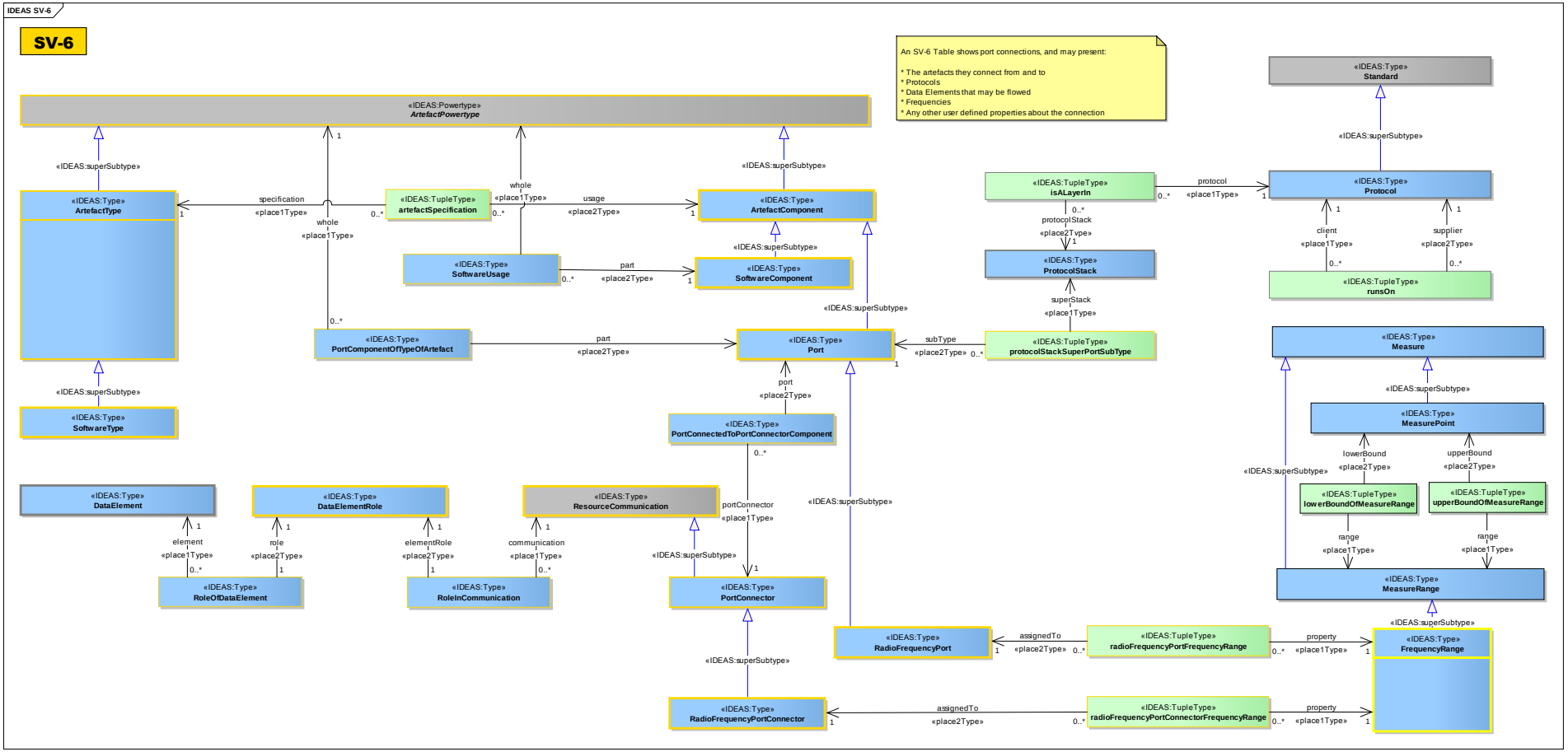


Figure 76 : SV-6

2.6.7 SV-7: Resource performance parameters matrix

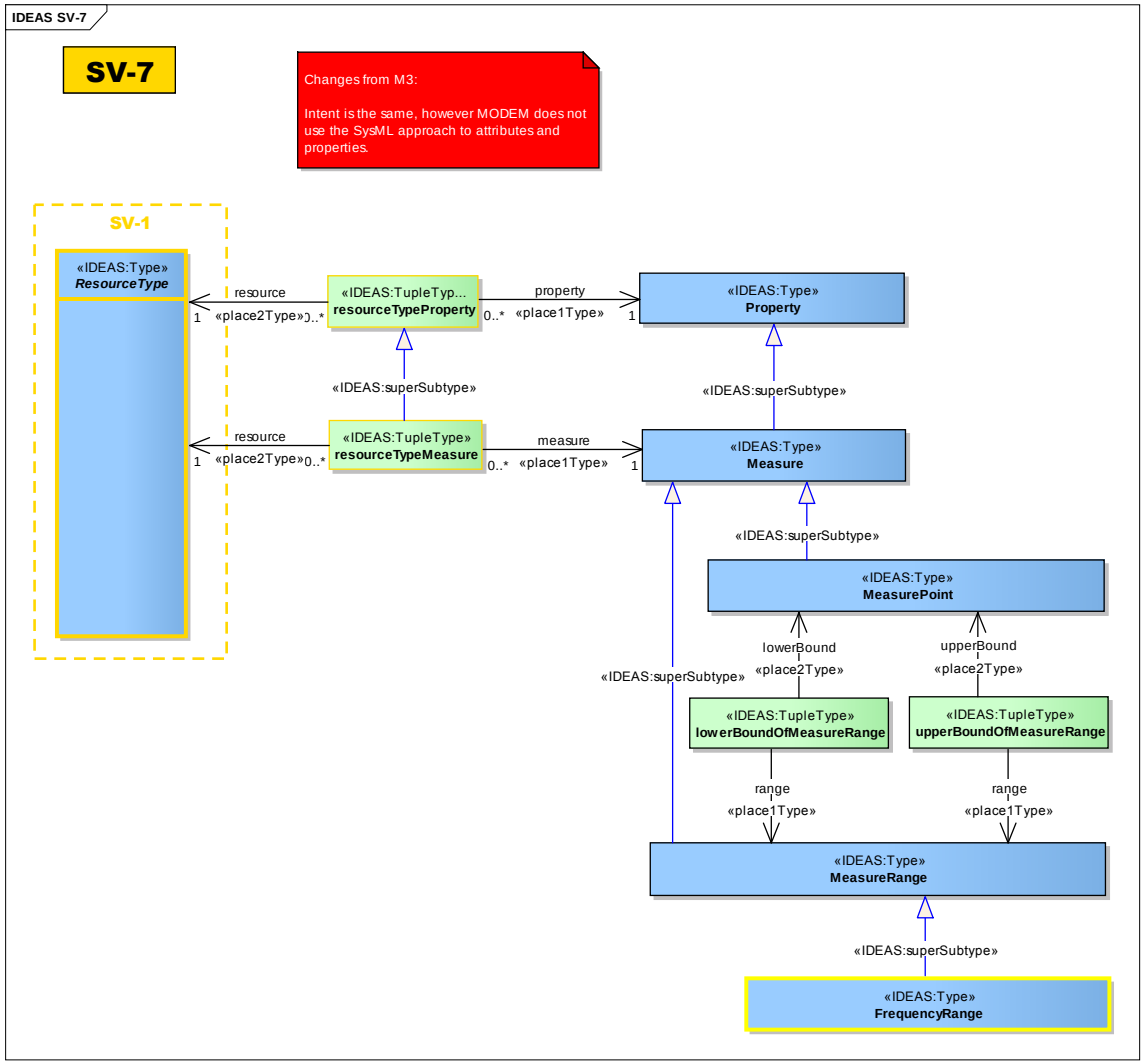


Figure 77 : SV-7

2.6.8 SV-8: Capability configuration management

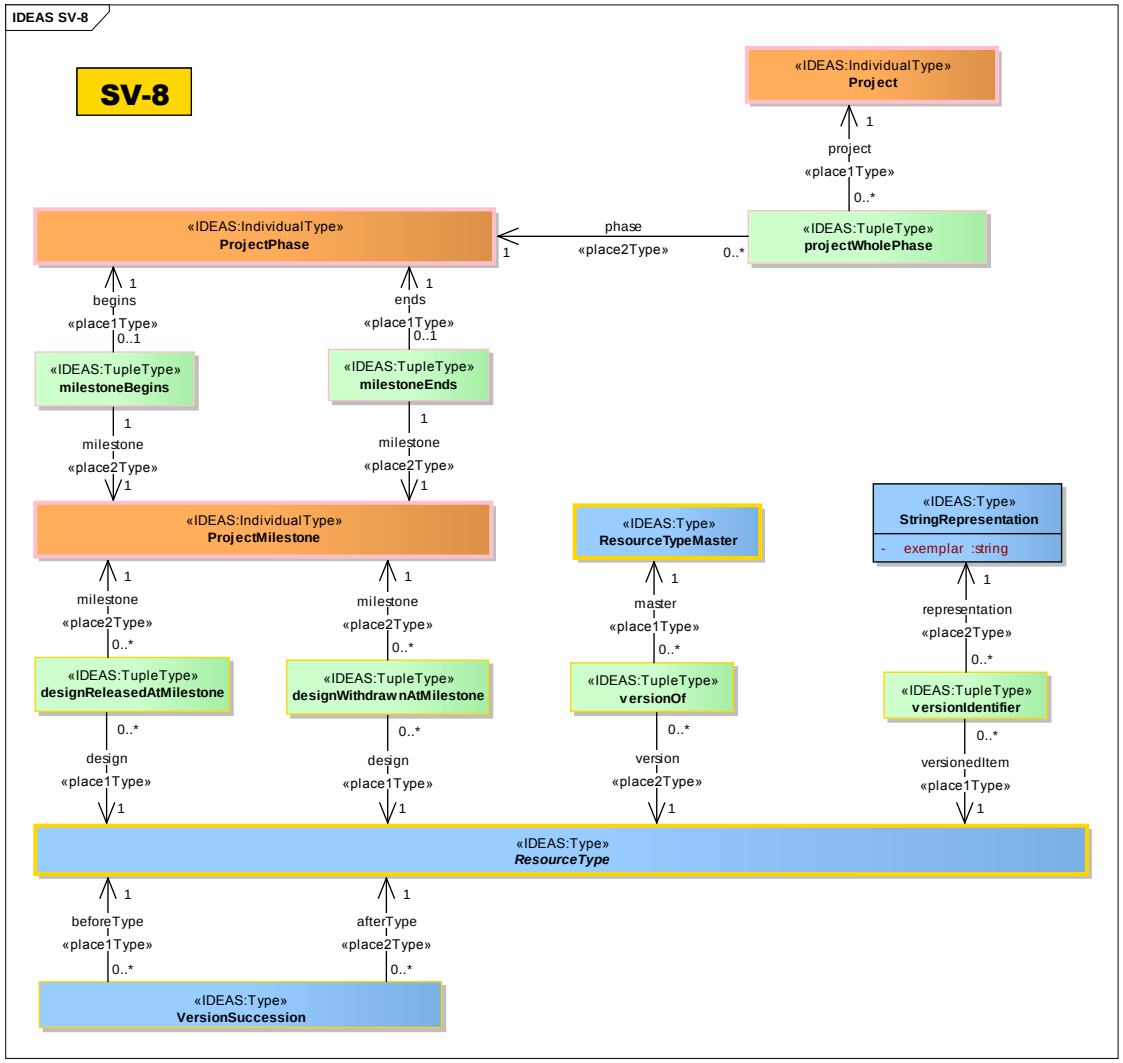


Figure 78 : SV-8

2.6.9 SV-9: Technology and skills forecast

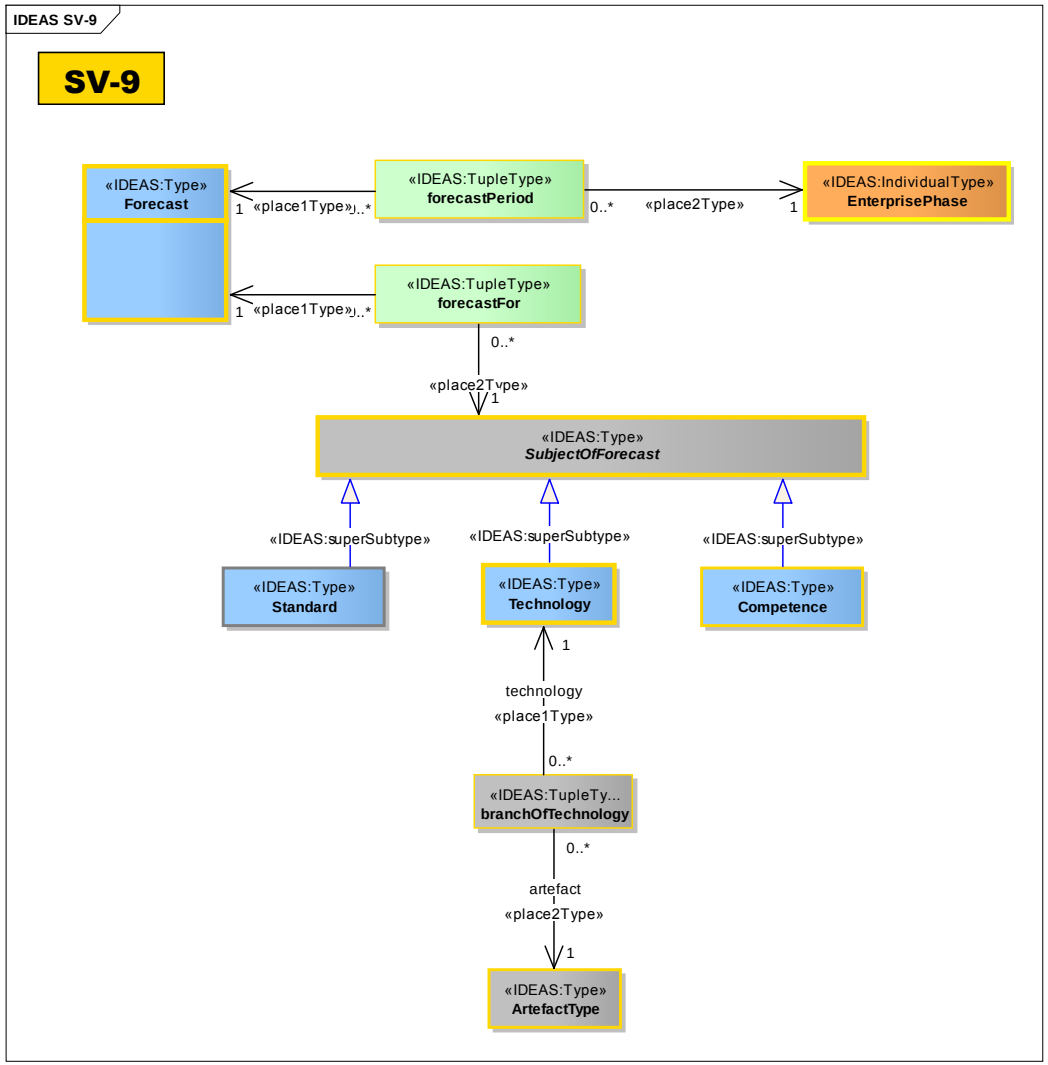


Figure 79 : SV-9

2.6.10 SV-10: Resource constraints, state transition and event-trace description

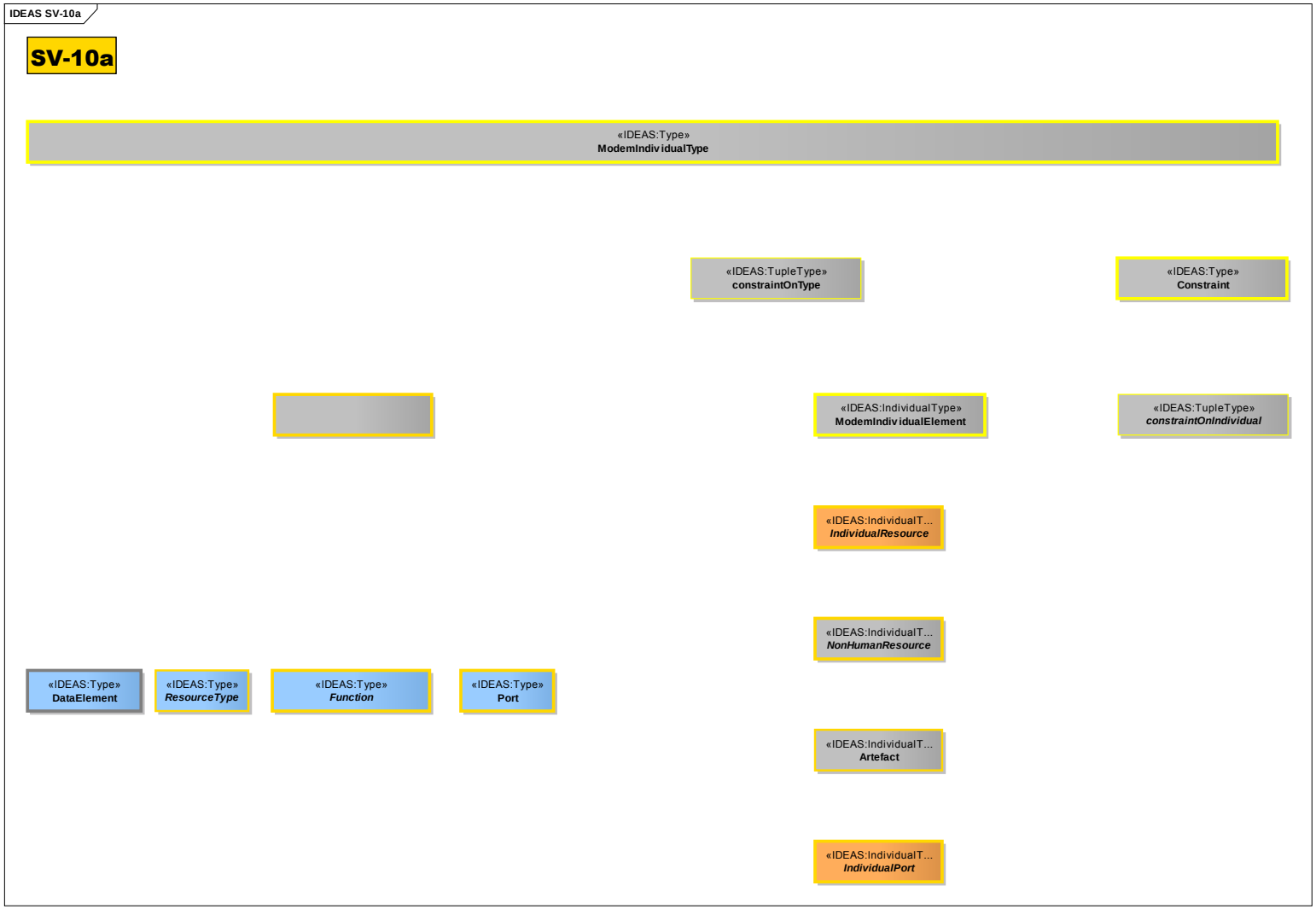


Figure 80 : SV-10a

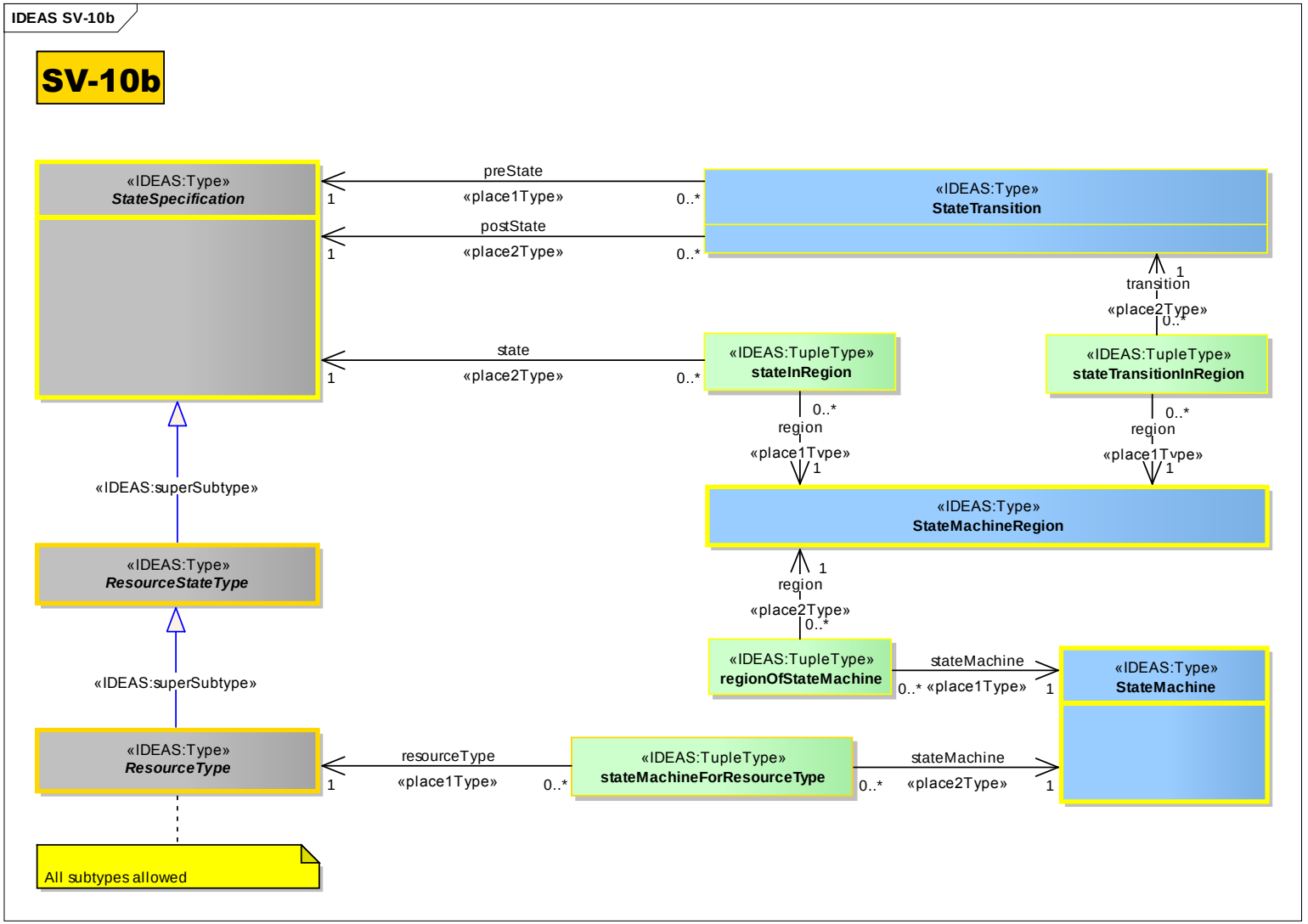


Figure 81 : SV-10b

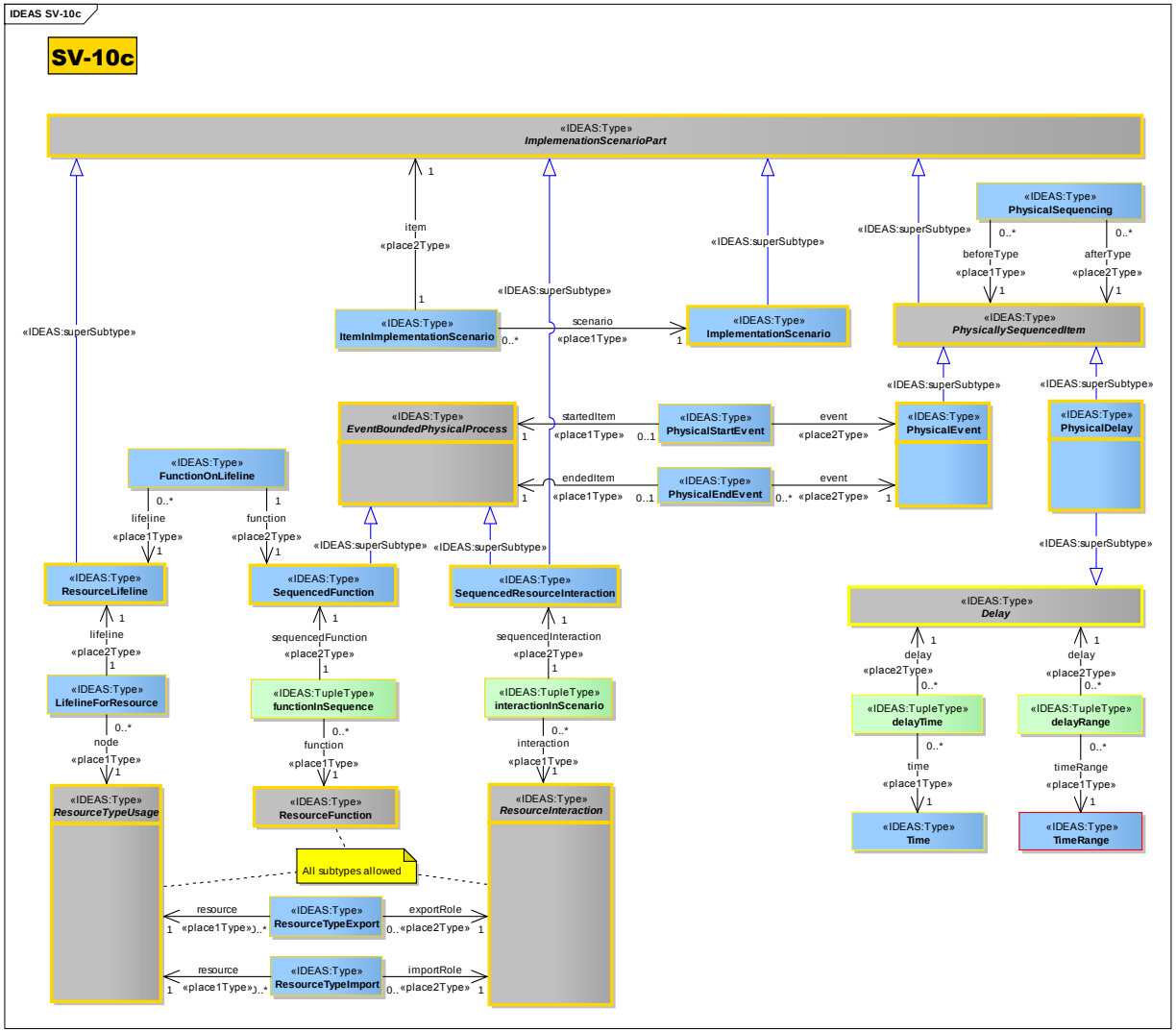


Figure 82 : SV-10c

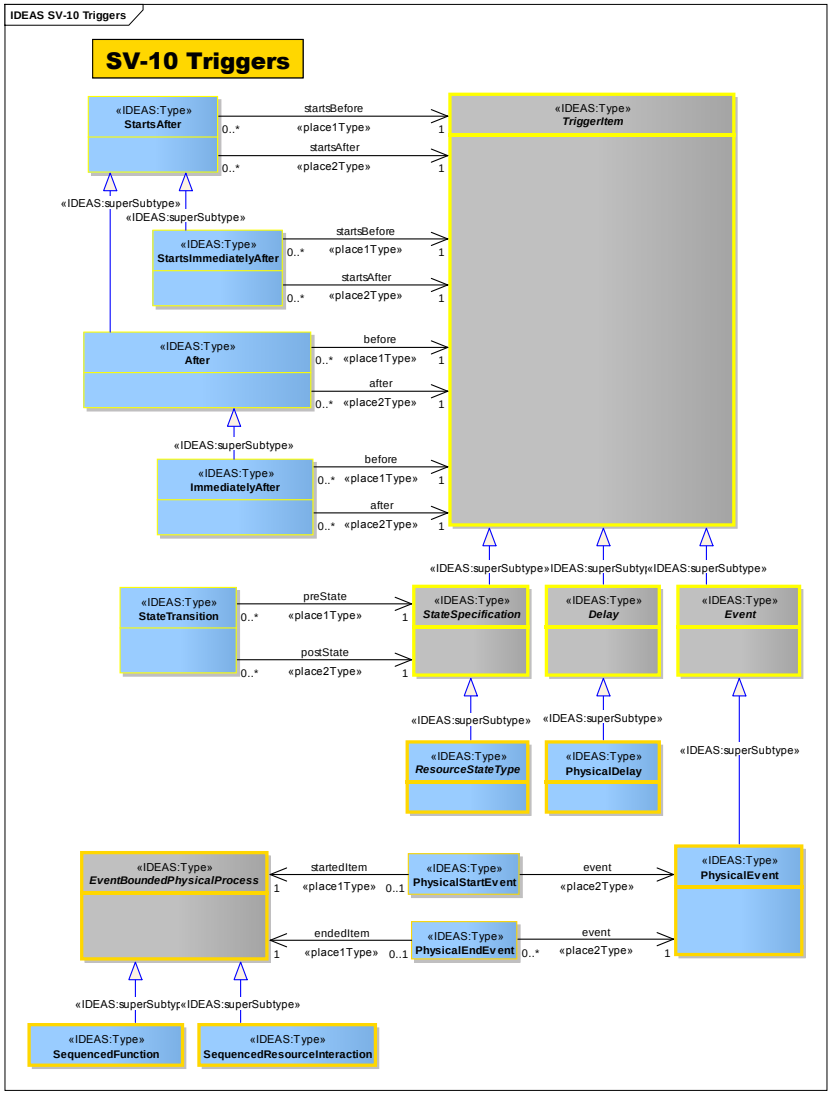


Figure 83 : SV-10 Triggers

2.6.11 SV-11: Physical schema

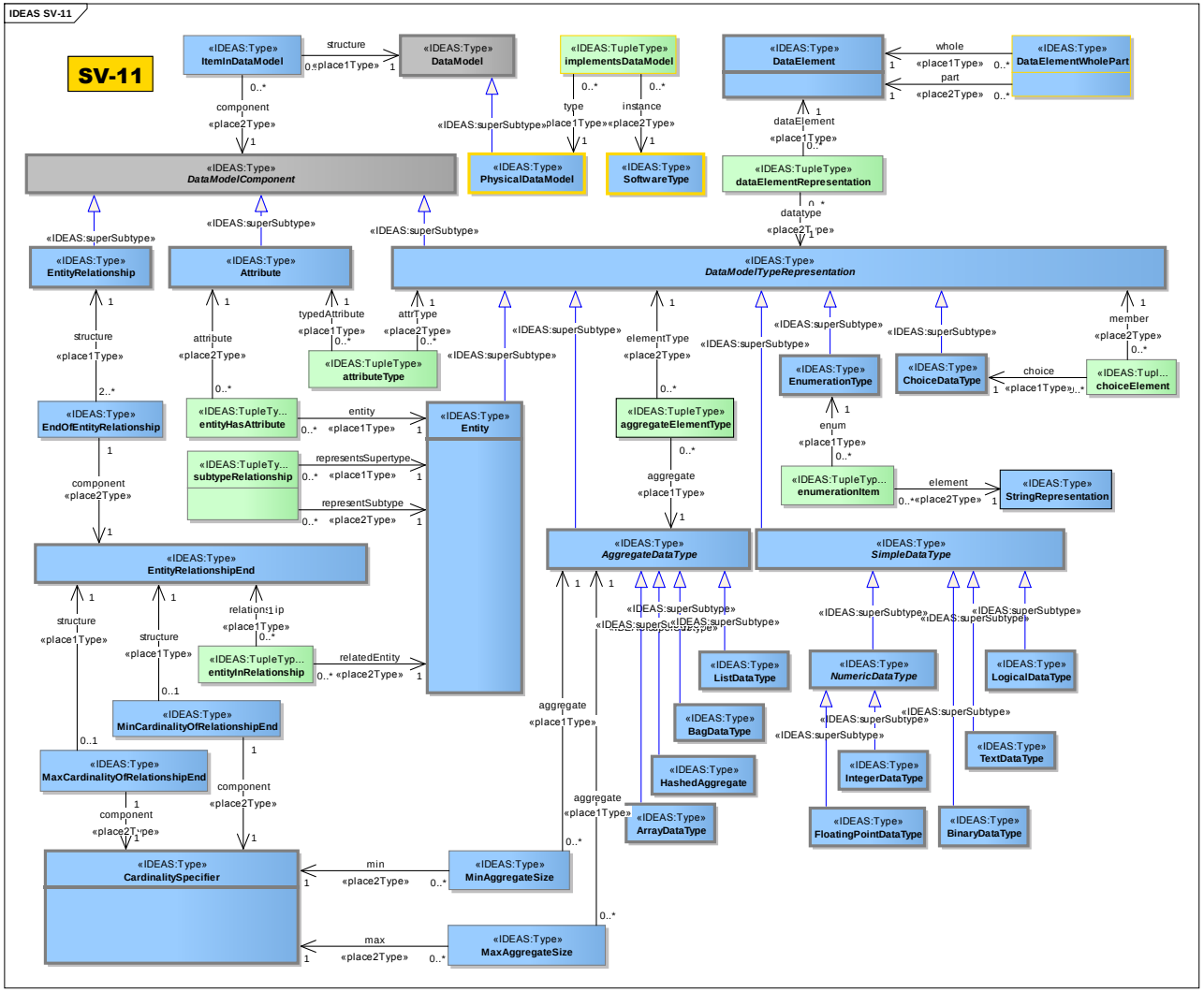


Figure 84 : SV-11

2.6.12 SV-12: Service provision and service composition

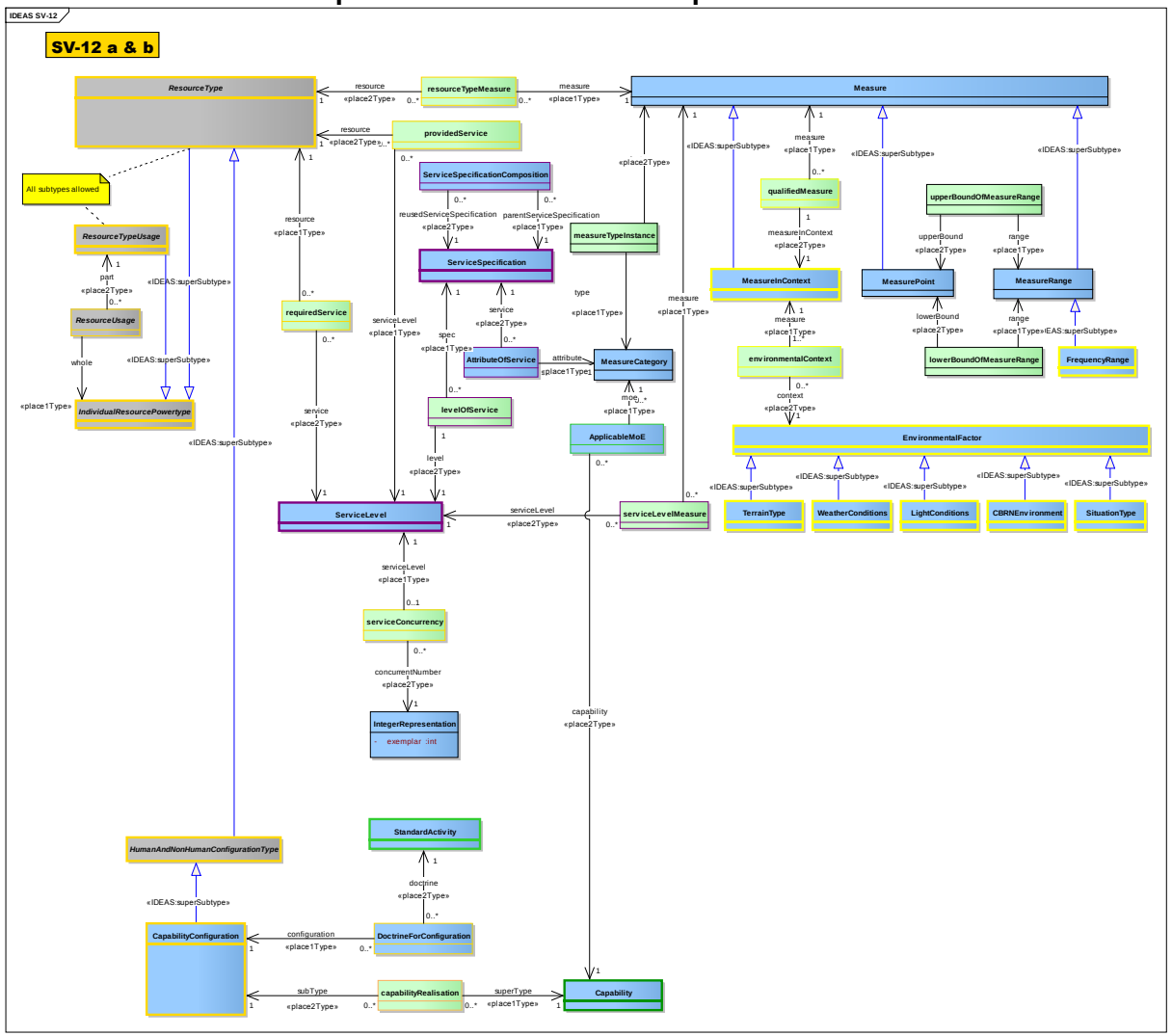


Figure 85 : SV-12

2.6.13 System Views elements list

System Views
AffectedResource «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AffectedResource - IndividualRoleType *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AffectedResource - TypicalWholePart *Association (source - target):*«place1Type» AffectedResource - ResourceType *Association (source - target):*«place2Type» AffectedResource - AffectedResourceRole *Attributes:* - An IndividualRoleType where the role extent is an AffectedResourceRole and the whole is a ResourceType.
AffectedResourceRole «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AffectedResourceRole - RoleExtentType *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AffectedResourceRole - ModemIndividualType *Attributes:* - A ModemIndividualType that is the role played by a ResourceType when it is acted upon by a Function.
Artefact «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Artefact - NonHumanResource *Dependency (element - is instance of):* «IDEAS:powertypeInstance» Artefact - ArtefactPowertype *Attributes:* - An IndividualResource that is non-human and man-made. Examples are "car", "radio", "diesel", etc.
ArtefactComponent «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArtefactComponent - ArtefactPowertype *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArtefactComponent - NonHumanResourceTypeConfigurationUsage *Attributes:* - A NonHumanResourceTypeConfigurationUsage that is a type of Artefact that is used as a component of an ArtefactType.
ArtefactInterface «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArtefactInterface - TypicalWholePart *Association (source - target):*«place1Type» ArtefactInterface - ArtefactType

Association (source - target):«place2Type»
ArtefactInterface - Interface
Attributes:
-
A TypicalWholePart that relates an ArtefactType to the Interface it provides or requires.

ArtefactType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArtefactType - ArtefactPowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ArtefactType - NonHumanResourceType
Attributes:
-
A type of man-made object. Examples are "car", "radio", "diesel", etc.
Note: It has no human components.

ArtefactUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ArtefactUsage - NonHumanResourceUsage
Association (source - target):«place1Type»
ArtefactUsage - ArtefactPowertype
Association (source - target):«place2Type»
ArtefactUsage - ArtefactComponent
Attributes:
-
A NonHumanResourceUsage that asserts a ArtefactComponent is used by an ArtefactType.

CapabilityConfiguration «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
CapabilityConfiguration - HumanAndNon-HumanConfigurationType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CapabilityConfiguration - FieldedCapabilityConfigurationPowertype
Attributes:
-
A composite structure representing the physical and human resources (and their interactions) that when brought together provide one or more Capabilities. A CapabilityConfiguration is a set of Resources configured to provide a capability, and should be guided by [doctrine] which may take the form of Standard or OperationalConstraint stereotypes.

CapabilityConfigurationConfigurationUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CapabilityConfigurationConfigurationUsage - HumanAndNonHumanResourceTypeConfigurationUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CapabilityConfigurationConfigurationUsage - FieldedCapabilityConfigurationPowertype
Attributes:
-
A HumanAndNonHumanResourceTypeConfigurationUsage that is a type of CapabilityConfiguration.

Competence «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Competence - ResponsibleHumanResourceStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Competence - SubjectOfForecast
Attributes:
-
A ResponsibleHumanResourceStateType where each instance is a state of a ResponsibleHumanResource that possesses a specific set of abilities defined by knowledge, skills and attitude.

ConfiguredHumanAndNonHumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ConfiguredHumanAndNonHumanResourceType - HumanAndNonHumanConfigurationTypeResourceUsage
Association (source - target):«place1Type»
ConfiguredHumanAndNonHumanResourceType - HumanAndNonHumanConfigurationPowertype
Association (source - target):«place2Type»
ConfiguredHumanAndNonHumanResourceType - HumanAndNonHumanResourceTypeConfigurationUsage
Attributes:
-
A resourceUsage that asserts that a state of a type of HumanAndNonHumanResouce is typically a component of a HumanAndNonHumanConfigurationType.

ConfiguredHumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ConfiguredHumanResourceType - HumanAndNonHumanConfigurationTypeResourceUsage
Association (source - target):«place1Type»
ConfiguredHumanResourceType - HumanAndNonHumanConfigurationPowertype
Association (source - target):«place2Type»
ConfiguredHumanResourceType - HumanResourceTypeConfigurationUsage
Attributes:
-
A resourceUsage that asserts that a state of a type of HumanResouce is typically a component of a HumanAndNonHumanConfigurationType.

ConfiguredNonHumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ConfiguredNonHumanResourceType - HumanAndNonHumanConfigurationTypeResourceUsage
Association (source - target):«place1Type»
ConfiguredNonHumanResourceType - HumanAndNonHumanConfigurationPowertype
Association (source - target):«place2Type»
ConfiguredNonHumanResourceType - NonHumanResourceTypeConfigurationUsage
Attributes:
-
A resourceUsage that asserts that a state of a type of NonHumanResouce is typically a component of a HumanAndNonHumanConfigurationType.

ConsumerFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConsumerFunction - ModemWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ConsumerFunction - IndividualExchangeRoleType

Association (source - target): «place1Type» ConsumerFunction - Function *Association (source - target):* «place2Type» ConsumerFunction - ResourceImport *Attributes:* - An IndividualExchangeRoleType where the role is a ResourceImport and the consumer is a Function.
Controls «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» Controls - ResourceCommunication *Attributes:* - A ResourceCommunication where one InteractionElement controls another.
DataElementRole «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» DataElementRole - InteractionElementRole *Attributes:* - An InteractionElementRole where the element is a DataElement.
DataElementWholePart «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» DataElementWholePart - TypicalWholePart *Association (source - target):*«place2Type» DataElementWholePart -DataElement *Association (source - target):*«place1Type» DataElementWholePart - DataElement *Attributes:* - A TypicalWholePart where one DataElement is a part of another.
DoctrineForConfiguration «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» DoctrineForConfiguration - ModemWholePartType *Association (source - target):*«place1Type» DoctrineForConfiguration - CapabilityConfiguration *Association (source - target):*«place2Type» DoctrineForConfiguration - StandardActivity *Attributes:* - A ModemWholePartType that asserts a StandardActivity is part of a CapabilityConfiguration - i.e. in order to deliver the Capability, the configuration must follow doctrinal processes.

EffectFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EffectFunction - ProcessWholeRoleExtentPartType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EffectFunction - TypicalWholePart
Association (source - target):«place1Type»
EffectFunction - ResourceFunction
Association (source - target):«place2Type»
EffectFunction - AffectedResourceRole
Attributes:
-
A TypicalWholePart that relates a ResourceFunction to the AffectedResourceRole played by a ResourceType when acted upon by the ResourceFunction.

EventBoundedPhysicalProcess «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EventBoundedPhysicalProcess - PhysicalArchitectureProcess
Attributes:
-
A PhysicalArchitectureProcess that can have PhysicalEvents marking its start and end points.

FieldedCapabilityConfiguration «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
FieldedCapabilityConfiguration - FieldedCapabilityConfigurationPowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FieldedCapabilityConfiguration - HumanAndNon-HumanConfiguration
Attributes:
-
An actual, fully-realised capability. A FieldedCapability must indicate its configuration (HumanAndNon-HumanConfiguration). Example: "HMS Iron Duke, configured and crewed, operating under the appropriate doctrine". Note - the CapabilityConfiguration that this realises would specify a UK Type 23 Frigate, the crew, the weapons systems, etc. Note: was called FieldedCapability in M3

FieldedCapabilityConfigurationPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FieldedCapabilityConfigurationPowertype - HumanAndNonHumanConfigurationPowertype
Attributes:
-
The powertype of FieldedCapabilityConfiguration.

Forecast «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Forecast - ModemIndividualType
Attributes:
-
A ModemIndividualType that is the forecasted future states of one or more SubjectOfForecasts for the forecast period.

Function «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Function - PhysicalArchitectureProcess
Attributes:
-
A PhysicalArchitectureProcess that is either carried out by a ResourceType or a ResourceTypeUsage.

FunctionComposition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FunctionComposition - TypicalWholePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FunctionComposition - ProcessWholeAndPartType
Association (source - target): «place2Type»
FunctionComposition - Function
Association (source - target): «place1Type»
FunctionComposition - Function
Attributes:
-
A TypicalWholePart that relates a parent (whole) Function to its child (part) Function.

FunctionGroup «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FunctionGroup - Function
Attributes:
-
A Function that is entirely composed of other Functions

FunctionGrouping «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FunctionGrouping - FunctionComposition
Association (source - target): «place1Type»
FunctionGrouping - FunctionGroup
Association (source - target): «place2Type»
FunctionGrouping - Function
Attributes:
-
A FunctionComposition where the parent is a FunctionGroup.

FunctionOnLifeline «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
FunctionOnLifeline - TypicalWholePart
Association (source - target): «place2Type»
FunctionOnLifeline - SequencedFunction
Association (source - target): «place1Type»
FunctionOnLifeline - ResourceLifeline
Attributes:
-

A TypicalWholePart where a SequencedFunction is part of a ResourceLifeline. Note: a given SequencedFunction may appear on one and only one ResourceLifeline.

HumanAndNonHumanConfiguration «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanAndNonHumanConfiguration - IndividualResource
Dependency (element - is instance of): «IDEAS:powertypeInstance»
HumanAndNonHumanConfiguration - HumanAndNonHumanConfigurationPowertype
Attributes:
-
An IndividualResource that is composed of both human and non-human resources.

HumanAndNonHumanConfigurationPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanAndNonHumanConfigurationPowertype - IndividualResourcePowertype
Attributes:
-
The powertype of HumanAndNonHumanConfiguration.

HumanAndNonHumanConfigurationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanAndNonHumanConfigurationType - ResourceType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanAndNonHumanConfigurationType - HumanAndNonHumanConfigurationPowertype
Attributes:
-
A ResourceType that has both Human and Non-Human components.

HumanAndNonHumanConfigurationTypeResourceUsage «IDEAS:Type»
Connectors:
Association (source - target):«place1Type»
HumanAndNonHumanConfigurationTypeResourceUsage - HumanAndNonHumanConfigurationPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanAndNonHumanConfigurationTypeResourceUsage - ResourceUsage
Attributes:
-
A resourceUsage that asserts that a state of a type of HumanResouce is typically a component of a HumanAndNonHumanConfigurationType.

HumanAndNonHumanResourceTypeConfigurationUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanAndNonHumanResourceTypeConfigurationUsage - ResourceTypeUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanAndNonHumanResourceTypeConfigurationUsage - HumanAndNonHumanConfigurationPowertype
Attributes:
-
ResourceTypeUsage that is a type of state or part of a HumanAndNonHumanResource that is used by (a part of, a component of) another HumanAndNonHumanResourceType .

HumanResourcePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanResourcePowertype - IndividualResourcePowertype

Attributes:
-
The powertype of HumanResource.

HumanResourceState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanResourceState - IndividualResourceState
Attributes:
-
An IndividualResourceState that is a temporal state of a HumanResource.

HumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanResourceType - ResourceType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
HumanResourceType - HumanResourcePowertype
Attributes:
-
A ResourceType that is a type of HumanResource. A PersonType, PostType, OrganisationType or OrganisationRoleType. [ABSTRACT]
Note: was called "OrganisationalResource" in M3 v1.2. Note: was called "OrganisationalResourceType" in M3.

HumanResourceTypeConfigurationUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanResourceTypeConfigurationUsage - HumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanResourceTypeConfigurationUsage - ResourceTypeUsage
Attributes:
-
A state or part of a HumanResourceType that is used by (a part of, a component of) another HumanResourceType.

HumanResourceTypeUsage «IDEAS:Type»
Connectors:
Association (source - target):«place1Type»
HumanResourceTypeUsage - HumanResourceType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HumanResourceTypeUsage - ResourceUsage
Attributes:
-
A resourceUsage that ia a type of humanResource which asserts a given HumanResourceType belongs to an ResourceTypeUsage.

ImplemenationScenarioPart «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ImplemenationScenarioPart - PhysicalArchitectureIndividualType
Attributes:
-
A ModemIndividualType that features in (i.e. is part of) an ImplemenationScenario.

ImplementationScenario «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ImplementationScenario - Scenario
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ImplementationScenario – ImplementationScenarioPart
Attributes:
-
A Scenario that features ResourceTypes, their Functions and Interactions.

IndividualInteractionElementRole «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualInteractionElementRole - ExchangedItemRole
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualInteractionElementRole - IndividualInteractionElementRolePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualInteractionElementRole - ModemIndividualElement
Attributes:
-
An ExchangedItemRole that is a role in an IndividualResourceInteraction.

IndividualInteractionElementRolePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualInteractionElementRolePowertype - ExchangedItemRoleType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualInteractionElementRolePowertype - ModemIndividualElementType
Attributes:
-
The powertype of IndividualInteractionElementRole.

IndividualPort «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualPort - IndividualPortPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualPort - Artefact
Attributes:
-
An Artefact that is a port or interface provided by (and part of) an Artefact. Note: subsumes "SystemPort" and "SoftwarePort" in M3.

IndividualPortConnectedToPortConnectorPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualPortConnectedToPortConnectorPowertype - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualPortConnectedToPortConnectorPowertype - CoupleType
Attributes:
-
The powertype of IndividualPortConnectedToPortConnector.

IndividualPortConnector «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualPortConnector - IndividualResourceInteraction
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualPortConnector - IndividualPortConnectorPowertype
Attributes:
-
An IndividualResourceInteraction that has a protocolStackTypeIndividualPortConnector to a ProtocolStack.

IndividualPortConnectorPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualPortConnectorPowertype - ExchangeType
Attributes:
-
The powertype of IndividualPortConnector.

IndividualPortPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualPortPowertype - IndividualResourcePartPowertype
Attributes:
-
The powertype of IndividualPort.

IndividualRadioFrequencyPort «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualRadioFrequencyPort - IndividualPort
Attributes:
-
An IndividualPort that use radio frequency.

IndividualRadioFrequencyPortConnector «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualRadioFrequencyPortConnector - IndividualRadioFrequencyPortConnectorPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualRadioFrequencyPortConnector - IndividualPortConnector
Attributes:
-
An IndividualPortConnector that connects two ports using a radio frequency.

IndividualRadioFrequencyPortConnectorPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualRadioFrequencyPortConnectorPowertype - IndividualPortConnectorPowertype
Attributes:
-
The powertype of IndividualRadioFrequencyPortConnector.

IndividualResource «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResource - IndividualResourceInteractionElement
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResource - Body
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResource - ModemIndividualElement
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualResource - IndividualResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResource - IndividualResourceState
Attributes:
-
A ModemIndividualElement that is an IndividualOrganisationalResource, an ItemOfMateriel or a ResourceConfiguration.

IndividualResourceElementRole «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualResourceElementRole - IndividualResourceElementRolePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceElementRole - IndividualInteractionElementRole
Attributes:
-
An IndividualInteractionElementRole that is a role in an IndividualResourceMovement.

IndividualResourceElementRolePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceElementRolePowertype - IndividualInteractionElementRolePowertype
Attributes:
-
The powertype of IndividualResourceElementRole.

IndividualResourceInteraction «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceInteraction - Exchange
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualResourceInteraction - IndividualResourceInteractionPowertype
Attributes:
-
An Exchange between IndividualResourceInteractionElements.

IndividualResourceInteractionElementPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceInteractionElementPowertype - ModemIndividualElementType
Attributes:
-
The powertype of IndividualResourceInteractionElement.

IndividualResourceInteractionPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceInteractionPowertype - ModemIndividualElementType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceInteractionPowertype - ExchangeType
Attributes:
-
The powertype of IndividualResourceInteraction.

IndividualResourceMovement «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualResourceMovement - IndividualResourceMovementPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceMovement - IndividualResourceInteraction
Attributes:
-
An IndividualResourceInteraction between IndividualResources.

IndividualResourceMovementPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceMovementPowertype - IndividualResourceInteractionPowertype
Attributes:
-
The powertype of IndividualResourceMovement.

IndividualResourcePart «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualResourcePart - IndividualResourcePartPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourcePart - ModemIndividualElement

Attributes:
-
A ModemIndividualElement that is a part of an IndividualResource. Note: an IndividualResource is an improper part of itself.

IndividualResourcePartPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourcePartPowertype - ModemIndividualElementType
Attributes:
-
The powertype of IndividualResourcePart.

IndividualResourcePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
IndividualResourcePowertype - IndividualResourceStatePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourcePowertype - BodyType

Attributes: - The powertype of IndividualResourceState
IndividualResourceState «IDEAS:IndividualType» *Connectors*: Generalization (element - is a subtype of): «IDEAS:superSubtype» IndividualResourceState – IndividualResourcePart Dependency (element - is instance of): «IDEAS:powertypeInstance» IndividualResourceState - IndividualResourceStatePowertype *Attributes*: - A ModemIndividualElement that is either a IndividualResource or a proper state of one.
IndividualResourceStatePowertype «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» IndividualResourceStatePowertype - IndividualResourcePartPowertype *Attributes:* - The powertype of IndividualResourceState.
IndividualResourceStateUsagePowertype «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» IndividualResourceStateUsagePowertype - ModemTemporalWholePartType *Association (source - target):*«place2Type» IndividualResourceStateUsagePowertype - IndividualResourceStatePowertype *Association (source - target):*«place1Type» IndividualResourceStateUsagePowertype - IndividualResourcePowertype *Attributes:* - The powertype of IndividualResourceStateUsage.
IndividualResourceStateWholeAndPartType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» IndividualResourceStateWholeAndPartType - ModemWholePartType *Attributes:* - The powertype of IndividualResourceStateWholeAndPart.
IndividualResourceUsagePowertype «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» IndividualResourceUsagePowertype - ModemWholePartType *Association (source - target):*«place1Type» IndividualResourceUsagePowertype - IndividualResourcePowertype *Association (source - target):*«place2Type» IndividualResourceUsagePowertype - IndividualResourcePartPowertype *Attributes:* - The powertype of individualResourceUsage.

IndividualResourceInteractionElement «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
IndividualResourceInteractionElement - IndividualResourceInteractionElementPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IndividualResourceInteractionElement - ModemIndividualElement
Attributes:
-
A ModemIndividualElement that is a exchanged in an IndividualResourceInteraction.

InteractionComposition «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InteractionComposition - TypicalWholePart
Association (source - target):«place2Type»
InteractionComposition - ResourceInteraction
Association (source - target):«place1Type»
InteractionComposition - InteractionGroup
Attributes:
-
A TypicalWholePart where one ResourceInteraction is part of an InteractionGroup.

InteractionElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InteractionElement - IndividualResourceInteractionElementPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InteractionElement - ModemIndividualType
Attributes:
-
A ModemIndividualType that can be flowed between Resources in a ResourceInteraction.
[ABSTRACT]

InteractionElementRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InteractionElementRole - IndividualInteractionElementRolePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InteractionElementRole - ModemIndividualType
Attributes:
-
A ModemIndividualType that is the Role played by an InteractionElement in a ResourceInteraction.
[ABSTRACT]

InteractionGroup «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InteractionGroup - ResourceInteraction
Attributes:
-
A ResourceInteraction that is composed of other ResourceInteractions.

ItemInImplementationScenario «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ItemInImplementationScenario - ItemInScenario
Association (source - target): «place2Type»
ItemInImplementationScenario - ImplemenationScenarioPart
Association (source - target): «place1Type»
ItemInImplementationScenario - ImplementationScenario
Attributes:
-
An ItemInScenario where the Scenario is an ImplementationScenario.

LifelineForResource «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LifelineForResource - TypicalTemporalWholePart
Association (source - target): «place2Type»
LifelineForResource - ResourceLifeline
Association (source - target): «place1Type»
LifelineForResource - ResourceType
Attributes:
-
A TypicalTemporalWholePart that asserts a ResourceLifeLine is a typical temporal part of a Resource.

NaturalResource «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NaturalResource - NonHumanResource
Dependency (element - is instance of): «IDEAS:powertypeInstance»
NaturalResource - NaturalResourcePowertype
Attributes:
-
An IndividualResource that is non-human and natural. Examples are "rock", "tree", "animal", etc.

NaturalResourceComponent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NaturalResourceComponent - NaturalResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NaturalResourceComponent - NonHumanResourceTypeConfigurationUsage
Attributes:
-
A NonHumanResourceTypeConfigurationUsage that is a type of NaturalResource, a specialisation of NaturalResourceType, that is used as a component of a NaturalResourceType.

NaturalResourcePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NaturalResourcePowertype - NonHumanResourcePowertype
Attributes:
-
The powertype of NaturalResource.

NaturalResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NaturalResourceType - NaturalResourcePowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NaturalResourceType - NonHumanResourceType
Attributes:
-
A NonHumanResourceType that is a type of NaturalResource.

NaturalResourceUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NaturalResourceUsage - NonHumanResourceUsage
Association (source - target):«place1Type»
NaturalResourceUsage - NaturalResourcePowertype
Association (source - target):«place2Type»
NaturalResourceUsage - NaturalResourceComponent
Attributes:
-
A NonHumanResourceUsage that asserts that a NaturalResourceComponent is used by a NaturalResourceType.

NonHumanResource «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NonHumanResource - NonHumanResourceState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NonHumanResource - IndividualResource
Dependency (element - is instance of): «IDEAS:powertypeInstance»
NonHumanResource - NonHumanResourcePowertype
Attributes:
-
An IndividualResource that is non-human.

NonHumanResourcePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NonHumanResourcePowertype - IndividualResourcePowertype
Attributes:
-
The powertype of NonHumanResource.

NonHumanResourceState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NonHumanResourceState - IndividualResourceState
Attributes:
-
A state of a NonHumanResource.

NonHumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NonHumanResourceType - ResourceType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NonHumanResourceType - NonHumanResourcePowertype
Attributes:
-
A ResourceType that is a type of NonHumanResource (i.e. an Artefact or NaturalResource). [ABSTRACT]

NonHumanResourceTypeConfigurationUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NonHumanResourceTypeConfigurationUsage - NonHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NonHumanResourceTypeConfigurationUsage - ResourceTypeUsage
Attributes:
-
A NonHumanResourceTypeConfigurationUsage that is a type of state or part of a NonHumanResource that is used by (a part of, a component of) another NonHumanResourceType.

NonHumanResourceUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NonHumanResourceUsage - ResourceUsage
Association (source - target):«place1Type»
NonHumanResourceUsage - NonHumanResourcePowertype
Attributes:
-
A ResourceUsage that asserts a type of NonHumanResource is used by a ResourceType.

OrganisationPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationPowertype - ResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationPowertype - OrganisationStatePowertype
Attributes:
-
The powertype of Organisation.

OrganisationRoleType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationRoleType - OrganisationalRolePowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»

OrganisationRoleType - HumanResourceType
Attributes:
-
A type of role a human resource may carry out in an organisation. This is not used as a component of a ResourceType. Note: was called "RoleType" in M3.

OrganisationRoleTypeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationRoleTypeUsage - HumanResourceTypeConfigurationUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationRoleTypeUsage - OrganisationalRolePowertype
Attributes:
-
A HumanResourceTypeConfigurationUsage that is a type of OrganisationRole , a specialisation of OrganisationRoleType, that is used as a component of a ResourceType.

OrganisationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
OrganisationType - OrganisationPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationType - ResponsibleHumanResourceType
Attributes:
-
A ResponsibleHumanResourceType and a ConstructedHumanResourceType that is a type of Organisation. This is not used as a component of a ResourceType. Examples: Government Department, Commercial Company, Accounting Department.

OrganisationTypeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationTypeUsage - ResponsibleHumanResourceTypeConfigurationUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationTypeUsage - OrganisationPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationTypeUsage - OrganisationStatePowertype
Attributes:
-
A ResponsibleHumanResourceTypeConfigurationUsage that is a type of Organisation, a specialisation of OrganisationType, that is used as a component of a ResourceType.

OrganisationalRolePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationalRolePowertype - HumanResourcePowertype
Attributes:
-
The powertype of OrganisationalRole.

PerformsFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PerformsFunction - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PerformsFunction - CapableOfType
Association (source - target):«place2Type»

PerformsFunction - ResourceFunction
Association (source - target):«place1Type»
PerformsFunction - ResourceType
Attributes:
-
A CapableOf that asserts a Function is conducted by a ResourceType.

PersonPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PersonPowertype - ResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PersonPowertype - AgentCapableOfResponsibilityStateType
Attributes:
-
The powertype of Person.

PersonType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PersonType - PersonPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PersonType - ResponsibleHumanResourceType
Attributes:
-
A ResponsibleHumanResourceType that is a type of person.

PhysicalArchitecture «IDEAS:IndividualType»
Connectors:

PhysicalEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicalEvent - Event
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicalEvent - PhysicallySequencedItem
Attributes:
-
An Event that marks the beginning or end of a EventBoundedPhysicalProcess.

PhysicalSequencing «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicalSequencing - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicalSequencing - ImmediateBeforeAfterType
Association (source - target):«place1Type»
PhysicalSequencing - PhysicallySequencedItem
Association (source - target):«place2Type»
PhysicalSequencing - PhysicallySequencedItem
Attributes:
-
An ImmediateBeforeAfterType that asserts one PhysicallySequencedItem occurs immediately after the other.

PhysicalStartEvent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicalStartEvent - StartBorderType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicalStartEvent - ModemWholePartType
Association (source - target):«place2Type»
PhysicalStartEvent - PhysicalEvent
Association (source - target):«place1Type»
PhysicalStartEvent - EventBoundedPhysicalProcess
Attributes:
-
A StartBorderType that relates an EventBoundedPhysicalProcess to the LogicalEvent that marks its start.
Note: there may be no more than one LogicalStartEvent for a given LogicallySequencedProcess.

PhysicallySequencedItem «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PhysicallySequencedItem - ImplemenationScenarioPart
Attributes:
-
An ImplemenationScenarioPart that is physically sequenced; i.e it has a PhysicalSequencing relation.

Port «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Port - ArtefactComponent
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Port - IndividualPortPowertype
Attributes:
-
An ArtefactComponent that is a type of IndividualPort. Note: was called "ResourcePort" in M3.

PortComponentOfArtefactPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PortComponentOfArtefactPowertype - ModemWholePartType
Attributes:
-
The powertype of portComponentOfArtefact.

PortComponentOfTypeOfArtefact «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PortComponentOfTypeOfArtefact - PortComponentOfArtefactPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PortComponentOfTypeOfArtefact - NonHumanResourceUsage
Association (source - target):«place2Type»
PortComponentOfTypeOfArtefact - Port
Association (source - target):«place1Type»
PortComponentOfTypeOfArtefact - ArtefactPowertype
Attributes:
-
A NonHumanResourceUsage that asserts a Port is a component of a type of Artefact.

PortConnectedToPortConnectorComponent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PortConnectedToPortConnectorComponent - IndividualPortConnectedToPortConnectorPowertype
Association (source - target):«place2Type»
PortConnectedToPortConnectorComponent - Port
Association (source - target):«place1Type»
PortConnectedToPortConnectorComponent - PortConnector
Attributes:
-
An PortConnectedToPortConnectorComponent that is a type of IndividualPortConnectedToPortConnector that asserts a Port is a part of a PortConnector.

PortConnector «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PortConnector - ResourceCommunication
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PortConnector - IndividualPortConnectorPowertype
Attributes:
-

A ResourceCommunication that has a protocolStackSuperResourcePortConnectorTypeSubType to a ProtocolStack. Note: was called "ResourcePortConnector" in M3.

PostInOrganisationPowertype «IDEAS:Powertype»
Connectors:
Association (source - target):«place2Type»
PostInOrganisationPowertype - PostPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostInOrganisationPowertype - IndividualResourceUsagePowertype
Association (source - target):«place1Type»
PostInOrganisationPowertype - OrganisationPowertype
Attributes:
-
The powertype of PostInOrganisation.

PostInOrganisationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostInOrganisationType - PostInOrganisationPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostInOrganisationType - HumanResourceTypeUsage
Association (source - target):«place1Type»
PostInOrganisationType - OrganisationPowertype
Association (source - target):«place2Type»
PostInOrganisationType - PostTypeUsage
Attributes:
-
A resourceUsage that asserts that a post exists in an OrganisationType of the type specified by the related PostType. Note: posts in organisations may or may not be filled. Note: was called "Post" in M3.

PostOccupyingResponsibleHumanResourceStateType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostOccupyingResponsibleHumanResourceStateType - ResponsibleHumanResourceStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostOccupyingResponsibleHumanResourceStateType - ResourceStateTypeUsage
Attributes:
-
A ResourceStateTypeUsage that has a type of state of a Post that is occupied by a type of ResponsibleHumanResource.

PostPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostPowertype - ResponsibleHumanResourcePowertype
Attributes:
-
The powertype of Post.

PostType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostType - ResponsibleHumanResourceType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostType - PostPowertype
Attributes:

-
An ConstructedHumanResourceType and ResponsibleHumanResourceType specifying a type of Post. This is not used as a component of a ResourceType. A type of point of contact or responsible person. Note that this is the type of post - e.g. Desk Officer, Commander, etc.

PostTypeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostTypeUsage - ResponsibleHumanResourceTypeConfigurationUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PostTypeUsage - PostPowertype
Attributes:
-
A ResponsibleHumanResourceTypeConfigurationUsage that is a type of Post, a specialisation of PostType, which is used as a component of a ResourceType. E.g. The specialisation of the PostType, Commander, may be a component of the Land Component - the Commander Land Component.

ProducerFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProducerFunction - IndividualExchangeRoleType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProducerFunction - ModemWholePartType
Association (source - target): «place2Type»
ProducerFunction - ResourceExport
Association (source - target): «place1Type»

RadioFrequencyPortConnector «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RadioFrequencyPortConnector - IndividualRadioFrequencyPortConnectorPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RadioFrequencyPortConnector - PortConnector
Attributes:
-
A ResourcePortConnector that is a type of IndividualRadioFrequencyPortConnector.

ResourceCommunication «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceCommunication - ResourceInteraction
Attributes:
-
A ResourceInteraction where DataElements are exchanged.

ResourceElementRole «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceElementRole - IndividualResourceElementRolePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceElementRole - InteractionElementRole
Attributes:
-
An InteractionElementRole where the flowed element is a ResourceType.

ResourceEnergyFlow «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceEnergyFlow - ResourceInteraction
Attributes:
-
A ResourceInteraction where energy is transferred between ResourceUsages.

ResourceExport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceExport - SendType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceExport -ModemIndividualType
Attributes:
-
A SendType where the sender is a ResourceType or Function

ResourceFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceFunction - Function
Attributes:
-
A Function carried out by a ResourceType.

ResourceImport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceImport - ReceiveType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceImport - ModemIndividualType
Attributes:
-
A ReceiveType where the receiver is a ResourceType or Function.

ResourceInteraction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInteraction - IndividualResourceInteractionPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInteraction - PhysicalArchitectureProcess
Attributes:
-
An ExchangeType where two ResourceTypes interact.
Examples: data exchange between systems, conversations between people, people using systems, flows of materiel from one resource to another, etc.

ResourceInteractionExport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInteractionExport - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceInteractionExport - SendInExchangeType
Association (source - target): «place2Type»
ResourceInteractionExport - ResourceExport
Association (source - target): «place1Type»
ResourceInteractionExport - ResourceInteraction
Attributes:
-
A SendInExchangeType where the sender is a ResourceType or Function.

ResourceInteractionImport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInteractionImport - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceInteractionImport - ReceiveInExchangeType
Association (source - target): «place2Type»
ResourceInteractionImport - ResourceImport
Association (source - target): «place1Type»
ResourceInteractionImport - ResourceInteraction
Attributes:
-
A RecieveInExchangeType where the receiver is a ResourceType or Function.

ResourceLifeline «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceLifeline - ImplemenationScenarioPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResourceLifeline - ResourceStateType
Attributes:
-
A ResourceStateType whose extent is defined by an ImplemenationScenario.

ResourceMovement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceMovement - ResourceInteraction
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceMovement - IndividualResourceMovementPowertype
Attributes:
-
A ResourceInteraction where the element that flows is a ResourceType.

ResourceStateType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceStateType - StateSpecification
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceStateType - IndividualResourceStatePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceStateType - ModemIndividualType
Attributes:
-
A type of state that a ResourceType may have.

ResourceStateTypeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceStateTypeUsage - IndividualResourceStatePowertype
Attributes:
-
An IndividualResourceStatePowertype that has a type of state of a Resource that is used by another type of Resource.

ResourceStateUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceStateUsage - IndividualResourceStateUsagePowertype
Association (source - target):«place2Type»
ResourceStateUsage - ResourceStateTypeUsage
Association (source - target): «place1Type»
ResourceStateUsage - IndividualResourceStatePowertype
Attributes:
-
A IndividualResourceStateUsagePowertype that is a type of resource state relation that asserts a type of resource state is used by a type of resource.

ResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceType - IndividualResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceType - FlowedElement
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceType - ResourceStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceType - InteractionElement
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceType - PhysicalArchitectureIndividualType
Attributes:
-
A PhysicalArchitectureIndividualType that is a type of IndividualResource. This is not used as a component of a ResourceType, but may use components. [ABSTRACT]

ResourceTypeExport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeExport - CapableOfType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeExport - ModemThing
Association (source - target):«place2Type»
ResourceTypeExport - ResourceInteraction
Association (source - target):«place1Type»
ResourceTypeExport - ResourceTypeUsage
Attributes:
-
A CapableOfType where a ResourceInteraction exports from a ResourceTypeUsage.

ResourceTypeImport «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeImport - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeImport - CapableOfType
Association (source - target):«place2Type»
ResourceTypeImport - ResourceInteraction
Association (source - target):«place1Type»
ResourceTypeImport - ResourceTypeUsage
Attributes:
-
A CapableOfType where a ResourceInteraction imports from a ResourceTypeUsage.

ResourceTypeMaster «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeMaster - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeMaster - IndividualResourcePowertype
Attributes:
-

A ModemIndividualType that is the master specification from which ResourceTypes are versioned.

ResourceTypeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeUsage - IndividualResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceTypeUsage - ModemIndividualType
Attributes:
-
A ModemIndividualType, that is a component of a ResourceType that is used by (a component of) another ResourceType.

ResourceUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceUsage - IndividualResourceUsagePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceUsage - ModemThing
Association (source - target):«place1Type»
ResourceUsage - IndividualResourcePowertype
Association (source - target):«place2Type»
ResourceUsage - ResourceTypeUsage
Attributes:
-
A ModemWholePartType that is a relationship between types of IndividualResources which asserts the ResourceTypeUsage is part of the ResourceType. The relationship is abstract, and one of its subtypes should be used to describe *how* a ResourceTypeUsage is part of a ResourceType. Note: was called "ResourceUsage" in M3.

ResponsibleHumanResourcePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourcePowertype - ResponsibleHumanResourceStatePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourcePowertype - AgentCapableOfResponsibilityType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourcePowertype - HumanResourcePowertype
Attributes:
-
The powertype of ResponsibleHumanResource.

ResponsibleHumanResourceStateOccupiesPostPowertype «IDEAS:Powertype»
Connectors:
Association (source - target):«place1Type»
ResponsibleHumanResourceStateOccupiesPostPowertype - PostPowertype
Association (source - target):«place2Type»
ResponsibleHumanResourceStateOccupiesPostPowertype - ResponsibleHumanResourceStatePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceStateOccupiesPostPowertype - IndividualResourceStateUsagePowertype
Attributes:
-
The powertype of responsibleHumanResourceStateOccupiesPost.

ResponsibleHumanResourceStateOccupiesPostType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceStateOccupiesPostType - ResourceStateUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceStateOccupiesPostType - ResponsibleHumanResourceStateOccupiesPostPowertype
Association (source - target):«place1Type»
ResponsibleHumanResourceStateOccupiesPostType - PostPowertype
Association (source - target):«place2Type»
ResponsibleHumanResourceStateOccupiesPostType - PostOccupyingResponsibleHumanResourceStateType
Attributes:
-
A ResourceStateUsage that asserts a ResponsibleHumanResourceTypeConfigurationUsage (i.e. a OrganisationRoleType, a PostTypeUsage or a OrganisationTypeUsage) occupies a PostType. Note it is a state of the ResponsibleHumanResourceTypeConfigurationUsage that occupies the PostType, as it can only occupy it for a period of time. Futhermore, many states of ResponsibleHumanResourceTypeConfigurationUsages can occupy the post across time. Typically, only one state occupies the post at a point in time.

ResponsibleHumanResourceStatePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceStatePowertype - IndividualResourceStatePowertype
Attributes:
-
The powertype of ResponsibleHumanResourceState.

ResponsibleHumanResourceStateType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceStateType - ResponsibleHumanResourceStatePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceStateType - ResourceStateType
Attributes:
-
A type of state that a ResponsibleHumanResourceType may have.

ResponsibleHumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceType - HumanResourceType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceType - ResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceType - ResponsibleHumanResourceStateType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceType - Stakeholder
Attributes:
-
A HumanResourceType that is a type of ResponsibleHumanResource. A PostType, OrganisationType or a PersonType.

ResponsibleHumanResourceTypeConfigurationUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ResponsibleHumanResourceTypeConfigurationUsage - ResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResponsibleHumanResourceTypeConfigurationUsage - HumanResourceTypeConfigurationUsage
Attributes:
-
A state or part of a ResponsibleHumanResourceType that is used by (a part of, a component of) another ResponsibleHumanResourceType.

RoleBourneByResponsibleHumanResourcePowertype «IDEAS:Powertype»
Connectors:
Association (source - target):«place1Type»
RoleBourneByResponsibleHumanResourcePowertype - ResponsibleHumanResourcePowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleBourneByResponsibleHumanResourcePowertype - IndividualResourceUsagePowertype
Association (source - target):«place2Type»
RoleBourneByResponsibleHumanResourcePowertype - OrganisationalRolePowertype
Attributes:
-
The powertype of roleBourneByResponsibleHumanResource.

RoleBourneByResponsibleHumanResourceType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleBourneByResponsibleHumanResourceType - HumanResourceTypeUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleBourneByResponsibleHumanResourceType - RoleBourneByResponsibleHumanResourcePowertype
Association (source - target):«place2Type»
RoleBourneByResponsibleHumanResourceType - OrganisationRoleTypeUsage
Association (source - target):«place1Type»
RoleBourneByResponsibleHumanResourceType - ResponsibleHumanResourcePowertype
Attributes:
-
A HumanResourceTypeUsage that asserts that a ResponsibleHumanResourcePowertype has an OrganisationRoleTypeUsage.

RoleInCommunication «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInCommunication - RoleInInteraction
Association (source - target):«place1Type»
RoleInCommunication - ResourceCommunication
Association (source - target):«place2Type»
RoleInCommunication - DataElementRole
Attributes:
-
A RoleInInteraction where the exchanged element is a DataElement exchanged over a ResourceCommunication.

RoleInIndividualInteractionPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInIndividualInteractionPowertype - ExchangedItemRoleInExchangeType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInIndividualInteractionPowertype - ModemIndividualElementType
Attributes:
-
The powertype of RoleInIndividualInteraction.

RoleInIndividualResourceMovementPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInIndividualResourceMovementPowertype - RoleInIndividualInteractionPowertype
Attributes:
-
The powertype of RoleInIndividualResourceMovement.

RoleInInteraction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInInteraction - TypicalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInInteraction - RoleInIndividualInteractionPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInInteraction - ModemWholePartType
Association (source - target):«place2Type»
RoleInInteraction - InteractionElementRole
Association (source - target):«place1Type»
RoleInInteraction - ResourceInteraction
Attributes:
-
An TypicalWholePart where the involving exchange is a ResourceInteraction. [ABSTRACT]

RoleInOrganisationPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInOrganisationPowertype - IndividualResourceUsagePowertype
Association (source - target):«place2Type»
RoleInOrganisationPowertype - OrganisationalRolePowertype
Association (source - target):«place1Type»
RoleInOrganisationPowertype - OrganisationPowertype
Attributes:
-
The powertype of roleInOrganisation.

RoleInOrganisationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInOrganisationType - HumanResourceTypeUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInOrganisationType - RoleInOrganisationPowertype
Association (source - target):«place1Type»
RoleInOrganisationType - OrganisationPowertype
Association (source - target):«place2Type»
RoleInOrganisationType - OrganisationRoleTypeUsage
Attributes:
-
A HumanResourceTypeUsage that is a type of roleInOrganisation which asserts that a given OrganisationRoleTypeUsage belongs to an OrganisationPowertype.

RoleInResourceMovement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInResourceMovement - RoleInInteraction
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleInResourceMovement - RoleInIndividualResourceMovementPowertype
Association (source - target):«place2Type»
RoleInResourceMovement - ResourceElementRole
Association (source - target):«place1Type»
RoleInResourceMovement - ResourceMovement
Attributes:
-
A RoleInInteraction where the interaction is a ResourceMovement.

RoleOfDataElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfDataElement - RoleOfInteractionElement
Association (source - target):«place1Type»
RoleOfDataElement - DataElement
Association (source - target):«place2Type»
RoleOfDataElement - DataElementRole
Attributes:
-
A RoleOfInteractionElement where the element is a DataElement.

RoleOfIndividualInteractionElementPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfIndividualInteractionElementPowertype - IndividualRoleAsExchangedItemType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfIndividualInteractionElementPowertype - ModemIndividualElementType
Attributes:
-
The powertype of RoleOfIndividualInteractionElement.

RoleOfIndividualResourceElementPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfIndividualResourceElementPowertype - RoleOfIndividualInteractionElementPowertype
Attributes:
-
The powertype of RoleOfIndividualResourceElement.

RoleOfInteractionElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfInteractionElement - TypicalWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfInteractionElement - ModemWholePartType
Association (source - target):«place1Type»
RoleOfInteractionElement - InteractionElement
Association (source - target):«place2Type»
RoleOfInteractionElement - InteractionElementRole
Attributes:
-
A TypicalWholePart relating an InteractionElement to its role in a ResourceInteraction. [ABSTRACT]

RoleOfResourceElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfResourceElement - RoleOfIndividualResourceElementPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfResourceElement - RoleOfInteractionElement
Association (source - target):«place1Type»
RoleOfResourceElement - ResourceType
Association (source - target):«place2Type»
RoleOfResourceElement - ResourceElementRole
Attributes:
-
A RoleOfInteractionElement where the element is a ResourceType.

SequencedFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SequencedFunction - EventBoundedPhysicalProcess
Attributes:
-
An EventBoundedPhysicalProcess that is the typical useage of a Function in a ResourceLifeLine.

SequencedResourceInteraction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SequencedResourceInteraction - ImplemenationScenarioPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SequencedResourceInteraction - EventBoundedPhysicalProcess
Attributes:
-
An ImplementationScenarioPart that is the typical occurance of a ResourceInteraction between two ResourceLifelines

Software «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Software - Artefact
Dependency (element - is instance of):«IDEAS:powertypeInstance»
Software - SoftwarePowertype
Attributes:
-
An executable computer programme, or fragment of an executable programme (e.g. a subroutine, class, etc.)

SoftwareComponent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SoftwareComponent - ArtefactComponent
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SoftwareComponent - SoftwarePowertype
Attributes:
-
A type of Software that is a hostingArtefactWholesoftwareTypePart of an ArtefactType. In other words, a type of Software that is hosted by an ArtefactType.

SoftwarePowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SoftwarePowertype - ArtefactPowertype
Attributes:
-
The powertype of Software.

SoftwareType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SoftwareType - ArtefactType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SoftwareType - SoftwarePowertype
Attributes:
-
An Artefact that is a type of Software.

SoftwareUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SoftwareUsage - NonHumanResourceUsage
Association (source - target):«place2Type»
SoftwareUsage - SoftwareComponent
Association (source - target):«place1Type»
SoftwareUsage - ArtefactPowertype
Attributes:
-
A NonHumanResourceUsage that asserts a SoftwareComponent is used by an ArtefactType.

StateUsedAsPostOccupationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StateUsedAsPostOccupationType - UsedResourceState
Association (source - target):«place1Type»
StateUsedAsPostOccupationType - ResponsibleHumanResourceType
Association (source - target):«place2Type»
StateUsedAsPostOccupationType - PostOccupyingResponsibleHumanResourceStateType
Attributes:
-
A UsedResourceState that asserts a ResponsibleHumanResourceType occupies a PostOccupyingResponsibleHumanResourceStateType.

SubOrganisationPowertype «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubOrganisationPowertype - IndividualResourceUsagePowertype
Association (source - target):«place2Type»
SubOrganisationPowertype - OrganisationStatePowertype
Association (source - target):«place1Type»
SubOrganisationPowertype - OrganisationPowertype
Attributes:
-
The powertype of subOrganisation.

SubOrganisationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubOrganisationType - HumanResourceTypeUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubOrganisationType - SubOrganisationPowertype
Association (source - target):«place2Type»
SubOrganisationType - SubOrganisationTypeUsage
Association (source - target):«place1Type»
SubOrganisationType - OrganisationPowertype
Attributes:
-
A HumanResourceTypeUsage that is a type of subOrganisation which asserts that a state of one type of Organisation, the OrganisationType, is typically the parent of another, the OrganisationTypeUsage. In other words, one type of Organisation is typically the parent of another for a period of time. E.g. a squadron may be part of a battalion. Note: was called "SubOrganisation" in M3.

SubOrganisationTypeUsage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubOrganisationTypeUsage - OrganisationTypeUsage
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubOrganisationTypeUsage - OrganisationStatePowertype
Attributes:
-
An OrganisatioTypeUsage that is type of Organisation that is a sub-organisation of another type of organisation.

SubjectOfForecast «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SubjectOfForecast - ModemIndividualType
Attributes:
-
A ModemIndividualType that is the subject of a Forecast.

Technology «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Technology - ArtefactPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Technology - SubjectOfForecast
Attributes:
-
An ArtefactPowertype that is a class of Artefact that defines a branch of engineering or computer science.

UsagePerformsFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
UsagePerformsFunction - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
UsagePerformsFunction - CapableOfType
Association (source - target):«place2Type»
UsagePerformsFunction - UsageSpecificFunction
Association (source - target):«place1Type»
UsagePerformsFunction - ResourceTypeUsage
Attributes:
-
A CapableOfType where a ResoureceTypeUsage is capable of conducting a UsageSpecificFunction.

UsageSpecificFunction «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
UsageSpecificFunction - Function
Association (source - target):«place2Type»
UsagePerformsFunction - UsageSpecificFunction
Attributes:
-
A PhysicalArchitectureProcess that is a particular usage of a Function. Note: this is used where there is a requirement to distinguish between two uses of a ResourceType which both have the same functionality, but put to different purposes. This is particularly important for tracing back to OV-5 Activities.

UsedResourceState «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
UsedResourceState - ResourceStateUsage
Association (source - target):«place2Type»
UsedResourceState - ResourceStateTypeUsage
Association (source - target):«place1Type»
UsedResourceState - ResourceType
Attributes:
-
A IndividualResourceStateUsagePowertype that is a type of IndividualResourceState Usage that asserts a type of resource state is used by a type of resource.

VersionSuccession «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
VersionSuccession - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
VersionSuccession - BeforeAfterType
Association (source - target):«place2Type»
VersionSuccession - ResourceType
Association (source - target):«place1Type»
VersionSuccession - ResourceType
Attributes:
-
A BeforeAfterType that asserts one ResourceType succeeds another. Note: both ResourceTypes must be versions of the same ResourceTypeMaster.

activityFunctionMapping «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
activityFunctionMapping - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»
activityFunctionMapping - OperationalActivity
Association (source - target): «place2Type»
activityFunctionMapping - Function
Attributes:
-
A modemIndividualTypeSpecialisation that relates an OperationalActivity or ActivityGroup to the Function or FunctionGroup that realises it.

artefactPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
artefactPart - individualResourceUsage
Association (source - target):«place2Type»
artefactPart - Artefact
Association (source - target):«place1Type»
artefactPart - Artefact
Attributes:
-
An individualResourceUsage that asserts one artefact is part of another.

artefactSpecification «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
artefactSpecification - usedNonHumanResourceTypeSpecialisation
Association (source - target):«place1Type»
artefactSpecification - ArtefactType
Association (source - target):«place2Type»
artefactSpecification - ArtefactComponent
Attributes:
-
An usedNonHumanResourceTypeSpecialisation that asserts that an ArtefactTypeUsage. is a specialisation of an ArtefactType.

branchOfTechnology «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
branchOfTechnology - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
branchOfTechnology - ArtefactType
Association (source - target):«place1Type»
branchOfTechnology - Technology
Attributes:
-
A modemIndividualTypeSpecialisation that asserts an ArtefactType belongs to a branch of Technology.

capabilityInstance «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
capabilityInstance - HumanAndNonHumanConfiguration
Association (source - target): «place1Type»
capabilityInstance - Capability
Generalization (element - is a subtype of): «IDEAS:superSubtype»
capabilityInstance - modemIndividualTypeInstance
Attributes:
-
A modemIndividualTypeInstance where the instance is a HumanAndNon-HumanConfiguration and the type is a Capability. This asserts that an individual configuration of people and equipment has a Capability.

capabilityRealisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilityRealisation - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
capabilityRealisation - CapabilityConfiguration
Association (source - target):«place1Type»
capabilityRealisation - Capability
Attributes:
-
A modemIndividualTypeSpecialisation that relates a CapabilityConfiguration to a Capability.

competenceForRole «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
competenceForRole - requiredCompetence
Association (source - target):«place1Type»
competenceForRole - Competence
Association (source - target):«place2Type»
competenceForRole - OrganisationRoleType
Attributes:
-
A requiredCompetence that asserts an OrganisationRoleType requires a Competence.

competenceToConduct «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
competenceToConduct - requiredCompetence
Association (source - target):«place1Type»
competenceToConduct - Competence
Association (source - target):«place2Type»
competenceToConduct - Function
Attributes:
-
A requiredCompetence that asserts a competence is required by the HumanResource to conduct the Function. Note: was called "toConduct" in M3.

configuredHumanAndNonHumanResource «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
configuredHumanAndNonHumanResource - individualResourceUsage
Association (source - target):«place1Type»
configuredHumanAndNonHumanResource - HumanAndNonHumanConfiguration
Association (source - target):«place2Type»
configuredHumanAndNonHumanResource - HumanAndNonHumanConfiguration
Attributes:
-
An individualResourceUsage that asserts a HumanAndNonHumanConfiguration uses another HumanAndNonHumanConfiguration.

configuredHumanResource «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
configuredHumanResource - individualResourceUsage
Association (source - target):«place2Type»
configuredHumanResource - HumanResource
Association (source - target):«place1Type»
configuredHumanResource - HumanAndNonHumanConfiguration
Attributes:
-
An individualResourceUsage that asserts that a HumanAndNonHumanConfiguration uses a HumanResource.

configuredNonHumanResource «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
configuredNonHumanResource - individualResourceUsage
Association (source - target):«place2Type»
configuredNonHumanResource - NonHumanResource
Association (source - target):«place1Type»
configuredNonHumanResource - HumanAndNonHumanConfiguration
Attributes:
-
n individualResourceUsage that asserts that a NonHumanResource uses a HumanAndNonHumanConfiguration.

connected «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
connected - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
connected - couple
Attributes:
-
A couple that asserts two things are connected.

fieldedConfiguration «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
fieldedConfiguration - resourceTypeInstance
Association (source - target): «place2Type»
fieldedConfiguration - FieldedCapabilityConfiguration
Association (source - target): «place1Type»
fieldedConfiguration - CapabilityConfiguration
Attributes:
-
A resourceTypeInstance where the type is a CapabilityConfiguration and the instance is a FieldedCapabilityConfiguration.

fieldedPhysicalArchitectureCapabilityConfigurationUsage «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
fieldedPhysicalArchitectureCapabilityConfigurationUsage - configuredHumanAndNonHumanResource
Association (source - target):«place2Type»
fieldedPhysicalArchitectureCapabilityConfigurationUsage - FieldedCapabilityConfiguration
Association (source - target):«place1Type»
fieldedPhysicalArchitectureCapabilityConfigurationUsage - PhysicalArchitecture
Attributes:
-
A configuredHumanAndNonHumanResource usage of a FieldedPhysicalArchitecture by a FieldedCapabilityConfiguration. Asserts that a FieldedCapabilityConfiguration is a component of a FieldedPhysicalArchitecture.

flowImplementation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
flowImplementation - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
flowImplementation - LogicalFlow
Association (source - target):«place2Type»
flowImplementation - ResourceInteraction
Attributes:
-
A modemIndividualTypeSpecialisation where a ResourceInteraction implements a LogicalFlow.

forecastFor «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
forecastFor - ModemThing
Association (source - target):«place2Type»
forecastFor - SubjectOfForecast
Generalization (element - is a subtype of):«IDEAS:superSubtype»
forecastFor - couple
Association (source - target):«place1Type»
forecastFor - Forecast
Attributes:
-
A couple that relates the Forecast to the SubjectOfForecast.

forecastPeriod «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
forecastPeriod - couple
Association (source - target):«place1Type»
forecastPeriod - Forecast
Association (source - target):«place2Type»
forecastPeriod - EnterprisePhase
Generalization (element - is a subtype of):«IDEAS:superSubtype»
forecastPeriod - ModemThing
Attributes:
-
A couple that relates the Forecast to the EnterprisePhase that it covers. Note: if a forecast does not correspond to an existing EnterprisePhase, new EnterprisePhases can be created to cover the period - i.e. you can have as many EnterprisePhases as are needed.

functionInSequence «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
functionInSequence - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»
functionInSequence - ResourceFunction
Association (source - target): «place2Type»
functionInSequence - SequencedFunction
Attributes:
-
A modemIndividualTypeSpecialisation that relates a ResourceFunction to its usage (as a SequencedFunction) on a ResourceLifeLine. Note: A SequencedFunction is based on only one Function

implementsDataModel «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
implementsDataModel - couple
Association (source - target):«place2Type»
implementsDataModel - SoftwareType
Association (source - target):«place1Type»
implementsDataModel - PhysicalDataModel
Attributes:
-
A couple that asserts that a SoftwareType implements a PhysicalDataModel.

individualPortConnectedToPortConnector «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
individualPortConnectedToPortConnector - IndividualPortConnector
Association (source - target):«place2Type»
individualPortConnectedToPortConnector - IndividualPort
Dependency (element - is instance of):«IDEAS:powertypeInstance»
individualPortConnectedToPortConnector - IndividualPortConnectedToPortConnectorPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualPortConnectedToPortConnector - connected
Attributes:
-
A connected relationship that asserts that an IndividualResourcePort is a part of an IndividualResourcePortConnection.

individualRadioFrequencyPortConnectedToPortConnector «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualRadioFrequencyPortConnectedToPortConnector - individualPortConnectedToPortConnector
Association (source - target):«place1Type»
individualRadioFrequencyPortConnectedToPortConnector - IndividualRadioFrequencyPortConnector
Association (source - target):«place2Type»
individualRadioFrequencyPortConnectedToPortConnector - IndividualRadioFrequencyPort
Attributes:
-
A individualPortConnectedToPortConnector that asserts that an IndividualRadioFrequencyPort is a part of an IndividualRadioFrequencyPortConnection.

individualResourceState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualResourceState - individualResourceWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualResourceState - modemTemporalWholePart
Association (source - target):«place2Type»
individualResourceState - IndividualResourceState
Attributes:
-
An individualResourceWholePart and a modemTemporalWholePart that links an IndividualResource to one of its states.

individualResourceStateUsage «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualResourceStateUsage - individualResourceState
Dependency (element - is instance of):«IDEAS:powertypeInstance»
individualResourceStateUsage - IndividualResourceStateUsagePowertype
Attributes:
-
An individualResourceState that is a usage relation between the used IndividualResourceState and individualResource.

individualResourceStateWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualResourceStateWholeAndPart - modemWholePart
Dependency (element - is instance of):«IDEAS:powertypeInstance»
individualResourceStateWholeAndPart - IndividualResourceStateWholeAndPartType
Association (source - target):«place2Type»
individualResourceStateWholeAndPart - IndividualResourceState
Association (source - target):«place1Type»
individualResourceStateWholeAndPart - IndividualResourceState
Attributes:
-
A modemWholePart where both the whole and part are IndividualResourceStates.

individualResourceUsage «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualResourceUsage - individualResourceWholePart
Dependency (element - is instance of):«IDEAS:powertypeInstance»
individualResourceUsage - IndividualResourceUsagePowertype
Association (source - target):«place1Type»
individualResourceUsage - IndividualResource
Association (source - target):«place2Type»
individualResourceUsage - IndividualResource
Attributes:
-
An individualResourceWholePart relationship where one IndividualResource uses a part (or all) of another.

individualResourceWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
individualResourceWholePart - modemWholePart
Association (source - target): «place2Type»
individualResourceWholePart - IndividualResourcePart
Association (source - target): «place1Type»
individualResourceWholePart - IndividualResource
Attributes:
-
A modemWholePart where the whole is an IndividualResource and the part is an IndividualResourcePart.

interactionInScenario «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
interactionInScenario - modemIndividualTypeSpecialisation
Association (source - target): «place1Type»
interactionInScenario - ResourceInteraction
Association (source - target): «place2Type»
interactionInScenario - SequencedResourceInteraction
Attributes:
-
A modemIndividualTypeSpecialisation that relates a ResourceInteraction to its usage (as a SequencedResourceInteraction) in an ImplementationScenario. Note: A SequencedResourceInteraction is based on only one ResourceInteraction

measureOfIndividualResourcePerformance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
measureOfIndividualResourcePerformance - modemIndividualTypeInstance
Generalization (element - is a subtype of):«IDEAS:superSubtype»
measureOfIndividualResourcePerformance - measureOfIndividual
Association (source - target):«place1Type»
measureOfIndividualResourcePerformance - Measure
Association (source - target):«place2Type»
measureOfIndividualResourcePerformance - IndividualResource
Attributes:
-
A measureOfIndividual that specifies the level of performance of an IndividualResource.

naturalResourcePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
naturalResourcePart - individualResourceUsage
Association (source - target):«place2Type»
naturalResourcePart - NaturalResource
Association (source - target):«place1Type»
naturalResourcePart - NaturalResource
Attributes:
-
An individualResourceUsage that asserts one NaturalResource is part of another.

naturalResourceSpecification «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
naturalResourceSpecification - usedNonHumanResourceTypeSpecialisation
Association (source - target):«place1Type»
naturalResourceSpecification - NaturalResourceType
Association (source - target):«place2Type»
naturalResourceSpecification - NaturalResourceComponent
Attributes:
-
A usedNonHumanResourceTypeSpecialisation that asserts that a NaturalResourceTypeUsageis a specialisation of a NaturalResourceType.

nodeRealisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
nodeRealisation - bodyTypeSuperSubType
Association (source - target):«place1Type»
nodeRealisation - Node
Association (source - target):«place2Type»
nodeRealisation - ResourceType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
nodeRealisation - modemIndividualTypeSpecialisation
Attributes:
-
A superSubtype that asserts that a ResourceType provides the functionality specified by an operational node.

portComponentOfArtefact «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
portComponentOfArtefact - individualResourceWholePart
Dependency (element - is instance of):«IDEAS:powertypeInstance»
portComponentOfArtefact - PortComponentOfArtefactPowertype
Association (source - target):«place2Type»
portComponentOfArtefact - IndividualPort
Association (source - target):«place1Type»
portComponentOfArtefact - Artefact
Attributes:
-
An individualResourceWholePart where the whole is an IndividualResource and the part is an ResourcePort.

protocolStackSuperResourcePortConnectorTypeSubType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
protocolStackSuperResourcePortConnectorTypeSubType - protocolStackSuperSubType
Association (source - target):«place2Type»
protocolStackSuperResourcePortConnectorTypeSubType - PortConnector
Attributes:
-
A superSubType relation with a superType ProtocolStack and a subType ResourcePortConnectorType.

protocolStackSuperPortSubType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
protocolStackSuperPortSubType - protocolStackSuperSubType
Association (source - target):«place2Type»
protocolStackSuperPortSubType - Port
Association (source - target):«place1Type»
protocolStackSuperPortSubType - ProtocolStack
Attributes:
-
A superSubType relation with a superType ProtocolStack and a subType Port.

protocolStackSuperSubType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
protocolStackSuperSubType - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
protocolStackSuperSubType - ProtocolStack
Attributes:
-
A modemIndividualTypeSpecialisation relation with a superType ProtocolStack.

protocolStackTypeIndividualPortConnector «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
protocolStackTypeIndividualPortConnector - protocolStackTypeInstance
Association (source - target):«place2Type»
protocolStackTypeIndividualPortConnector - IndividualPortConnector
Attributes:
-
A typeInstance relation between the type ProtocolStack and the instance IndividualResourcePortConnection.

protocolStackTypeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
protocolStackTypeInstance - modemIndividualTypeInstance
Association (source - target):«place1Type»
protocolStackTypeInstance - ProtocolStack
Attributes:
-
A modemIndividualTypeInstance that asserts that a ProtocolStack has a ResoucePort as an instance.

protocolStackTypePortInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
protocolStackTypePortInstance - protocolStackTypeInstance
Association (source - target):«place2Type»
protocolStackTypePortInstance - IndividualPort
Attributes:
-
A ProtocolStackTypeInstance that asserts that a ProtocolStack has a ResourcePort as an instance.

providedService «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
providedService - modemIndividualTypeSpecialisation
Generalization (element - is a subtype of):«IDEAS:superSubtype»
providedService - bodyTypeSuperSubType
Association (source - target):«place2Type»
providedService - ResourceType
Association (source - target):«place1Type»
providedService - ServiceLevel
Attributes:
-

A superSubtype that asserts that a Resource delivers a Service to a specified ServiceLevel.
radioFrequencyPortConnectorFrequencyRange «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» radioFrequencyPortConnectorFrequencyRange - radioFrequencyRangeAssignment *Association (source - target):*«place2Type» radioFrequencyPortConnectorFrequencyRange - RadioFrequencyPortConnector *Association (source - target):*«place1Type» radioFrequencyPortConnectorFrequencyRange - FrequencyRange *Attributes:* - A radioFrequencyRangeAssignment that asserts a radio frequency range has been assigned to a RadioFrequencyPortConnector.
radioFrequencyPortFrequencyRange «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» radioFrequencyPortFrequencyRange - radioFrequencyRangeAssignment *Association (source - target):*«place2Type» radioFrequencyPortFrequencyRange - RadioFrequencyPort *Association (source - target):*«place1Type» radioFrequencyPortFrequencyRange - FrequencyRange *Attributes:* - A radioFrequencyRangeAssignment that asserts a radio frequency range has been assigned to a RadioFrequencyPort.
radioFrequencyRangeAssignment «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» radioFrequencyRangeAssignment - measureOfType *Association (source - target):*«place1Type» radioFrequencyRangeAssignment - FrequencyRange *Attributes:* - A measureOfType that asserts a radio frequency range has been assigned.
realisationAsFieldedCapability «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» realisationAsFieldedCapability - modemIndividualTypeInstance *Association (source - target):*«place2Type» realisationAsFieldedCapability - FieldedCapabilityConfiguration *Association (source - target):*«place1Type» realisationAsFieldedCapability - CapabilityConfiguration *Attributes:* - A modemIndividualTypeInstance that relates a CapabilityConfiguration to a FieldedCapabilityConfiguration.

requiredCompetence «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
requiredCompetence - Competence
Generalization (element - is a subtype of):«IDEAS:superSubtype»
requiredCompetence - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
requiredCompetence -ModemThing
Attributes:
-
A couple that asserts a Competence is required.

requiredService «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
requiredService - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
requiredService - couple
Association (source - target):«place1Type»
requiredService - ResourceType
Association (source - target):«place2Type»
requiredService - ServiceLevel
Attributes:
-
A couple that asserts a ResourceType requires a Service (to a given ServiceLevel) in order to function.

resourceTypeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
resourceTypeInstance - modemIndividualTypeInstance
Association (source - target):«place2Type»
resourceTypeInstance - IndividualResource
Association (source - target):«place1Type»
resourceTypeInstance - ResourceType
Attributes:
-
A modelIndividualTypeInstance that relates an IndividualResource to its ResourceType.

resourceTypeMeasure «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
resourceTypeMeasure - measureOfType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
resourceTypeMeasure - resourceTypeProperty
Association (source - target):«place1Type»
resourceTypeMeasure - Measure
Association (source - target):«place2Type»
resourceTypeMeasure - ResourceType
Attributes:
-
A measureOfType where the type is a ResourceType.

resourceTypeProperty «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
resourceTypeProperty - propertyOfType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
resourceTypeProperty - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
resourceTypeProperty - Property
Association (source - target):«place2Type»
resourceTypeProperty - ResourceType
Attributes:
-
A propertyOfType where the type is a ResourceType.

roleInIndividualInteraction «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
roleInIndividualInteraction - exchangedItemRoleInExchange
Dependency (element - is instance of):«IDEAS:powertypeInstance»
roleInIndividualInteraction - RoleInIndividualInteractionPowertype
Association (source - target):«place2Type»
roleInIndividualInteraction - IndividualInteractionElementRole
Association (source - target):«place1Type»
roleInIndividualInteraction -IndividualResourceInteraction
Attributes:
-
n exchangedItemRoleInExchange that asserts the IndividualInteractionElementRole is a component of the exchange.

roleOfIndividualResourceElement «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RoleOfIndividualResourceElement - roleOfIndividualInteractionElement
Dependency (element - is instance of):«IDEAS:powertypeInstance»
RoleOfIndividualResourceElement - RoleOfIndividualResourceElementPowertype
Association (source - target):«place1Type»
RoleOfIndividualResourceElement - IndividualResource
Association (source - target):«place2Type»
RoleOfIndividualResourceElement - IndividualResourceElementRole
Attributes:
-
An roleOfIndividualInteractionElement that asserts the IndividualResourceElementRole is the thing being exchanged by the IndividualResource.

roleInIndividualResourceMovement «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
roleInIndividualResourceMovement - RoleInIndividualResourceMovementPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
roleInIndividualResourceMovement - roleInIndividualInteraction
Association (source - target):«place2Type»
roleInIndividualResourceMovement - IndividualResourceElementRole
Association (source - target):«place1Type»
roleInIndividualResourceMovement - IndividualResourceMovement

Attributes:
-
A roleInIndividualInteraction that asserts the IndividualResourceElementRole is a component of the exchange.

roleOfIndividualInteractionElement «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
roleOfIndividualInteractionElement - RoleOfIndividualInteractionElementPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
roleOfIndividualInteractionElement - individualRoleAsExchangedItem
Generalization (element - is a subtype of):«IDEAS:superSubtype»
roleOfIndividualInteractionElement - modemWholePart
Association (source - target):«place1Type»
roleOfIndividualInteractionElement - IndividualResourceInteractionElement
Association (source - target):«place2Type»
roleOfIndividualInteractionElement - IndividualInteractionElementRole
Attributes:
-
An individualRoleAsExchangedItem that asserts the IndividualResourceInteractionElement is the thing being exchanged by the IndividualResourceInteractionElement.

serviceConcurrency «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceConcurrency - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
serviceConcurrency - representedBy
Association (source - target):«place2Type»
serviceConcurrency - IntegerRepresentation
Association (source - target):«place1Type»
serviceConcurrency - ServiceLevel
Attributes:
-
A representedBy that assigns an IntegerRepresentation for the number of concurrent ServiceSpecifications required for a ServiceLevel.

serviceFunctionFunctionMapping «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
serviceFunctionFunctionMapping - ServiceFunction
Association (source - target): «place2Type»
serviceFunctionFunctionMapping - Function
Generalization (element - is a subtype of): «IDEAS:superSubtype»
serviceFunctionFunctionMapping - modemIndividualTypeSpecialisation
Attributes:
-
A modemIndividualTypeSpecialisation that relates an OperationalActivity or ActivityGroup to the Function or FunctionGroup that realises it.

softwareComponent «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
softwareComponent - artefactPart
Association (source - target):«place2Type»
softwareComponent - Software

Association (source - target):«place1Type»
softwareComponent - Software
Attributes:
-
An artefactPart that asserts that a Software is component of an Artefact.

softwareSpecification «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
softwareSpecification - artefactSpecification
Association (source - target):«place1Type»
softwareSpecification - SoftwareType
Association (source - target):«place2Type»
softwareSpecification - SoftwareComponent
Attributes:
-
An artefactSpecification that asserts that a SoftwareComponent is a specialisation of a SoftwareType.

stateMachineForResourceType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateMachineForResourceType - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
stateMachineForResourceType - appliedStateMachine
Association (source - target):«place1Type»
stateMachineForResourceType - ResourceType
Association (source - target):«place2Type»
stateMachineForResourceType - StateMachine
Attributes:
-
A appliedStateMachine that relates a ResourceType to its state machine.

usedCapabilityConfigurationSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedCapabilityConfigurationSpecialisation - usedHumanAndNonHumanConfigurationTypeSpecialisation
Association (source - target):«place1Type»
usedCapabilityConfigurationSpecialisation - CapabilityConfiguration
Association (source - target):«place2Type»
usedCapabilityConfigurationSpecialisation - CapabilityConfigurationConfigurationUsage
Attributes:
-
An usedHumanAndNonHumanConfigurationTypeSpecialisation between CapabilityConfiguration and CapabilityConfigurationConfigurationUsage.

usedHumanAndNonHumanConfigurationTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedHumanAndNonHumanConfigurationTypeSpecialisation - usedResourceTypeSpecialisation
Association (source - target):«place1Type»
usedHumanAndNonHumanConfigurationTypeSpecialisation - HumanAndNonHumanConfigurationType
Association (source - target):«place2Type»
usedHumanAndNonHumanConfigurationTypeSpecialisation - HumanAndNonHumanResourceTypeConfigurationUsage

Attributes:
-
A usedResourceTypeSpecialisation that asserts that an HumanAndNonHumanConfigurationTypeis a superType of a UsedHumanAndNonHumanConfigurationType.

usedNonHumanResourceTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedNonHumanResourceTypeSpecialisation - usedResourceTypeSpecialisation
Association (source - target):«place1Type»
usedNonHumanResourceTypeSpecialisation - NonHumanResourceType
Association (source - target):«place2Type»
usedNonHumanResourceTypeSpecialisation - NonHumanResourceTypeConfigurationUsage
Attributes:
-
An usedNonHumanResourceTypeSpecialisation between NonHumanResourceType and NonHumanResourceTypeConfigurationUsage.

usedOrganisationRoleTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedOrganisationRoleTypeSpecialisation - usedResourceTypeSpecialisation
Association (source - target):«place1Type»
usedOrganisationRoleTypeSpecialisation - OrganisationRoleType
Association (source - target):«place2Type»
usedOrganisationRoleTypeSpecialisation - OrganisationRoleTypeUsage
Attributes:
-
An usedResourceTypeSpecialisation that asserts a OrganisationRoleTypeUsage is a specialisation of a OrganisationRoleType.

usedOrganisationTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedOrganisationTypeSpecialisation - usedResourceTypeSpecialisation
Association (source - target):«place1Type»
usedOrganisationTypeSpecialisation - OrganisationType
Association (source - target):«place2Type»
usedOrganisationTypeSpecialisation - OrganisationTypeUsage
Attributes:
-
An usedResourceTypeSpecialisation that asserts a OrganisationTypeUsage is a specialisation of a OrganisationType.

usedPhysicalArchitectureSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedPhysicalArchitectureSpecialisation - usedHumanAndNonHumanConfigurationTypeSpecialisation
Association (source - target):«place1Type»
usedPhysicalArchitectureSpecialisation - PhysicalArchitecture
Association (source - target):«place2Type»
usedPhysicalArchitectureSpecialisation - PhysicalArchitectureConfigurationUsage
Attributes:
-
An usedHumanAndNonHumanConfigurationTypeSpecialisation between PhysicalArchitecture and PhysicalArchitectureConfigurationUsage.

usedPostTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedPostTypeSpecialisation - usedResourceTypeSpecialisation
Association (source - target):«place1Type»
usedPostTypeSpecialisation - PostType
Association (source - target):«place2Type»
usedPostTypeSpecialisation - PostTypeUsage
Attributes:
-
An usedResourceTypeSpecialisation that asserts a PostTypeUsagei s a specialisation of a PostType.

usedResourceTypeSpecialisation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
usedResourceTypeSpecialisation - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
usedResourceTypeSpecialisation - ResourceType
Association (source - target):«place2Type»
usedResourceTypeSpecialisation - ResourceTypeUsage
Attributes:
-
A superSubtype that asserts that a ResourceType is a superType of ResourceTypeUsage.

versionIdentifier «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
versionIdentifier - representedBy
Generalization (element - is a subtype of):«IDEAS:superSubtype»
versionIdentifier - ModemThing
Association (source - target):«place2Type»
versionIdentifier - StringRepresentation
Association (source - target):«place1Type»
versionIdentifier - ResourceType
Attributes:
-
A representedBy that asserts a StringRepresentation represents the version identifier of a ResourceType.

versionOf «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
versionOf - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
versionOf - ResourceTypeMaster
Association (source - target):«place2Type»
versionOf - ResourceType
Attributes:
-
A modemIndividualTypeSpecialisation that asserts a ResourceType is a version of a ResourceTypeMaster.

2.6.14 System Views additional diagrams.

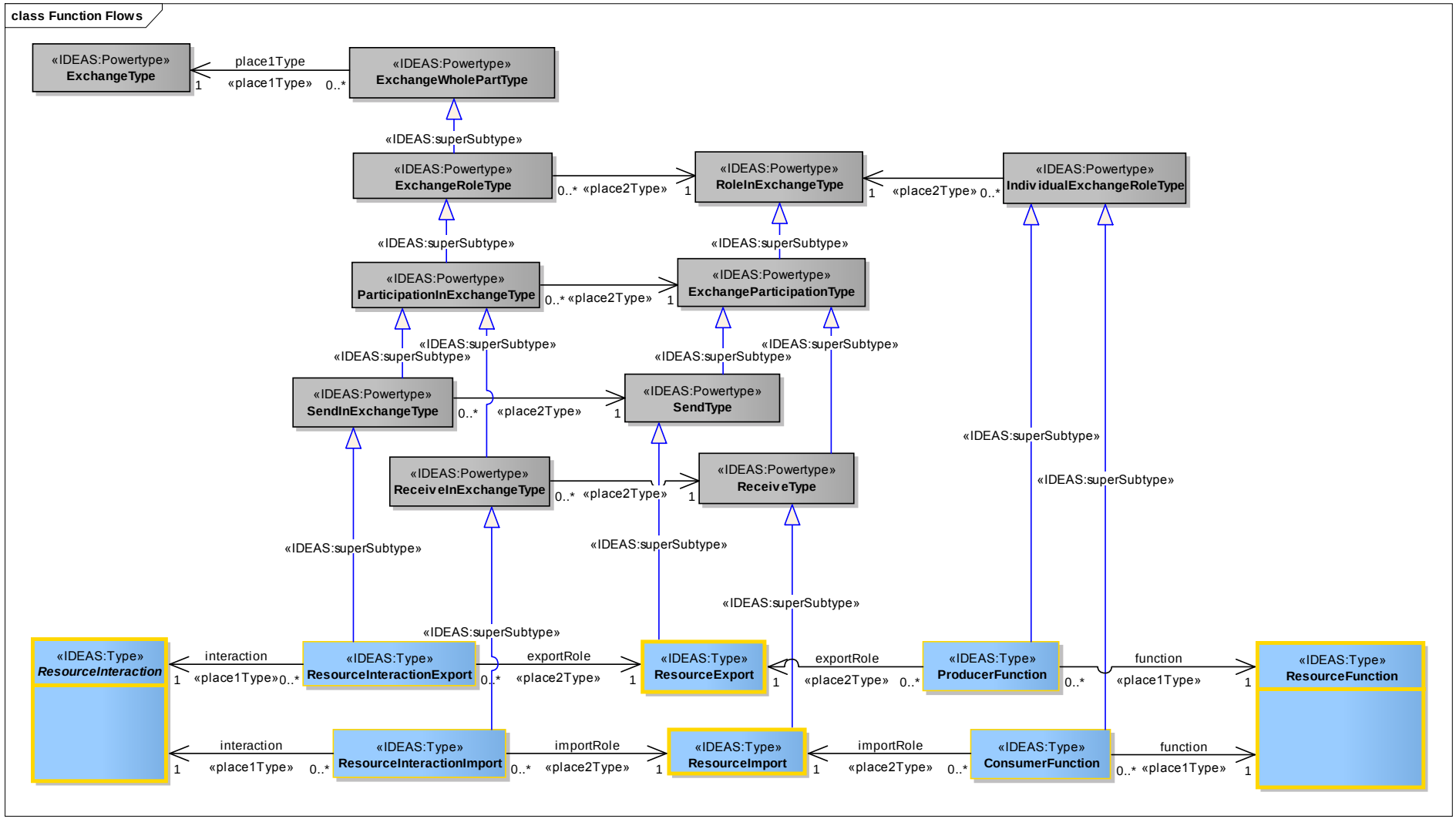


Figure 86 : Function Flows

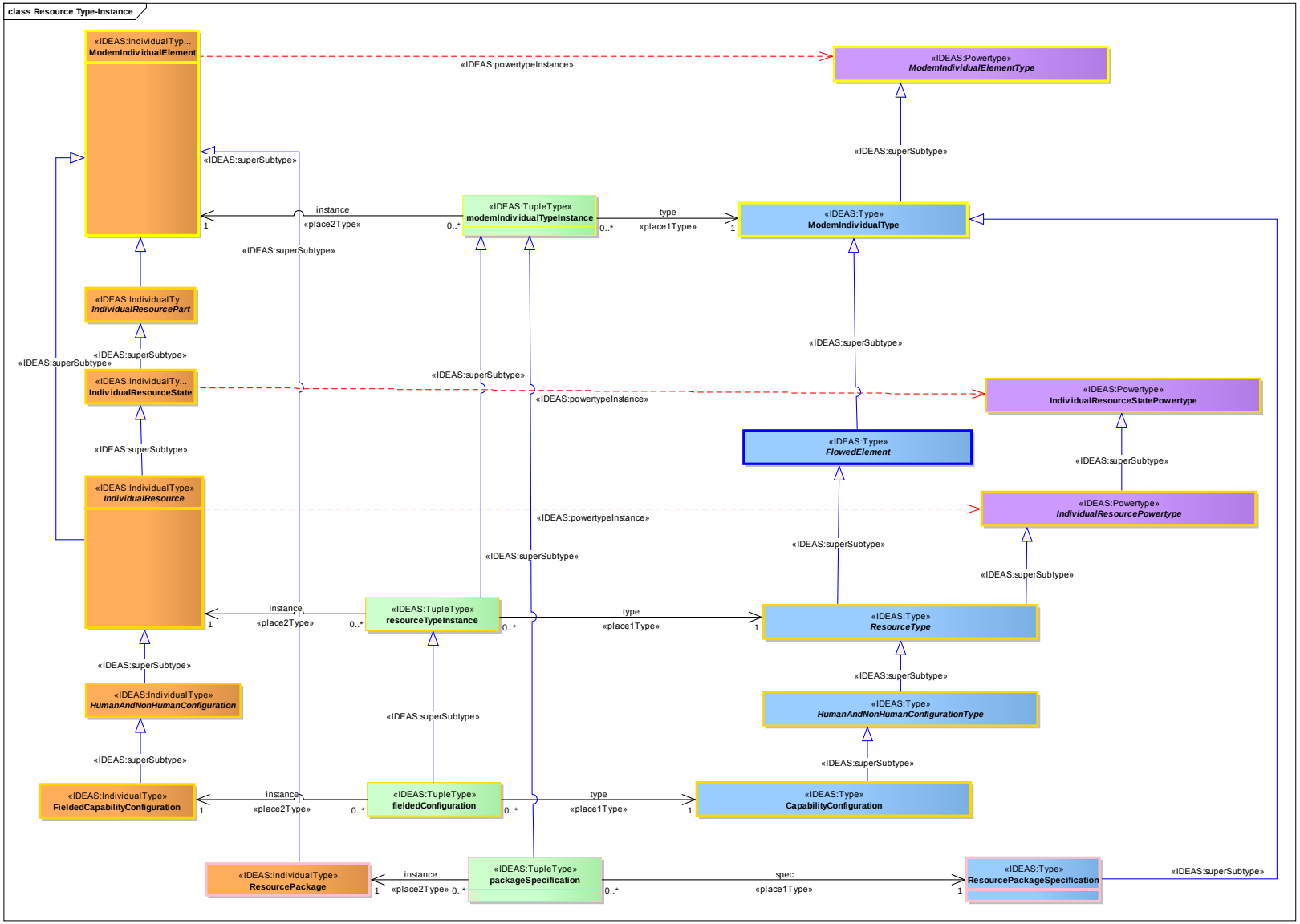


Figure 87 : Resource Type - Instance

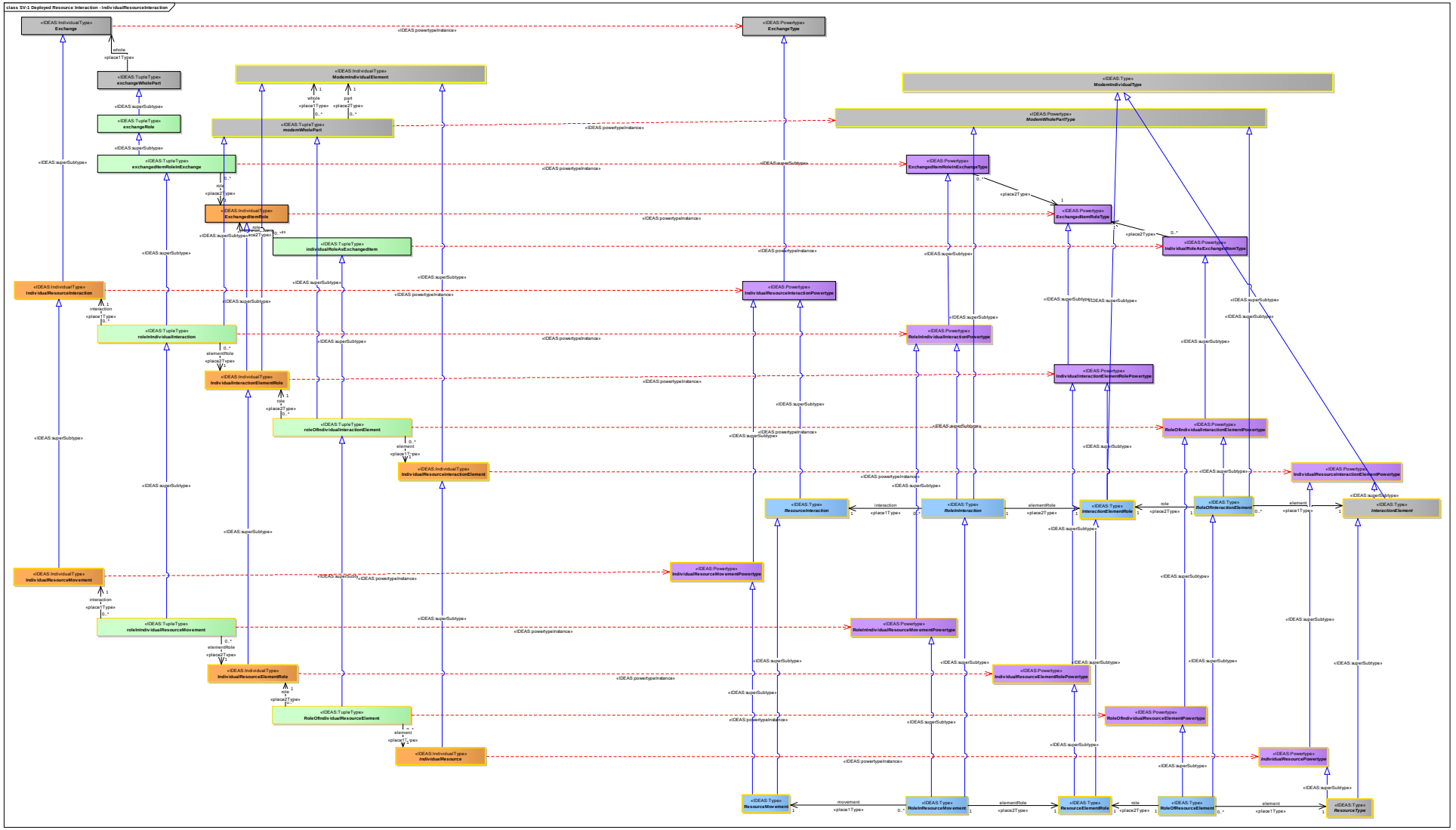


Figure 88 : SV-1 Deployed Resource Interaction - IndividualResourceInteraction

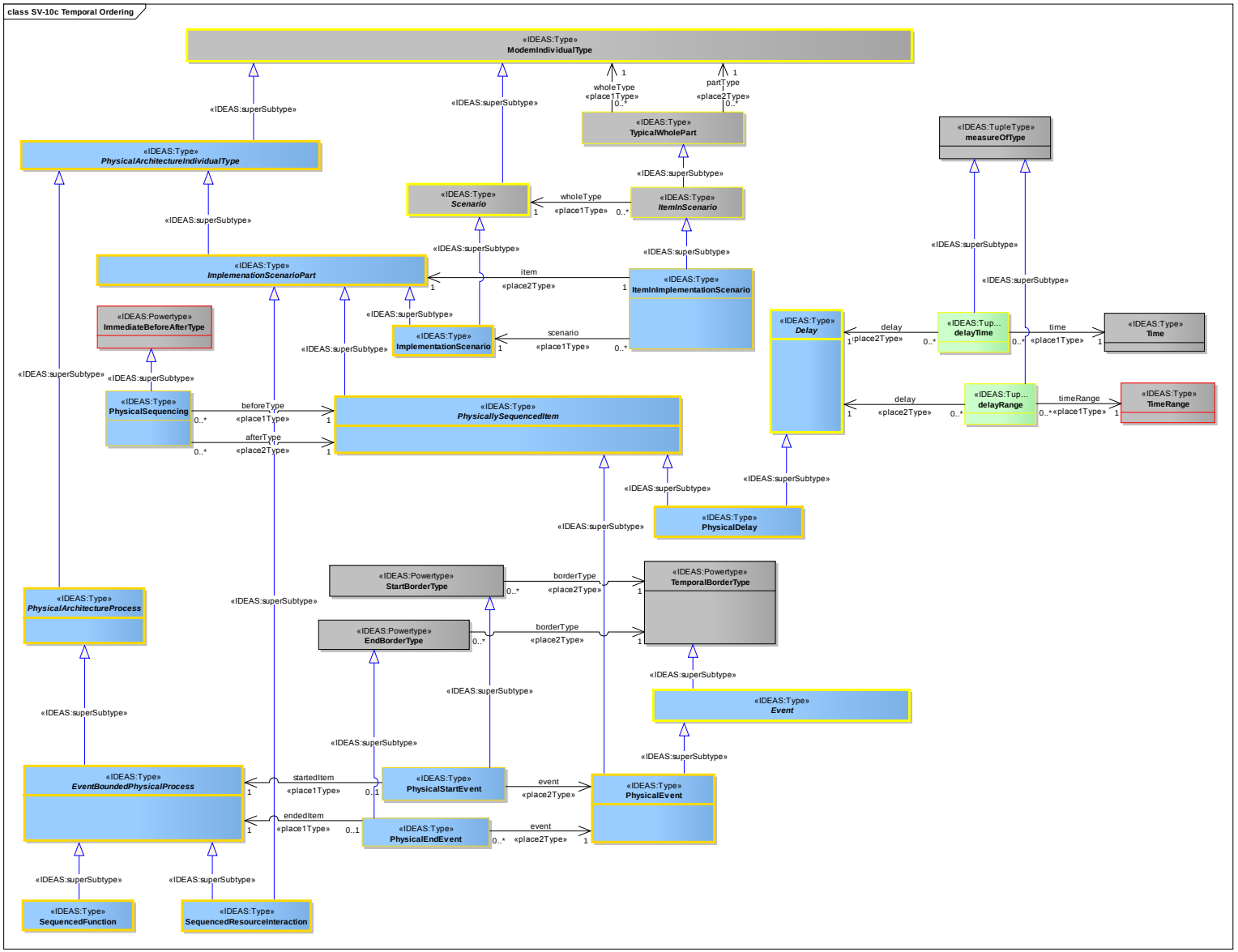


Figure 89 : SV-10c Temporal Ordering

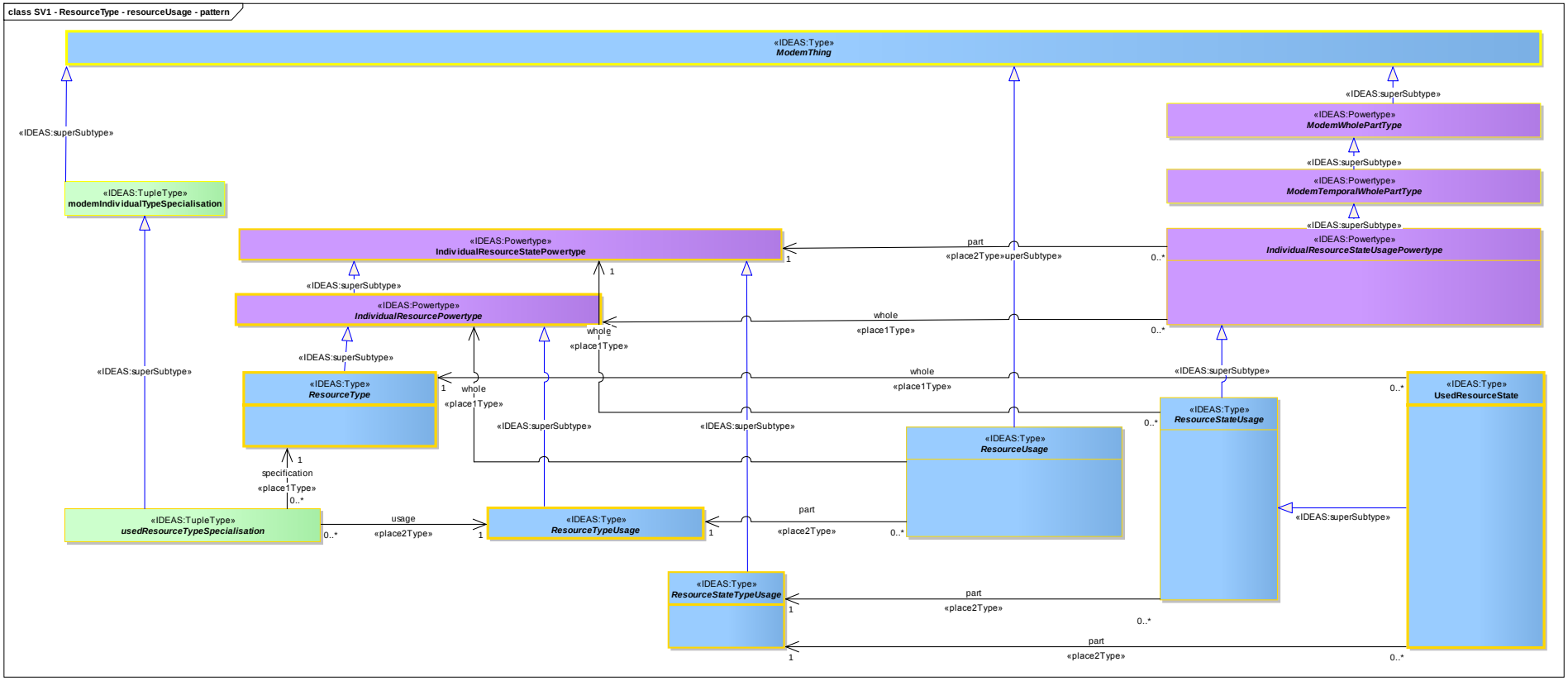


Figure 90 : SV1 - ResourceType - resourceUsage - pattern

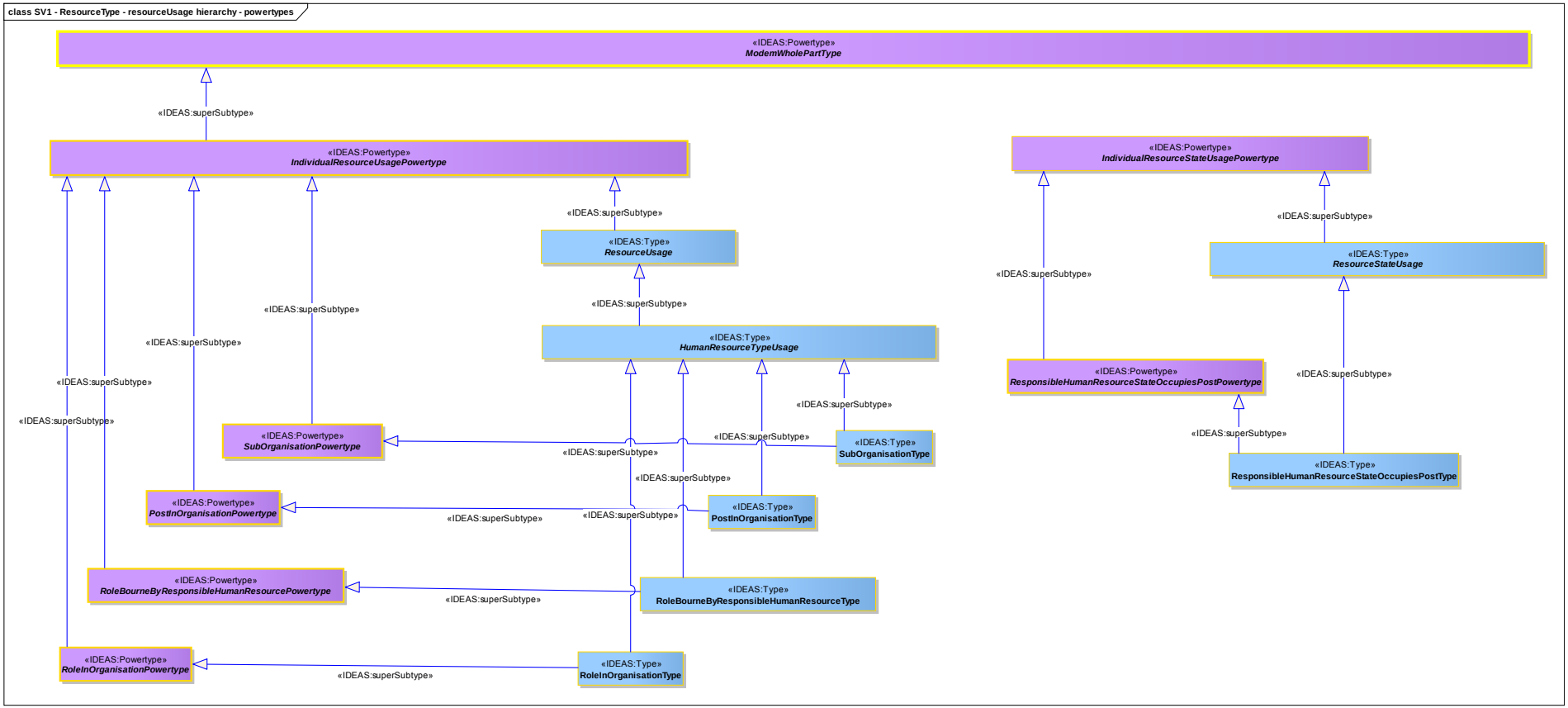


Figure 91 : SV1 - ResourceType - resourceUsage hierarchy - powertypes

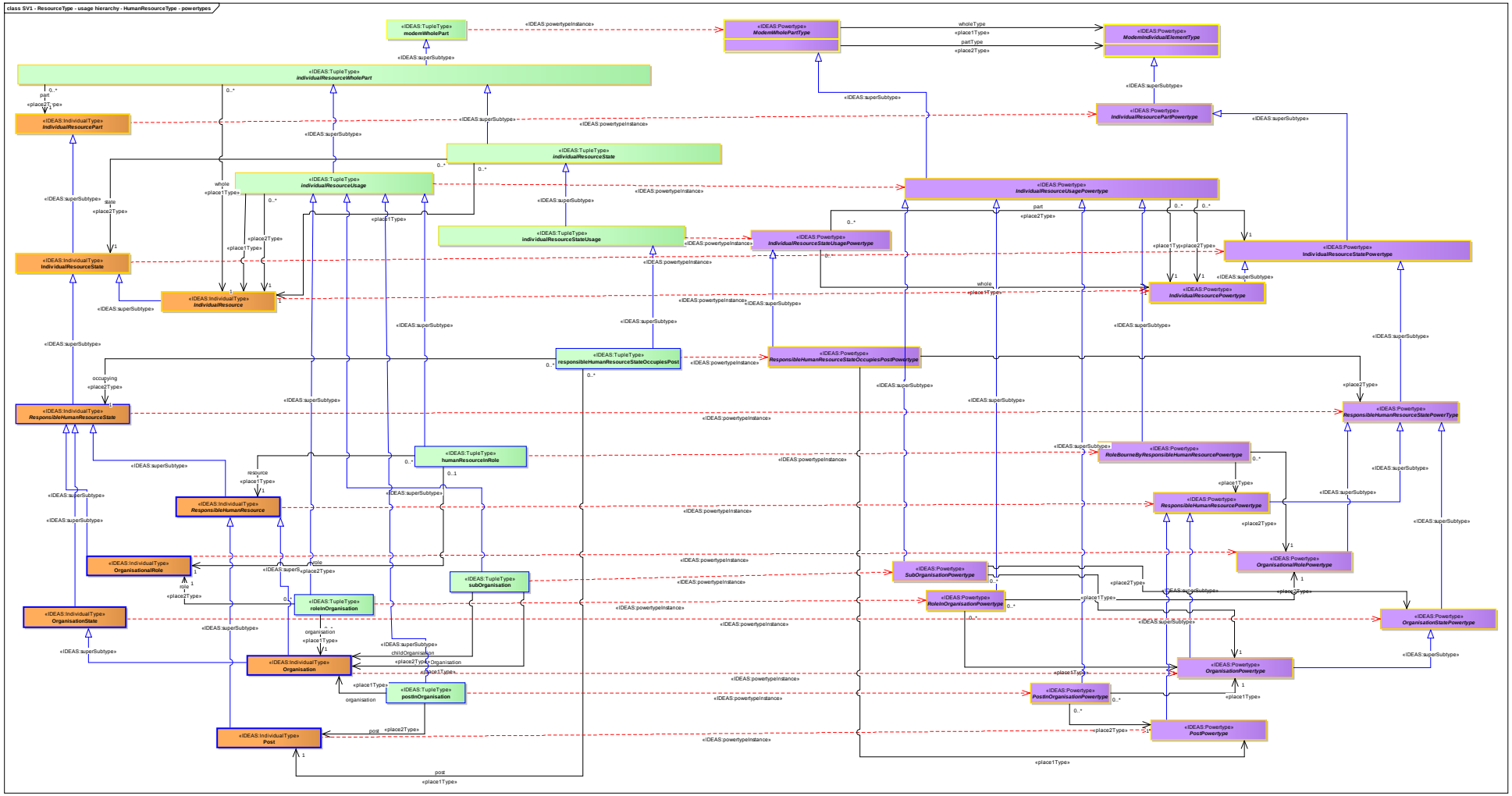


Figure 92 : SV1 - ResourceType - usage hierarchy - HumanResourceType - powertypes

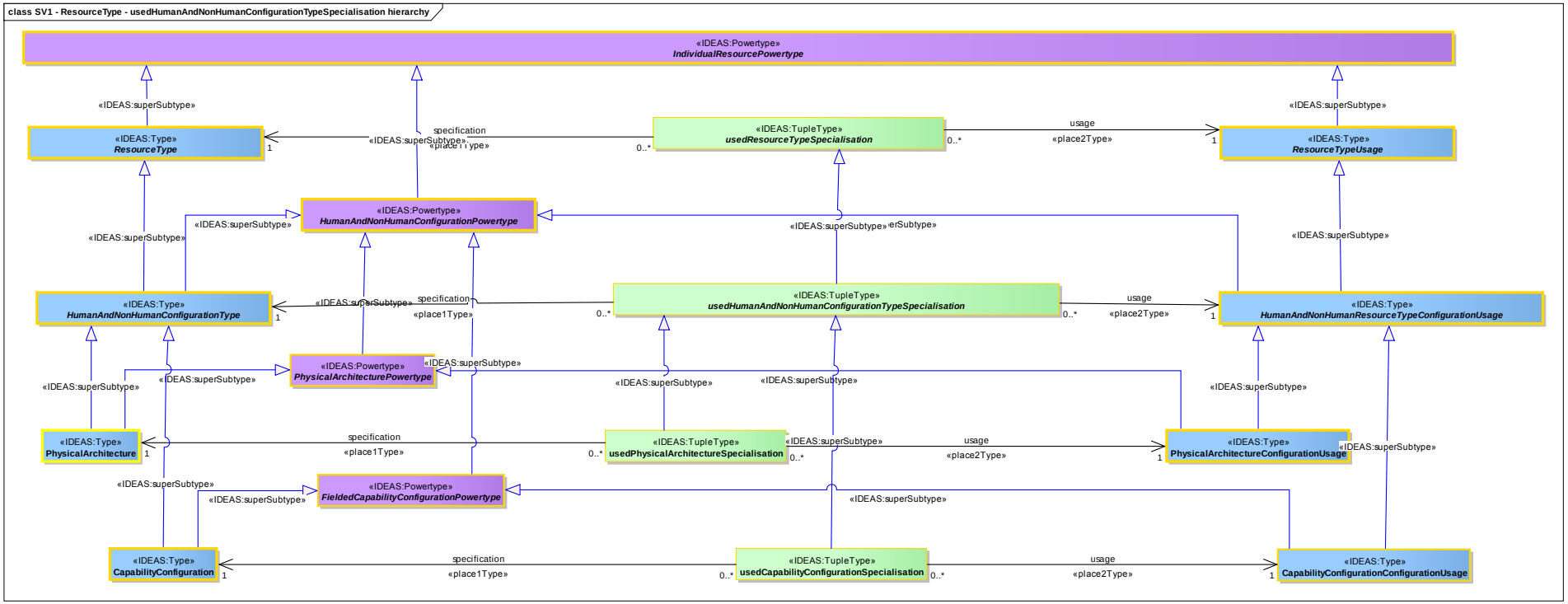


Figure 93 : SV1 - ResourceType - usedHumanAndNonHumanConfigurationTypeSpecialisation hierarchy

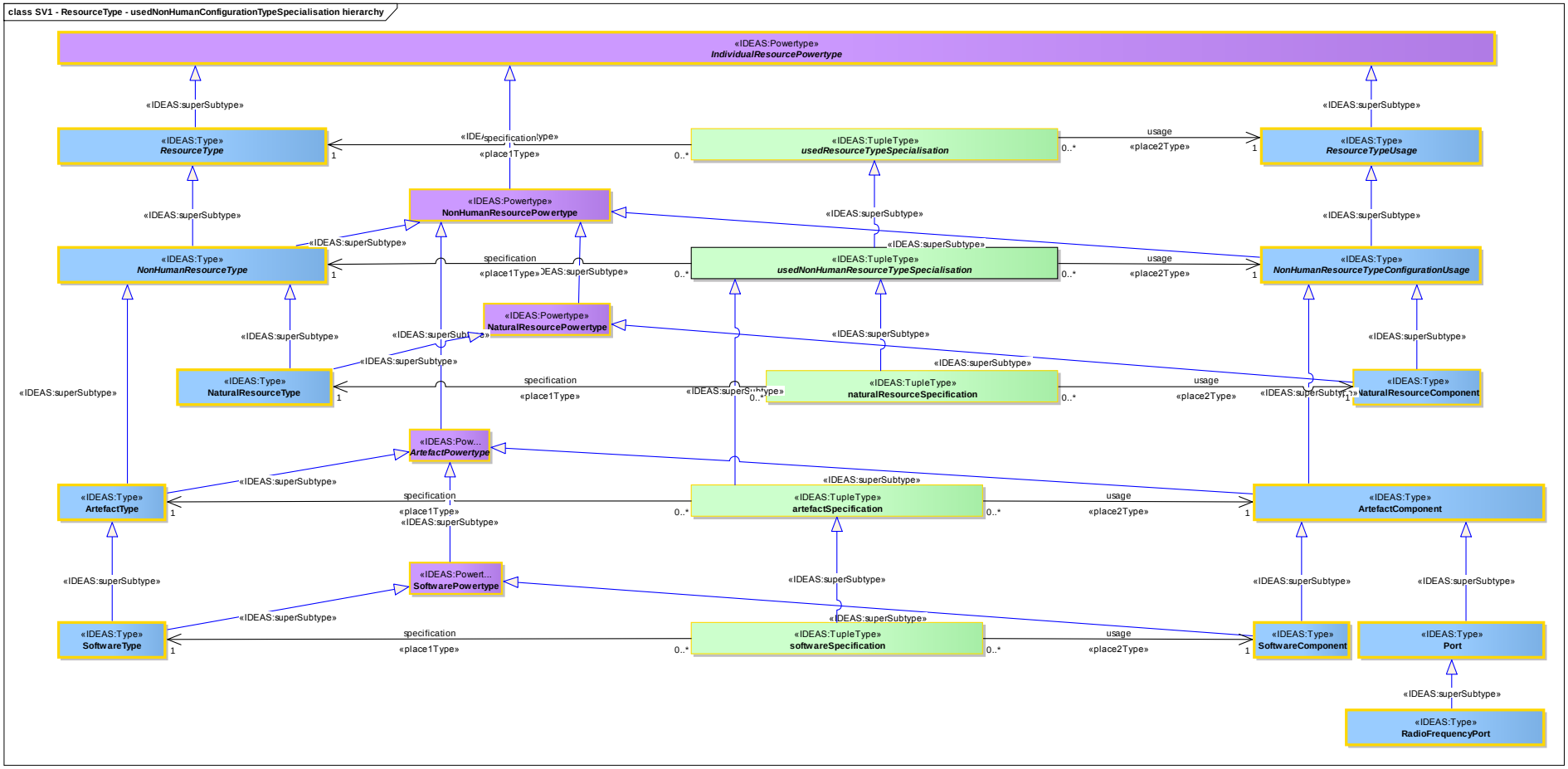


Figure 94 : SV1 - ResourceType - usedNonHumanConfigurationTypeSpecialisation hierarchy

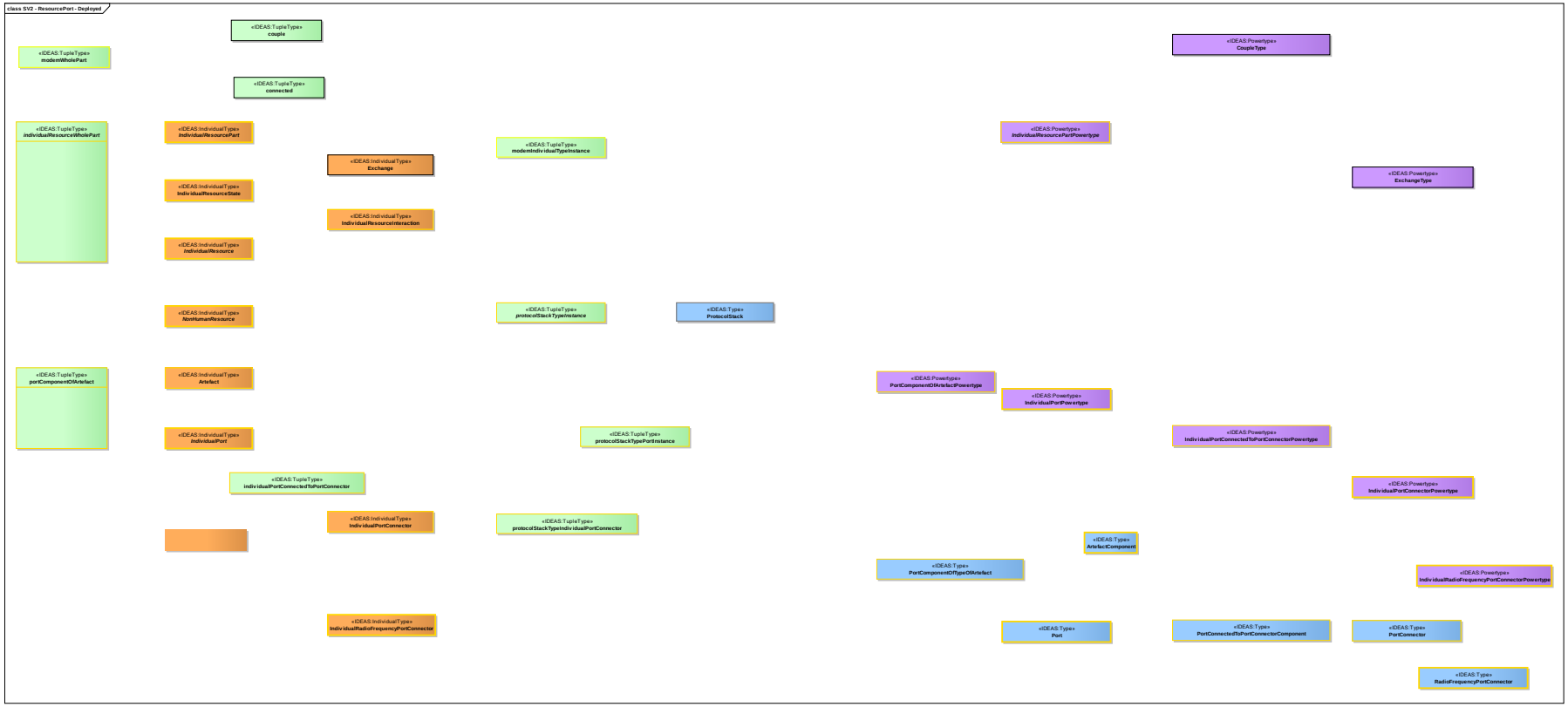


Figure 95 : SV2 - ResourcePort - Deployed

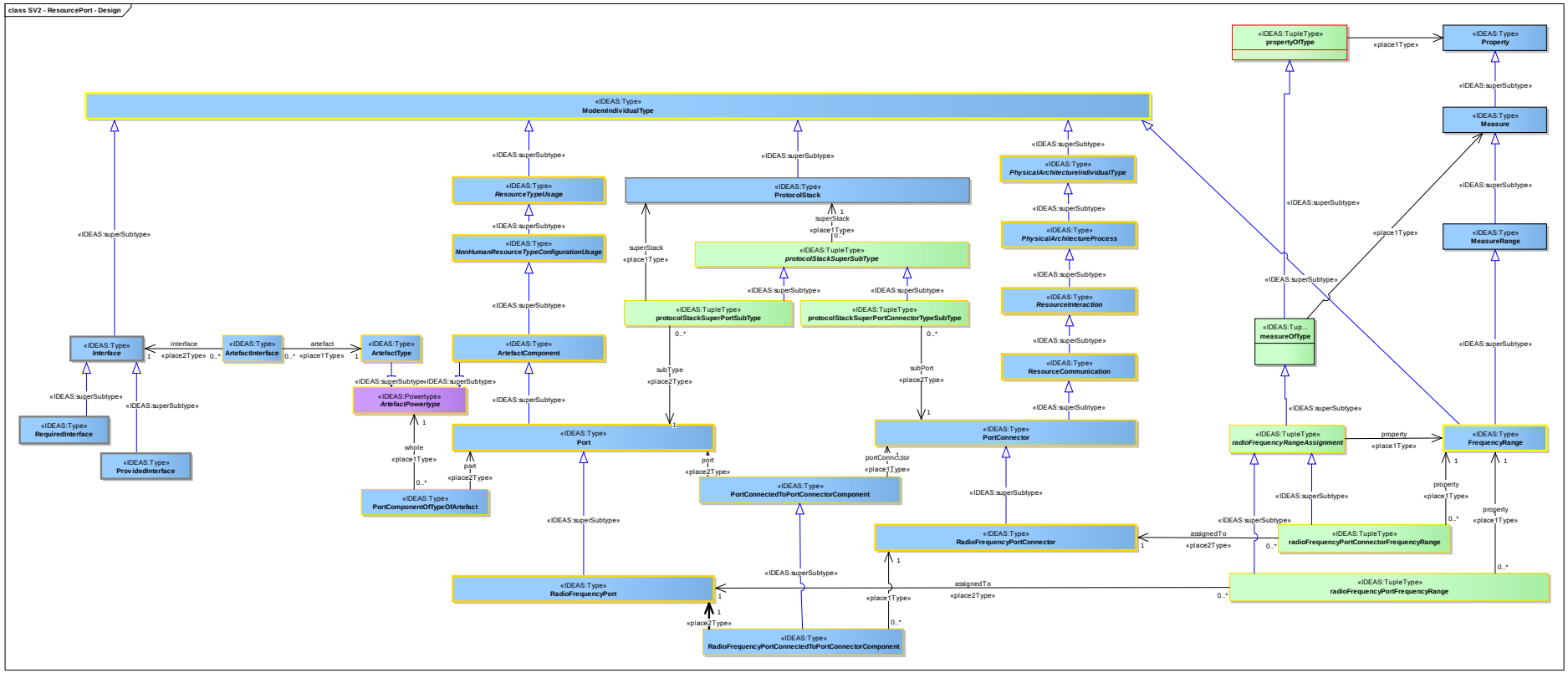


Figure 96 : SV2 - ResourcePort - Design

2.7 Technical standards views

2.7.1 TV-1: Standards profile, TV-2: Standards forecast

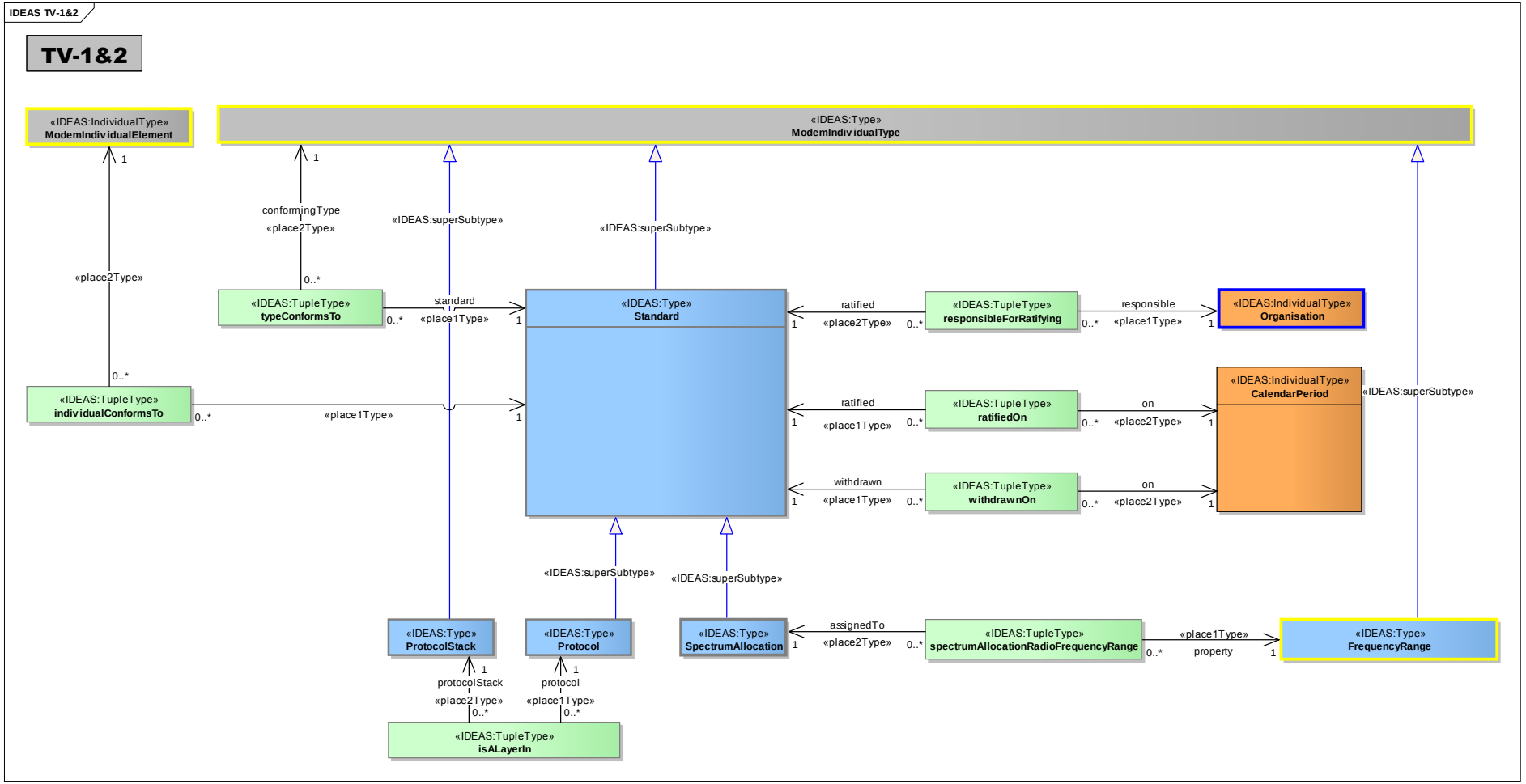


Figure 97 : TV-1&2

2.7.2 TV-3: Standard configuration

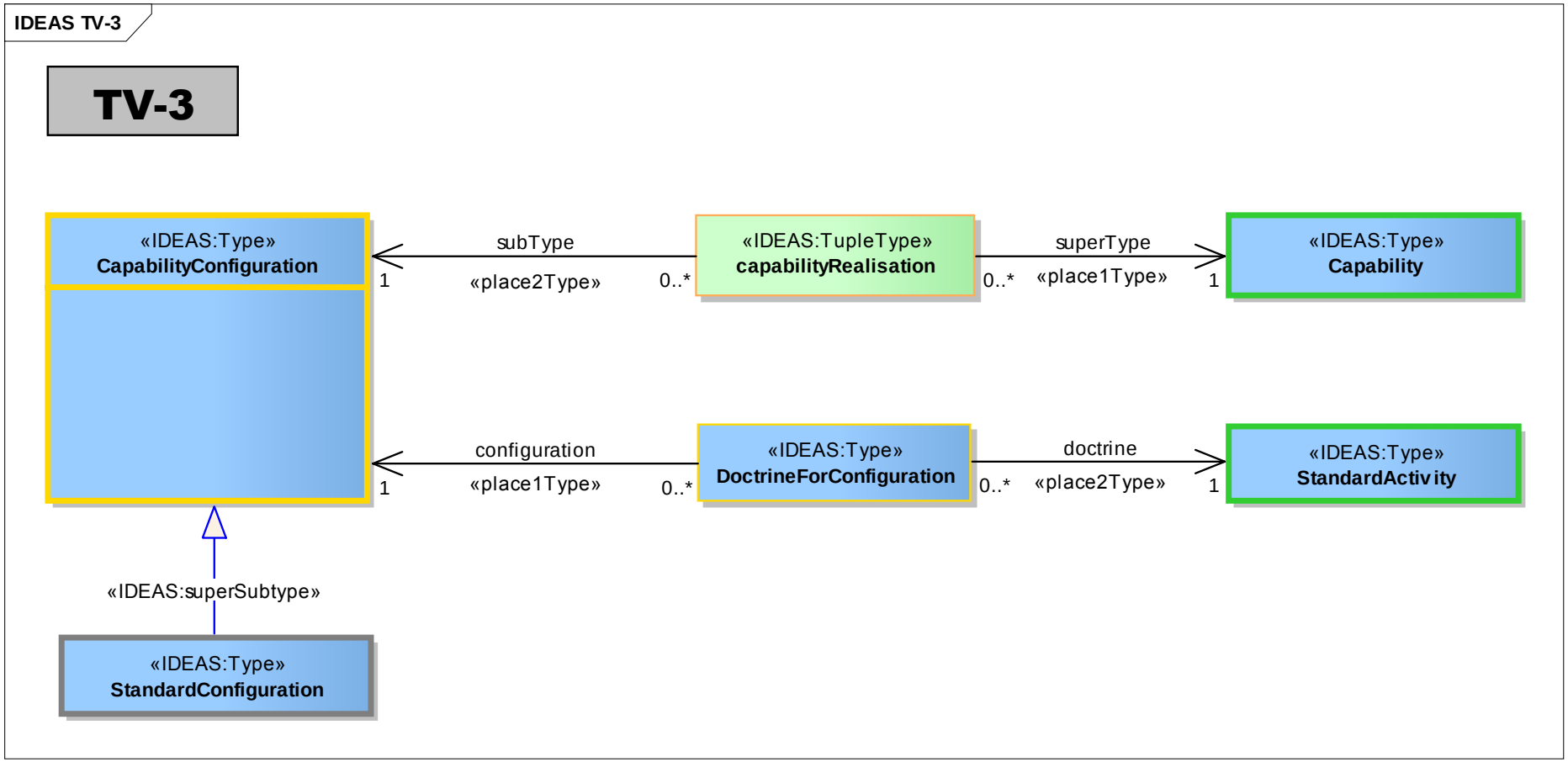


Figure 98 : TV-3

2.7.3 Protocols

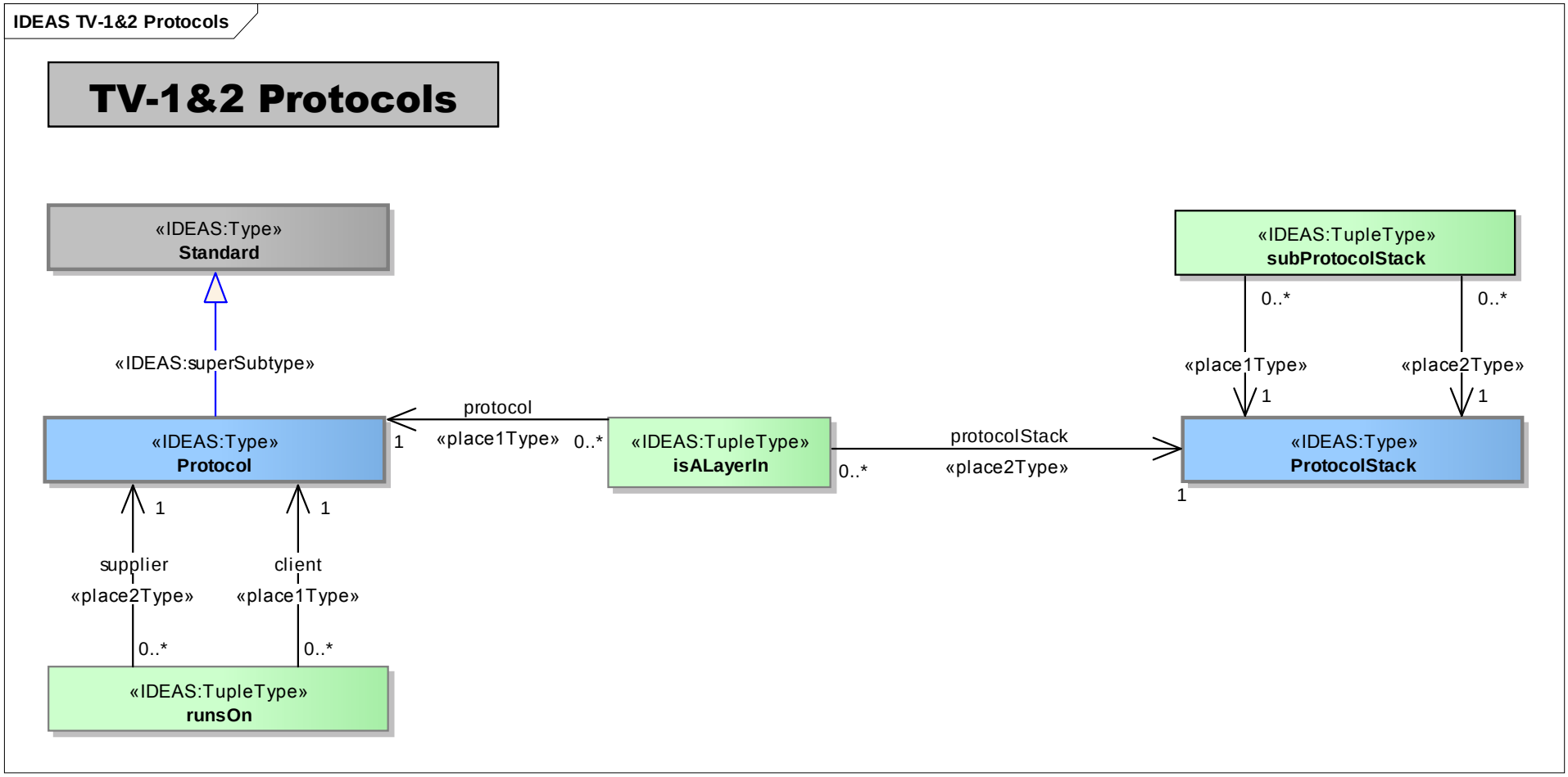


Figure 99 : TV-1&2 Protocols

2.7.4 Technical standards Views elements list

Technical Standard Views
AggregateDataType «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AggregateDataType - DataModelTypeRepresentation *Attributes:* - A DataModelTypeRepresentation which is an aggregate of other DataModelTypeRepresentations.
ArrayDataType «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ArrayDataType - AggregateDataType *Attributes:* - An AggregateDataType whose members are addressed using a numeric index.
AsynchronousOperation «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» AsynchronousOperation - OperationSpecification *Attributes:* - An OperationSpecification where the caller and called do not wait for each other to complete the communication.
Attribute «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Attribute - DataModelComponent *Attributes:* - A DataModelComponent that is a defined property of an Entity.
BagDataType «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» BagDataType - AggregateDataType *Attributes:* - An AggregateDataType whose members are not kept in any particular order - i.e. there is no way to address a particular member.
BinaryDataType «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» BinaryDataType - SimpleDataType *Attributes:* - A SimpleDataType whose instances are binary objects. Note: Data Models may instantiate several different BinaryDataTypes - e.g. "BLOB", "MPEG", "varbinary", etc.

CardinalitySpecifier «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
CardinalitySpecifier - ModemIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
CardinalitySpecifier - IntegerRepresentation
Attributes:
exemplar
An IntegerRepresentation that specifies the cardinality of an EntityRelationshipEnd.

ChoiceDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ChoiceDataType - DataModelTypeRepresentation
Attributes:
-
A DataModelTypeRepresentation which represents a choice of datatypes, restricted by the architect to a list. Note: Also known as a SELECT in some data modelling languages (e.g. ISO10303-11).

DataElement «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DataElement - InteractionElement
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DataElement – SymbolOrSymbolStringType
Attributes:
-
A SymbolOrSymbolStringType that represents interactions between resource elements.

DataModel «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
DataModel - StructuredRepresentation
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DataModel - ModemIndividualType
Attributes:
-
A StructuredRepresentation defining the structure of data, showing classifications of data elements and relationships between them.

DataModelComponent «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
DataModelComponent - Representation
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DataModelComponent - ModemIndividualType
Attributes:
-
A Representation that can be part of a DataModel.

DataModelTypeRepresentation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DataModelTypeRepresentation - DataModelComponent
Attributes:

-
A DataModelComponent that can be used to represent the type of something.

EndOfEntityRelationship «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EndOfEntityRelationship - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EndOfEntityRelationship - RepresentationInStructure
Association (source - target): «place2Type»
EndOfEntityRelationship - EntityRelationshipEnd
Association (source - target): «place1Type»
EndOfEntityRelationship - EntityRelationship
Attributes:
-
A RepresentationStructure where an EntityRelationship has 2 or more EntityRelationshipEnds.

Entity «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Entity - DataModelTypeRepresentation
Attributes:
-
A DataModelComponent that defines an item of interest.

EntityRelationship «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EntityRelationship - StructuredRepresentation
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EntityRelationship - DataModelComponent
Attributes:
-
A DataModelComponent that represents a relationship between two or more Entities.

EntityRelationshipEnd «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EntityRelationshipEnd - StructuredRepresentation
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EntityRelationshipEnd - ModemIndividualType
Attributes:
-
A DataModelComponent that is one end of an EntityRelationship.

EnumerationType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EnumerationType - DataModelTypeRepresentation
Attributes:
-
A DataModelTypeRepresentation which consists of named values.

FloatingPointDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
FloatingPointDataType - NumericDataType
Attributes:
-
A NumericDataType whose instances are real numbers. Note: Data Models may instantiate several different IntegerDataTypes - e.g. "float", "double", "real", etc.

HashedAggregate «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
HashedAggregate - AggregateDataType
Attributes:
-
An AggregateDataType whose members are indexed using an identifier.

IntegerDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IntegerDataType - NumericDataType
Attributes:
-
A NumericDataType whose instances are integer numbers. Note: Data Models may instantiate several different IntegerDataTypes - e.g. "LongInt", "short", "word", etc.

Interface «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Interface - ModemIndividualType
Attributes:
-
A ModemIndividualType that is an interface either provided or required by another ModemIndividualType.

InterfaceOperation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InterfaceOperation - ProcessPartOfBodyType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InterfaceOperation - TypicalWholePart
Association (source - target): «place1Type»
InterfaceOperation - InterfaceSpecification
Association (source - target): «place2Type»
InterfaceOperation - OperationSpecification
Attributes:
-
A TypicalWholePart that relates an OperationSpecification to its InterfaceSpecification.

InterfaceSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InterfaceSpecification - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
InterfaceSpecification - BodyType
Attributes:

-
A ModemIndividualType that is a part of another ModemIndividualType that defines how it communicates.

ItemInDataModel «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ItemInDataModel - ModemThing
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ItemInDataModel - RepresentationInStructure
Association (source - target): «place2Type»
ItemInDataModel - DataModelComponent
Association (source - target): «place1Type»
ItemInDataModel - DataModel
Attributes:
-
A RepresentationInStructure where a DataModelComponent is part of a DataModel.

ListDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ListDataType - AggregateDataType
Attributes:
-
An AggregateDataType whose members are stored and accessed as an ordered list.

LogicalDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
LogicalDataType - SimpleDataType
Attributes:
-
A SimpleDataType whose instances are true/false or true/false/unknown. Note: Data Models may instantiate several different LogicalDataTypes - e.g. "Boolean", "YesNo", "BOOL", etc.

MaxAggregateSize «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MaxAggregateSize - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MaxAggregateSize - RepresentationInStructure
Association (source - target):«place1Type»
MaxAggregateSize - AggregateDataType
Association (source - target):«place2Type»
MaxAggregateSize - CardinalitySpecifier
Attributes:
-
A RepresentationInStructure that specifies the maximum size of an AggregateDataType.

MaxCardinalityOfRelationshipEnd «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
MaxCardinalityOfRelationshipEnd - RepresentationInStructure
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MaxCardinalityOfRelationshipEnd - ModemThing
Association (source - target): «place2Type»
MaxCardinalityOfRelationshipEnd - CardinalitySpecifier
Association (source - target): «place1Type»
MaxCardinalityOfRelationshipEnd - EntityRelationshipEnd
Attributes:
-
A RepresentationInStructure that asserts a CardinalitySpecifier is part of an EntityRelationshipEnd, and that it represents the maximum cardinality value of that end.
Note: If no Maximum Cardinality is specified (i.e. there is no instance of this tuple type related to the EntityRelationshipEnd) then the default is "many" or "*".

MessageSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MessageSpecification - DataElement
Attributes:
-
A DataElement that specifies the content of a message.

MinAggregateSize «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MinAggregateSize - RepresentationInStructure
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MinAggregateSize - ModemThing
Association (source - target):«place1Type»
MinAggregateSize - AggregateDataType
Association (source - target):«place2Type»
MinAggregateSize - CardinalitySpecifier
Attributes:
-
A RepresentationInStructure that specifies the minimum size of an AggregateDataType.

MinCardinalityOfRelationshipEnd «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
MinCardinalityOfRelationshipEnd - RepresentationInStructure
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MinCardinalityOfRelationshipEnd - ModemThing
Association (source - target): «place2Type»
MinCardinalityOfRelationshipEnd - CardinalitySpecifier
Association (source - target): «place1Type»
MinCardinalityOfRelationshipEnd - EntityRelationshipEnd
Attributes:
-
A RepresentationInStructure that asserts a CardinalitySpecifier is part of an EntityRelationshipEnd, and that it represents the minimum cardinality value of that end. Note: If no Minimum Cardinality is specified (i.e. there is no instance of this tuple type related to the EntityRelationshipEnd) then the default is zero.

NumericDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
NumericDataType - SimpleDataType
Attributes:
-
A SimpleDataType whose instances are numbers.

OperationInputParameter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OperationInputParameter - TypicalWholePart
Association (source - target):«place1Type»
OperationInputParameter - OperationSpecification
Association (source - target):«place2Type»
OperationInputParameter - OperationParameter
Attributes:
-
A TypicalWholePart where an OperationParameter is passed into a OperationSpecification.

OperationParameter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OperationParameter - ModemIndividualType
Attributes:
-
A ModemIndividualType that is a part of an OperationSpecification
OperationParameters are passed in and out of OperationSpecifications.

OperationReadWriteParameter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OperationReadWriteParameter - TypicalWholePart
Association (source - target):«place1Type»
OperationReadWriteParameter - OperationSpecification
Association (source - target):«place2Type»
OperationReadWriteParameter - OperationParameter
Attributes:
-
A TypicalWholePart where an OperationParameter is passed into a OperationSpecification that can then be modified and the result read after processing.

OperationReturnParameter «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OperationReturnParameter - TypicalWholePart
Association (source - target):«place1Type»
OperationReturnParameter - OperationSpecification
Association (source - target):«place2Type»
OperationReturnParameter - OperationParameter
Attributes:
-
A TypicalWholePart where an OperationParameter is passed out of an OperationSpecification.

OperationSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OperationSpecification - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OperationSpecification - ProcessType
Attributes:
-
A ModemIndividualType that is an invokable part of an InterfaceSpecification.
[ABSTRACT]

Protocol «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Protocol - Standard
Attributes:
-
A Standard for communication.

ProtocolStack «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProtocolStack - ModemIndividualType
Attributes:
-
A ModemIndividualType that is all the Individuals which conform to one or more specified protocols (ordered into a stack) that may be implemented by one or more ResourcePorts. Note: was called "ImplementedProtocol" in M3.

ProvidedInterface «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProvidedInterface - Interface
Attributes:
-
An Interface describing what a ServiceSpecification or a ResourceType is capable of providing when invoked by an external element.

RequiredInterface «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
RequiredInterface - Interface
Attributes:
-
An Interface describing what a ServiceSpecification or a ResourceType requires from an external element.

SecurityPolicy «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SecurityPolicy - Constraint
Attributes:
-
A Constraint that is concerned with security.

SimpleDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SimpleDataType - DataModelTypeRepresentation
Attributes:
-
A DataModelTypeRepresentation that is used to specify the type of a literal (e.g. text, integer, floating point number, etc.)

SpectrumAllocation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SpectrumAllocation - Standard
Attributes:
-
A Standard specifying a particular frequency range of the electromagnetic spectrum that is allotted to a particular usage.

Standard «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Standard - SubjectOfForecast
Generalization (element - is a subtype of):«IDEAS:superSubtype»
Standard - ModemIndividualType
Attributes:
-
A ratified and peer-reviewed specification that is used to guide or constrain the architecture. A Standard may be applied to any element in the architecture.

StandardConfiguration «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StandardConfiguration - CapabilityConfiguration
Attributes:
-
A CapabilityConfiguration that has been designated as a standard configuration.

SupportedMessageFormat «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
supportedMessageFormat - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
supportedMessageFormat - couple
Association (source - target):«place1Type»
supportedMessageFormat - Interface
Association (source - target):«place2Type»
supportedMessageFormat - MessageSpecification
Attributes:
-
A couple that relates an Interface to a MessageSpecification that it can support.

SynchronousOperation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SynchronousOperation - OperationSpecification
Attributes:

-
An OperationSpecification where the caller and called wait for each other to complete the communication.

TextDataType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
TextDataType - SimpleDataType
Attributes:
-
A SimpleDataType whose instances are text literals. Note: Data Models may instantiate several different TextDataTypes - e.g. "String", "XML Text", "WideString", etc.

aggregateElementType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
aggregateElementType - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
aggregateElementType - couple
Association (source - target):«place1Type»
aggregateElementType - AggregateDataType
Association (source - target):«place2Type»
aggregateElementType - DataModelTypeRepresentation
Attributes:
-
A couple that relates an AggregateDataType to the DataModelTypeRepresentation that specifies the data type of each of its elements.

attributeType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
attributeType - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
attributeType - ModemThing
Association (source - target):«place2Type»
attributeType - DataModelTypeRepresentation
Association (source - target):«place1Type»
attributeType - Attribute
Attributes:
-
A couple that relates an Attribute to the DataModelTypeRepresentation that specifies its type.

choiceElement «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
choiceElement - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
choiceElement - ModemThing
Association (source - target):«place1Type»
choiceElement - ChoiceDataType
Association (source - target):«place2Type»
choiceElement - DataModelTypeRepresentation
Attributes:
-
A couple that asserts a DataModelTypeRepresentation is a valid choice in a ChoiceDataType.

conformsTo «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
conformsTo - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
conformsTo - ModemThing
Association (source - target):«place1Type»
conformsTo - Standard
Attributes:
-
A couple that asserts a thing in the architecture model conforms to a standard.

dataElementRepresentation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
dataElementRepresentation - representedByDataType
Association (source - target):«place1Type»
dataElementRepresentation - DataElement
Association (source - target):«place2Type»
dataElementRepresentation - DataModelTypeRepresentation
Attributes:
-
A representedByDataType that asserts an DataElement is represented by a DataModelTypeRepresentation.

entityHasAttribute «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
entityHasAttribute - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
entityHasAttribute - ModemThing
Association (source - target):«place2Type»
entityHasAttribute - Attribute
Association (source - target):«place1Type»
entityHasAttribute - Entity
Attributes:
-
A couple asserting that an Entity has an Attribute.

entityInRelationship «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
entityInRelationship - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
entityInRelationship - couple
Association (source - target):«place2Type»
entityInRelationship - Entity
Association (source - target):«place1Type»
entityInRelationship - EntityRelationshipEnd
Attributes:
-
A couple relating a RelationshipInDataModel to one of the Entities it relates.

enumerationItem «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enumerationItem - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
enumerationItem - couple
Association (source - target):«place2Type»
enumerationItem - StringRepresentation
Association (source - target):«place1Type»
enumerationItem - EnumerationType
Attributes:
-
A Couple that relates a StringRepresentation to an EnumerationType of which it is an element.

individualConformsTo «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
individualConformsTo - Standard
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualConformsTo - conformsTo
Association (source - target):«place2Type»
individualConformsTo - ModemIndividualElement
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualConformsTo - modemIndividualTypeInstance
Attributes:
-
A modemIndividualTypeInstance that asserts that an element in the architecture conforms to a Standard.

isALayerIn «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IsALayerIn - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
IsALayerIn - superSubtype
Association (source - target):«place2Type»
IsALayerIn - Protocol
Association (source - target):«place1Type»
IsALayerIn - ProtocolStack
Attributes:
-
A superSubtype that asserts that a ProtocolStack is a kind of Protocol. The Protocol is a layer in the ProtocolStack. The order of the layering is determined by the Protocols' runsOn relations. Note: amalgamates "ProtocolLayer" and "ImplementedOn" in M3.

parameterRepresentation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
parameterRepresentation - representedByDataType
Association (source - target):«place2Type»
parameterRepresentation - DataModelTypeRepresentation
Association (source - target):«place1Type»
parameterRepresentation - OperationParameter
Attributes:

-
A representedBy that links an OperationParameter to its datatype (DataModelTypeRepresentation).

ratifiedOn «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ratifiedOn - ModemThing
Association (source - target):«place2Type»
ratifiedOn - CalendarPeriod
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ratifiedOn - couple
Association (source - target):«place1Type»
ratifiedOn - Standard
Attributes:
-
A couple that asserts a Standard has been ratified on a date.

representedByDataType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
representedByDataType - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
representedByDataType - representedBy
Association (source - target):«place2Type»
representedByDataType - DataModelTypeRepresentation
Attributes:
-
A representedBy that asserts a Thing is represented by a DataModelTypeRepresentation.

responsibleForRatifying «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
responsibleForRatifying - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
responsibleForRatifying - ModemThing
Association (source - target):«place2Type»
responsibleForRatifying - Standard
Association (source - target):«place1Type»
responsibleForRatifying - Organisation
Attributes:
-
A couple that asserts than an Organisation is responsible for the ratification of a standard.
Note: was called "RatificationBody" in M3.

runsOn «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
runsOn - couple
Generalization (element - is a subtype of):«IDEAS:superSubtype»
runsOn - ModemThing
Association (source - target):«place1Type»
runsOn - Protocol

Association (source - target):«place2Type»
runsOn - Protocol
Attributes:
-
A couple that asserts that one Protocol (client) may be implemented on another (supplier). This determines the layer order in the ProtocolStack.

specForInterface «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SpecForInterface - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
SpecForInterface - Interface
Association (source - target):«place1Type»
SpecForInterface - InterfaceSpecification
Attributes:
-
A modemIndividualTypeSpecialisation that relates an Interface to the InterfaceSpecification that specifies it.

spectrumAllocationRadioFrequencyRange «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
spectrumAllocationRadioFrequencyRange - FrequencyRange
Generalization (element - is a subtype of):«IDEAS:superSubtype»
spectrumAllocationRadioFrequencyRange - radioFrequencyRangeAssignment
Association (source - target):«place2Type»
spectrumAllocationRadioFrequencyRange - SpectrumAllocation
Attributes:
-
A radioFrequencyRangeAssignment that asserts a spectrum allocation has been assigned to a frequency range.

subProtocolStack «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
subProtocolStack - ProtocolStack
Association (source - target):«place2Type»
subProtocolStack - ProtocolStack
Generalization (element - is a subtype of):«IDEAS:superSubtype»
subProtocolStack - superSubtype
Attributes:
-
A superSubtype that asserts that one ProtocolStack is a superType of another.

subtypeRelationship «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
subtypeRelationship - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
subtypeRelationship - couple
Association (source - target):«place1Type»
subtypeRelationship - Entity
Association (source - target):«place2Type»
subtypeRelationship - Entity

Attributes:
-
A couple that asserts that the type represented by one Entity is a subtype of the type represented by the other Entity.

typeConformsTo «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
typeConformsTo - modemIndividualTypeSpecialisation
Association (source - target):«place2Type»
typeConformsTo - ModemIndividualType
Association (source - target):«place1Type»
typeConformsTo - Standard
Generalization (element - is a subtype of):«IDEAS:superSubtype»
typeConformsTo - conformsTo
Attributes:
-
A modemIndividualTypeSpecialisation that asserts a type in the architecture conforms to a Standard.

withdrawnOn «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
withdrawnOn - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
WithdrawnOn - couple
Association (source - target):«place2Type»
WithdrawnOn - CalendarPeriod
Association (source - target):«place1Type»
WithdrawnOn - Standard
Attributes:
-
A couple that asserts a Standard has been withdrawn on a date.

2.7.5 Technical standards Views additional diagrams.

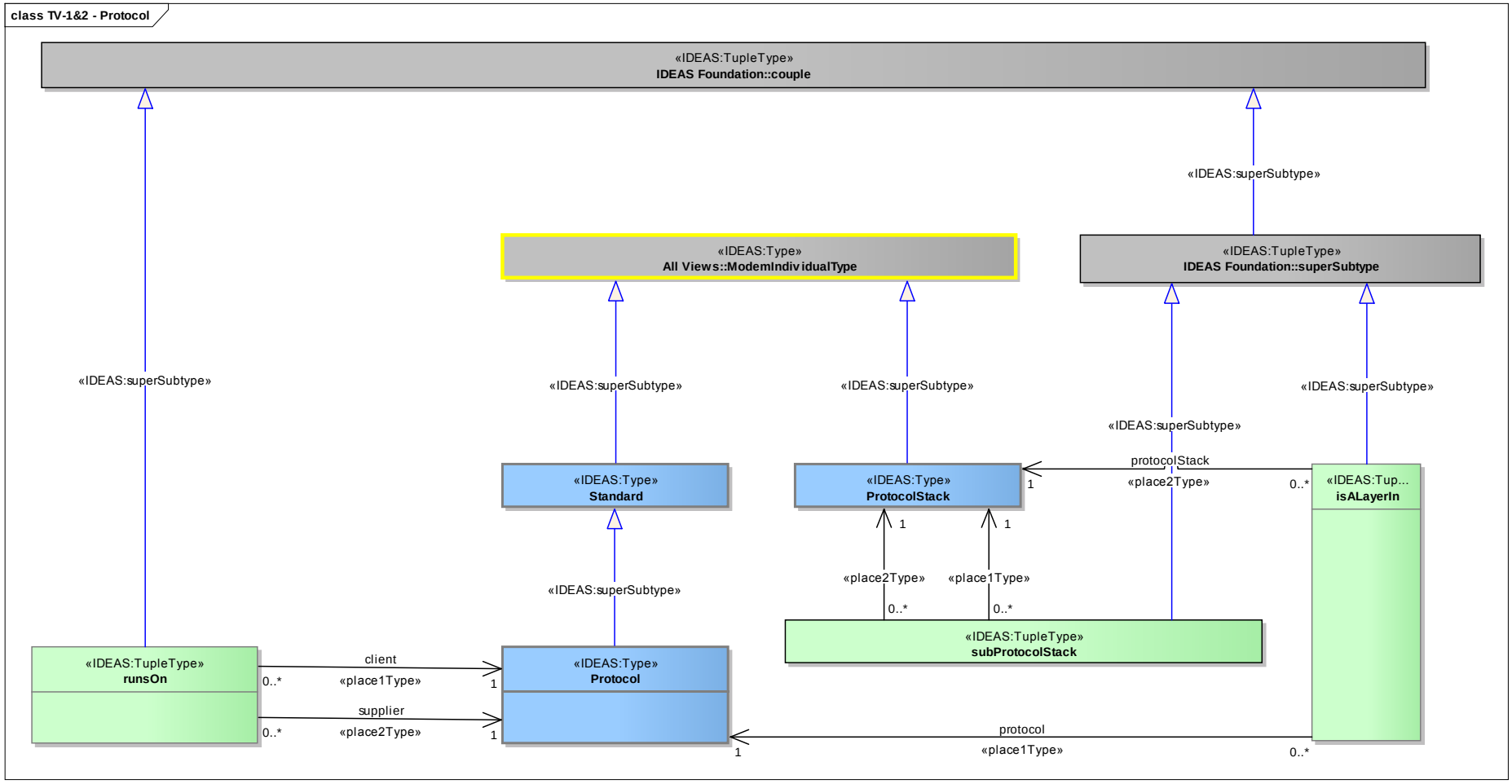


Figure 100 : TV-1&2 - Protocol

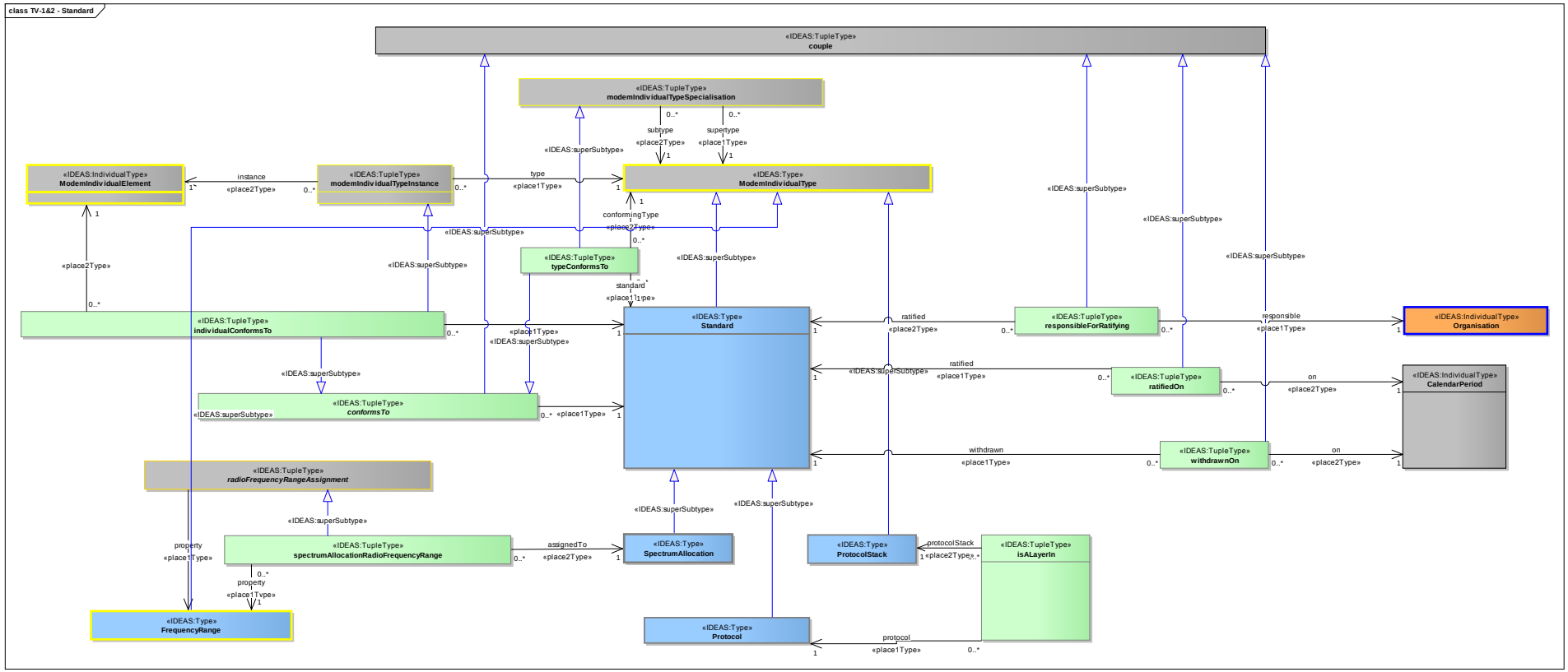


Figure 101 : TV-1&2 - Standard

2.8 Acquisition views

2.8.1 AcV-1: Acquisition clusters

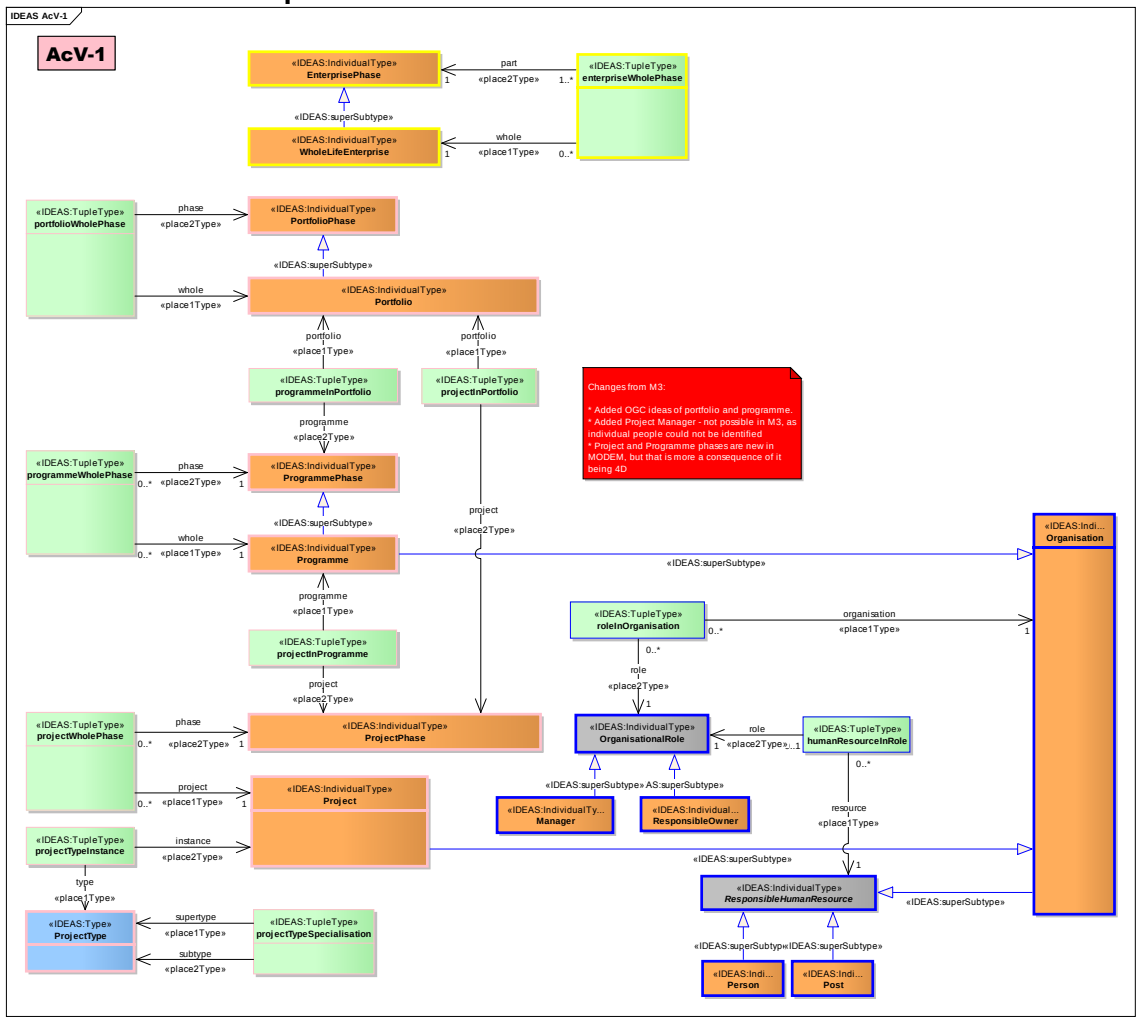


Figure 102 : AcV-1

2.8.2 AcV-2: Programme timelines

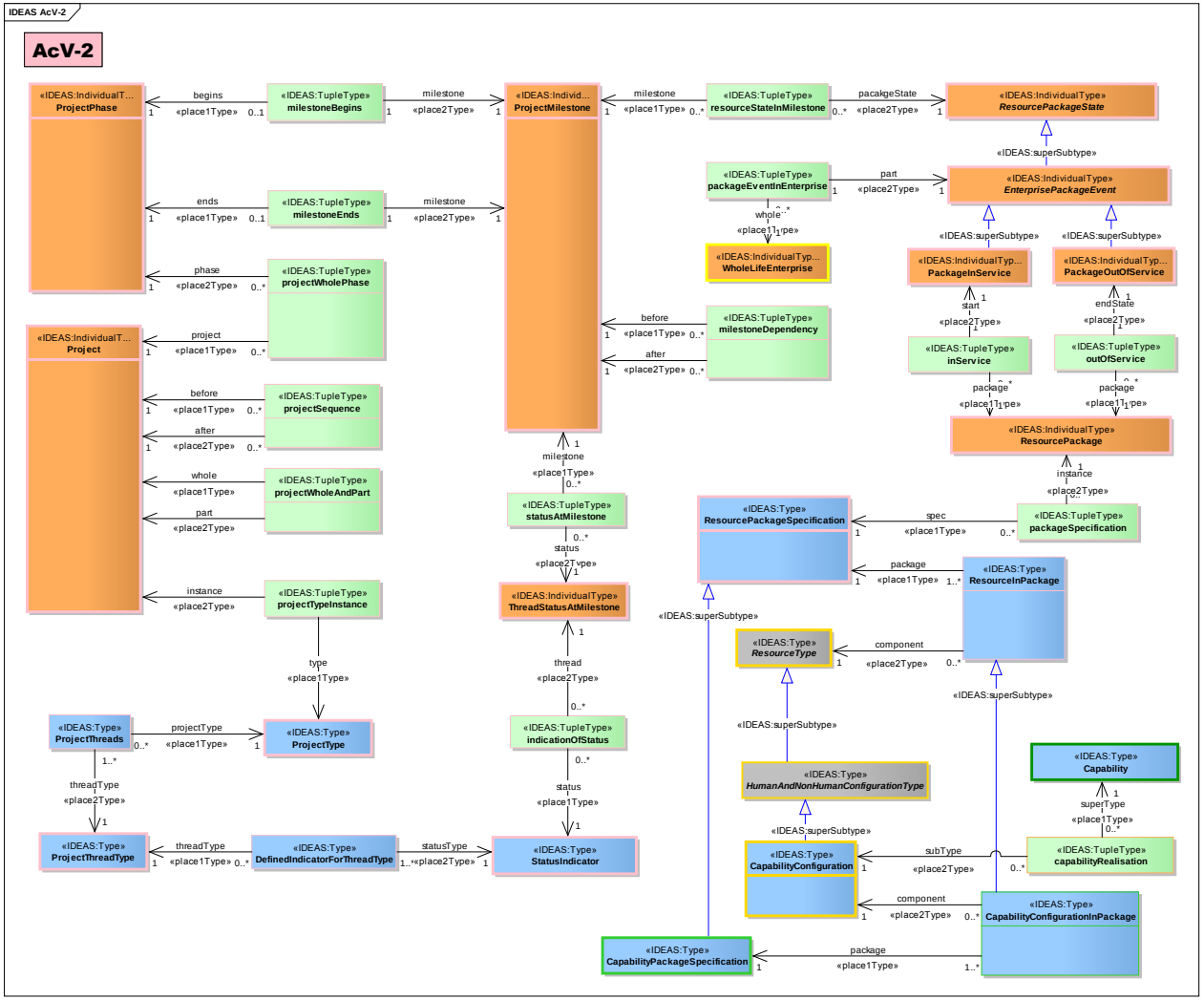


Figure 103 : AcV-2

2.8.3 Acquisition Views elements list

Acquisition Views

DefinedIndicatorForThreadType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
DefinedIndicatorForThreadType - StatusOfThreadType
Association (source - target):«place2Type»
DefinedIndicatorForThreadType - StatusIndicator
Association (source - target):«place1Type»
DefinedIndicatorForThreadType - ProjectThreadType
Attributes:
-
A StatusOfThreadType that specifies a StatusIndicator may be used to classify ProjectThreads of a particular ProjectThreadType.

EnterprisePackageEvent «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EnterprisePackageEvent- ResourcePackageState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
EnterprisePackageEvent - EnterprisePart
Attributes:
-
A ResourcePackageState that is an event that occurs in a WholeLifeEnterprise - e.g. the introduction of a new Capability at the point of a PackageInService.

OrganisationPackageEvent «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationPackageEvent - OrganisationPart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
OrganisationPackageEvent - ResourcePackageState
Attributes:
-
A ResourcePackageState that is an event that occurs in a WholeLifeEnterprise - e.g. the introduction of a new Capability at the point of a PackageInService.

PackageInService «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
PackageInService - EnterprisePackageEvent
Attributes:
-
An IndividualResourceState that marks in in-service point for a ResourcePackage.

PackageOutOfService «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
PackageOutOfService - EnterprisePackageEvent
Attributes:
-
An IndividualResourceState which marks the point at which a ResourcePackage ceases to be in service. Note: the components of the package may go on in service in some other configuration, but the package itself is retired.

Portfolio «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Portfolio - Undertaking
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Portfolio - PortfolioPhase
Attributes:
-
An Undertaking comprised of the Projects and Programmes that are the totality of an Organisation's investment (or segment thereof) in the changes required to achieve its strategic objectives.
[Based on OGC Definition]
OGC defines a portfolio as the totality of an organisation's investment (or segment thereof) in the changes required to achieve its strategic objectives. Portfolio Management is a coordinated collection of strategic processes and decisions which enable the most effective balance of organisational change and business as usual/operations.

PortfolioPart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
PortfolioPart - UndertakingPart
Attributes:
-
An UndertakingPart that is part of a Portfolio.

PortfolioPhase «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
PortfolioPhase - UndertakingState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
PortfolioPhase - PortfolioPart
Attributes:
-
A PortfolioPart that is a temporal part of a Portfolio.

Programme «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Programme - ProgrammePhase
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Programme - Organisation
Attributes:
-
An Undertaking that is a temporary, flexible organisation created to co-ordinate, direct and oversee the implementation of a set of related Projects and Tasks in order to deliver outcomes and benefits related to the organisation's strategic objectives.
A programme is likely to have a lifespan of several years. During a programme lifecycle, projects are initiated, executed, and closed. Programmes provide an umbrella under which these projects can be co-ordinated. The programme integrates the projects so that it can deliver an outcome greater than the sum of its parts.
[Adapted from OGC definition)

ProgrammePart «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProgrammePart - OrganisationPart
Attributes:
-
An OrganisationPart that is a part of a Programme.

ProgrammePhase «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProgrammePhase - ProgrammePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProgrammePhase - OrganisationState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProgrammePhase - PortfolioPart
Attributes:
-
A ProgrammePart that is a temporal part of a Programme

Project «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Project - ProjectState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Project - ProjectPowertype
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Project - Organisation
Attributes:
-
An Undertaking that is a time-limited endeavour to create a specific set of products or services.

ProjectMilestone «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProjectMilestone - ProjectState
Attributes:
-
A ProjectPart that marks the end of one ProjectPhase and possibly the beginning of another.
Note: the temporal extent of a ProjectMilestone is likely to be finite - e.g. there may be milestone meetings, funding reviews, etc. before another Project or ProjectPhase can start.

ProjectPart «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ProjectPart - ProjectPartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProjectPart - UndertakingPart
Attributes:
-
A ProcessPart that is a part of a Project - i.e, an Individual whose entire spatio-temporal extent is within the extent of a Project.
Example - a document that is created for a project and only used within that project that is destroyed before the project completes.
Example - a milestone review meeting for a Project
Example - a ProjectPhase

ProjectPhase «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProjectPhase - ProjectState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ProjectPhase - ProjectPhaseType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProjectPhase - ProgrammePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProjectPhase - PortfolioPart
Attributes:
-
An UndertakingState that is a temporal part of a Project and has been nominated as a phase of a Project.

ProjectState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectState - OrganisationState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectState - ProjectPart
Attributes:
-
A ProjectPart that is a temporal part of a Project.

ProjectThread «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProjectThread - ProjectThreadState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ProjectThread - ProjectThreadPowertype
Attributes:
-
A ProjectPart that is an aspect of the Project used for refining the measurement of progress of the project. In UK MOD, this could be one of the defence lines of development, or DOTMLP in the US.
Example: The Training aspect of a helicopter acquisition project.

ProjectThreadState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectThreadState - ProjectPart
Attributes:
-
A temporal part of a ProjectThread.

ProjectThreadType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectThreadType - ProjectThreadPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectThreadType - ModemIndividualType
Attributes:
-
A ProjectThreadPowertype that is used to classify ProjectThreads.

ProjectThreads «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectThreads - ThreadInProjectType
Association (source - target):«place1Type»
ProjectThreads - ProjectType
Association (source - target):«place2Type»
ProjectThreads - ProjectThreadType
Attributes:
-
A ThreadInProjectType that relates a ProjectType to its ProjectThreadTypes.

ProjectType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectType - ProjectPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ProjectType - ModemIndividualType
Attributes:
-
A ProjectType that is used to classify Projects.

ResourceInPackage «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourceInPackage - ModemWholePartType
Association (source - target):«place2Type»
ResourceInPackage - ResourceType
Association (source - target):«place1Type»
ResourceInPackage - ResourcePackageSpecification
Attributes:
-
A ResourceUsage that specifies that a ResourceType is part of a DeliveryPackageSpecification.

ResourcePackage «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourcePackage - ModemIndividualElement
Dependency (element - is instance of):«IDEAS:powertypeInstance»
ResourcePackage - ResourcePackageType
Attributes:
-
A HumanAndNon-HumanConfiguration that is a collection of IndividualResources for a purpose.
Example: All the fully configured aircraft delivered in an acquisition programme.
Example: A force element package put together for a particular operation.
Example: A tranche of new assets delivered into an enterprise.

ResourcePackageSpecification «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourcePackageSpecification - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»

ResourcePackageSpecification - ResourcePackageType
Attributes:
-
A ResourcePackageType that specifies the types of Resource (i.e. ResourceTypes) that make up a ResourcePackage.

ResourcePackageState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourcePackageState - IndividualResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ResourcePackageState - TemporalBorder
Attributes:
-
A temporal part of a ResourcePackage.

StatusIndicator «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StatusIndicator - ModemIndividualType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
StatusIndicator - ThreadStatusType
Attributes:
-
A ThreadStatusType that classifies a ThreadStatusAtMilestone to indicate its status.

ThreadInProjectType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ThreadInProjectType - ProcessWholePartType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ThreadInProjectType - ModemWholePartType

Attributes:
-
The powertype of threadInProject.

ThreadStatusAtMilestone «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
ThreadStatusAtMilestone - ProjectThreadState
Dependency (element - is instance of):«IDEAS:powertypeInstance»
ThreadStatusAtMilestone - ThreadStatusType
Attributes:
-
A ProjectThreadState that is part of a ProjectMilestone.

capabilityPackageDeliversCapability «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
capabilityPackageDeliversCapability - modemIndividualTypeSpecialisation
Association (source - target):«place1Type»
capabilityPackageDeliversCapability - Capability
Association (source - target):«place2Type»

capabilityPackageDeliversCapability - CapabilityPackageSpecification
Attributes:
-
A modemIndividualTypeSpecialisation where a CapabilityPackageSpecification provides a Capability.

inService «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
inService - individualResourceState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
inService - startBorder
Association (source - target):«place1Type»
inService - ResourcePackage
Association (source - target):«place2Type»
inService - PackageInService
Attributes:
-
A startBorder that indicates that an PackageInService marks the introduction into service of a ResourcePackage.

indicationOfStatus «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
indicationOfStatus - modemIndividualTypeInstance
Association (source - target):«place1Type»
indicationOfStatus - StatusIndicator
Association (source - target):«place2Type»
indicationOfStatus - ThreadStatusAtMilestone
Attributes:
-
A modemIndividualTypeInstance where a ThreadStatusAtMilestone is classified by a StatusIndicator.

milestoneBegins «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
milestoneBegins - startBoundary
Generalization (element - is a subtype of): «IDEAS:superSubtype»
milestoneBegins - milestoneInProject
Association (source - target): «place1Type»
milestoneBegins - ProjectPhase
Association (source - target): «place2Type»
milestoneBegins - ProjectMilestone
Attributes:
-
A startBoundary that asserts a ProjectMilestone marks the beginning of a Project or ProjectPhase.

milestoneDependency «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
milestoneDependency - ModemThing
Generalization (element - is a subtype of):«IDEAS:superSubtype»
milestoneDependency - beforeAfter
Association (source - target):«place2Type»

milestoneDependency - ProjectMilestone
Association (source - target):«place1Type»
milestoneDependency - ProjectMilestone
Attributes:
-
A beforeAfter that asserts one ProjectMilestone shall occur before the other. Note: This is intended to related milestones from different projects where progress in one project depends on the other.

milestoneEnds «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
milestoneEnds - endBoundary
Generalization (element - is a subtype of): «IDEAS:superSubtype»
milestoneEnds - milestoneInProject
Association (source - target): «place1Type»
milestoneEnds - ProjectPhase
Association (source - target): «place2Type»
milestoneEnds - ProjectMilestone
Attributes:
-
An endBoundary that asserts a ProjectMilestone marks the end of a Project or ProjectPhase.

milestoneInProject «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
milestoneInProject - projectTemporalPart
Association (source - target):«place2Type»
milestoneInProject - ProjectMilestone
Association (source - target):«place1Type»
milestoneInProject - ProjectPhase
Attributes:
-
A projectWholePart that asserts that a ProjectMilestone is part of a Project or ProjectPhase.

outOfService «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
outOfService - endBorder
Generalization (element - is a subtype of):«IDEAS:superSubtype»
outOfService - individualResourceState
Association (source - target):«place2Type»
outOfService - PackageOutOfService
Association (source - target):«place1Type»
outOfService - ResourcePackage
Attributes:
-
An endBorder that indicates that an PackageOutOfService marks the termination of service of a ResourcePackage.

packageEventInEnterprise «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
packageEventInEnterprise - enterpriseWholePart
Association (source - target):«place2Type»

packageEventInEnterprise - EnterprisePackageEvent
Association (source - target):«place1Type»
packageEventInEnterprise - WholeLifeEnterprise
Attributes:
-
An enterpriseWholePart where a EnterprisePackageEvent is part of a WholeLifeEnterprise - e.g. the package is rolled-out into the enterprise.

packageSpecification «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
packageSpecification - modemIndividualTypeInstance
Association (source - target): «place2Type»
packageSpecification - ResourcePackage
Association (source - target): «place1Type»
packageSpecification - ResourcePackageSpecification
Attributes:
-
A modemIndividualTypeInstance that relates a ResourcePackage to the ResourcePackageSpecification that specifies the sos:

portfolioWholePhase - PortfolioPhase
Association (source - target): «place1Type»
portfolioWholePhase - Portfolio
Attributes:
-
A portfolioWholePart where the part is a temporal part and is a PortfolioPhase.

programmeInPortfolio «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
programmeInPortfolio - portfolioWholePart
Association (source - target): «place1Type»
programmeInPortfolio - Portfolio
Association (source - target): «place2Type»
programmeInPortfolio - ProgrammePhase
Attributes:
-
A portfolioWholePart where the part is a ProgrammePhase.

programmeWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
programmeWholeAndPart - programmeWholePhase
Association (source - target): «place2Type»
programmeWholeAndPart - Programme
Association (source - target): «place1Type»
programmeWholeAndPart - Programme
Attributes:
-
A programmeWholePhase where both the whole and part are Programmes.

programmeWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
programmeWholePart - undertakingWholePart
Association (source - target): «place2Type»
programmeWholePart - ProgrammePart
Association (source - target): «place1Type»
programmeWholePart - Programme
Attributes:
-
An undertakingWholePart where the whole is a Programme and the part is a ProgrammePart.

programmeWholePhase «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
programmeWholePhase programmeWholePart
Association (source - target): «place2Type»
programmeWholePhase - ProgrammePhase
Association (source - target): «place1Type»
programmeWholePhase - Programme
Attributes:

-
A programmeWholePart where the part is temporal part and is a ProgrammePhase.

projectInPortfolio «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
projectInPortfolio - portfolioWholePart
Association (source - target): «place1Type»
projectInPortfolio - Portfolio
Association (source - target): «place2Type»
projectInPortfolio - ProjectPhase
Attributes:
-
A portfolioWholePart where the part is a ProjectPhase.

projectInProgramme «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
projectInProgramme - programmeWholePart
Association (source - target): «place1Type»
projectInProgramme - Programme
Association (source - target): «place2Type»
projectInProgramme - ProjectPhase
Attributes:
-
A programmeWholePart where the part is a ProjectPhase.
Note that because projects may move from one programme to another (e.g. in a re-organisation), it is a ProjectPhase that is part of the Programme. ProjectPhase is a supertype of Project, so this still allows for the limit case where a Project is part of a Programme for its whole life.

projectSequence «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectSequence - beforeAfter
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectSequence - ModemThing
Association (source - target):«place2Type»
projectSequence - Project
Association (source - target):«place1Type»
projectSequence - Project
Attributes:
-
A beforeAfter that asserts one Project cannot start until another has finished.

projectTemporalPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectTemporalPart - undertakingWholeState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectTemporalPart - projectWholePart
Association (source - target):«place2Type»
projectTemporalPart - ProjectState
Association (source - target):«place1Type»

projectTemporalPart - Project
Attributes:
-
An undertakingWholeState that relates a Project to another ProjectState that is a temporal part of it.

projectTypeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
projectTypeInstance - modemIndividualTypeInstance
Association (source - target): «place2Type»
projectTypeInstance - Project
Association (source - target): «place1Type»
projectTypeInstance - ProjectType
Attributes:
-
A modafIndividualTypeInstance that asserts a Project is an instance of a ProjectType.

projectWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
projectWholeAndPart - projectWholePart
Association (source - target): «place2Type»
projectWholeAndPart - Project
Association (source - target): «place1Type»
projectWholeAndPart - Project
Attributes:
-
A projectWholePart where both the whole and part are Projects.

projectWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectWholePart - processWholePart
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectWholePart - modemWholePart
Association (source - target):«place2Type»
projectWholePart - ProjectPart
Association (source - target):«place1Type»
projectWholePart - Project
Attributes:
-
A processWholePart that relates a Project to a ProjectPart that is entirely within the extent of the Project.

projectWholePhase «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
projectWholePhase - projectTemporalPart
Association (source - target):«place2Type»
projectWholePhase - ProjectPhase
Association (source - target):«place1Type»
projectWholePhase - Project
Attributes:

-
A projectPhaseTemporalPart where the whole is a Project.

resourceStateInMilestone «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
resourceStateInMilestone - modemWholePart
Association (source - target): «place1Type»
resourceStateInMilestone - ProjectMilestone
Association (source - target): «place1Type»
resourceStateInMilestone - ResourcePackageState
Attributes:
-
A modemWholePart that asserts a ResourcePackageState occurs within a ProjectMilestone.

statusAtMilestone «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
statusAtMilestone - modemWholePart
Association (source - target): «place1Type»
statusAtMilestone - ProjectMilestone
Association (source - target):«place2Type»
statusAtMilestone - ThreadStatusAtMilestone
Attributes:
-
A modemWholePart which relates a ThreadStatusAtMilestone to the ProjectMilestone it is part of.

statusOfThread «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
statusOfThread - StatusOfThreadType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
statusOfThread - threadTemporalWholePart
Association (source - target):«place2Type»
statusOfThread - ThreadStatusAtMilestone
Association (source - target):«place1Type»
statusOfThread - ProjectThread
Attributes:
-
A threadTemporalWholePart that relates a ProjectThread to a ThreadStatus that is temporal part of the thread.

threadInProject «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
threadInProject - ThreadInProjectType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
threadInProject - projectWholePart
Association (source - target):«place1Type»
threadInProject - Project
Association (source - target):«place2Type»
threadInProject - ProjectThread
Attributes:

-
A projectWholePart that relates a Project to a ProjectThread that is part of it.

threadTemporalWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
threadTemporalWholePart - modemTemporalWholePart
Association (source - target):«place2Type»
threadTemporalWholePart - ProjectThreadState
Association (source - target):«place1Type»
threadTemporalWholePart - ProjectThread
Attributes:
-
A modemTemporalWholePart that relates a ProjectThread to one of its states.

2.8.4 Acquisition Views additional diagrams.

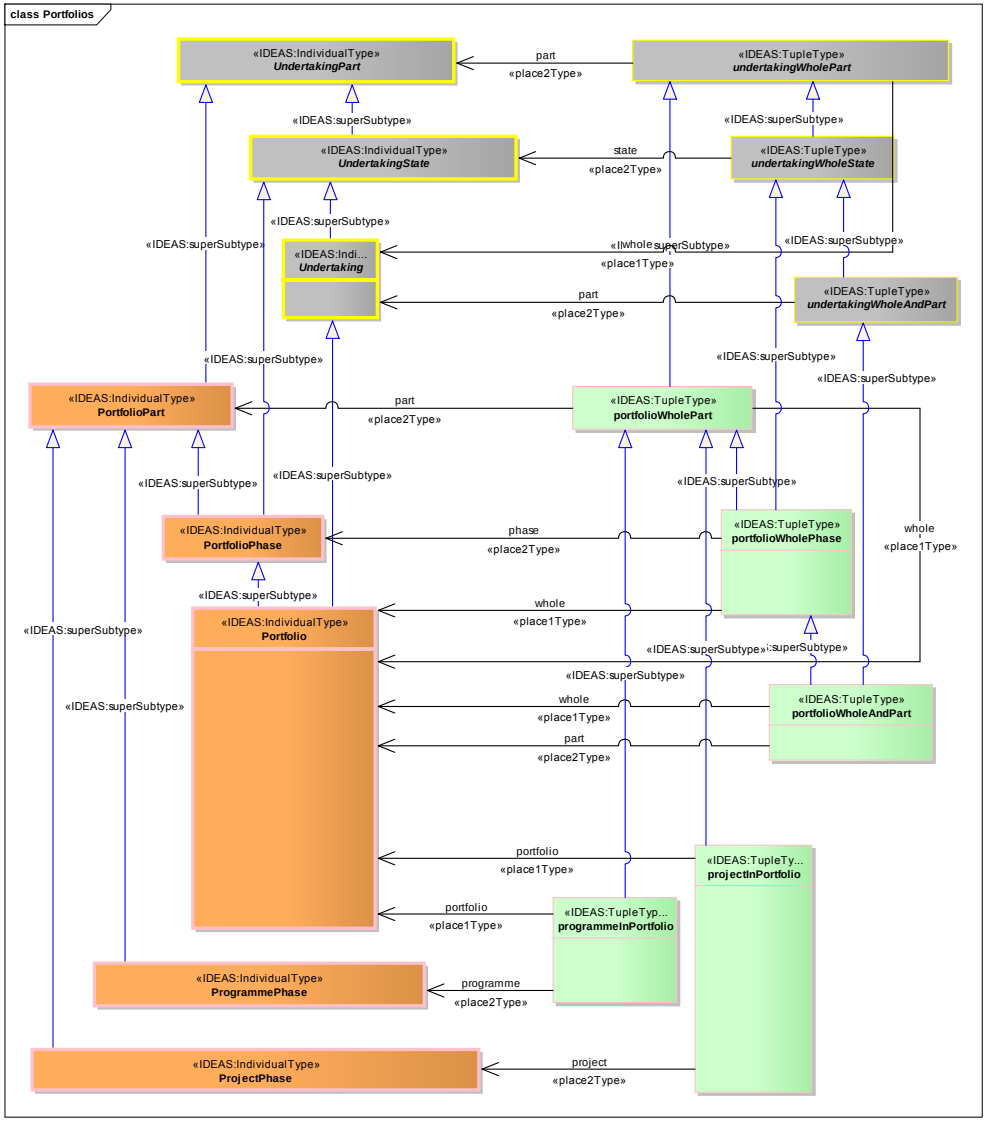


Figure 104 : Portfolios

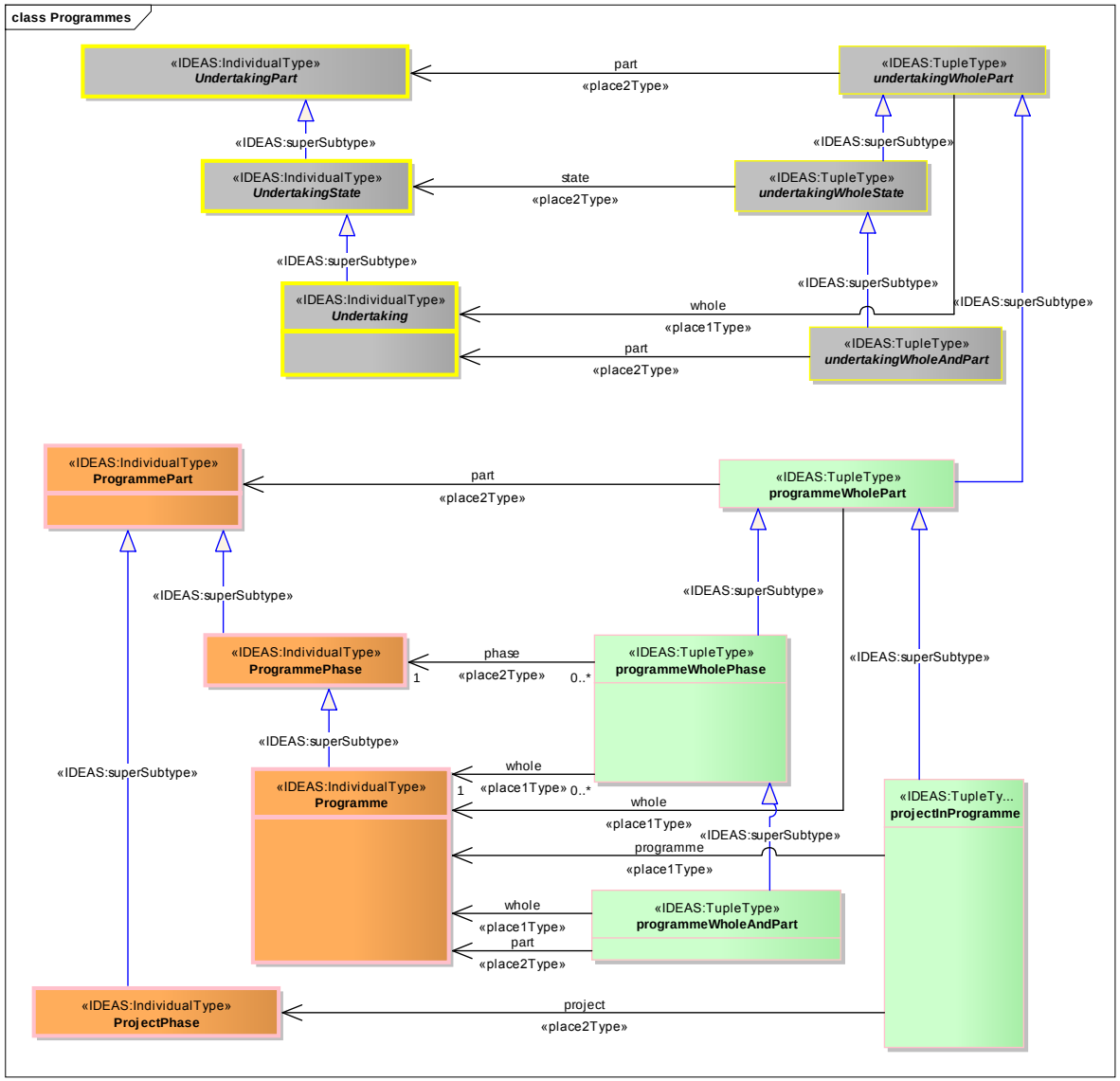


Figure 105 : Programmes

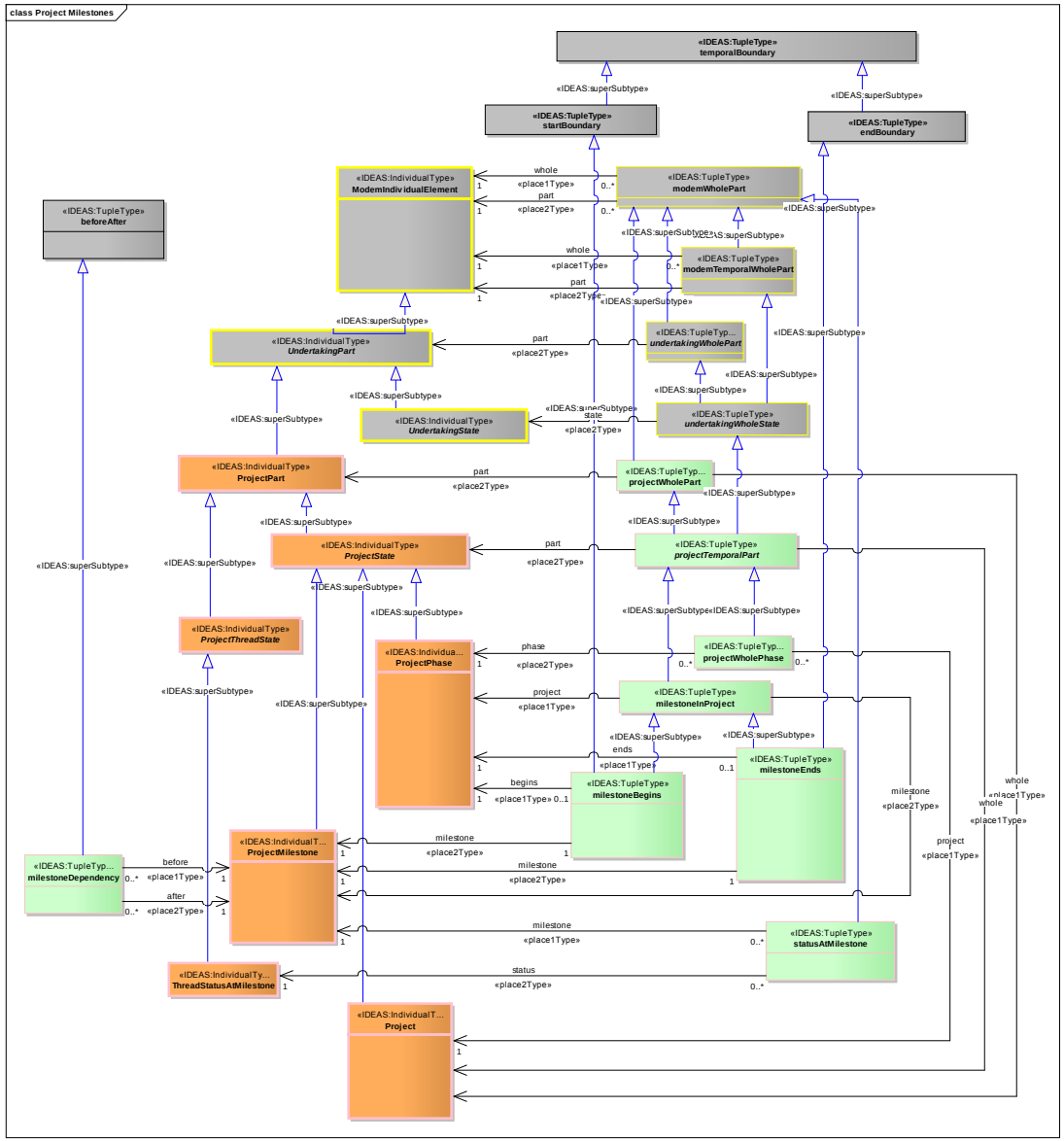


Figure 106 : Project Milestones

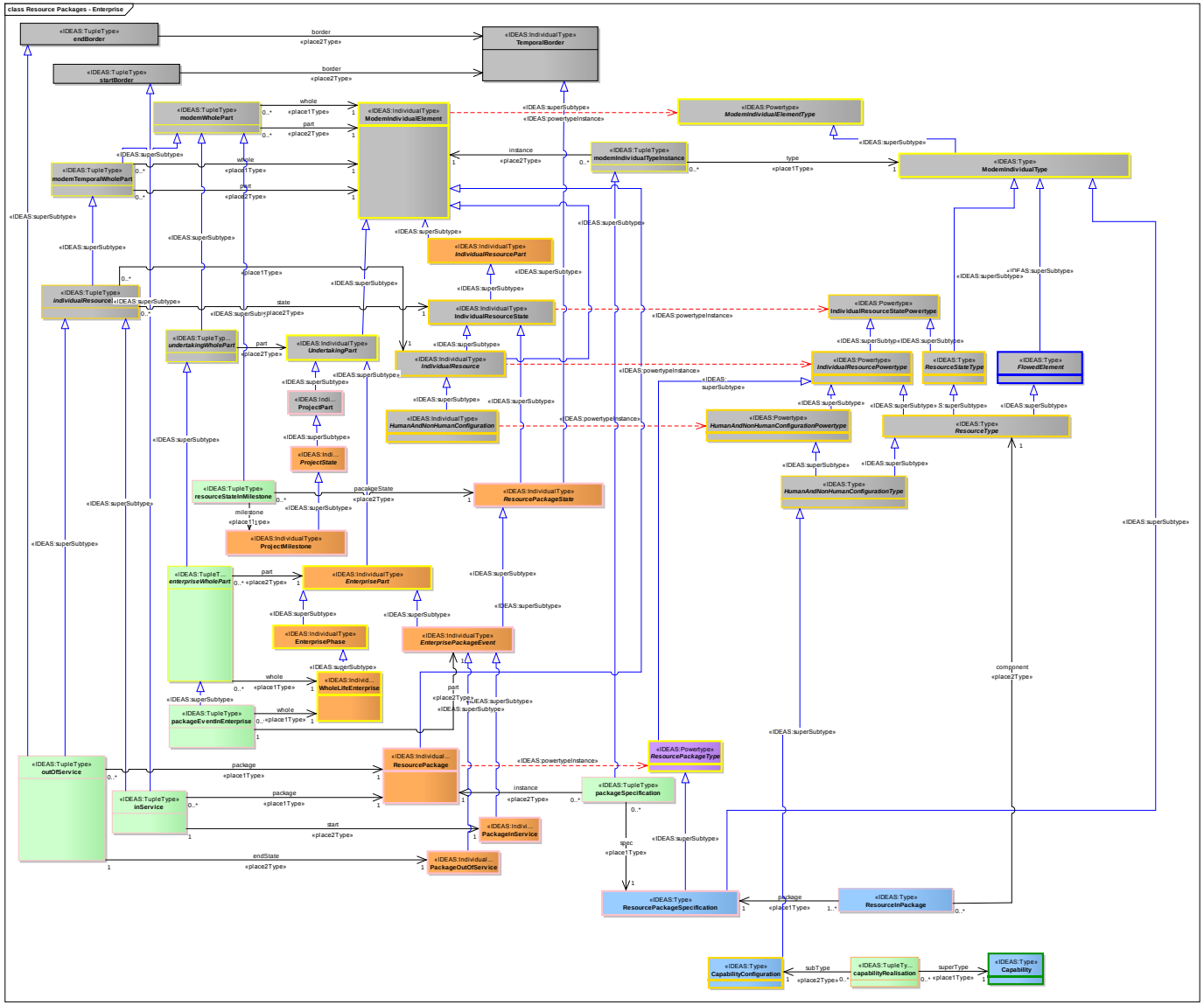


Figure 107 : Resource Packages - Enterprise

3. Additional information

3.1 Introduction

This section contains the following:

- The IDEAS foundation
- Additions made to the IDEAS foundation in order to deal with MODAF
- IDEAS patterns used to bridge the gap between the IDEAS foundation and MODAF

3.2 The IDEAS foundation

3.2.1 IDEAS Foundation elements list

IDEAS Foundation
couple «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place2Type» couple - Thing *Dependency (element - is an instance of):*«IDEAS:powertypeInstance» couple - CoupleType *Association (source - target):*«place1Type» couple - Thing *Generalization (element - is a specialization of):*«IDEAS:superSubtype» couple - tuple *Attributes:* - A tuple with two places.
Individual «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Individual - Thing *Dependency (element - is an instance of):*«IDEAS:powertypeInstance» Individual - IndividualType *Attributes:* - A Thing that has spatio-temporal extent. Note1 - this may be some that existed in the past, exists now, or may exist in some future possible world. Note2 - the Individual may be scattered - i.e. it is the fusion of several disconnect parts. Examples: * Earth * The Eiffel Tower * Me, You * Me and You * France * Sir Isaac Newton

IndividualType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
IndividualType - Type
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
IndividualType - IndividualTypeType
Dependency (element - is an instance of):«IDEAS:typeInstance»
IndividualType - Powertype
Attributes:
-
The Powertype of Individual.
Examples:
* Cars
* Boats
* Mountains
* Planets
* Deliveries
* Organisations

Powertype «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Powertype - Type
Attributes:
-
A Powertype is a set of all the subsets of a given Type.

powertypeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
powertypeInstance - typeInstance
Association (source - target):«place1Type»
powertypeInstance - Powertype
Association (source - target):«place2Type»
powertypeInstance - Type
Attributes:
-
A typeInstance that asserts that the a Type is the **Powertype** of the type which is the instance.
Note: A powertype is the set of all subsets of a given type.

superSubtype «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
superSubtype - Type
Association (source - target):«place2Type»
superSubtype - Type
Generalization (element - is a specialization of):«IDEAS:superSubtype»
superSubtype - couple
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
superSubtype - SuperSubtypeType
Attributes:
-

A couple relating two Types which asserts that one type is a subset of the other.
Note - This is the standard super-sub-type relation where the sub-type can either be a proper sub-type or an improper sub-type of the super-type. (An improper sub-type of a type is the type itself. A proper sub-type of a type is not the type itself.) This relation holds between types, where every instance of the sub-type is also an instance of the super-type. Hence the type, Humans, is a sub-type of the type, Animals, because every instance of a Human is also an instance of Animal.
Examples:
* VW Golfs is a subtype of Cars
* People over 2m tall is a subtype of People
* Types of Pump is a subtype Types of Equipment

Thing «IDEAS:Type»
Connectors:
-
Attributes:
-
The union of Individual, Type, and tuple.

tuple «IDEAS:TupleType»
Connectors:
Association (source - target):«placeType»
tuple - Thing
Generalization (element - is a specialization of):«IDEAS:superSubtype»
tuple - Thing
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
tuple - TupleType
Attributes:
-
A relationship between two or more things.
Note: Tuples are identified by their places (i.e. the ends of the relationship).
Examples:
* The year 2004 is after the year 2001
* My car is and instance of the type "VW Golfs"
* The type "VW Golfs" is a subtype of the type "Cars"

TupleType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
TupleType - PlaceableType
Dependency (element - is an instance of):«IDEAS:typeInstance»
TupleType- Powertype
Generalization (element - is a specialization of):«IDEAS:superSubtype»
TupleType - Type
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
TupleType - TupleTypeType
Association (source - target):«placeType»
TupleType - Type
Attributes:
-
The Powertype of tuple.
Examples:
* wholePart
* beforeAfter
* typeInstance

* superSubtype
Type «IDEAS:Type» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Type - Thing *Attributes:* - A set (or class) of Things. Note1: Types are identified by their members (i.e. all the things of that type). Note2: The IDEAS Foundation is a higher-order ontology, so Types may have members that are also Types. Examples: * Cars * Volkswagen Beetles * Red Things * Naval Commanders * Ranks
typeInstance «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place2Type» typeInstance - Thing *Association (source - target):*«place1Type» typeInstance - Type *Generalization (element - is a specialization of):*«IDEAS:superSubtype» typeInstance - couple *Attributes:* - A couple that asserts that a Thing is a member of a Type. Examples: * I am a member of the type People (hard to believe, but true) * Field Marshal is of the type Rank * Viscount Bernard Montgomery is of the type Person * The state of Viscount Bernard Montgomery from 1945 to his death is of type Field Marshal
wholePart «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place1Type» wholePart - Individual *Association (source - target):*«place2Type» wholePart - Individual *Generalization (element - is a specialization of):*«IDEAS:superSubtype» wholePart - couple *Dependency (element - is an instance of):*«IDEAS:powertypeInstance» wholePart - WholePartType *Attributes:* - A couple that asserts one (part) Individual is part of another (whole) Individual. Note - This is the standard whole-part relation where the whole can either be a proper part or an improper part of the part. (An improper part of a whole is the whole itself. Whereas a chocolate chip would be a proper part of a cookie, only the entire cookie is an improper part of itself) Examples: * My head is part of Me

* January 2004 is part of 2004
* The opening shot of the First World War is part of the First World War
* The Border of France is part of France
* The Border of France and Belgium is part of the Border of France
* The Border of France and Belgium is part of the Border of Belgium

IDEAS Disjoint

SetOfDisjointIndividuals «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of): «IDEAS:superSubtype»
SetOfDisjointIndividuals - SetOfDisjointThings
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfDisjointIndividuals - IndividualType
Attributes:
-
An IndividualType and a SetOfDisjointThings whose instances are types all of whose instances are Individuals and each instance is disjoint, i.e. has no common part.
This mirrors at the mereological level the mathematical notion of disjoint sets - see a DisjointType.

SetOfDisjointThings «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfDisjointThings - Type
Attributes:
-
A Type whose instances are types, where for each of them, all their instances are pairwise disjoint.

SetOfDisjointTypes «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfDisjointTypes - SetOfDisjointThings
Attributes:
-
A SetOfDisjointThings whose instances are Types(i) all of whose instances are Types(ii) where each Type(ii) is disjoint, i.e. has no common instance, from all others.
This corresponds to the mathematical notion of disjoint sets - see a definition below:
Two or more sets which have no elements in common. For example, the sets A = {a,b,c} and B = {d,e,f} are disjoint. (from http://www.mathwords.com/d/disjoint_sets.htm)

IDEAS Intersection

IntersectionOfSetOfOverlappingIndividuals «IDEAS:Type»
Connectors:
Association (source - target):«place1Type»
IntersectionOfSetOfOverlappingIndividuals - SetOfOverlappingIndividuals
Generalization (element - is a specialization of):«IDEAS:superSubtype»
IntersectionOfSetOfOverlappingIndividuals - IntersectionOfSetOfOverlappingThings
Association (source - target):«place2Type»
IntersectionOfSetOfOverlappingIndividuals -

IntersectionOfSetOfOverlappingThings «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
IntersectionOfSetOfOverlappingThings - CoupleType
Association (source - target):«place1Type»
IntersectionOfSetOfOverlappingThings - SetOfOverlappingThings
Association (source - target):«place2Type»
IntersectionOfSetOfOverlappingThings - Singleton
Attributes:
-
A CoupleType whose instances are couples that link the intersection of a set of Things with its sum.
For more details, see its sub-types: IntersectionOfSetOfOverlappingTypes and IntersectionOfSetOfOverlappingIndividuals

IntersectionOfSetOfOverlappingTypes «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
IntersectionOfSetOfOverlappingTypes - IntersectionOfSetOfOverlappingThings
Generalization (element - is a specialization of):«IDEAS:superSubtype»
IntersectionOfSetOfOverlappingTypes - SuperSubtypeType
Association (source - target):«place1Type»
IntersectionOfSetOfOverlappingTypes - SetOfOverlappingTypes
Attributes:
-
A SuperSubtypeType and an IntersectionOfSetOfOverlappingThings that contains all superSubtype couples that link a SetOfOverlappingTypes with the Types formed by their intersection.

IDEAS Numbers

RealNumberType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
RealNumberType - Type
Attributes:
-
The Powertype of RealNumber

Integer «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Integer - Number
Attributes:
-
A RationalNumber that can be written without a fractional or decimal component.
Example: 65, 7, and −56 are integers; 1.6 and 1½ are not integers.
Note: In other terms, integers are the numbers one can count with items such as apples or fingers, and their negatives, as well as 0.

Number «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Number - Type
Attributes:
-
A Type that is a number - i.e. a RealNumber or an Integer

RealNumber «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
RealNumber - Number
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
RealNumber - RealNumberType
Attributes:
-
A Type that is a Dedekind cut of the set of rational numbers.
Note: There are different definitions for Real Number in mathematics

ScaleMapping «IDEAS:Type»
Connectors:
Association (source - target):«place1Type»
ScaleMapping - MeasureCategory
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ScaleMapping - CoupleType
Association (source - target):«place2Type»
ScaleMapping - RealNumberType
Attributes:
-
A CoupleType whose members are all the couples linking MeasurePoints to RealNumbers. The CoupleType (i.e. the set of couples) represents the scale.

IDEAS Overlap

SetOfOverlappingIndividuals «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfOverlappingIndividuals - SetOfOverlappingThings
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfOverlappingIndividuals - IndividualType
Attributes:
-
An IndividualType and a SetOfOverlappingThings whose instances are types all of whose instances are Individuals and each instance overlaps all others, i.e. there is a part common to all instances.

SetOfOverlappingThings «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfOverlappingThings - Type
Attributes:
-
A Type whose instances are types, where for each of them, all their instances overlap all other instances.

SetOfOverlappingTypes «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfOverlappingTypes - SetOfOverlappingThings
Attributes:
-
A SetOfOverlappingThings whose instances are Types(i) all of whose instances are Types(ii) where each Type(ii) overlaps all others, i.e. there is at least one instance common to all Types(ii)

SetOfProperOverlappingIndividuals «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfProperOverlappingIndividuals - SetOfProperOverlappingThings
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfProperOverlappingIndividuals - SetOfOverlappingIndividuals
Attributes:
-
A SetOfOverlappingIndividuals and a SetOfProperOverlappingThings whose instances are types, where for each of them, none of their instances is a strict part of any other.

SetOfProperOverlappingThings «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfProperOverlappingThings - SetOfOverlappingThings
Attributes:
-
A SetOfOverlappingThings whose instances are types, where for each of them, none of their instances is a strict subtype or part of any other.

SetOfProperOverlappingTypes «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfProperOverlappingTypes - SetOfOverlappingTypes
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SetOfProperOverlappingTypes - SetOfProperOverlappingThings
Attributes:
-
A SetOfOverlappingTypes and a SetOfProperOverlappingThings whose instances are types, where for each of them, none of their instances is a strict subtype of any other.

SingletonIndividualType «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SingletonIndividualType - IndividualType
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SingletonIndividualType - Singleton
Attributes:
-
A Singleton and an IndividualType - i.e. a set containing exactly one Individual.

IDEAS Periods of time

April «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
April - CalendarMonth
Attributes:
-
A CalendarMonth that is an April

August «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
August - CalendarMonth
Attributes:
-

A CalendarMonth that is an August

CalendarPeriod «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
CalendarPeriod - Period
Attributes:
-
A Period that corresponds to a recognised date or time
Examples:
1st June 1974
1885
14:44:01 on 2nd June 1974
December 2008

Day «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Day - CalendarPeriod
Attributes:
-
A CalendarPeriod that is a named day in a SevenDayWeek.

December «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
December - CalendarMonth
Attributes:
-
A CalendarMonth that is a December

February «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
February - CalendarMonth
Attributes:
-
A CalendarMonth that is a February

FractionOfASecond «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
FractionOfASecond - CalendarPeriod
Attributes:
-
A CalendarPeriod that is shorter than a Second

Friday «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Friday - Day
Attributes:
-
A Day that is a Friday

Hour «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Hour - CalendarPeriod
Attributes:
-
A Calendar Period that is 60 minutes. An hour roughly corresponds to 1/24th of a median earth day

January «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
January - CalendarMonth
Attributes:
-
A CalendarMonth that is a January
Examples:
* January 2008
* January 1592

July «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
July - CalendarMonth
Attributes:
-
A CalendarMonth that is a July

June «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
June - CalendarMonth
Attributes:
-
A CalendarMonth that is a June

March «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
March - CalendarMonth
Attributes:
-
A CalendarMonth that is a March

May «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
May - CalendarMonth
Attributes:
-
A CalendarMonth that is a May

Millisecond «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Millisecond - FractionOfASecond
Attributes:
-
A CalendarPeriod that corresponds to one thousandth of a Second.

Minute «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Minute - CalendarPeriod
Attributes:
-
A CalendarPeriod that corresponds to sixty seconds

Monday «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Monday - Day
Attributes:
-
A Day that is a Monday

CalendarMonth «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
CalendarMonth - CalendarPeriod
Attributes:
-
A CalendarPeriod that is a Month in the Gregorian Calendar

Nanosecond «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Nanosecond - FractionOfASecond
Attributes:
-
A CalendarPeriod that corresponds to one billionth of a Second

November «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
November - CalendarMonth
Attributes:
-
A CalendarMonth that is a November

October «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
October - CalendarMonth
Attributes:
-

A CalendarMonth that is an October
Saturday «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Saturday - Day *Attributes:* - A Day that is a Saturday
Second «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Second - CalendarPeriod *Attributes:* - A CalendarPeriod that corresponds to the period of time defined as one second by the International System of Units (SI).
September «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» September - CalendarMonth *Attributes:* - A CalendarMonth that is a September
SevenDayWeek «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» SevenDayWeek - CalendarPeriod *Attributes:* - A CalendarPeriod that is a grouping of seven Days
Sunday «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Sunday - Day *Attributes:* - A Day that is a Sunday
TenthOfSecond «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):* «IDEAS:superSubtype» TenthOfSecond - FractionOfASecond *Attributes:* - A CalendarPeriod that corresponds to 0.1 Seconds
Thursday «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Thursday - Day *Attributes:*

-
A Day that is a Thursday

Tuesday «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of): «IDEAS:superSubtype»
Tuesday - Day
Attributes:
-
A Day that is a Tuesday

Wednesday «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Wednesday - Day
Attributes:
-
A Day that is a Wednesday

Year «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Year - CalendarPeriod
Attributes:
-
A CalendarPeriod that corresponds to the time between two recurrences of an event related to the orbit of the Earth around the Sun

YearQuarter «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
YearQuarter - CalendarPeriod
Attributes:
-
A CalendarPeriod that corresponds to three Months

timeSuperTypeDurationSubtype «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
timeSuperTypeDurationSubtype - Duration
Association (source - target):«place2Type»
timeSuperTypeDurationSubtype - Time
Generalization (element - is a specialization of):«IDEAS:superSubtype»
timeSuperTypeDurationSubtype - superSubtype
Attributes:
-
Asserts that a given Time is the supertype of a Duration.

Duration «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
Duration - PeriodType
Attributes:
-
A PeriodType that is an arbitrary period of time

Examples:
20 Minutes - the set of all 20 minute periods
3.345 Nanoseconds- the set of all 3.345 Nanosecond periods
7000 Years- the set of all 7000 Year periods

IDEAS Namespace

DateTimeName «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
DateTimeName - Name
Attributes:
-
A Name that represents a CalendarPeriod

ISO8601-YYYY «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY - NamingScheme
Attributes:
-
An ISO8601DateTime that is a a Representation of a Year
Example:
1994

ISO8601-YYYY-MM «IDEAS:NamingScheme»
Connectors:
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY-MM - NamingScheme
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-MM - ISO8601DateTime
Attributes:
-
An ISO8601DateTime that is a a Representation of a Month.

ISO8601-YYYY-MM-DDThh «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-MM-DDThh - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY-MM-DDThh - NamingScheme
Attributes:
-
An ISO8601DateTime that is a a Representation of a SevenDayWeek.
Example:
1994-W20

ISO8601-YYYY-MM-DDThh:mm «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-MM-DDThh:mm - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY-MM-DDThh:mm - NamingScheme
Attributes:
-
An ISO8601DateTime that is a a Representation of a SevenDayWeek.
Example:
1994-W20

ISO8601-YYYY-MM-DDThh:mm:ss «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-MM-DDThh:mm:ss - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY-MM-DDThh:mm:ss - NamingScheme
Attributes:
-
An ISO8601DateTime that is a a Representation of a SevenDayWeek.
Example:
1994-W20

ISO8601-YYYY-MM-DDThh:mm:ss. «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-MM-DDThh:mm:ss. - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY-MM-DDThh:mm:ss. - NamingScheme
Attributes:
An ISO8601DateTime that is a a Representation of a SevenDayWeek.
Example:
1994-W20

ISO8601-YYYY-Qq «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-Qq - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»

ISO8601-YYYY-Qq - NamingScheme
Attributes:
-
An ISO8601DateTime that is a a Representation of a Quarter.
Note: this is a non-standard extension of the ISO8601 format
Example:
1994-Q2

ISO8601-YYYY-Www «IDEAS:NamingScheme»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601-YYYY-Www - ISO8601DateTime
Dependency (element - is an instance of):«IDEAS:typeInstance»
ISO8601-YYYY-Www - NamingScheme
Attributes:
-
An ISO8601DateTime that is a a Representation of a SevenDayWeek.
Example:
1994-W20

ISO8601DateTime «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
ISO8601DateTime - DateTimeName
Attributes:
-
A DateTimeName that represents a CalendarPeriod in the extended format recommended by ISO8601

dayNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
dayNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
dayNamedByISO8601 - ISO8601-YYYY-MM-DD
Association (source - target):«place1Type»
dayNamedByISO8601 - Day
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-MM-DD names a Day

fractionOfASecondNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
fractionOfASecondNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
fractionOfASecondNamedByISO8601 - ISO8601-YYYY-MM-DDThh:mm:ss.
Association (source - target):«place1Type»
fractionOfASecondNamedByISO8601 - FractionOfASecond
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-MM-DDThh:mm:ss. names a FractionOfASecond

hourNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
hourNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
hourNamedByISO8601 - ISO8601-YYYY-MM-DDThh
Association (source - target):«place1Type»
hourNamedByISO8601 - Hour
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-MM-DDThh names an hour

minuteNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
minuteNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
minuteNamedByISO8601 - ISO8601-YYYY-MM-DDThh:mm
Association (source - target):«place1Type»
minuteNamedByISO8601 - Minute
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-MM-DDThh:mm names a Minute

monthNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
monthNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
monthNamedByISO8601 - ISO8601-YYYY-MM
Association (source - target):«place1Type»
monthNamedByISO8601 - CalendarMonth
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-MM names a Month

namedByDateTimeName «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
namedByDateTimeName - namedBy
Association (source - target):«place1Type»
namedByDateTimeName - CalendarPeriod
Association (source - target):«place2Type»
namedByDateTimeName - DateTimeName
Attributes:
-
A namedBy that asserts a CalendarPeriod is represented by a DateTimeName

quarterNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
quarterNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
quarterNamedByISO8601 - ISO8601-YYYY-Qq
Association (source - target):«place1Type»
quarterNamedByISO8601 - YearQuarter
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-Qq names a YearQuarter

secondNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
secondNamedByISO8601 - namedByDateTimeName
Association (source - target):«place2Type»
secondNamedByISO8601 - ISO8601-YYYY-MM-DDThh:mm:ss
Association (source - target):«place1Type»
secondNamedByISO8601 - Second
Attributes:
-
A namedBy that asserts that a nISO8601-YYYY-MM-DDThh:mm:ss names a Second

weekNamedByISO8601 «IDEAS:TupleType»
Connectors:
Association (source - target):«place2Type»
weekNamedByISO8601 - ISO8601-YYYY-Www
Generalization (element - is a specialization of):«IDEAS:superSubtype»
weekNamedByISO8601 - namedByDateTimeName
Association (source - target):«place1Type»
weekNamedByISO8601 - SevenDayWeek
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY-Www names a SevenDayWeek

yearNamedByISO8601 «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
yearNamedByISO8601 - namedByDateTimeName
Association (source - target):«place1Type»
yearNamedByISO8601 - Year
Association (source - target):«place2Type»
yearNamedByISO8601 - ISO8601-YYYY
Attributes:
-
A namedBy that asserts that an ISO8601-YYYY names a Year

IDEAS Powertypes
BeforeAfterType «IDEAS:Powertype» *Connectors:* *Association (source - target):*«place1Type» BeforeAfterType - IndividualType *Association (source - target):*«place2Type» BeforeAfterType - IndividualType *Generalization (element - is a specialization of):*«IDEAS:superSubtype» BeforeAfterType - CoupleType *Attributes:* - A TupleType that is Powertype of beforeAfter
CoupleType «IDEAS:Powertype» *Connectors:* *Dependency (element - is an instance of):*«IDEAS:typeInstance» CoupleType - Powertype *Generalization (element - is a specialization of):*«IDEAS:superSubtype» CoupleType - TupleType *Association (source - target):*«place2Type» CoupleType - Type *Association (source - target):*«place1Type» CoupleType - Type *Attributes:* - The TupleType that is the Powertype of couple
DescriptionType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» DescriptionType - RepresentationType *Attributes:* - A RepresentationType that is the Powertype of Description
IndividualTypeType «IDEAS:Powertype» *Connectors:* *Dependency (element - is an instance of):*«IDEAS:powertypeInstance» IndividualTypeType - IndividualTypeTypeType *Dependency (element - is an instance of):*«IDEAS:typeInstance» IndividualTypeType - Powertype *Generalization (element - is a specialization of):*«IDEAS:superSubtype» IndividualTypeType - Type *Attributes:* - A PlaceableType that is the Powertype of IndividualType
IndividualTypeTypeType «IDEAS:Powertype» *Connectors:* *Dependency (element - is an instance of):*«IDEAS:typeInstance» IndividualTypeTypeType - Powertype *Generalization (element - is a specialization of):*«IDEAS:superSubtype»

IndividualTypeTypeType - Type
Attributes:
-
A PlaceableType that is the Powertype of IndividualTypeType

InstantType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
InstantType - IndividualType
Attributes:
-
An IndividualType that is the Powertype of Instant

NameType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
NameType - RepresentationType
Attributes:
-
A RepresentationType that is the Powertype of Name

PeriodType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PeriodType - IndividualType
Attributes:
-
An IndividualType that is the Powertype of Period

PlaceableType «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PlaceableType - Type
Attributes:
-
A Type which has placeTypes defined for it - e.g. TupleType and its powertype levels.

RepresentationType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
RepresentationType - Type
Attributes:
-
A Type that is the Powertype of Representation

SuperSubtypeType «IDEAS:Powertype»
Connectors:
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
SuperSubtypeType - SuperSubtypeTypeType
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SuperSubtypeType - CoupleType
Attributes:
-
A CoupleType that is the Powertype of superSubtype

SuperSubtypeTypeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SuperSubtypeTypeType - TupleTypeType
Attributes:
-
A TupleTypeType that is the Powertype of SuperSubtypeType

TemporalWholePartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
TemporalWholePartType - WholePartType
Attributes:
-
A WholePartType that is the Powertype of temporalWholePart

TupleTypeType «IDEAS:Powertype»
Connectors:
Association (source - target):«placeType»
TupleTypeType - Type
Generalization (element - is a specialization of):«IDEAS:superSubtype»
TupleTypeType - PlaceableType
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
TupleTypeType - TupleTypeTypeType
Dependency (element - is an instance of):«IDEAS:typeInstance»
TupleTypeType - Powertype
Attributes:
-
A PlaceableType that is the Powertype of TupleType

TupleTypeTypeType «IDEAS:Powertype»
Connectors:
Association (source - target):«placeType»
TupleTypeTypeType - Type
Generalization (element - is a specialization of):«IDEAS:superSubtype»
TupleTypeTypeType - PlaceableType
Dependency (element - is an instance of):«IDEAS:typeInstance»
TupleTypeTypeType - Powertype
Dependency (element - is an instance of):«IDEAS:typeInstance»
TupleTypeTypeType - PlaceableType
Attributes:
-
A PlaceableType that is the Powertype of TupleTypeType

WholePartType «IDEAS:Powertype»
Connectors:
Association (source - target):«place1Type»
WholePartType - IndividualType
Association (source - target):«place2Type»
WholePartType - IndividualType
Generalization (element - is a specialization of):«IDEAS:superSubtype»
WholePartType - CoupleType

Attributes: - A CoupleType that is the Powertype of wholePart
IDEAS Singletons, Doubletons etc
doubletonTypeInstance «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place1Type» doubletonTypeInstance - Doubleton *Generalization (element - is a specialization of):*«IDEAS:superSubtype» doubletonTypeInstance - typeInstance *Attributes:* - A typeInstance that asserts a Thing is an instance of a Doubleton
quadrupletonTypeInstance «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place1Type» quadrupletonTypeInstance - Quadrupleton *Generalization (element - is a specialization of):*«IDEAS:superSubtype» quadrupletonTypeInstance - typeInstance *Attributes:* - A typeInstance that asserts a Thing is an instance of a Quadrupleton
quintupletonTypeInstance - «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place1Type» quintupletonTypeInstance - Quintupleton *Generalization (element - is a specialization of):*«IDEAS:superSubtype» quintupletonTypeInstance - typeInstance *Attributes:* - A typeInstance that asserts a Thing is an instance of a Quintupleton
singletonTypeInstance «IDEAS:TupleType» *Connectors:* *Association (source - target):*«place1Type» singletonTypeInstance - Singleton *Generalization (element - is a specialization of):*«IDEAS:superSubtype» singletonTypeInstance - typeInstance *Attributes:* - A typeInstance that asserts a Thing is an instance of a Singleton
tripletonTypeInstance «IDEAS:TupleType» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» tripletonTypeInstance - typeInstance *Association (source - target):*«place1Type» tripletonTypeInstance - Tripleton *Attributes:* -

A typeInstance that asserts a Thing is an instance of a Tripleton
Doubleton «IDEAS:Type» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Doubleton - Type *Attributes:* - A Type that has exactly two instances
Quadrupleton «IDEAS:Type» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Quadrupleton - Type *Attributes:* - A Type that has exactly four instances
Quintupleton «IDEAS:Type» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Quintupleton - Type *Attributes:* - A Type that has exactly five instances
Singleton «IDEAS:Type» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Singleton - Type *Attributes:* - A Type with only one instance
Tripleton «IDEAS:Type» *Connectors:* *Generalization (element - is a specialization of):*«IDEAS:superSubtype» Tripleton - Type *Attributes:* - A Type that has exactly three instances
IDEAS Sum, fusion and union
FusionOfSetOfIndividuals «IDEAS:Type» *Connectors:* *Association (source - target):*«place2Type» FusionOfSetOfIndividuals - IndividualType *Generalization (element - is a specialization of):*«IDEAS:superSubtype» FusionOfSetOfIndividuals - WholePartType *Generalization (element - is a specialization of):*«IDEAS:superSubtype» FusionOfSetOfIndividuals - SumOfSetOfThings *Association (source - target):*«place1Type» FusionOfSetOfIndividuals - SingletonIndividualType *Attributes:*

-
A SumOfSetOfThings and a WholePartType whose instances link the fusion of a set of Individuals with its fused whole.
In other words, this relates an IndividualType (a collection of Individuals) to its mereological sum.
See http://plato.stanford.edu/entries/mereology/#Sum

PartitionOfSetOfDisjointIndividuals «IDEAS:Type»
Connectors:
Association (source - target):«place2Type»
PartitionOfSetOfDisjointIndividuals - SetOfDisjointIndividuals
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PartitionOfSetOfDisjointIndividuals - PartitionOfSetOfDisjointThings
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PartitionOfSetOfDisjointIndividuals - FusionOfSetOfIndividuals
Attributes:
-
A FusionOfSetOfIndividuals whose fusioned Type is a SetOfDisjointIndividuals.
This is a division of a Individual into disjoint parts.

PartitionOfSetOfDisjointThings «IDEAS:Type»
Connectors:
Association (source - target):«place2Type»
PartitionOfSetOfDisjointThings - SetOfDisjointThings
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PartitionOfSetOfDisjointThings - SumOfSetOfThings
Attributes:
-
A SumOfSetOfThings whose summed Type is a SetOfDisjointThings.
This is a division of a Thing into disjoint parts/sub-types.

PartitionOfSetOfDisjointTypes «IDEAS:Type»
Connectors:
Association (source - target):«place2Type»
PartitionOfSetOfDisjointTypes - SetOfDisjointTypes
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PartitionOfSetOfDisjointTypes - UnionOfSetOfTypes
Generalization (element - is a specialization of):«IDEAS:superSubtype»
PartitionOfSetOfDisjointTypes - PartitionOfSetOfDisjointThings
Attributes:
-
A UnionOfSetOfTypes whose unioned Type is a SetOfDisjointTypes
This is a division of a Type into disjoint subTypes.
See http://en.wikipedia.org/wiki/Partition_of_a_set and http://en.wikipedia.org/wiki/Partition_of_a_set.

SumOfSetOfThings «IDEAS:Type»
Connectors:
Association (source - target):«place2Type»
SumOfSetOfThings - Type
Generalization (element - is a specialization of):«IDEAS:superSubtype»
SumOfSetOfThings - CoupleType
Association (source - target):«place1Type»
SumOfSetOfThings - Singleton
Attributes:

-
A CoupleType whose instances are couples that link the sum of a set of Things with its sum.
For more detail, see its subTypes - UnionOfSetOfTypes and FusionOfSetOfIndividuals.

UnionOfSetOfTypes «IDEAS:Type»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
UnionOfSetOfTypes - SuperSubtypeType
Generalization (element - is a specialization of):«IDEAS:superSubtype»
UnionOfSetOfTypes - SumOfSetOfThings
Association (source - target):«place2Type»
UnionOfSetOfTypes - Type
Attributes:
-
A SumOfSetOfThings and a SuperSubtypeType whose instances are couples that link the union of a set of Types with these Types.
In other words, this relates a Type (a collection of Types) to its set-theoretic sum (aka union).

IDEAS Tuples

quadruple «IDEAS:TupleType»
Connectors:
Association (source - target):«place2Type»
quadruple - Thing
Association (source - target):«place1Type»
quadruple - Thing
Association (source - target):«place4Type»
quadruple - Thing
Association (source - target):«place3Type»
quadruple - Thing
Generalization (element - is a specialization of):«IDEAS:superSubtype»
quadruple - tuple
Attributes:
-
A tuple which has four places

quintuple «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
quintuple - tuple
Association (source - target):«place2Type»
quintuple - Thing
Association (source - target):«place3Type»
quintuple - Thing
Association (source - target):«place5Type»
quintuple - Thing
Association (source - target):«place1Type»
quintuple - Thing
Association (source - target):«place4Type»
quintuple - Thing
Attributes:
-
A tuple which has five places

triple «IDEAS:TupleType»
Connectors:
Association (source - target):«place3Type»
triple - Thing
Association (source - target):«place2Type»
triple - Thing
Association (source - target):«place1Type»
triple - Thing
Generalization (element - is a specialization of):«IDEAS:superSubtype»
triple - tuple
Attributes:
-
A tuple which has three places

IDEAS Temporal whole part

endBoundary «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
endBoundary - temporalBoundary
Attributes:
-
A temporalBoundary where the boundary is a end boundary of the whole.

startBoundary «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
startBoundary - temporalBoundary
Attributes:
-
A temporalBoundary where the boundary is a start boundary of the whole.

temporalBoundary «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
temporalBoundary - temporalWholePart
Association (source - target):«place2Type»
temporalBoundary - Individual
Attributes:
-
A temporalWholePart where the part is a temporal boundary of the whole.

temporalWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialization of):«IDEAS:superSubtype»
temporalWholePart - wholePart
Dependency (element - is an instance of):«IDEAS:powertypeInstance»
temporalWholePart - TemporalWholePartType
Attributes:
-
A wholePart that asserts the spatial extent of the (whole) individual is co-extensive with the spatial extent of the (part) individual for a particular period of time.

IDEAS BeforeAfter
beforeAfter «IDEAS:TupleType» *Connectors:* *Association (source - target):* «place1Type» beforeAfter - Individual *Association (source - target):* «place2Type» beforeAfter - Individual *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» beforeAfter - couple *Dependency (element - is instance of):* «IDEAS:powertypeInstance» beforeAfter - BeforeAfterType *Attributes:* - A couple that asserts one Individual's temporal extent is completely before the temporal extent of another.
IDEAS Intentional construction
IntentionallyConstructedIndividual «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» IntentionallyConstructedIndividual - Individual *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» IntentionallyConstructedIndividual - IntentionallyConstructedThing *Attributes:* - An Individual that is an IntentionallyConstructedThing
intentionallyConstructedTuple «IDEAS:TupleType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» intentionallyConstructedTuple - IntentionallyConstructedThing *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» intentionallyConstructedTuple - tuple *Attributes:* - A tuple that is an IntentionallyConstructedThing
IntentionallyConstructedThing «IDEAS:Type» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» IntentionallyConstructedThing - Thing *Attributes:* - A Thing that is intentionally constructed. These are Things that society has come to identify as significant - e.g. money, marriage For more information on this refer to "The Construction of Social Reality" by John Searle ISBN 01402.35906
IntentionallyConstructedType «IDEAS:Type» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» IntentionallyConstructedType - Type *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» IntentionallyConstructedType - IntentionallyConstructedThing

Attributes:
-
A Type that is an IntentionallyConstructedThing

IDEAS Properties & Measures

propertyOfIndividual «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
propertyOfIndividual - typeInstance
Association (source - target): «place1Type»
propertyOfIndividual - Property
Association (source - target): «place2Type»
propertyOfIndividual - Individual
Attributes:
-
A typeInstance that asserts an Individual is an instance of a Property - i.e. the Individual "has" a property
Examples:
A product being "expensive"
A laptop weighing 2.2kg
A car travelling between 40 and 50 km/h

propertyOfType «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
propertyOfType - Property
Association (source - target): «place2Type»
propertyOfType - IndividualType
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
propertyOfType - superSubtype
Attributes:
-
A superSubtype that asserts an IndividualType is a subtype of a Property - i.e. it asserts all members of the Individual type "have" a property
Examples:
All London Buses are red
All Porsche 911 2.2S have a mass between 900 and 960 kg
All atoms of mercury have an atomic weight of 200.59 ·mol−

MeasureType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
MeasureType - IndividualTypeType
Attributes:
-
The IndividualTypeType that is the powertype of Measure

dispositionManifestation «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
dispositionManifestation - CategoricalProperty
Association (source - target): «place1Type»
dispositionManifestation - DispositionalProperty
Generalization (element - is a specialisation of): «IDEAS:superSubtype»

dispositionManifestation - couple
Attributes:
-
A couple that asserts a CategoricalProperty has members that manifest a DispositionalProperty

electricCurrentInAmperes «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
electricCurrentInAmperes - ElectricCurrent
Association (source - target): «place2Type»
electricCurrentInAmperes - ValueInAmperes
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
electricCurrentInAmperes - measureNamedNumericallyBy
Attributes:
-
A measureNamedNumericallyBy that names an ElectricCurrent with its ValueInAmperes

frequencyInHertz «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
frequencyInHertz - measureNamedNumericallyBy
Association (source - target): «place1Type»
frequencyInHertz - Frequency
Association (source - target): «place2Type»
frequencyInHertz - ValueInHertz
Attributes:
-
A measureNamedNumericallyBy that names a Frequency with its ValueInHertz

lengthInMetres «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
lengthInMetres - Length
Association (source - target): «place2Type»
lengthInMetres - ValueInMetres
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
lengthInMetres - measureNamedNumericallyBy
Attributes:
-
A measureNamedNumericallyBy that names a Length with its ValueInMetres

lowerBoundOfMeasureRange «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
lowerBoundOfMeasureRange - MeasurePoint
Association (source - target): «place1Type»
lowerBoundOfMeasureRange - MeasureRange
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
lowerBoundOfMeasureRange - superSubtype
Attributes:
-
A superSubtype that asserts the MeasureInstance that is the lower bound (i.e. minimum measure) of a MeasureRange

luminousIntensityInCandela «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
luminousIntensityInCandela - ValueInCandela
Association (source - target): «place1Type»
luminousIntensityInCandela - LuminousIntensity
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
luminousIntensityInCandela - measureNamedNumericallyBy
Attributes:
-
A measureNamedNumericallyBy that names a LuminousIntensity with its ValueInCandela

massInKilograms «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
massInKilograms - Mass
Association (source - target): «place2Type»
massInKilograms - ValueInKilograms
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
massInKilograms - measureNamedNumericallyBy
Attributes:
-
A measureNamedNumericallyBy that names a Mass with its ValueInKilograms

measureNamedNumericallyBy «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
measureNamedNumericallyBy - MeasurePoint
Association (source - target): «place2Type»
measureNamedNumericallyBy - NumericMeasureRepresentation
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
measureNamedNumericallyBy - namedBy
Attributes:
-
A namedBy that asserts that a MeasureInstance has a NumericMeasureRepresentation

measureOfIndividual «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
measureOfIndividual - Measure
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
measureOfIndividual - propertyOfIndividual
Attributes:
-
A propertyOfIndividual that asserts an Individual is an instance of a Measure - i.e. the Individual "has" a property corresponding to the Measure.
Examples:
A laptop weighing 2.2kg
A car travelling between 40 and 50 km/h

measureOfType «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
measureOfType - propertyOfType
Association (source - target): «place1Type»
measureOfType - Measure
Attributes:
-
A propertyOfType that asserts an IndividualType is a subtype of a Measure - i.e. it asserts all members of the Individual type have a property corresponding to the Measure
Examples:
All Porsche 911 2.2S have a mass between 900 and 960 kg
All atoms of mercury have an atomic weight of 200.59 ·mol−

measureTypeInstance «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
measureTypeInstance - Measure
Association (source - target): «place1Type»
measureTypeInstance - MeasureCategory
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
measureTypeInstance - typeInstance
Attributes:
-
A typeInstance that asserts a Measure is an instance of a MeasureCategory.
Examples:
2kg is a mass
40m/s is a velocity

temperatureInKelvin «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
temperatureInKelvin - ThermodynamicTemperature
Association (source - target): «place2Type»
temperatureInKelvin - ValueInKelvin
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
temperatureInKelvin - measureNamedNumericallyBy
Attributes:
-
A measureNamedNumericallyBy that names a ThermodynamicTemperature with its ValueInKelvin

timeInSeconds «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
timeInSeconds - ValueInSeconds
Association (source - target): «place1Type»
timeInSeconds - Time
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
timeInSeconds - measureNamedNumericallyBy
Attributes:
-
A measureNamedNumericallyBy that names a Time with its ValueInSeconds

upperBoundOfMeasureRange «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
upperBoundOfMeasureRange - superSubtype
Association (source - target): «place2Type»
upperBoundOfMeasureRange - MeasurePoint
Association (source - target): «place1Type»
upperBoundOfMeasureRange - MeasureRange
Attributes:
-
A superSubtype that asserts the MeasureInstance that is the upper bound (i.e. maximum measure) of a MeasureRange

CategoricalMeasure «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
CategoricalMeasure - CategoricalProperty
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
CategoricalMeasure - Measure
Attributes:
-
A CategoricalProperty and a Measure - i.e. a CategoricalProperty that is measureable

CategoricalProperty «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
CategoricalProperty - Property
Attributes:
-
A Property that is always exhibited by its instances (Individuals). Formally, a CategoricalProperty is the set of things that have a property simpliciter without reference to the capability to manifest another property (as is the case with DispositionalProperties (qv.)). So, for example, the property of 'flying at Mach 2' is a CategoricalProperty, whereas 'being capable of flying at Mach 2' is not.
Examples:
10kg
40cm
Fitting in an ISO container

DispositionalMeasure «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
DispositionalMeasure - DispositionalProperty
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
DispositionalMeasure - Measure
Attributes:
-
A MeasureableProperty and a DispositionalProperty whose members share a common property that is measurable.

DispositionalProperty «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
DispositionalProperty - Property
Attributes:
-
A Property whose members are Individuals that have a property of being capable to manifest a CategoricalProperty under certain conditions other things being equal. It is critical when describing the disposition to

specify the conditions both for the dispositional and the categorical property that is capable of being manifested. These can range from quite stringent conditions, the DispositionalProperty of 'being capable of flying at Mach 2, at a moment's notice' to the more lax, the property of 'being capable of flying at Mach 2, once suitably configured'. Note that these have the same manifestation - the categorical property of 'flying at Mach 2'. Similarly, it is often critical to describe in detail the conditions that apply to the CategoricalProperty that can be manifested, so, for example, 'flying at Mach 2, in good weather'.
Example:
Ability to fly at Mach 2
Ability to strike a target 10km away
Ability to dissolve in water

ElectricCurrent «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ElectricCurrent - MeasurePoint
Dependency (element - is instance of): «IDEAS:typeInstance»
ElectricCurrent - MeasureCategory
Attributes:
-
A MeasureInstance whose members are Individuals that all have the same electric current flowing through them
Examples:
5 Amps
13 Amps

Frequency «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
Frequency - MeasureCategory
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Frequency - MeasurePoint
Attributes:
-
A MeasureInstance whose instances are Individuals that all oscillate at the same frequency
Examples:
100Hz
60GHz

Length «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
Length - MeasureCategory
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Length - MeasurePoint
Attributes:
-
A MeasureInstance whose instances are Individuals that all have the same length
Examples:
2mm
8 miles

LuminousIntensity «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
LuminousIntensity - MeasureCategory
Generalization (element - is a specialisation of): «IDEAS:superSubtype»

LuminousIntensity - MeasurePoint
Attributes:
-
A MeasureInstance whose members are Individuals that all have the same luminous intensity
Examples:
5 Candela
20 Candle Power
4 hefnerkerze

Mass «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Mass - MeasurePoint
Dependency (element - is instance of): «IDEAS:typeInstance»
Mass - MeasureCategory
Attributes:
-
A MeasureInstance whose members are Individuals that all have the same mass
Examples:
2kg
2.8lbs

MeasureCategory «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
MeasureCategory - MeasureType
Attributes:
-
A MeasureType whose members are recognised types of MeasureInstance.
Examples:
Mass (incluibutes:

-
A NamingScheme used for numerically representing MeasureInstances

MeasureRange «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
MeasureRange - Measure
Attributes:
-
A Measure that is characterised by two MeasurePoints that define its upper and lower bounds.

Measure «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Measure - MeasureType
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Measure - Property
Attributes:
-
An Property whose members are Individuals that all share a common, measurable property, or whose properties lie within a MeasureRange.
Examples:
2kg
4 weeks

A MeasureNamingScheme whose members are representations of SI Units.

ThermodynamicTemperature «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ThermodynamicTemperature - MeasurePoint
Dependency (element - is instance of): «IDEAS:typeInstance»
ThermodynamicTemperature - MeasureCategory
Attributes:
-
A MeasureInstance whose members are Individuals that all have the same thermodynamic temperature
Examples:
4 deg K
12 deg F
22 deg C

Time «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
Time - MeasureCategory
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Time - MeasurePoint
Attributes:
-
A MeasureInstance whose members are Individuals that have a particular temporal dimension of the same length.
Examples:
22 seconds
14 weeks
The time taken for light to travel 2km in a vacuum

ValueInAmperes «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
ValueInAmperes - SiUnitRepresentationScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ValueInAmperes - NumericMeasureRepresentation
Attributes:
-
A NumericMeasureRepresentation that represents an ElectricCurrent in amperes

ValueInCandela «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
ValueInCandela - SiUnitRepresentationScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ValueInCandela - NumericMeasureRepresentation
Attributes:
-
A NumericMeasureRepresentation that represents a LuminousIntensity in candela

ValueInHertz «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
ValueInHertz - SiUnitRepresentationScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ValueInHertz - NumericMeasureRepresentation
Attributes:
-
A NumericMeasureRepresentation that represents a Frequency in hertz

ValueInKelvin «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ValueInKelvin - NumericMeasureRepresentation
Dependency (element - is instance of): «IDEAS:typeInstance»
ValueInKelvin - SiUnitRepresentationScheme
Attributes:
-
A NumericMeasureRepresentation that represents a ThermodynamicTemperature in degrees kelvin

ValueInKilograms «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
ValueInKilograms - SiUnitRepresentationScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ValueInKilograms - NumericMeasureRepresentation
Attributes:
A NumericMeasureRepresentation that represents a Mass in kilograms

ValueInMetres «IDEAS:Type»
Connectors:
Dependency (element - is instance of): «IDEAS:typeInstance»
ValueInMetres - SiUnitRepresentationScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
ValueInMetres - NumericMeasureRepresentation
Attributes:
-
A NumericMeasureRepresentation that represents a Length in metres

ValueInSeconds «IDEAS:Type»
Connectors:
Dependency (element - is instance of):«IDEAS:typeInstance»
ValueInSeconds - SiUnitRepresentationScheme
Generalization (element - is a specialisation of):«IDEAS:superSubtype»
ValueInSeconds - NumericMeasureRepresentation
Attributes:
-
A NumericMeasureRepresentation that represents a Time in seconds

IDEAS Period or Instant
Instant «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» Instant - PeriodOrInstant *Dependency (element - is instance of):* «IDEAS:powertypeInstance» Instant - InstantType *Attributes:* - A PeriodOrInstant whose temporal extent tends towards zero.
Period «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» Period - PeriodOrInstant *Dependency (element - is instance of):* «IDEAS:powertypeInstance» Period - PeriodType *Attributes:* - A PeriodOrInstant whose temporal extent is greater than zero
PeriodOrInstant «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» PeriodOrInstant - Individual *Attributes:* - An Individual whose spatial extent is infinite, but whose temporal extent is finite or zero.
happensIn «IDEAS:TupleType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» happensIn - wholePart *Association (source - target):* «place1Type» happensIn - PeriodOrInstant *Attributes:* - A wholePart that asserts that an Individual is a part of a PeriodOrInstant. Note: IDEAS is 4D, so this means the individual [part] is entirely within the extent of the PeriodOrInstant [in]
IDEAS Representation
NumericSign «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a specialisation of):* «IDEAS:superSubtype» NumericSign - Sign *Dependency (element - is instance of):* «IDEAS:powertypeInstance» NumericSign - NumericSignType *Attributes:* - A Sign that signifies a number. Also known as a numeral.

Sign «IDEAS:IndividualType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Sign - Individual
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Sign - SignType
Attributes:
-
An Individual that signifies a Thing.
Example: 'BOSTON' signifies BOSTON

IDEASName «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
IDEASName - StringRepresentation
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
IDEASName - Name
Dependency (element - is instance of): «IDEAS:typeInstance»
IDEASName - UniqueNamingScheme
Attributes:
-
A Name used by the IDEAS model to uniquely identify a Thing.

NumericSignType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
NumericSignType - SignType
Attributes:
-
A SignType that is the Powertype of NumericSign

SignType «IDEAS:Powertype»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
SignType - IndividualType
Attributes:
-
An IndividualType that is the Powertype of Sign.

describedBy «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
describedBy - representedBy
Association (source - target): «place2Type»
describedBy - Description
Attributes:
-
A representedBy that asserts that a Description describes a Thing.

descriptionSchemeInstance «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
descriptionSchemeInstance - Description
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
descriptionSchemeInstance - representationSchemeInstance
Association (source - target): «place1Type»
descriptionSchemeInstance - DescriptionScheme
Attributes:
-
A representationSchemeInstance that asserts a Description is a member of a DescriptionScheme.

namedBy «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
namedBy - Name
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
namedBy - representedBy
Association (source - target): «place1Type»
namedBy - Thing
Attributes:
-
A couple that asserts that a Name describes a Thing.

namingSchemeInstance «IDEAS:TupleType»
Connectors:
Association (source - target): «place2Type»
namingSchemeInstance - Name
Association (source - target): «place1Type»
namingSchemeInstance - NamingScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
namingSchemeInstance - representationSchemeInstance
Attributes:
-
A representationSchemeInstance that asserts a Name is a member of a NamingScheme.

representationSchemeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
representationSchemeInstance - typeInstance
Association (source - target): «place1Type»
representationSchemeInstance - RepresentationScheme
Association (source - target): «place2Type»
representationSchemeInstance - Representation
Attributes:
-
A typeInstance that asserts a Representation is a member of a RepresentationScheme.

representedBy «IDEAS:TupleType»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
representedBy - couple
Association (source - target): «place2Type»
representedBy - Representation
Attributes:
-
A couple that asserts that a Representation represents a Thing.

Description «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Description - Representation
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Description - DescriptionType
Attributes:
-
A Representation that describes a Thing

DescriptionScheme «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
DescriptionScheme - DescriptionType
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
DescriptionScheme - RepresentationScheme
Attributes:
A RepresentationScheme and DescriptionType whose members are intentionally descriptions

FloatingPointRepresentation «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
FloatingPointRepresentation - NumericSignType
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
FloatingPointRepresentation - Representation
Attributes:
exemplar
A NumericSignType and Representation in which a string of digits (or bits) represents a rational number.
See http://en.wikipedia.org/wiki/Floating_point

IntegerRepresentation «IDEAS:Type»
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
IntegerRepresentation - NumericSignType
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
IntegerRepresentation - Representation
Attributes:
exemplar
A NumericSignType and Representation in which a string of digits (or bits) represents an integer.

Name «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Name - Representation
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Name - NameType
Generalization (element - is a specialisation of): «superSubtype»
Name - IndividualType
Attributes:
-
A Representation that identifies a Thing.
Implementation note: The inherited exemplarString provides a written example of the uttered name.
Example:
'USA' names the United States of America

NamingScheme «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
NamingScheme - Type
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
NamingScheme - RepresentationScheme
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
NamingScheme - NameType
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
NamingScheme - IndividualTypeType
Attributes:
-
An NameType and a RepresentationScheme whose members are intentionally Names.
Examples:
ISO 3166 Country Codes
IdeasNames

Representation «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
Representation - SignType
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Representation - RepresentationType
Attributes:
exemplar
A SignType where all the individual Signs are intended to signify the same Thing.

RepresentationScheme «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
RepresentationScheme - RepresentationType
Attributes:
-
A RepresentationType that is a collection of Representations that are intended to be the preferred Representations in certain contexts.

StringDescription «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
StringDescription - StringRepresentation
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
StringDescription - Description
Attributes:
-
A Description and a StringRepresentations that is a description expressed as text

StringName «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
StringName - StringRepresentation
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
StringName - Name
Attributes:
-
A Name and a StringRepresentations that is a name expressed as text

StringRepresentation «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
StringRepresentation - Representation
Attributes:
exemplar
A Representation whose all members are all strings.

UniqueNamingScheme «IDEAS:Type»
Connectors:
Generalization (element - is a specialisation of): «IDEAS:superSubtype»
UniqueNamingScheme - NamingScheme
Attributes:
-
A NamingScheme where different Names will not contain tokens of the same Representation Type.
Example:
A UniqueNamingScheme would not have two names that used tokens of the character string 'USA' to name two different things. This does not exclude the possibility that the same thing may have two names within the scheme. For example, a scheme could contain the Names 'US' and 'USA', which both name the United States of America."

3.3 IDEAS Foundation additions

3.3.1 IDEAS foundation addition diagrams

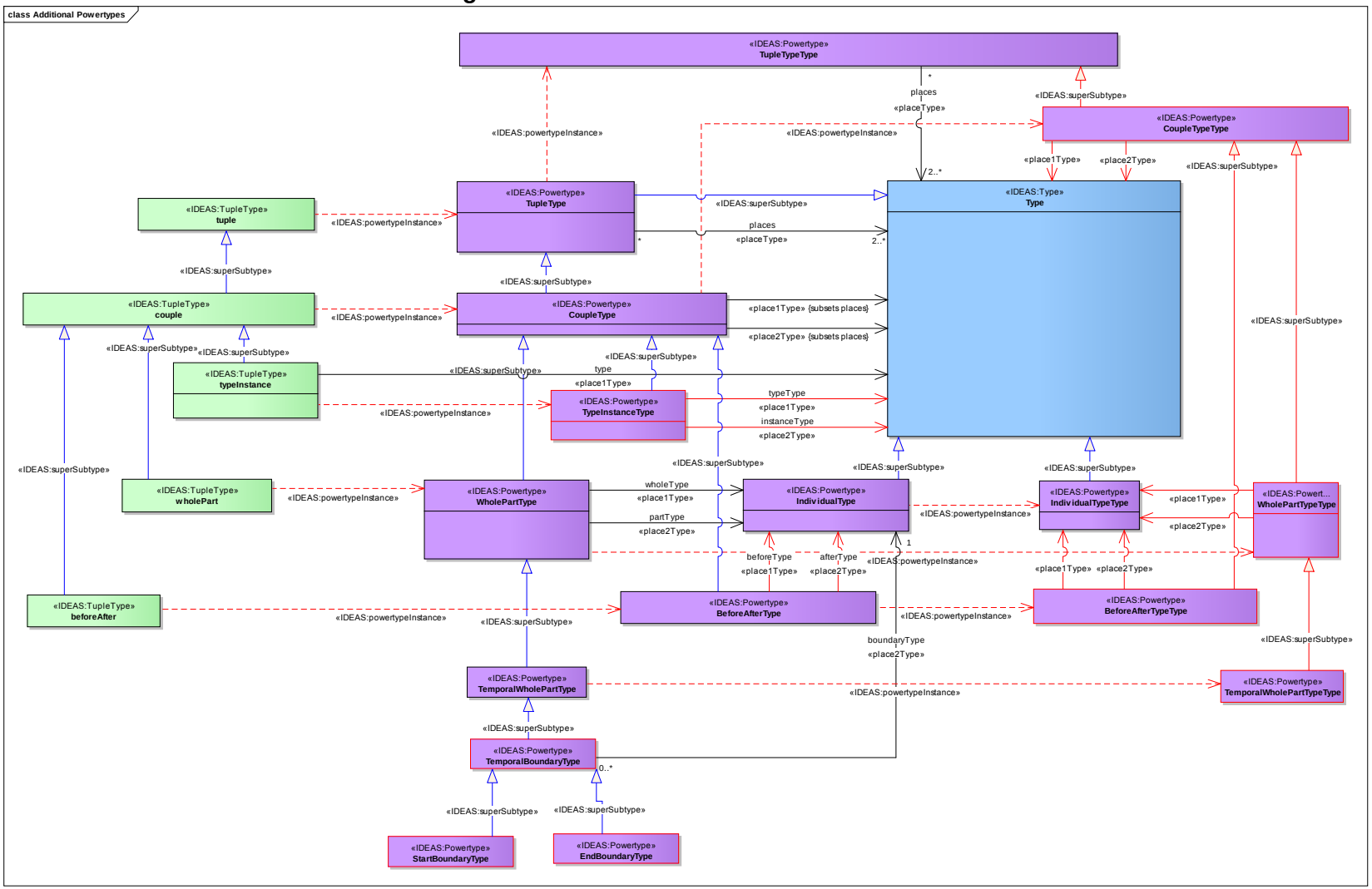


Figure 108 : Additional Powertypes

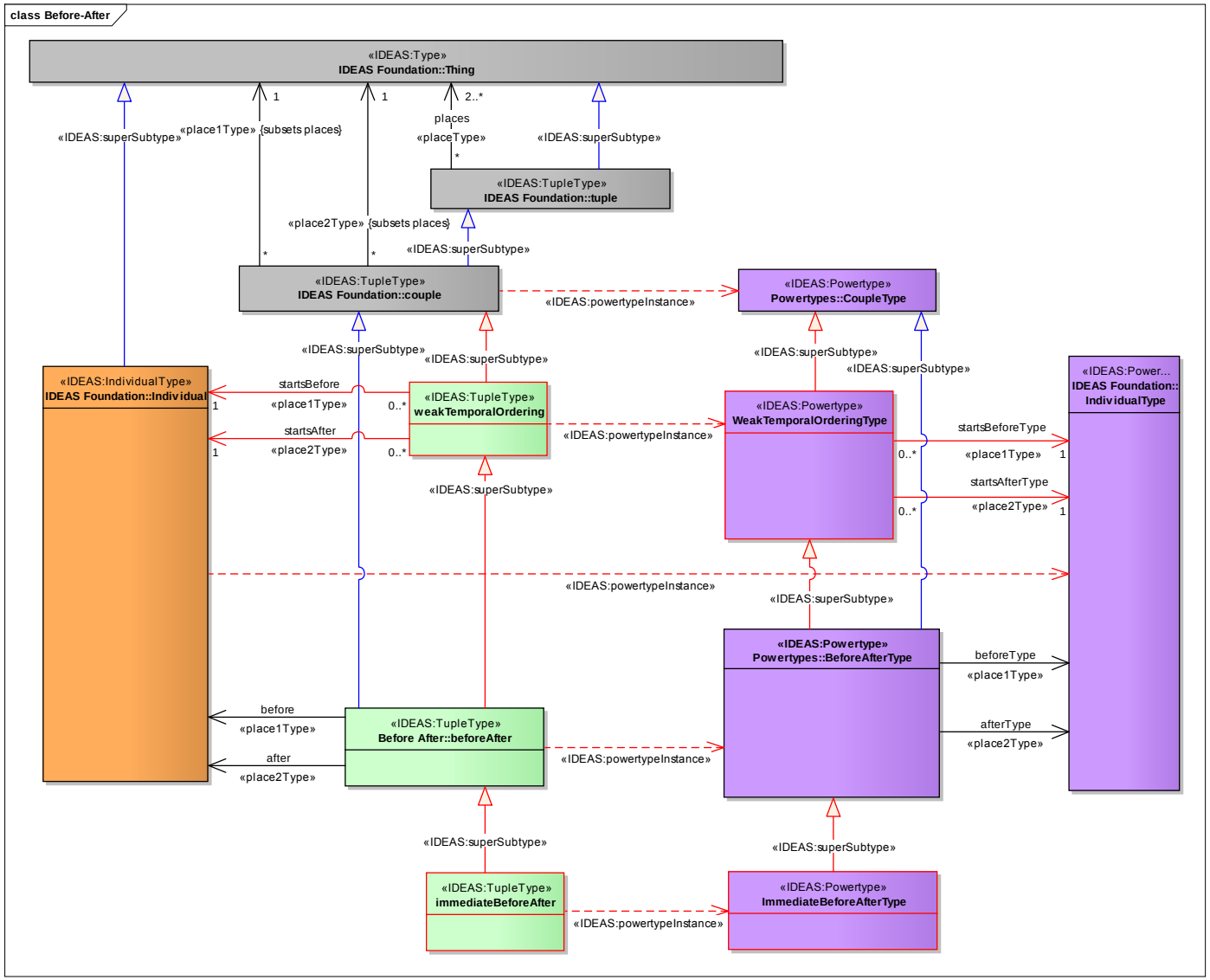


Figure 109 : Before - After

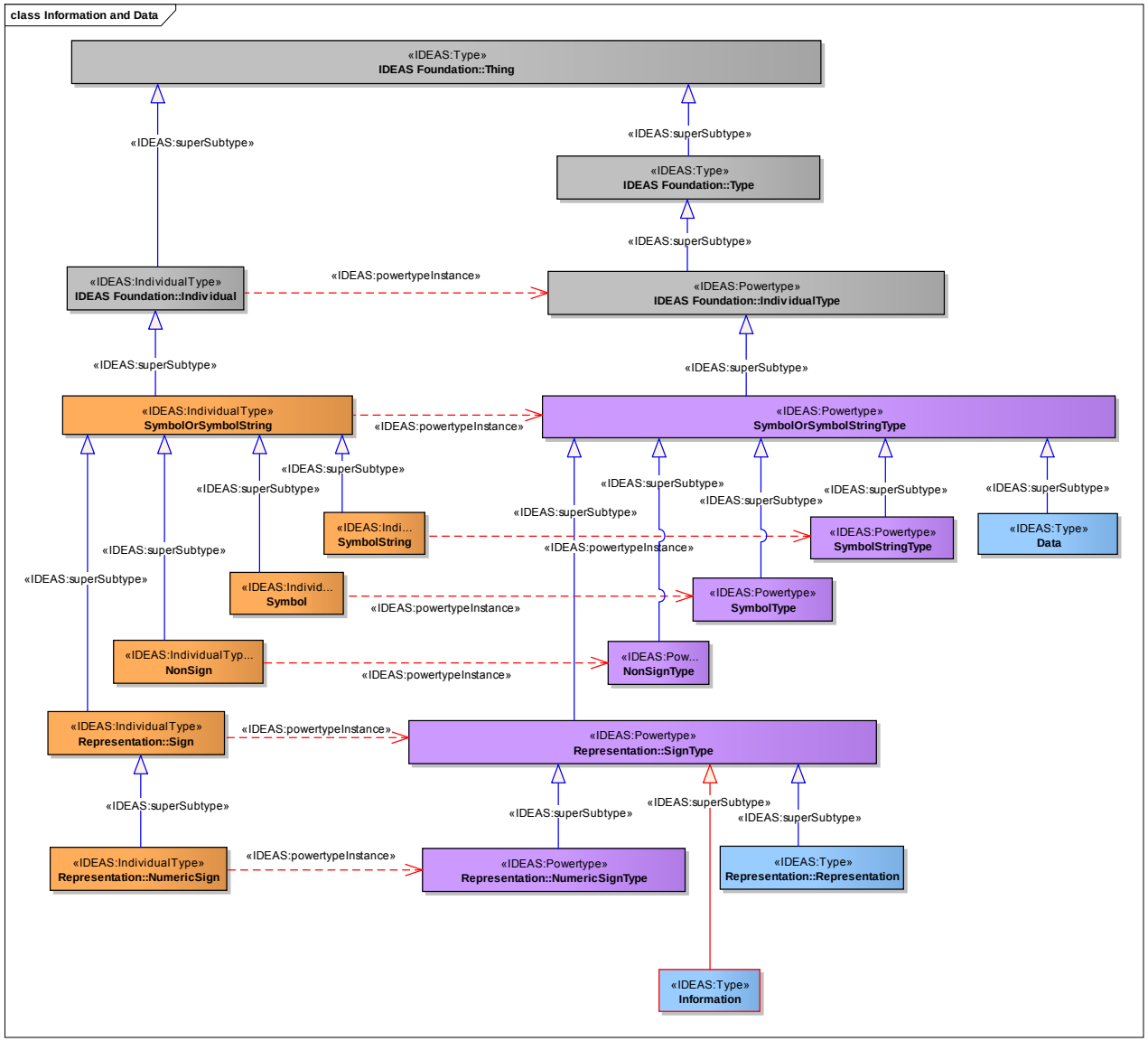


Figure 110 : Information and Data

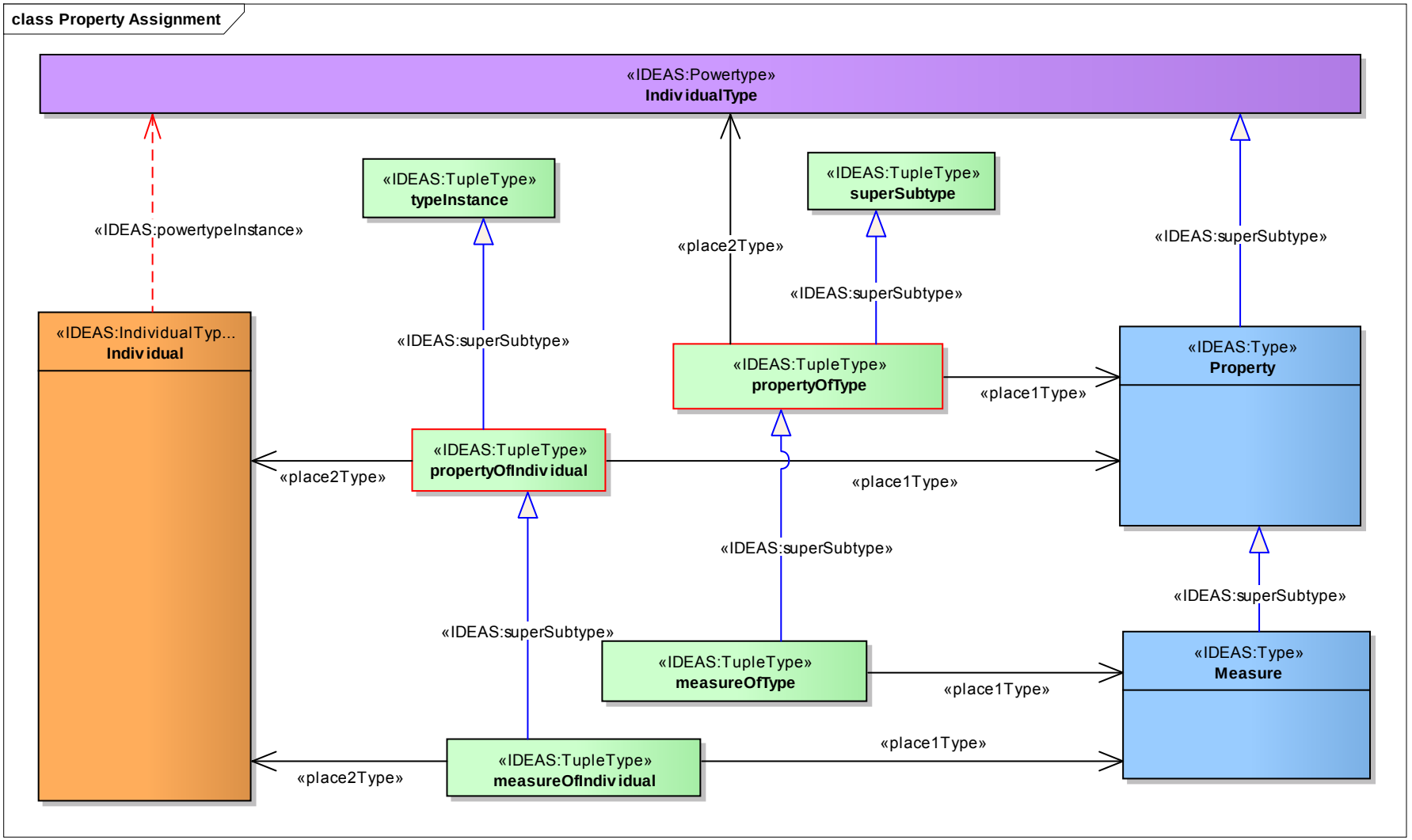


Figure 111 : Property Assignment

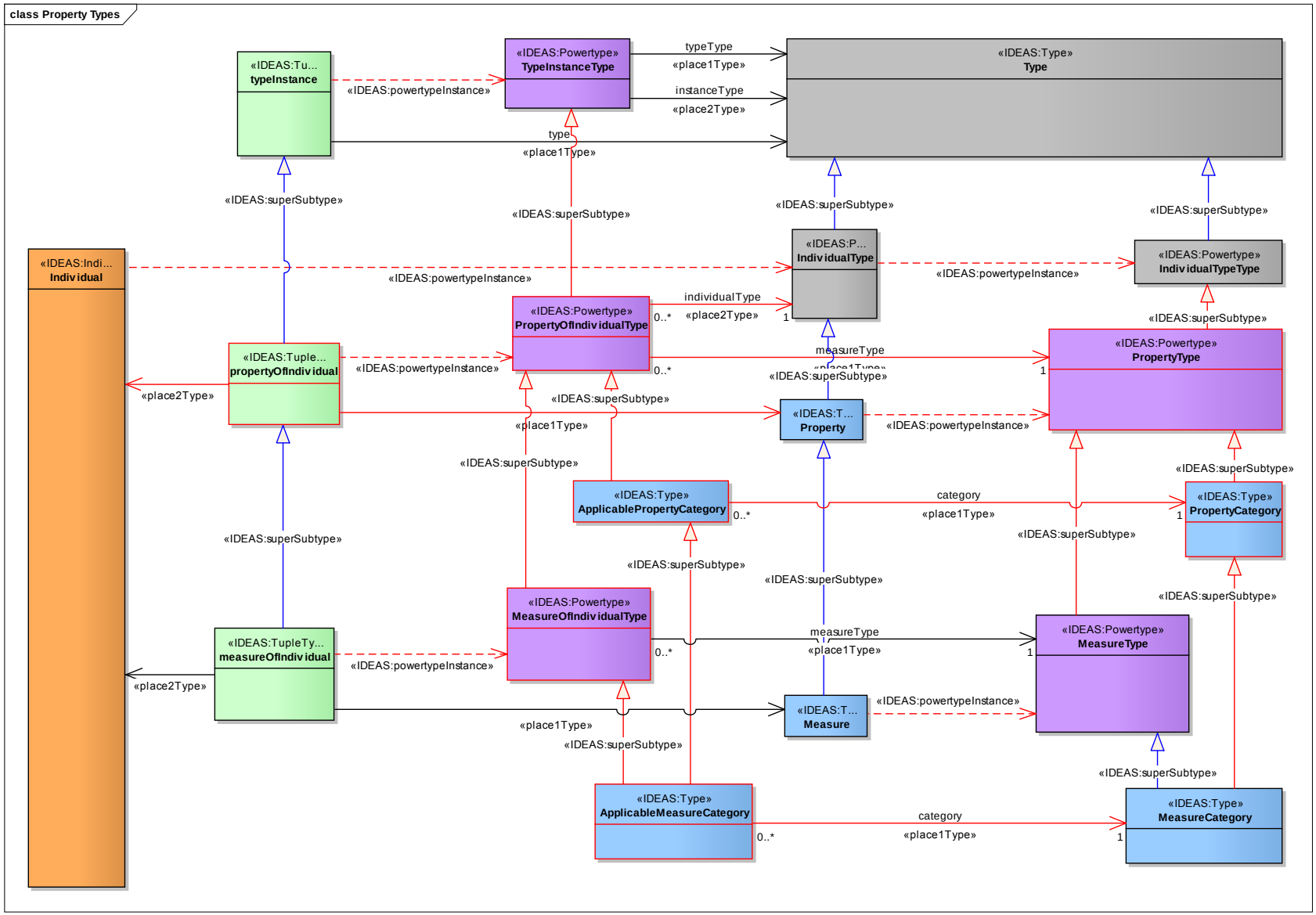


Figure 112 : Property types

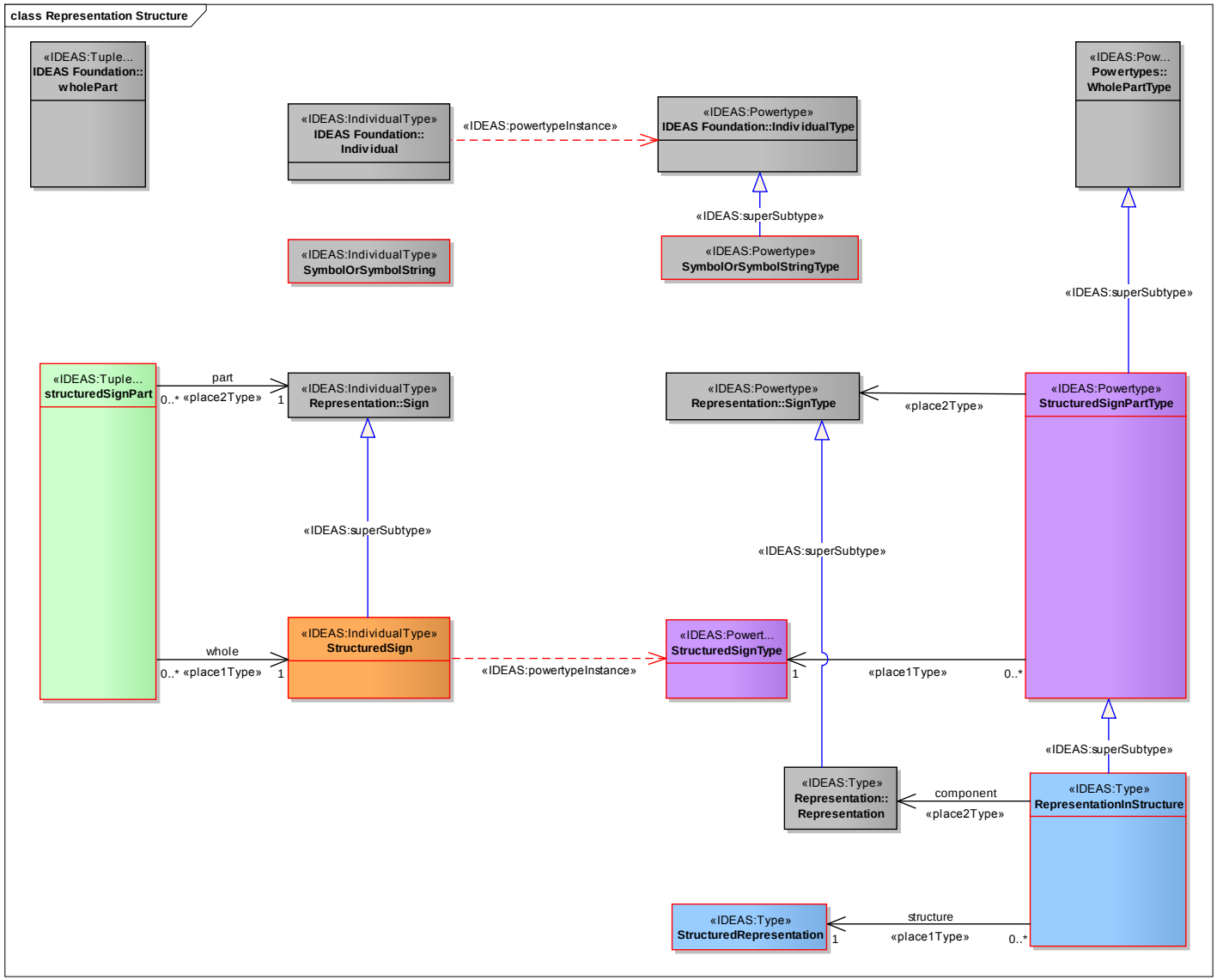


Figure 113 : Representation Structure

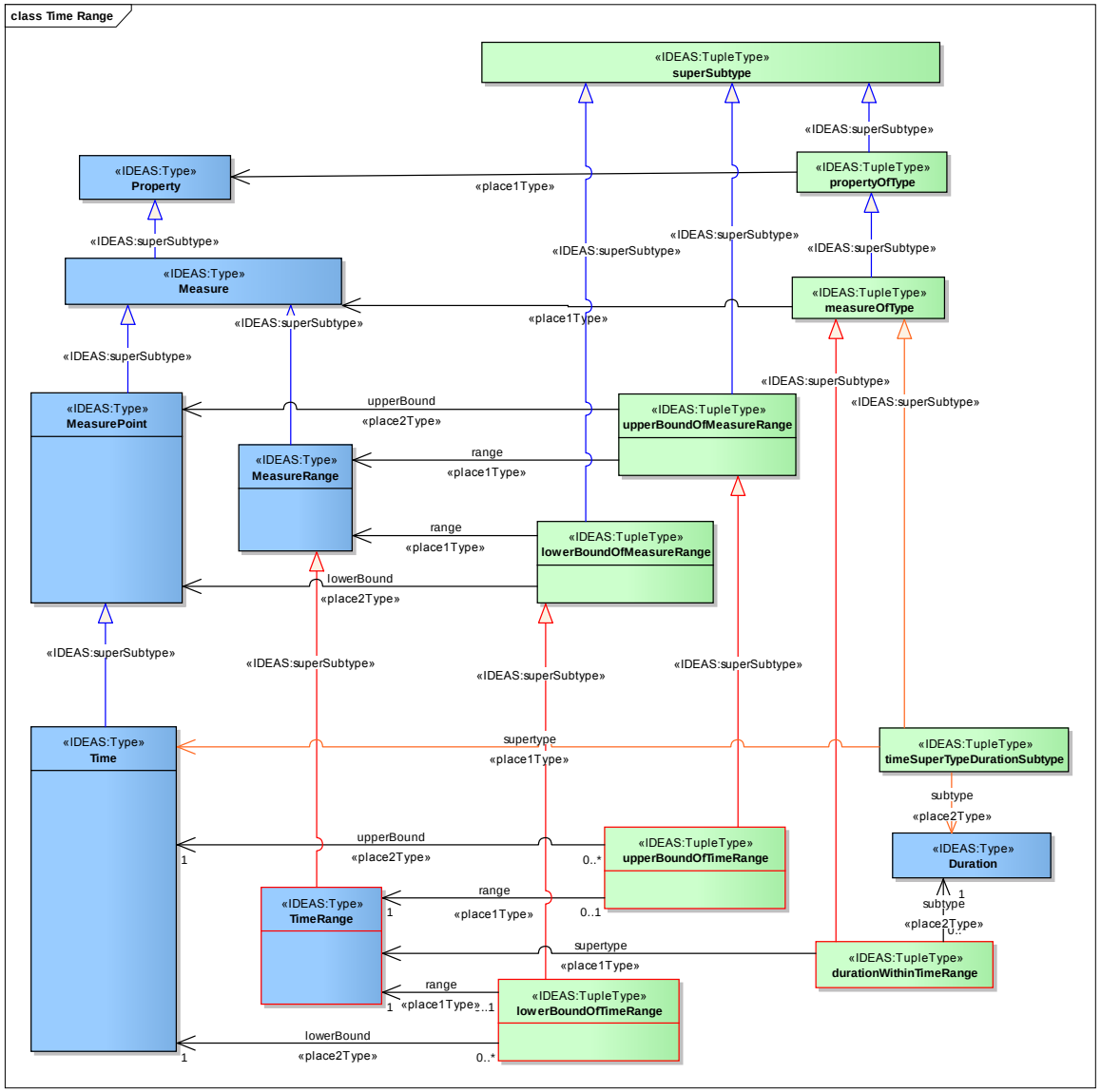


Figure 114 : Time Range

3.3.2 IDEAS Foundation addition elements list

Naming schemes
ModemName «IDEAS:UniqueNamingScheme» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ModemName - StringRepresentation *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ModemName - Name *Attributes:* - A Name that originates in the MODEM Architecture Framework (MODEM).
IDEAS Foundation additions
ApplicableMeasureCategory «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ApplicableMeasureCategory - MeasureOfIndividualType *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ApplicableMeasureCategory - ApplicablePropertyCategory *Association (source - target):*«place1Type» ApplicableMeasureCategory - MeasureCategory *Attributes:* - A MeasureOfIndividualType that asserts a given IndividualType has instances which may have properties that are instances of a MeasureCategory.
ApplicablePropertyCategory «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):*«IDEAS:superSubtype» ApplicablePropertyCategory - PropertyOfIndividualType *Association (source - target):*«place1Type» ApplicablePropertyCategory - PropertyCategory *Attributes:* - A PropertyOfIndividualType that asserts a given IndividualType has instances which may have properties that are instances of a PropertyCategory.
BeforeAfterTypeType «IDEAS:Powertype» *Connectors:* *Association (source - target):* «place1Type» BeforeAfterTypeType - IndividualTypeType *Association (source - target):* «place2Type» BeforeAfterTypeType - IndividualTypeType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» BeforeAfterTypeType - CoupleTypeType *Attributes:* - The powertype of BeforeAfterType
CoupleTypeType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» CoupleTypeType - TupleTypeType *Association (source - target):* «place1Type»

CoupleTypeType - Type
Association (source - target): «place2Type»
CoupleTypeType - Type
Attributes:
-
The powertype of CoupleType

Data «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Data - SymbolOrSymbolStringType
Attributes:
-
A SymbolOrSymbolStringType that is a non arbitrary set of Symbols which may or may not convey meaning.

EndBoundaryType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
EndBoundaryType - TemporalBoundaryType
Attributes:
-
The powertype of endBoundary.

ImmediateBeforeAfterType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ImmediateBeforeAfterType - BeforeAfterType
Attributes:
-
The powertype of immediateBeforeAfter.

Information «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Information - SignType
Attributes:
-
A SignType that is a non-arbitrary set of Signs which together convey meaning.

MeasureOfIndividualType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
MeasureOfIndividualType - PropertyOfIndividualType
Association (source - target):«place1Type»
MeasureOfIndividualType - MeasureType
Attributes:
-
The powertype of measureOfIndividual.

NonSign «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NonSign - SymbolOrSymbolString
Dependency (element - is instance of): «IDEAS:powertypeInstance»

NonSign - NonSignType
Attributes:
A SymbolOrSymbolString does not refer to anything. An example of this would be the symbol string that makes up an encryption key.

NonSignType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
NonSignType - SymbolOrSymbolStringType
Attributes:
-
The powertype of NonSign.

PropertyCategory «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PropertyCategory - PropertyType
Attributes:
-
A PropertyType that specifies a recognised type of Property.

PropertyOfIndividualType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PropertyOfIndividualType - TypeInstanceType
Association (source - target):«place2Type»
PropertyOfIndividualType - IndividualType
Association (source - target):«place1Type»
PropertyOfIndividualType - PropertyType
Attributes:
-
The powertype of propertyOfIndividual.

PropertyType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
PropertyType - IndividualTypeType
Attributes:
-
The powertype of Property.

RepresentationInStructure «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RepresentationInStructure - StructuredSignPartType
Association (source - target): «place2Type»
RepresentationInStructure - Representation
Association (source - target): «place1Type»
RepresentationInStructure - StructuredRepresentation
Attributes:
-
A StructuredSignPartType that asserts a Representation is part of a StructuredRepresentation.

StartBoundaryType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StartBoundaryType - TemporalBoundaryType
Attributes:
-
The powertype of startBoundary.

StructuredRepresentation «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StructuredRepresentation - StructuredSignType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StructuredRepresentation - Representation
Attributes:
-
A Representation that has parts that are also Representations.

StructuredSign «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StructuredSign - Sign
Dependency (element - is instance of): «IDEAS:powertypeInstance»
StructuredSign - StructuredSignType
Attributes:
-
A Sign that has parts that are also Signs.

StructuredSignPartType «IDEAS:Powertype»
Connectors:
Association (source - target): «place2Type»
StructuredSignPartType - SignType
Association (source - target): «place1Type»
StructuredSignPartType - StructuredSignType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StructuredSignPartType - WholePartType
Attributes:
-
The powertype of structuredSignPart.

StructuredSignType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StructuredSignType - SignType
Attributes:
-
The powertype of StructuredSign.

Symbol «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Symbol - SymbolOrSymbolString
Dependency (element - is instance of): «IDEAS:powertypeInstance»

Symbol - SymbolType
Attributes:
-
A SymbolOrSymbolType that is a single Symbol.

SymbolOrSymbolString «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SymbolOrSymbolString - Individual
Dependency (element - is instance of): «IDEAS:powertypeInstance»
SymbolOrSymbolString - SymbolOrSymbolStringType
Attributes:
-
An Individual that collects either symbols or strings.

SymbolOrSymbolStringType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SymbolOrSymbolStringType - IndividualType
Attributes:
-
The powertype of SymbolOrSymbolString.

SymbolString «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SymbolString - SymbolOrSymbolString
Dependency (element - is instance of): «IDEAS:powertypeInstance»
SymbolString - SymbolStringType
Attributes:
-
A SymbolOrSymbolString whose extent is the fusion of two or more Symbols.

SymbolStringType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SymbolStringType - SymbolOrSymbolStringType
Attributes:
-
The powertype of SymbolString.

SymbolType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SymbolType - SymbolOrSymbolStringType
Attributes:
-
The powertype of Symbol.

TemporalBoundaryType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
TemporalBoundaryType - TemporalWholePartType
Association (source - target): «place2Type»

TemporalBoundaryType - IndividualType
Attributes:
-
The powertype of temporalBoundary.

TemporalWholePartTypeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
TemporalWholePartTypeType - WholePartTypeType
Attributes:
-
The powertype of TemporalWholePartType.

TimeRange «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
TimeRange - MeasureRange
Attributes:
-
A MeasureRange where the bounds are Times.

TypeInstanceType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
TypeInstanceType - CoupleType
Association (source - target): «place2Type»
TypeInstanceType - Type
Association (source - target): «place1Type»
TypeInstanceType - Type
Attributes:
-
The powertype of typeInstance.

WeakTemporalOrderingType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
WeakTemporalOrderingType - CoupleType
Association (source - target): «place2Type»
WeakTemporalOrderingType - IndividualType
Association (source - target): «place1Type»
WeakTemporalOrderingType - IndividualType
Attributes:
-
The powertype of weakTemporalOrdering.

WholePartTypeType «IDEAS:Powertype»
Connectors:
Association (source - target): «place1Type»
WholePartTypeType - IndividualTypeType
Association (source - target): «place2Type»
WholePartTypeType - IndividualTypeType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
WholePartTypeType - CoupleTypeType

Attributes:
-
The powertype of WholePartType.

ZeroDurationIndividual «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ZeroDurationIndividual - ZeroDurationIndividualType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ZeroDurationIndividual - Individual
Attributes:
-
An Individual whose temporal extent is zero, but whose spatial extent is not zero (i.e. finite or infinite).

ZeroDurationIndividualType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ZeroDurationIndividualType - IndividualType
Attributes:
-
The powertype of ZeroDurationIndividual.

durationWithinTimeRange «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
durationWithinTimeRange - measureOfType
Association (source - target): «place2Type»
durationWithinTimeRange - Duration
Association (source - target): «place1Type»
durationWithinTimeRange - TimeRange
Attributes:
-
A superSubtype where the subtype is the set of all periods that fall within the lower and upper bounds of the TimeRange which is the supertype.

immediateBeforeAfter «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
immediateBeforeAfter - ImmediateBeforeAfterType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
immediateBeforeAfter - beforeAfter
Attributes:
-
A beforeAfter where the preceding Individual's temporal end adjoins the temporal start of the following Individual - i.e. one individual immediate follows the other.

lowerBoundOfTimeRange «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
lowerBoundOfTimeRange - lowerBoundOfMeasureRange
Association (source - target): «place2Type»
lowerBoundOfTimeRange - Time
Association (source - target): «place1Type»
lowerBoundOfTimeRange - TimeRange
Attributes:

-
An upperBoundOfMeasureRange where the upper bound is a Time and the range is s TimeRange.

weakTemporalOrdering «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
weakTemporalOrdering - WeakTemporalOrderingType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
weakTemporalOrdering - couple
Association (source - target): «place2Type»
weakTemporalOrdering - Individual
Association (source - target): «place1Type»
weakTemporalOrdering - Individual
Attributes:
-
A couple that asserts one Individual starts before another - i.e. the start temporal boundary of one occurs before the start temporal boundary of the other. Note: this includes cases where one Individual starts *and* ends before the other (see beforeAfter).

SingletonIndividualTypeType «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SingletonIndividualTypeType -Singleton
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SingletonIndividualTypeType - IndividualTypeType
Generalization (element - is a subtype of):«IDEAS:superSubtype»
SingletonIndividualTypeType - SetOfOwnedStateSets
Attributes:
-
An IndividualTypeType that has only one member.

singletonIndividualTypeTypeInstance «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
singletonIndividualTypeTypeInstance - singletonTypeInstance
Association (source - target):«place2Type»
singletonIndividualTypeTypeInstance - IndividualType
Association (source - target):«place1Type»
singletonIndividualTypeTypeInstance - SingletonIndividualTypeType
Attributes:
-
A singletonTypeInstance where the type is a SingletonIndividualTypeType and the instance is an IndividualType.

incompletePartitionOfSetsOfIndividuals «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
incompletePartitionOfSetsOfIndividuals - WholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
incompletePartitionOfSetsOfIndividuals - incompletePartitionOfSetsOfThings
Association (source - target): «place2Type»
incompletePartitionOfSetsOfIndividuals - SetOfDisjointIndividuals
Attributes:
-

An instance of this type contains all wholes-parts couples that link an element with each type in a set of disjoint elements that incompletely partition it. The ontic cardinalities are different from the epistemic ones.

incompletePartitionOfSetsOfThings «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
incompletePartitionOfSetsOfThings - CoupleType
Association (source - target): «place1Type»
incompletePartitionOfSetsOfThings - Singleton
Association (source - target): «place2Type»
incompletePartitionOfSetsOfThings - SetOfDisjointThings
Attributes:
-
An instance of this type contains all couples that link the incomplete partition of a set of objects with the object being partitioned. In this framework, an complete partition (AKA a partition) is the limiting case of a partition. For more detail, see its sub-types.

incompletePartitionOfSetsOfTypes «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
incompletePartitionOfSetsOfTypes - SuperSubtypeType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
incompletePartitionOfSetsOfTypes - incompletePartitionOfSetsOfThings
Association (source - target): «place2Type»
incompletePartitionOfSetsOfTypes - SetOfDisjointTypes
Attributes:
-
An instance of this type contains all super-sub-types couples that link a type with each type in a set of disjoint types that incompletely partition it. The ontiististypition iten-ldisjoint types t y

3.4 Patterns

3.4.1 Body capable of process diagrams

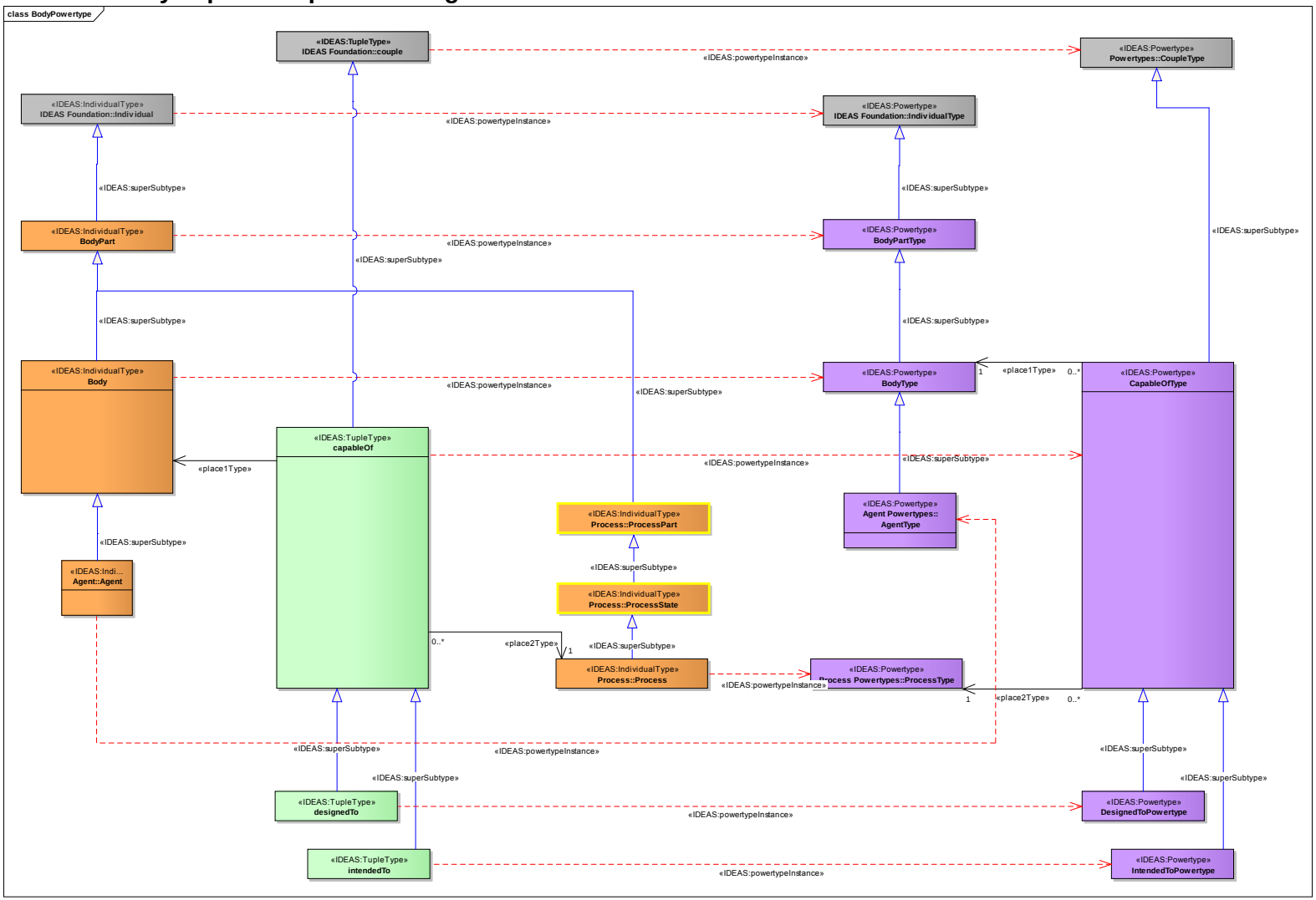


Figure 115 : BodyPowertype

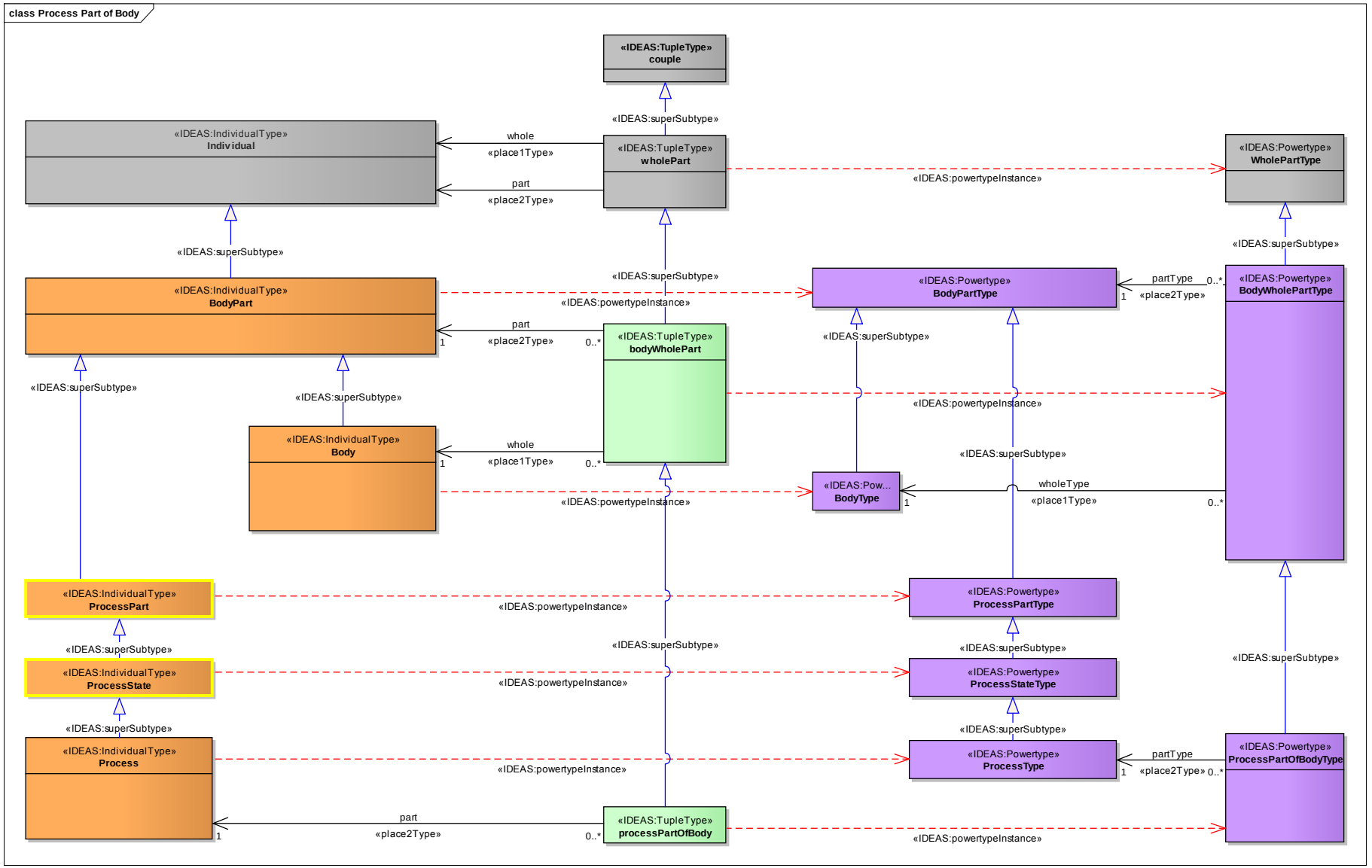


Figure 116 : Process Part of Body

3.4.2 Body capable of process elements list

Foundation additions Body capable of process
Body «IDEAS:IndividualType» *Connectors:* *Dependency (element - is instance of):* «IDEAS:powertypeInstance» Body - BodyType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Body - BodyPart *Attributes:* - An Individual that is capable of performing a Process.
BodyPart «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» BodyPart - Individual *Dependency (element - is instance of):* «IDEAS:powertypeInstance» BodyPart - BodyPartType *Attributes:* - An Individual that is a part of a Body.
BodyPartType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» BodyPartType - IndividualType *Attributes:* - The powertype of BodyPart.
BodyType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» BodyType - BodyPartType *Attributes:* - The powertype of Body.
BodyWholePartType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» BodyWholePartType - WholePartType *Association (source - target):* «place2Type» BodyWholePartType - BodyPartType *Association (source - target):* «place1Type» BodyWholePartType - BodyType *Attributes:* - The powertype of bodyWholePart.

CapableOfType «IDEAS:Powertype»
Connectors:
Association (source - target): «place2Type»
CapableOfType - ProcessType
Association (source - target): «place1Type»
CapableOfType - BodyType
Generalization (element -

bodyWholePart «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
bodyWholePart - BodyWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
bodyWholePart - wholePart
Association (source - target): «place2Type»
bodyWholePart - BodyPart
Association (source - target): «place1Type»
bodyWholePart - Body
Attributes:
-
A wholePart that asserts an Individual is part of a Body.

capableOf «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
capableOf - couple
Dependency (element - is instance of): «IDEAS:powertypeInstance»
capableOf - CapableOfType
Association (source - target): «place1Type»
capableOf - Body
Association (source - target): «place2Type»
capableOf - Process
Attributes:
-
A couple that asserts that a Body is capable of having a Process as part of it.

designedTo «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of):«IDEAS:powertypeInstance»
designedTo - DesignedToPowertype
Generalization (element - is a subtype of):«IDEAS:superSubtype»
designedTo - capableOf
Attributes:
-
A capableOf that asserts that a Body is designed to have a Process as part of it.

intendedTo «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
intendedTo - capableOf
Dependency (element - is instance of):«IDEAS:powertypeInstance»
intendedTo - IntendedToPowertype
Attributes:
-
A capableOf that asserts that a Body is intended to have a Process as part of it.

processPartOfBody «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
processPartOfBody - bodyWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
processPartOfBody - ProcessPartOfBodyType
Association (source - target): «place2Type»
processPartOfBody - Process
Attributes:
-
A bodyWholePart where the part is an entire Process - i.e. the process is entirely within the extent of the Body.

3.4.3 Temporal border diagrams

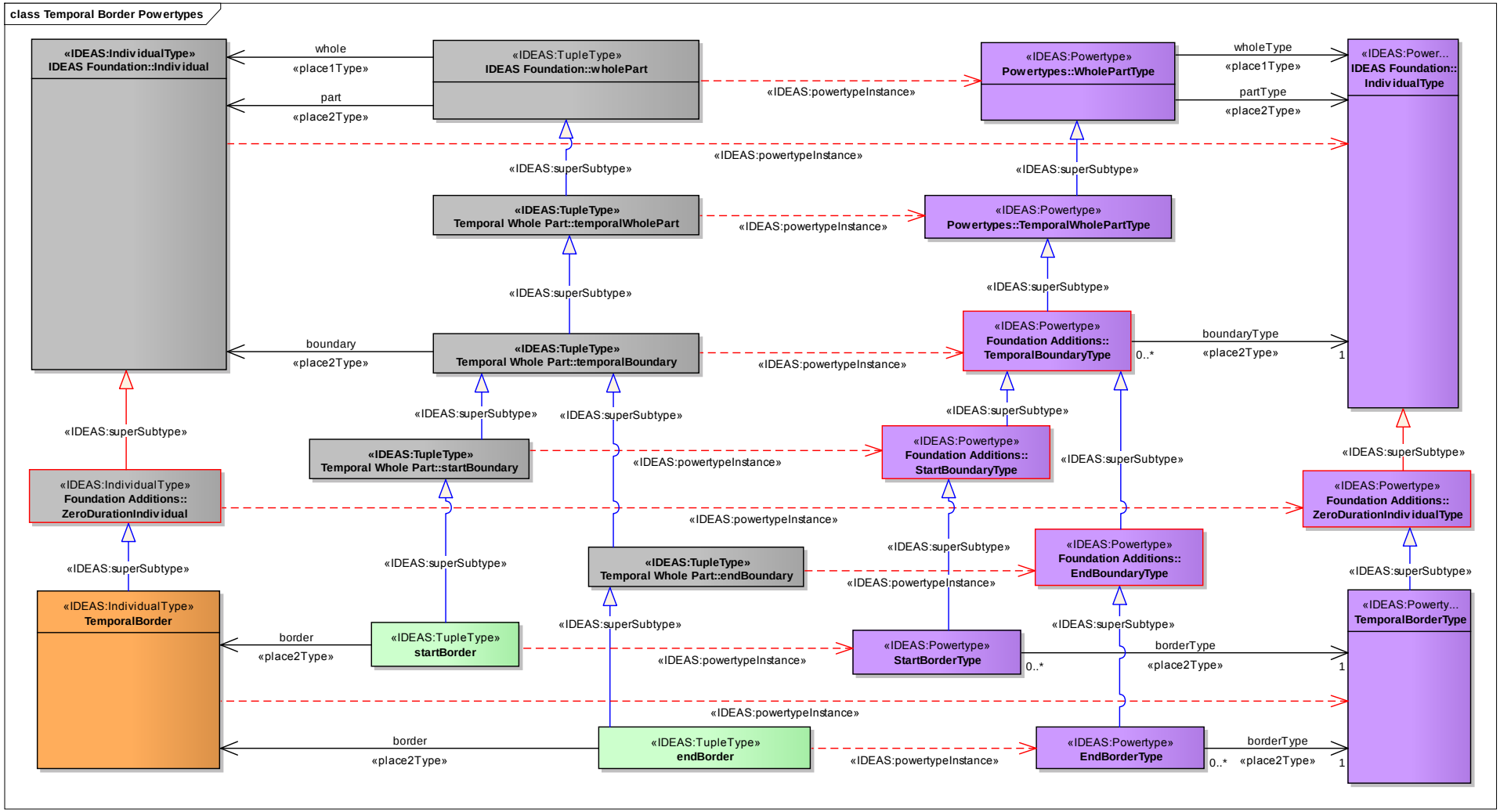


Figure 117: Temporal Border Powertypes

3.4.4 Temporal border elements list

Temporal border
EndBorderType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» EndBorderType - EndBoundaryType *Association (source - target):* «place2Type» EndBorderType - TemporalBorderType *Attributes:* - The powertype of endBorder.
StartBorderType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» StartBorderType - StartBoundaryType *Association (source - target):* «place2Type» StartBorderType - TemporalBorderType *Attributes:* - The powertype of startBorder.
TemporalBorder «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» TemporalBorder - ZeroDurationIndividual *Dependency (element - is instance of):* «IDEAS:powertypeInstance» TemporalBorder - TemporalBorderType *Attributes:* - An Individual whose temporal extent is instantaneous, and whose spatial extent corresponds with the start or end of the Individuals for which it is the temporal border.
TemporalBorderType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» TemporalBorderType - ZeroDurationIndividualType *Attributes:* - The powertype of TemporalBorder.
endBorder «IDEAS:TupleType» *Connectors:* *Dependency (element - is instance of):* «IDEAS:powertypeInstance» endBorder - EndBorderType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» endBorder - endBoundary *Association (source - target):* «place2Type» endBorder - TemporalBorder *Attributes:* - An endBoundary where the boundary is a TemporalBorder.

startBorder «IDEAS:TupleType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
startBorder - StartBorderType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
startBorder - startBoundary
Association (source - target): «place2Type»
startBorder - TemporalBorder
Attributes:
-
A startBoundary where the boundary is a TemporalBorder.

3.4.5 State and interaction diagrams

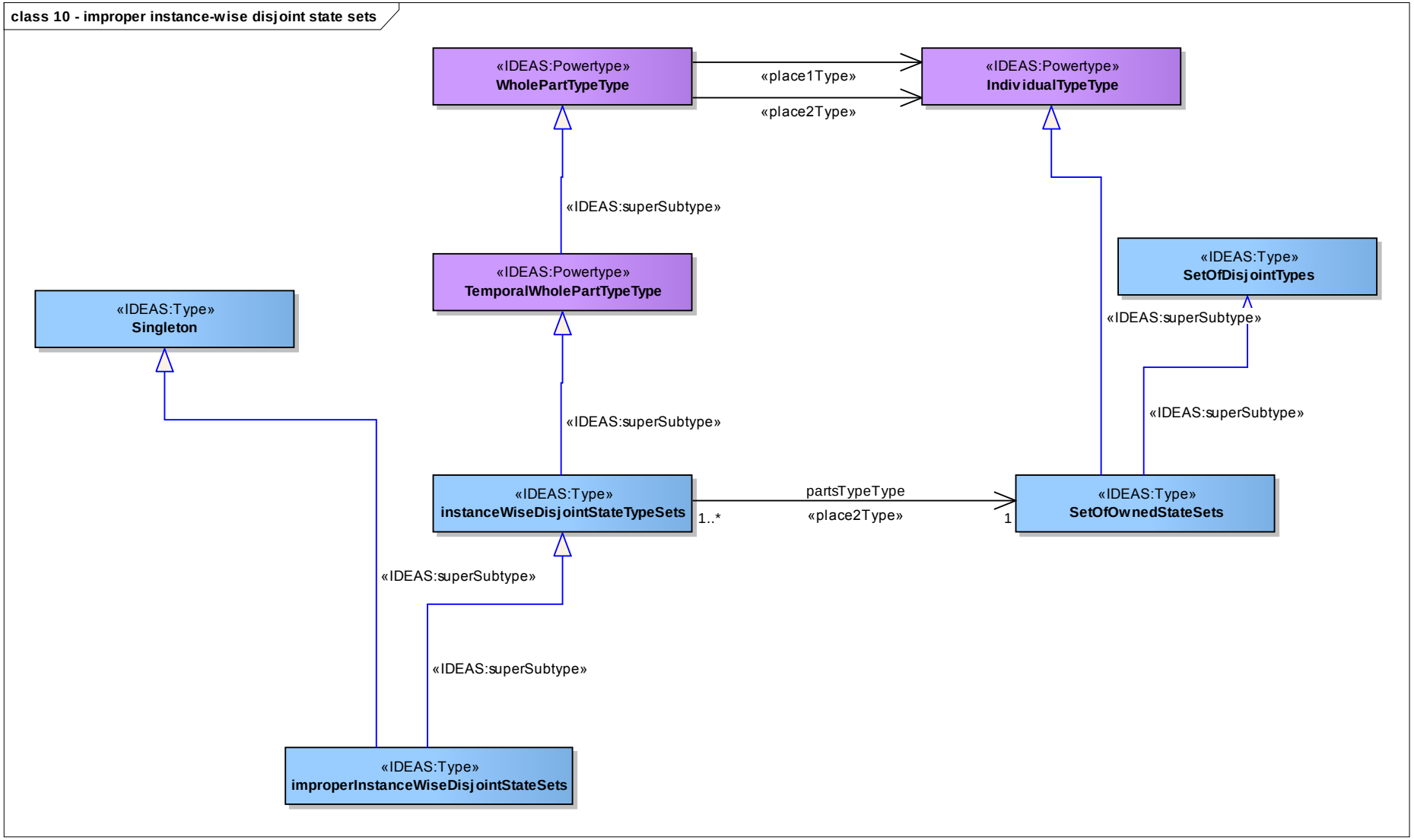


Figure 118 : improper instance-wise disjoint state sets

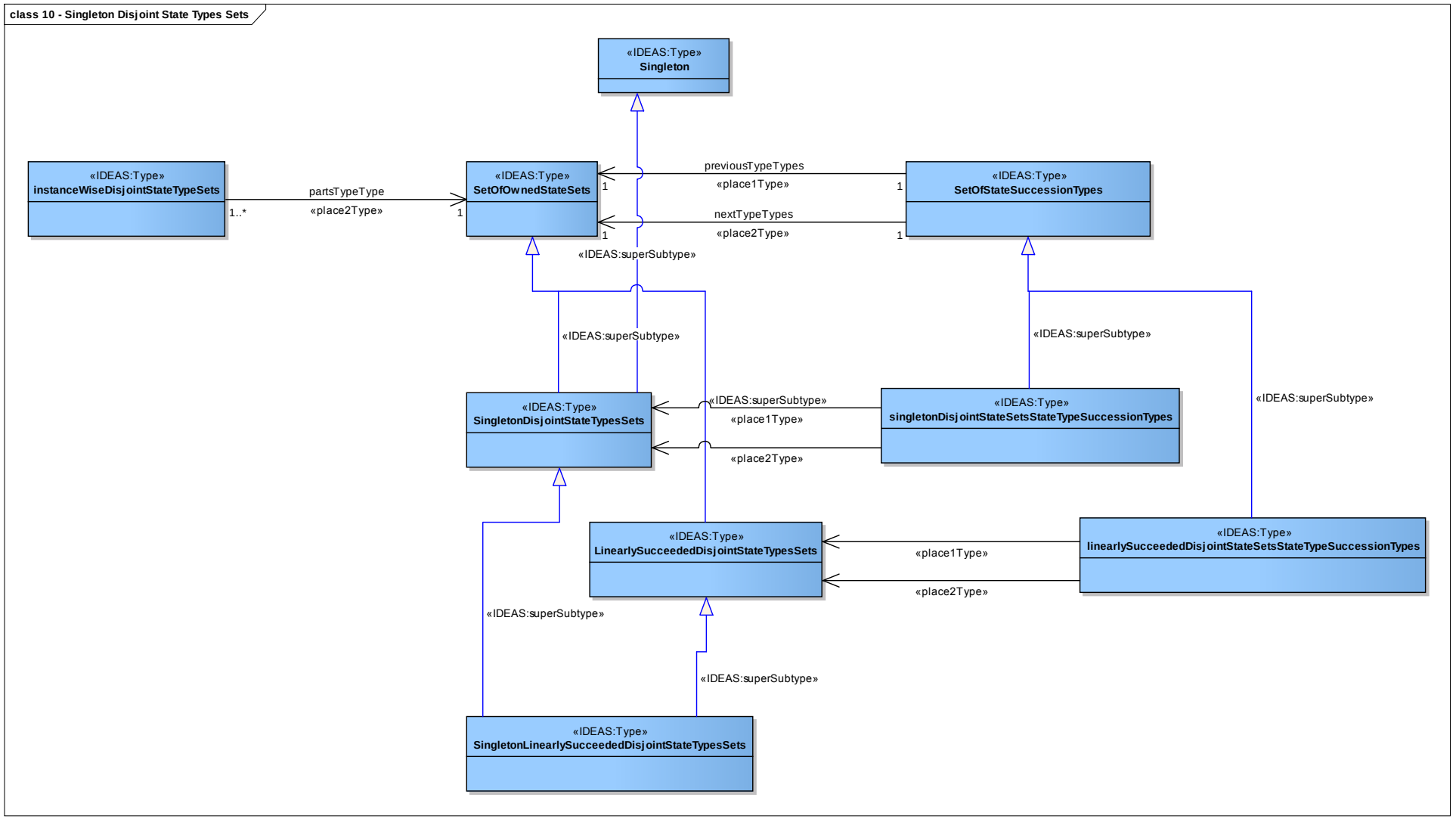


Figure 119 : Singleton Disjoint State Types Sets

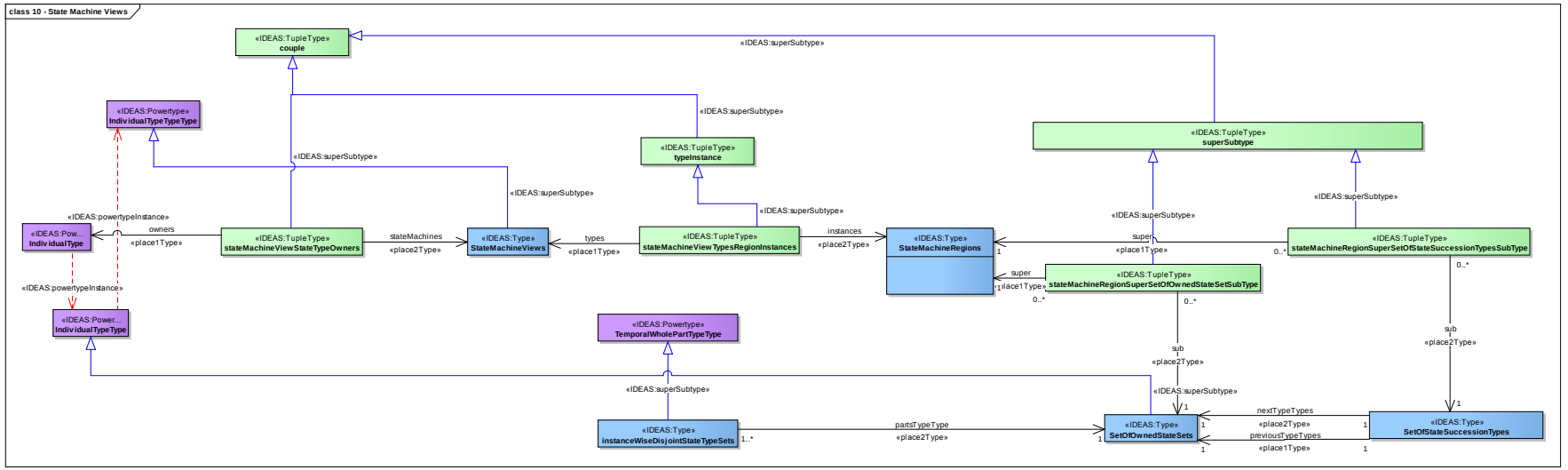


Figure 120 : State Machine Views

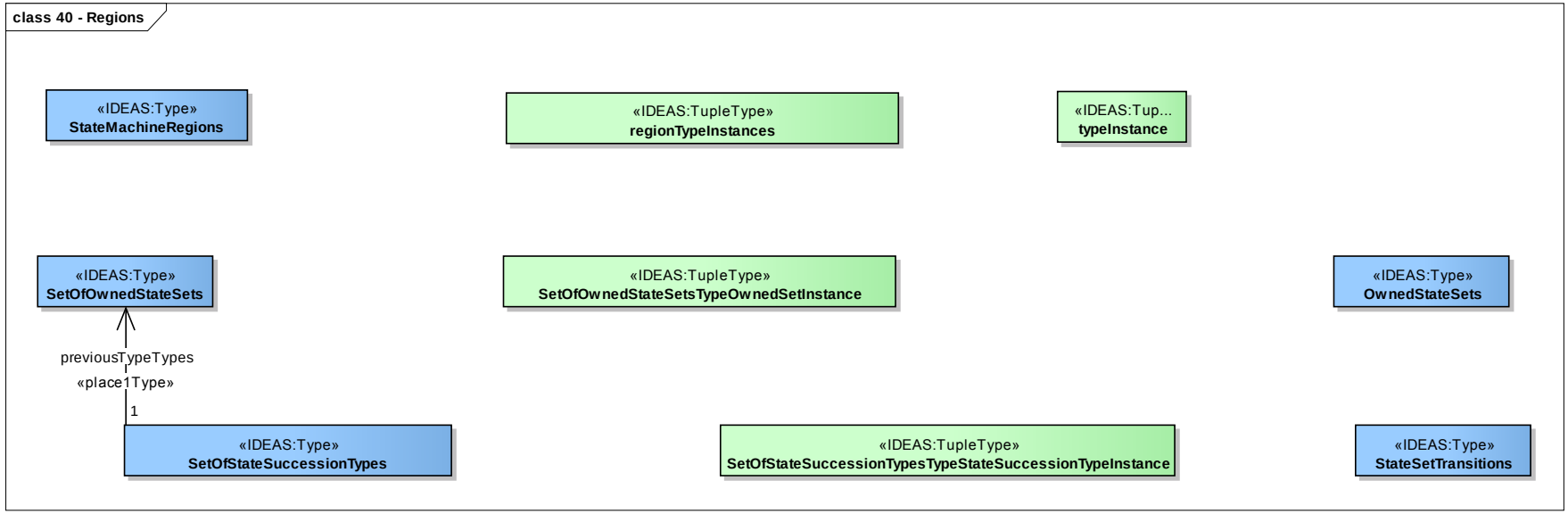


Figure 121 : Regions

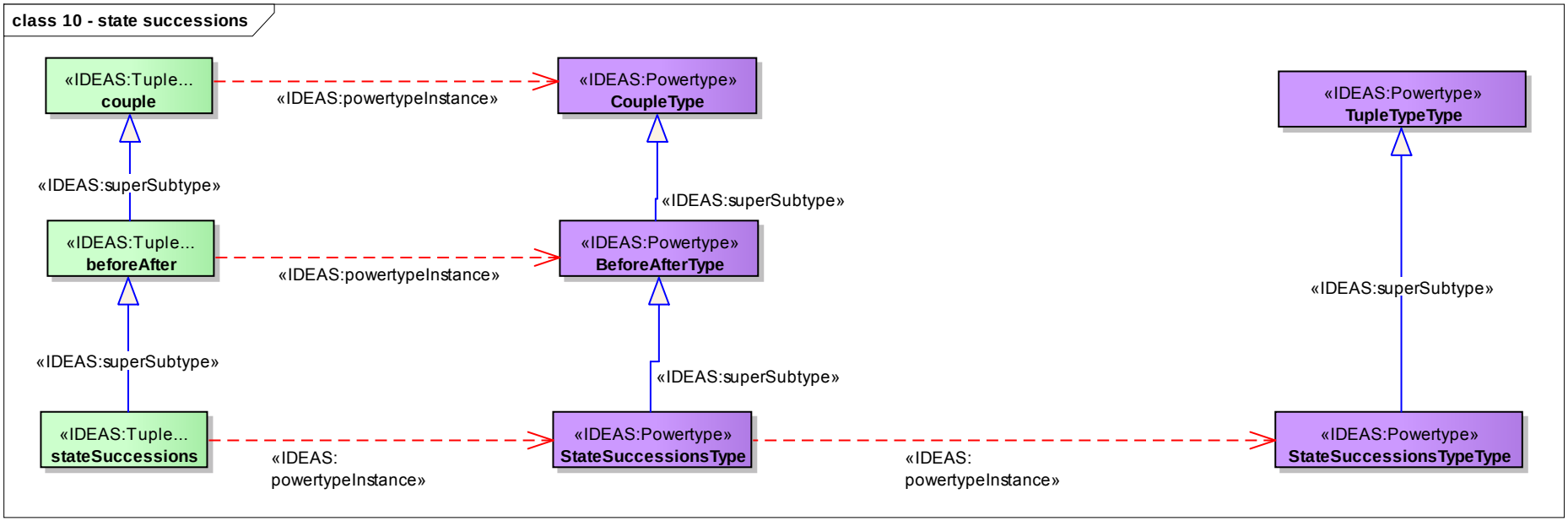


Figure 122 : state successions

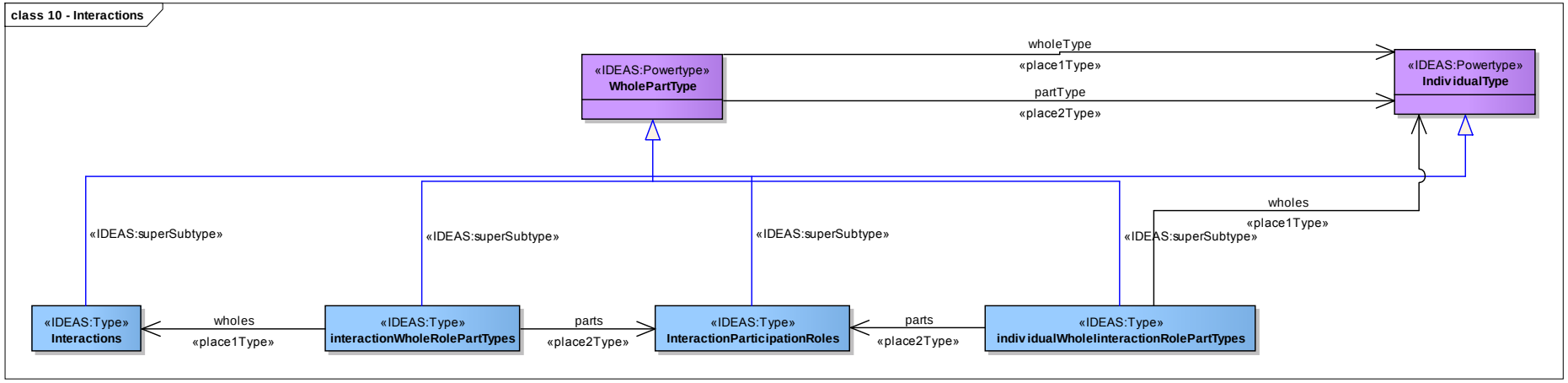


Figure 123 : Interactions

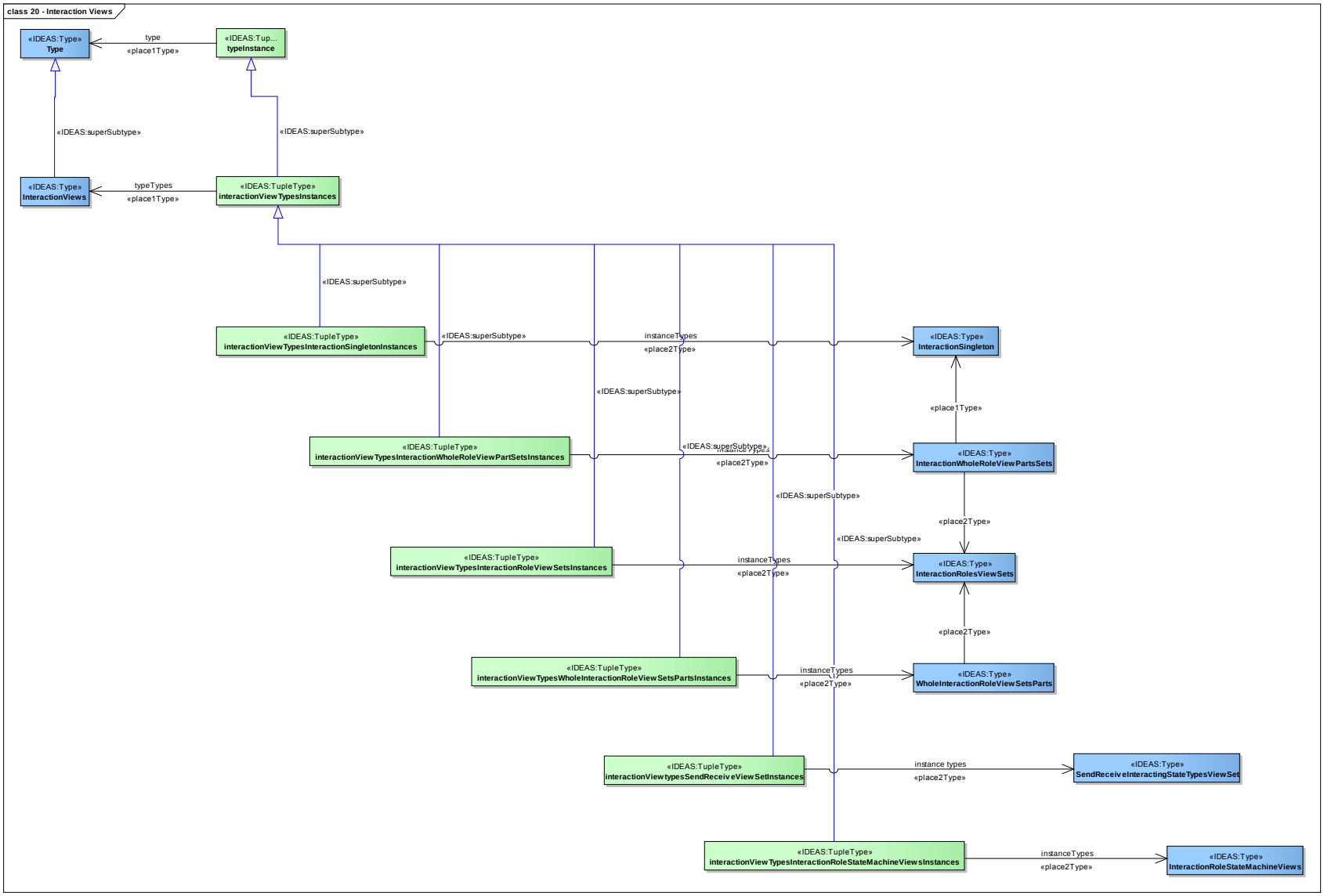


Figure 124 : Interaction Views

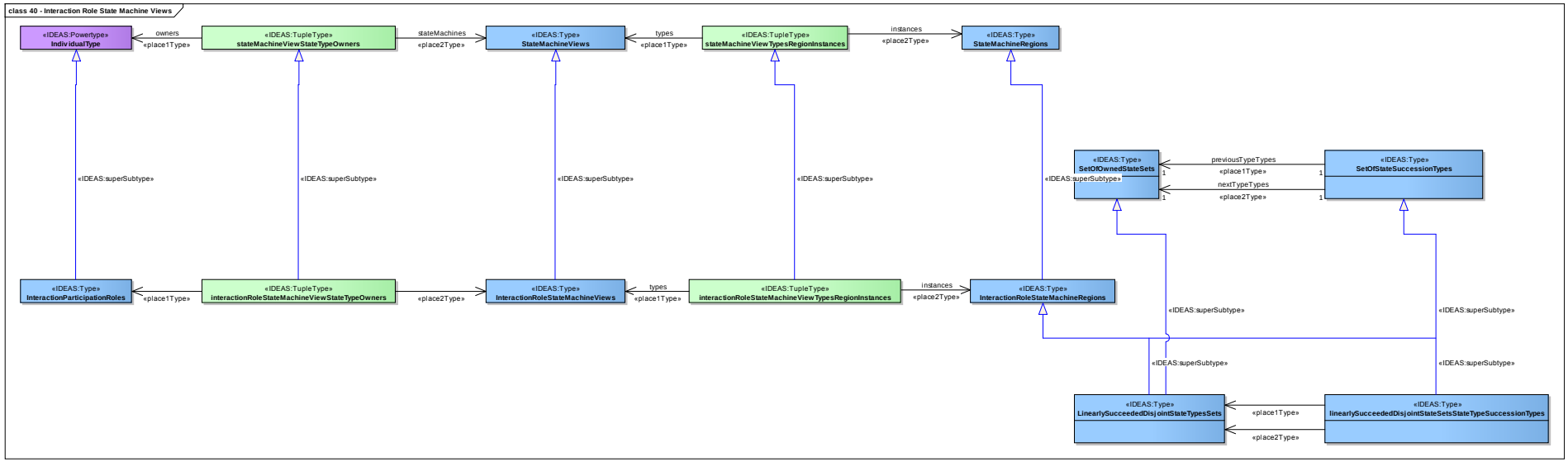


Figure 125 : Interaction Role State Machine Views

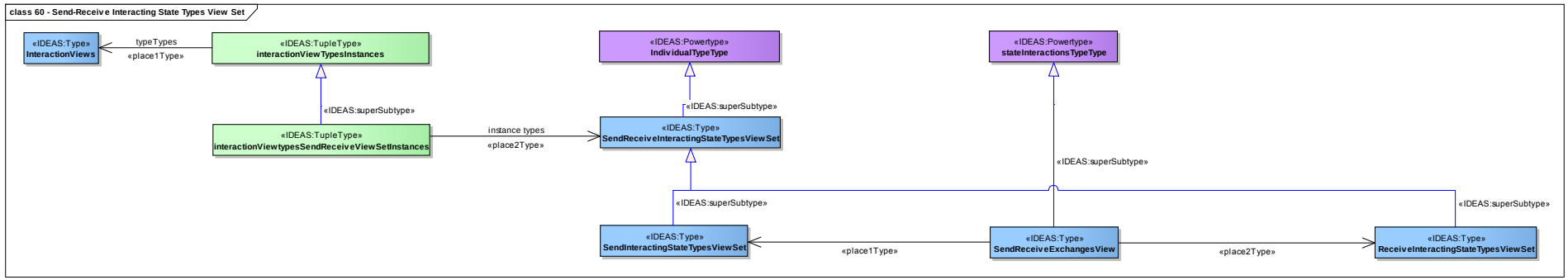


Figure 126 : Send-Receive Interacting State Types View Set

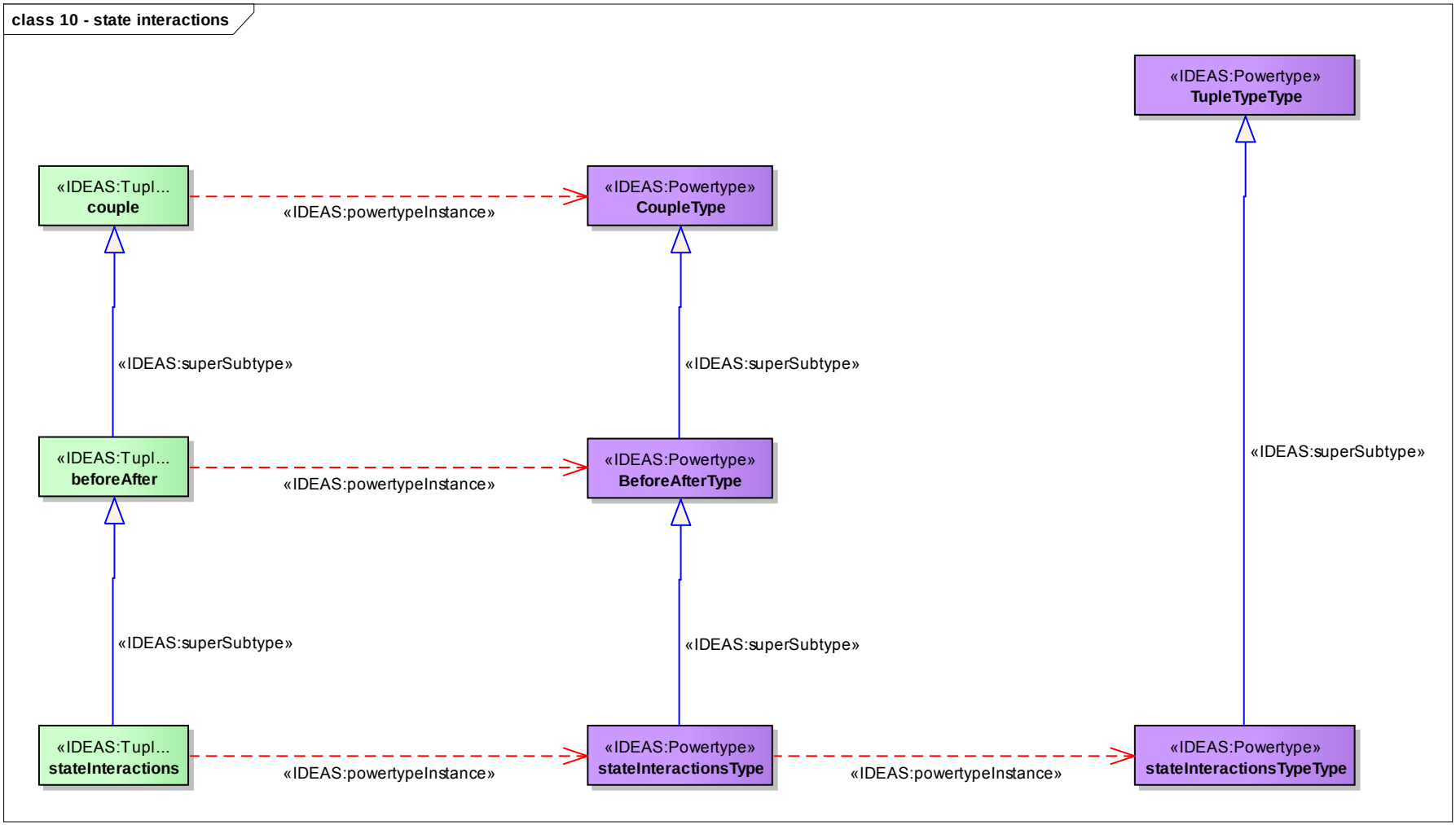


Figure 127 : state interactions

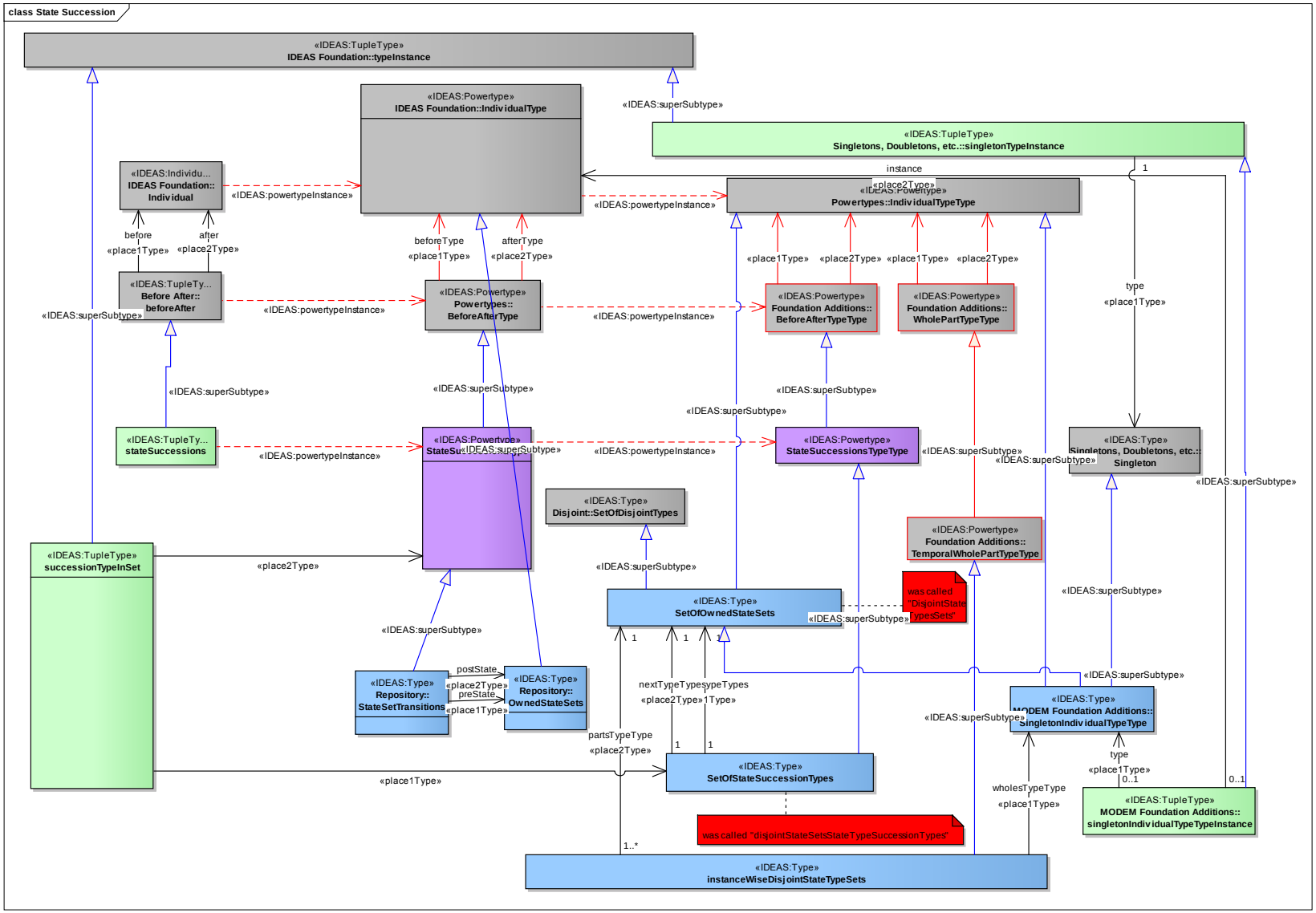


Figure 128 : State Succession

3.4.6 State and interactions elements list

State machine views
OwnedStateSets «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» OwnedStateSets - IndividualType *Dependency (element - is instance of):* «IDEAS:powertypeInstance» OwnedStateSets - OwnedStateSetsType *Attributes:* - An IndividualType that contains all the state for an owning IndividualType.
OwnedStateSetsType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» OwnedStateSetsType - IndividualTypeType *Attributes:* - A powertype of OwnedStateSetsType.
SetOfOwnedStateSetsTypeOwnedSetInstance «IDEAS:TupleType» *Connectors:* *Association (source - target):* «place1Type» SetOfOwnedStateSetsTypeOwnedSetInstance - SetOfOwnedStateSets *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SetOfOwnedStateSetsTypeOwnedSetInstance - RegionTypeInstances *Association (source - target):* «place2Type» SetOfOwnedStateSetsTypeOwnedSetInstance - OwnedStateSets *Attributes:* - A regionTypeInstance that asserts an OwnedStateSet is an instance of a SetOfOwnedStateSets.
SetOfStateSuccessionTypesTypeStateSuccessionTypeInstance «IDEAS:TupleType» *Connectors:* *Association (source - target):* «place2Type» SetOfStateSuccessionTypesTypeStateSuccessionTypeInstance - StateSetTransitions *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SetOfStateSuccessionTypesTypeStateSuccessionTypeInstance - RegionTypeInstances *Association (source - target):* «place1Type» SetOfStateSuccessionTypesTypeStateSuccessionTypeInstance - SetOfStateSuccessionTypes *Attributes:* - A regionTypeInstance that asserts an StateSetTransitions is an instance of a SetOfStateSuccessionTypes.
StateMachineRegions «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» StateMachineRegions - Type *Attributes:* - A Type that has a subTypes one SetOfOwnedStateSets and its associated SetOfStateSuccessionTypes.

StateMachineViews «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StateMachineViews - IndividualTypeTypeType
Attributes:
-
An IndividualTypeTypeType that contains one or more StateMachineViews as instances.

StateSetTransitions «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StateSetTransitions - transitions
Dependency (element - is instance of): «IDEAS:powertypeInstance»
StateSetTransitions - stateSetTransitionsType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
StateSetTransitions - stateSuccessionsType
Association (source - target): «place2Type»
StateSetTransitions - OwnedStateSets
Association (source - target): «place1Type»
StateSetTransitions - OwnedStateSets
Attributes:
-
A Transitions and a stateSuccessionType that asserts transitions for an OwnedStateSets.

regionTypeInstances «IDEAS:TupleType»
Connectors:
Association (source - target):«place1Type»
regionTypeInstances - StateMachineRegions
Generalization (element - is a subtype of):«IDEAS:superSubtype»
regionTypeInstances - typeInstance
Attributes:
-
A typeInstance that asserts a StateMachineRegion is a type of some instance.

stateMachineRegionSuperSetOfOwnedStateSetSubType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateMachineRegionSuperSetOfOwnedStateSetSubType - superSubtype
Association (source - target): «place2Type»
stateMachineRegionSuperSetOfOwnedStateSetSubType - SetOfOwnedStateSets
Association (source - target): «place1Type»
stateMachineRegionSuperSetOfOwnedStateSetSubType - StateMachineRegions
Attributes:
-
A superSubType that asserts a SetOfOwnedStateSets is a subType of a StateMachineRegion.

stateMachineRegionSuperSetOfStateSuccessionTypesSubType «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateMachineRegionSuperSetOfStateSuccessionTypesSubType - superSubtype
Association (source - target): «place2Type»
stateMachineRegionSuperSetOfStateSuccessionTypesSubType - SetOfStateSuccessionTypes

Association (source - target): «place1Type»
stateMachineRegionSuperSetOfStateSuccessionTypesSubType - StateMachineRegions
Attributes:
-
A superSubType that asserts a SetOfStateSuccessionTypes is a subType of a StateMachineRegion.

stateMachineViewStateTypeOwners «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateMachineViewStateTypeOwners - couple
Association (source - target): «place1Type»
stateMachineViewStateTypeOwners - IndividualType
Association (source - target): «place2Type»
stateMachineViewStateTypeOwners - StateMachineViews
Attributes:
-
A couple that asserts that an IndividualType is a type of owner of a StateMachine View.

stateMachineViewTypesRegionInstances «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateMachineViewTypesRegionInstances - typeInstance
Association (source - target): «place2Type»
stateMachineViewTypesRegionInstances - StateMachineRegions
Association (source - target): «place1Type»
stateMachineViewTypesRegionInstances - StateMachineViews
Attributes:
-
A typeInstance that asserts a StateMachineRegion is an instance of a StateMachineViews.

stateSetOwners «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateSetOwners - TemporalWholePartType
Dependency (element - is instance of): «IDEAS:powertypeInstance»
stateSetOwners - stateSetOwnersType
Association (source - target): «place2Type»
stateSetOwners - OwnedStateSets
Association (source - target): «place1Type»
stateSetOwners - IndividualType
Attributes:
-
A TemporalWholeParttype that asserts an OwnedStateSet is a type of temporal part of the owning IndividualType.

stateSetOwnersType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateSetOwnersType - TemporalWholePartTypeType
Attributes:
-
A powertype of StateSetOwners.

stateSetTransitionsType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateSetTransitionsType - StateSuccessionsTypeType
Attributes:
-
A powertype of StateSetTransitions.

transitions «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
transitions - CoupleType
Attributes:
-
A CoupleType.

Interaction view patterns

InteractionParticipationRoles «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionParticipationRoles - IndividualType
Dependency (element - is instance of): «IDEAS:powertypeInstance»
InteractionParticipationRoles - InteractionParticipationRolesType
Attributes:
-
An IndividualType that is the type of participation of an Individual in an Interaction. For example, 'Waiter role in Eat Restaurant Meal'.

InteractionParticipationRolesType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionParticipationRolesType - IndividualTypeType
Attributes:
-
The powertype of InteractionParticipationRoles.

InteractionRoleStateMachineRegions «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionRoleStateMachineRegions - StateMachineRegions
Attributes:
-
A StateMachineRegions for a InteractionRoleStateMachineView.

InteractionRoleStateMachineViews «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionRoleStateMachineViews - StateMachineViews
Attributes:
-
A StateMachineView for a InteractionParticipationRole.

InteractionRolesViewSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionRolesViewSets - InteractionParticipationRolesType
Attributes:
-
A InteractionParticipationRolesType that is a set of the InteractionParticipationRoles in the InteractionView.

InteractionSingleton «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionSingleton - InteractionsType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionSingleton - Singleton
Attributes:
-
An InteractionType and a Singleton.

InteractionViews «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionViews - Type
Attributes:
-
A Type that contains as instances all the elements of the view.

InteractionWholeRoleViewPartsSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionWholeRoleViewPartsSets - WholePartTypeType
Association (source - target): «place1Type»
InteractionWholeRoleViewPartsSets - InteractionSingleton
Association (source - target): «place2Type»
InteractionWholeRoleViewPartsSets - InteractionRolesViewSets
Attributes:
-
A WholePartTypeType that asserts the InteractionParticipationRolesType is a type of part of the InteractionSingleton.

Interactions «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Interactions - IndividualType
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Interactions - InteractionsType
Attributes:
-
An IndividualType that is composed of types of particpating Individuals. For example, 'Eat Restaurant Meal'.

InteractionsType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
InteractionsType - IndividualTypeType
Attributes:

-
The powertype of Interactions.

ReceiveInteractingStateTypesViewSet «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ReceiveInteractingStateTypesViewSet - SendReceiveInteractingStateTypesViewSet
Attributes:
-
A SendReceiveInteractingStateTypesViewSet that contains a type of receiving state in the view.

SendInteractingStateTypesViewSet «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SendInteractingStateTypesViewSet - SendReceiveInteractingStateTypesViewSet
Attributes:
-
A SendReceiveInteractingStateTypesViewSet that contains a type of sending state in the view.

SendReceiveExchangesView «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SendReceiveExchangesView - stateInteractionsTypeType
Association (source - target): «place2Type»
SendReceiveExchangesView - ReceiveInteractingStateTypesViewSet
Association (source - target): «place1Type»
SendReceiveExchangesView - SendInteractingStateTypesViewSet
Attributes:
-
A stateInteractionsTypeType that asserts one type of state sends an exchange and another receives the exchange.

SendReceiveInteractingStateTypesViewSet «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SendReceiveInteractingStateTypesViewSet - IndividualTypeType
Attributes:
-
An IndividualTypeType that contains a type of sending or receiving state in the view.

WholeInteractionRoleViewSetsParts «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
WholeInteractionRoleViewSetsParts - WholePartTypeType
Association (source - target): «place2Type»
WholeInteractionRoleViewSetsParts - InteractionRolesViewSets
Attributes:
-
A WholePartTypeType that asserts the InteractionParticipationRolesType is a type of part of the type Individual participating in the Interaction.

individualWholeIinteractionRolePartTypes «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
individualWholeIinteractionRolePartTypes - WholePartType
Association (source - target):«place2Type»
individualWholeIinteractionRolePartTypes - InteractionParticipationRoles
Association (source - target):«place1Type»
individualWholeIinteractionRolePartTypes - IndividualType
Attributes:
-
A WholePartType that asserts an InteractionParticipationRole is a type of part of an IndividualType.

interactionRoleStateMachineViewStateTypeOwners «IDEAS:TupleType»
Connectors:
Association (source - target): «place1Type»
interactionRoleStateMachineViewStateTypeOwners - InteractionParticipationRoles
Association (source - target): «place2Type»
interactionRoleStateMachineViewStateTypeOwners - InteractionRoleStateMachineViews
Generalization (element - is a subtype of): «IDEAS:superSubtype»
interactionRoleStateMachineViewStateTypeOwners - stateMachineViewStateTypeOwners
Attributes:
-
A stateMachineViewStateTypeOwner that asserts a InteractionRoleStateMachineView is owned by a InteractionParticipationRole.

interactionRoleStateMachineViewTypesRegionInstances «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
interactionRoleStateMachineViewTypesRegionInstances - stateMachineViewTypesRegionInstances
Association (source - target): «place2Type»
interactionRoleStateMachineViewTypesRegionInstances - InteractionRoleStateMachineRegions
Association (source - target): «place1Type»
interactionRoleStateMachineViewTypesRegionInstances - InteractionRoleStateMachineViews
Attributes:
-
A stateMachineViewTypesRegionInstances that asserts a InteractionRoleStateMachineRegion is an instance of a InteractionRoleStateMachineView.

interactionViewTypesInstances «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
interactionViewTypesInstances - typeInstance
Association (source - target): «place1Type»
interactionViewTypesInstances - InteractionViews
Attributes:
-
A typeInstance that asserts something is an instance of an InteractionView.

interactionViewTypesInteractionRoleStateMachineViewsInstances «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
interactionViewTypesInteractionRoleStateMachineViewsInstances - interactionViewTypesInstances
Association (source - target): «place2Type»
interactionViewTypesInteractionRoleStateMachineViewsInstances - InteractionRoleStateMachineViews

Attributes: - A interactionViewTypesInstances that asserts a InteractionRoleStateMachineViews is an instance of an InteractionView.
interactionViewTypesInteractionRoleViewSetsInstances «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» interactionViewTypesInteractionRoleViewSetsInstances - interactionViewTypesInstances *Association (source - target):* «place2Type» interactionViewTypesInteractionRoleViewSetsInstances - InteractionRolesViewSets *Attributes:* - A interactionViewTypesInstances that asserts an InteractionRolesViewSets is an instance of an InteractionView.
interactionViewTypesInteractionSingletonInstances «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» interactionViewTypesInteractionSingletonInstances - interactionViewTypesInstances *Association (source - target):* «place2Type» interactionViewTypesInteractionSingletonInstances - InteractionSingleton *Attributes:* - A interactionViewTypesInstances that asserts a InteractionSingleton is an instance of an InteractionView.
interactionViewTypesInteractionWholeRoleViewPartSetsInstances «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» interactionViewTypesInteractionWholeRoleViewPartSetsInstances - interactionViewTypesInstances *Association (source - target):* «place2Type» interactionViewTypesInteractionWholeRoleViewPartSetsInstances - InteractionWholeRoleViewPartsSets *Attributes:* - A interactionViewTypesInstances that asserts an InteractionWholeRoleViewPartsSets is an instance of an InteractionView.
interactionViewTypesWholeInteractionRoleViewSetsPartsInstances «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» interactionViewTypesWholeInteractionRoleViewSetsPartsInstances - interactionViewTypesInstances *Association (source - target):* «place2Type» interactionViewTypesWholeInteractionRoleViewSetsPartsInstances - WholeInteractionRoleViewSetsParts *Attributes:* - A interactionViewTypesInstances that asserts an WholeInteractionRoleViewSetsPart is an instance of an InteractionView.
interactionViewtypesSendReceiveViewSetInstances «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» interactionViewtypesSendReceiveViewSetInstances - interactionViewTypesInstances *Association (source - target):* «place2Type» interactionViewtypesSendReceiveViewSetInstances - SendReceiveInteractingStateTypesViewSet *Attributes:* - A interactionViewTypesInstances that asserts a SendReceiveInteractingStateTypesViewSet is an instance of an InteractionView.

interactionWholeRolePartTypes «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
interactionWholeRolePartTypes - WholePartType
Association (source - target): «place2Type»
interactionWholeRolePartTypes - InteractionParticipationRoles
Association (source - target): «place1Type»
interactionWholeRolePartTypes - Interactions
Attributes:
-
A WholePartType that asserts an InteractionParticipationRole is a type of part of an Interaction.

State interactions

stateInteractions «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateInteractions - beforeAfter
Dependency (element - is instance of): «IDEAS:powertypeInstance»
stateInteractions - stateInteractionsType
Attributes:
-
A beforeAfter that asserts that one state is before another.

stateInteractionsType «IDEAS:Powertype»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
stateInteractionsType - stateInteractionsTypeType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateInteractionsType - BeforeAfterType
Attributes:
-
The powertype of stateInteractions.

stateInteractionsTypeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
stateInteractionsTypeType - TupleTypeType
Attributes:
-
The powertype of stateInteractionsType.

LinearlySucceededDisjointStateTypesSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LinearlySucceededDisjointStateTypesSets - SetOfOwnedStateSets
Generalization (element - is a subtype of): «IDEAS:superSubtype»
LinearlySucceededDisjointStateTypesSets - InteractionRoleStateMachineRegions
Attributes:
-
A SetOfOwnedStateSets where each state type is succeeded by no more than one other state type.

SingletonDisjointStateTypesSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SingletonDisjointStateTypesSets - SetOfOwnedStateSets
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SingletonDisjointStateTypesSets - Singleton
Attributes:
-
A SetOfOwnedStateSets and a Singleton.

SingletonLinearlySucceededDisjointStateTypesSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SingletonLinearlySucceededDisjointStateTypesSets - SingletonDisjointStateTypesSets
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SingletonLinearlySucceededDisjointStateTypesSets - LinearlySucceededDisjointStateTypesSets
Attributes:
-
A SingletonDisjointStateTypesSets and a LinearlySucceededDisjointStateTypesSets. Note: 'Singleton Linearly Succeeded Disjoint State Types Sets' cannot have any succession, as any such successions would not be linear.

disjointStateTypesSetsSuperSubTypeHierarchy «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
disjointStateTypesSetsSuperSubTypeHierarchy - superSubtype
Association (source - target): «place2Type»
disjointStateTypesSetsSuperSubTypeHierarchy - SetOfOwnedStateSets
Association (source - target): «place1Type»
disjointStateTypesSetsSuperSubTypeHierarchy - SetOfOwnedStateSets
Attributes:
-
A superSubType that asserts one SetofOwnedStateSets is a subType of another.

improperInstanceWiseDisjointStateSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
improperInstanceWiseDisjointStateSets - Singleton
Generalization (element - is a subtype of): «IDEAS:superSubtype»
improperInstanceWiseDisjointStateSets - instanceWiseDisjointStateTypeSets
Attributes:
-
A Singleton and an InstanceWiseDisjointStateTypeSets that asserts a SetOfOwnedStateSets is an improper part of an IndividualTypeType. At the limit, each instance of an Element Powertype has itself as an improper temporal stage. The union of these is the instance of the Element Powertype.

instanceWiseCompletePartitionStateTypeSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
instanceWiseCompletePartitionStateTypeSets - instanceWiseDisjointStateTypeSets
Attributes:
-
A instanceWiseDisjointStateTypeSets where the state types completely partition the whole.

linearlySucceededDisjointStateSetsStateTypeSuccessionTypes «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
linearlySucceededDisjointStateSetsStateTypeSuccessionTypes - InteractionRoleStateMachineRegions
Association (source - target): «place1Type»
linearlySucceededDisjointStateSetsStateTypeSuccessionTypes - LinearlySucceededDisjointStateTypesSets
Association (source - target): «place2Type»
linearlySucceededDisjointStateSetsStateTypeSuccessionTypes - LinearlySucceededDisjointStateTypesSets
Generalization (element - is a subtype of): «IDEAS:superSubtype»
linearlySucceededDisjointStateSetsStateTypeSuccessionTypes - SetOfStateSuccessionTypes
Attributes:
-
A SetOfStateSuccessionTypes that asserts that a one LinearlySucceededDisjointStateTypesSets is succeed by only one other.

singletonDisjointStateSetsStateTypeSuccessionTypes «IDEAS:Type»
Connectors:
Association (source - target): «place2Type»
singletonDisjointStateSetsStateTypeSuccessionTypes - SingletonDisjointStateTypesSets
Association (source - target): «place1Type»
singletonDisjointStateSetsStateTypeSuccessionTypes - SingletonDisjointStateTypesSets
Generalization (element - is a subtype of): «IDEAS:superSubtype»
singletonDisjointStateSetsStateTypeSuccessionTypes - SetOfStateSuccessionTypes
Attributes:
-
A SetOfStateSuccessionTypes where the successions are between SingletonDisjointStateTypesSets.

State successions

SetOfOwnedStateSets «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SetOfOwnedStateSets - IndividualTypeType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SetOfOwnedStateSets - OwnedStateSetsType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SetOfOwnedStateSets - SetOfDisjointTypes
Attributes:
-
A SetOfDisjointTypes and IndividualTypeType where each instance is a disjoint set of state types, whose union is instance-wise disjoint relative to the related instance of Element Powertype.

SetOfStateSuccessionTypes «IDEAS:Type»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SetOfStateSuccessionTypes - StateSuccessionsTypeType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
SetOfStateSuccessionTypes - stateSetTransitionsType
Association (source - target): «place2Type»
SetOfStateSuccessionTypes - SetOfOwnedStateSets
Association (source - target): «place1Type»
SetOfStateSuccessionTypes - SetOfOwnedStateSets
Attributes:
-

A StateSuccessionsTypeType that asserts a type of succession between instances of SetOfOwnedStateSets.
instanceWiseDisjointStateTypeSets «IDEAS:Type» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» instanceWiseDisjointStateTypeSets - stateSetOwnersType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» instanceWiseDisjointStateTypeSets - TemporalWholePartTypeType *Association (source - target):* «place2Type» instanceWiseDisjointStateTypeSets - SetOfOwnedStateSets *Association* (source - target): «place1Type» instanceWiseDisjointStateTypeSets - SingletonIndividualTypeType *Attributes:* - A TemporalWholePartTypeType that asserts a SetOfOwnedStateSets is a type of type of part of an IndividualTypeType.
stateSuccessions «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» stateSuccessions - beforeAfter *Dependency (element - is instance of):* «IDEAS:powertypeInstance» stateSuccessions - stateSuccessionsType *Attributes:* - A beforeAfter that asserts a succession between states.
StateSuccessionsType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» stateSuccessionsType - BeforeAfterType *Dependency (element - is instance of):* «IDEAS:powertypeInstance» stateSuccessionsType - StateSuccessionsTypeType *Attributes:* - A powertype of stateSuccessions.
StateSuccessionsTypeType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» StateSuccessionsTypeType - TupleTypeType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» StateSuccessionsTypeType - BeforeAfterTypeType *Attributes:* - A powertype of stateSuccessionType.

successionTypeInSet «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
successionTypeInSet - typeInstance
Association (source - target): «place2Type»
successionTypeInSet - stateSuccessionsType
Association (source - target): «place1Type»
successionTypeInSet - SetOfStateSuccessionTypes
Attributes:
-
A typeInstance that asserts a StateSuccessionType is a member of a SetOfStateSuccessionTypes.

3.4.7 Exchange diagrams

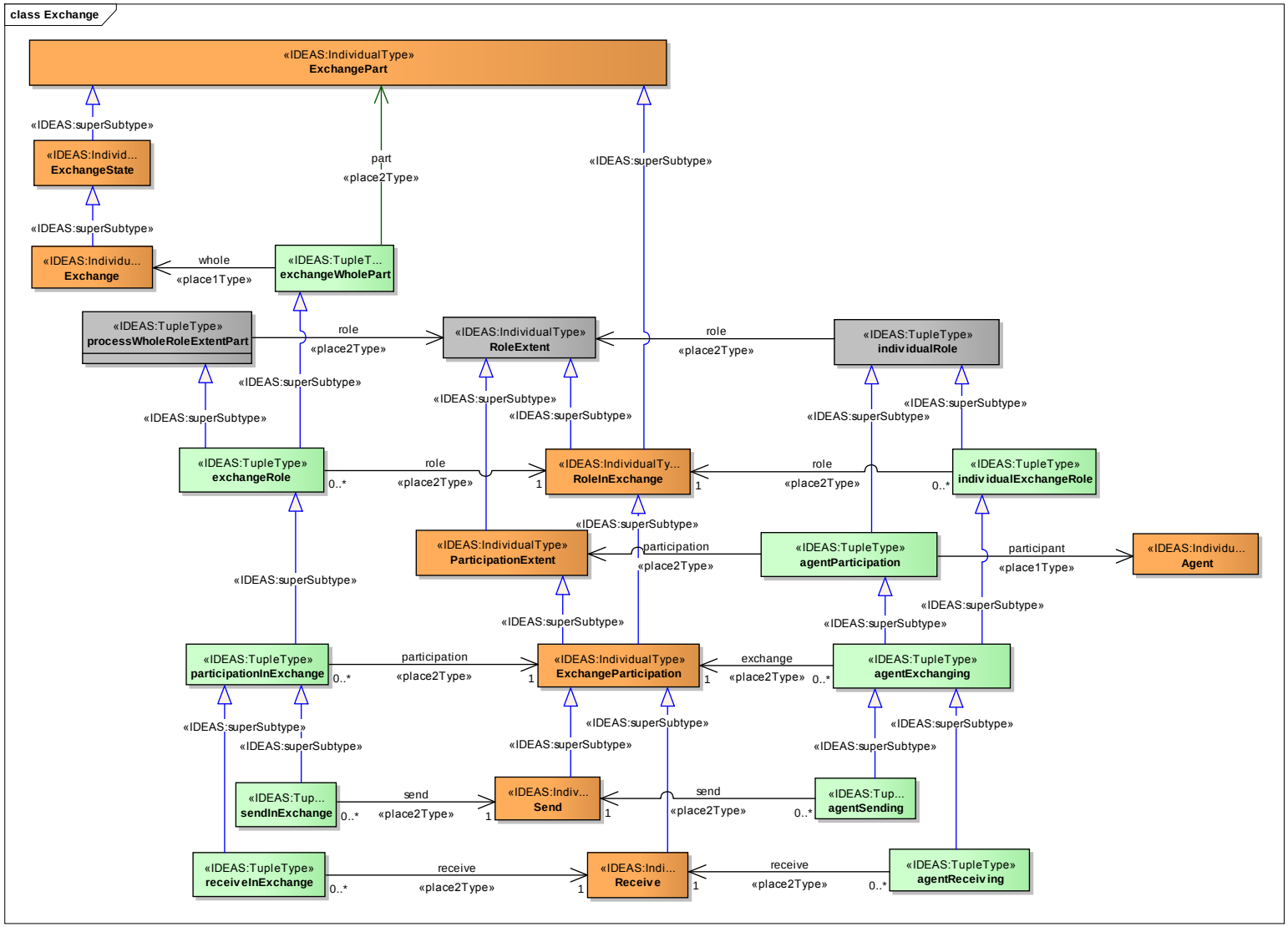


Figure 129 : Exchange

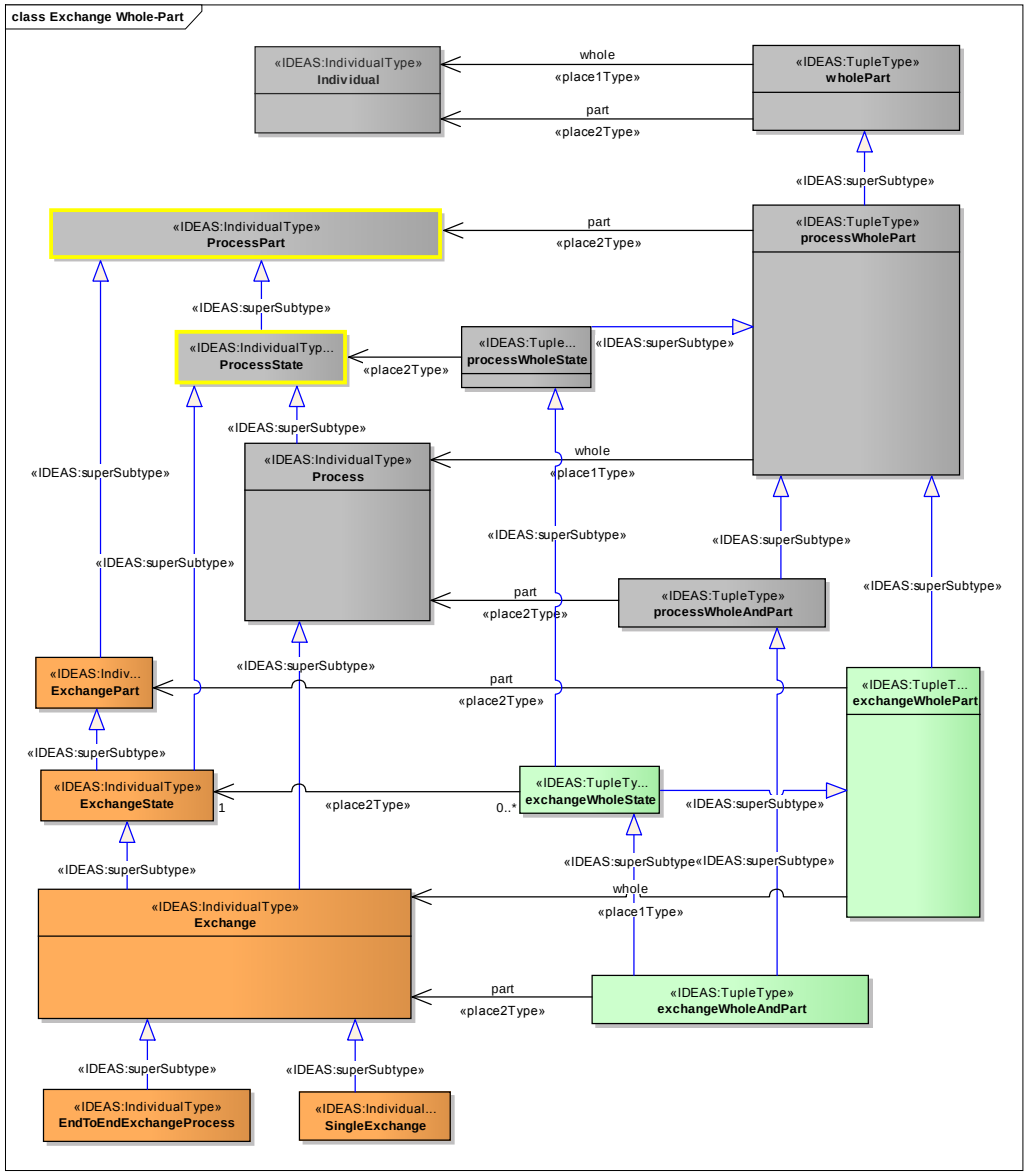


Figure 130 : Exchange Whole-Part

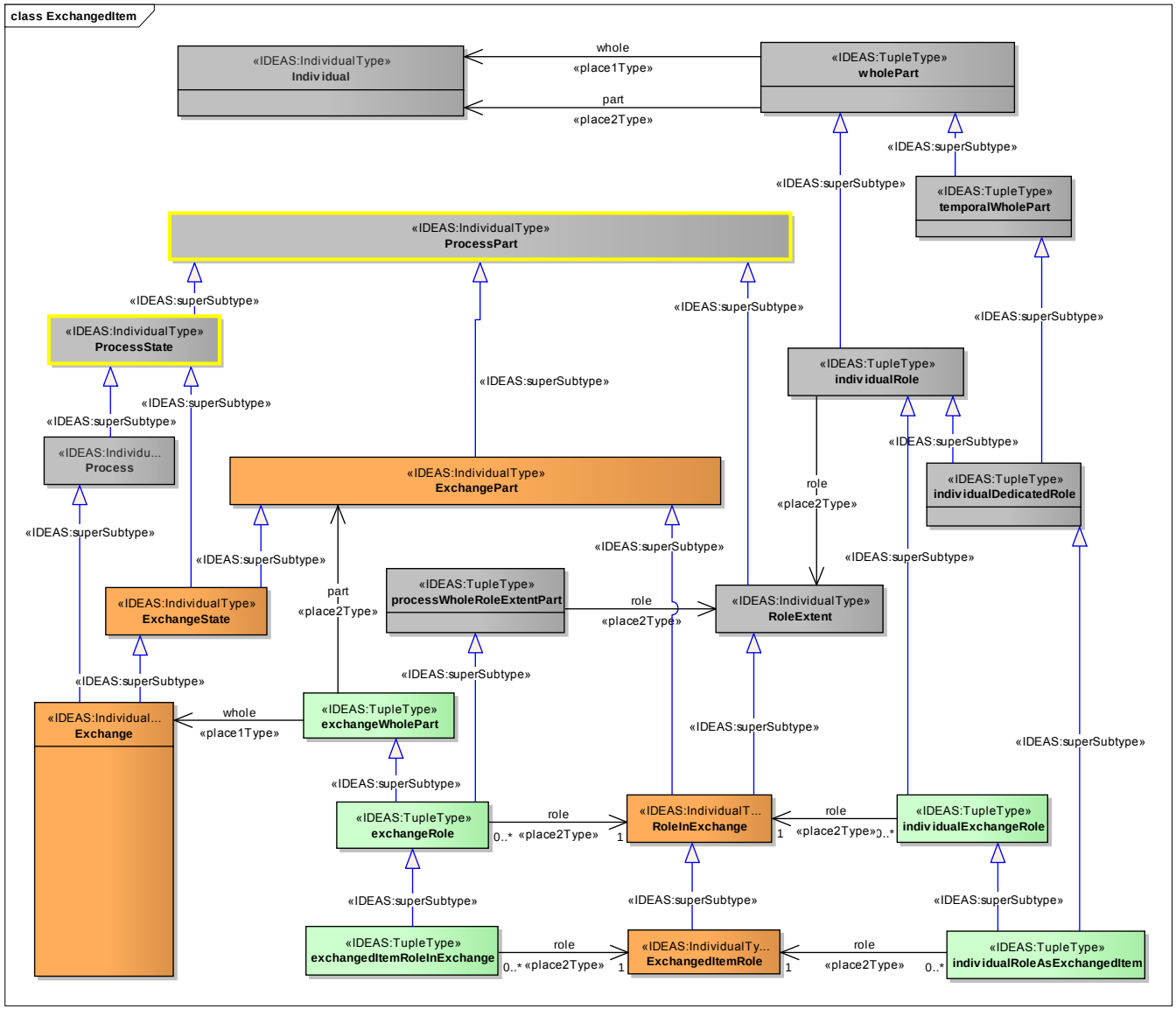


Figure 131 : ExchangedItem

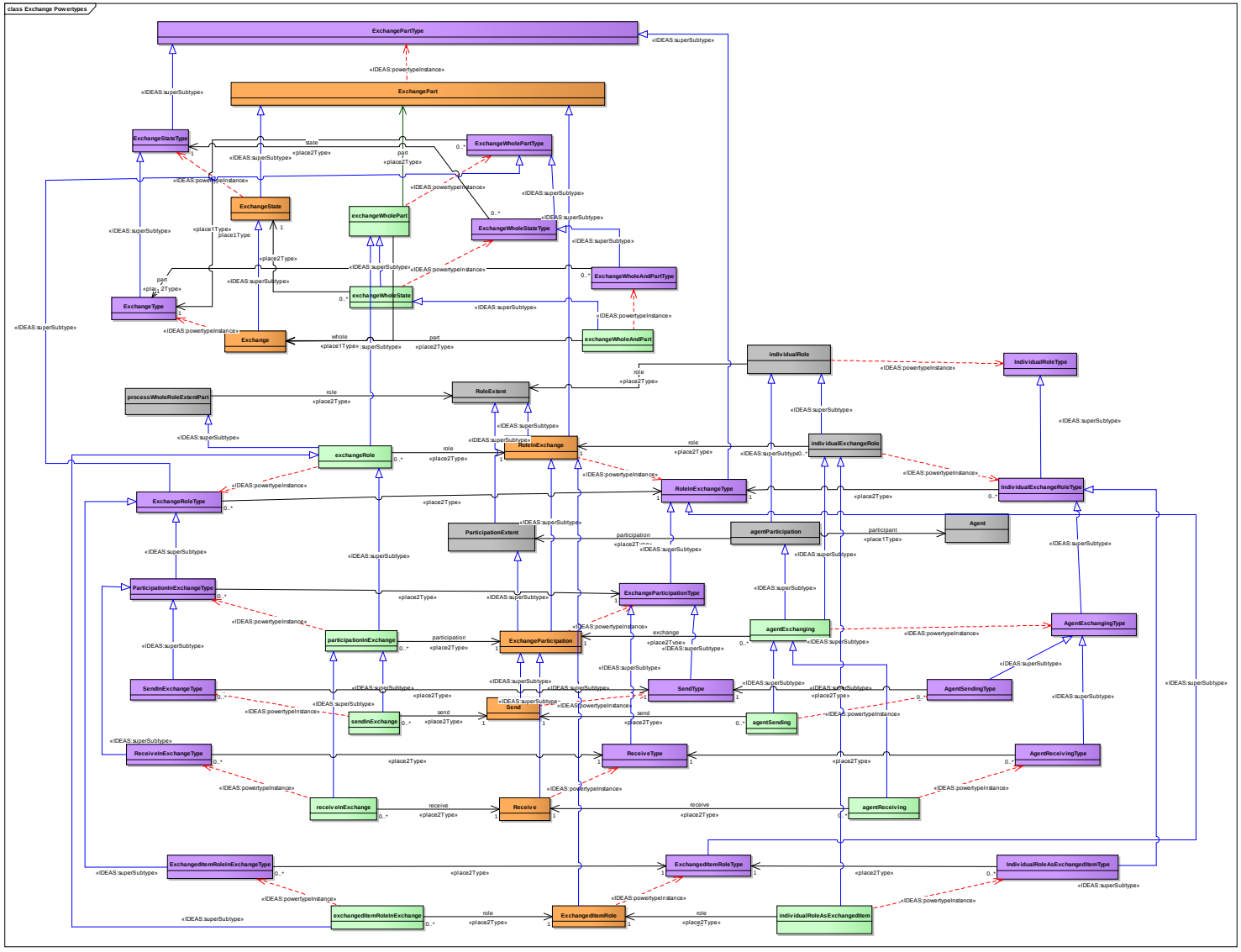


Figure 132 : Exchange Powertypes

3.4.8 Exchange elements list

Exchange
DirectedExchange «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» DirectedExchange - Exchange *Attributes:* - An Exchange where the exchanged Individuals all flow in one direction.
EndToEndExchangeProcess «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» EndToEndExchangeProcess - Exchange *Attributes:* - An Exchange which consists of other Exchanges. Note: the Exchanges which are part of the EndToEndExchangeProcess need not be a sequence - some may run in parallel.
Exchange «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Exchange - Process *Generalization (element - is a subtype of):* «IDEAS:superSubtype» Exchange - ExchangeState *Dependency (element - is instance of):* «IDEAS:powertypeInstance» Exchange - ExchangeType *Attributes:* - A Process where one Agent exchanges one or more Individuals with another Agent.
ExchangePart «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ExchangePart - ProcessPart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ExchangePart - ExchangePartType *Attributes:* - A ProcessPart that is part of an Exchange.
ExchangeParticipation «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ExchangeParticipation - ParticipationExtent *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ExchangeParticipation - RoleInExchange *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ExchangeParticipation - ExchangeParticipationType *Attributes:* - A RoleInExchange and a ParticipationExtent that is an Agent's participation in an Exchange.

ExchangeState «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeState - ExchangePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeState - ProcessState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ExchangeState - ExchangeStateType
Attributes:
-
An ExchangePart that is a temporal part of an Exchange.

ExchangedItemRole «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangedItemRole - RoleInExchange
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ExchangedItemRole - ExchangedItemRoleType
Attributes:
-
A RoleinExchange where the Process is an Exchange and the Individual's role is as the thing being exchanged.

Receive «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Receive - ExchangeParticipation
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Receive - ReceiveType
Attributes:
-
An ExchangeParticipation that is the receiving Agent's participation in an Exchange.

RoleInExchange «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInExchange - ExchangePart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleInExchange - RoleExtent
Dependency (element - is instance of): «IDEAS:powertypeInstance»
RoleInExchange - RoleInExchangeType
Attributes:
-
An ExchangePart that is an Individual's role in the Exchange.

Send «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Send - ExchangeParticipation
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Send - SendType
Attributes:
-

An ExchangePart and a ParticipationExtent that is the sending Agent's participation in an Exchange.
SingleExchange «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SingleExchange - Exchange *Attributes:* - An Exchange that has no parts that are also Exchanges. Example: One person handing another a document.
agentExchanging «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentExchanging - agentParticipation *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentExchanging - individualExchangeRole *Dependency (element - is instance of):* «IDEAS:powertypeInstance» agentExchanging - AgentExchangingType *Association (source - target):* «place2Type» agentExchanging - ExchangeParticipation *Attributes:* - An agentParticipation where the participation is in an Exchange.
agentReceiving «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentReceiving - agentExchanging *Dependency (element - is instance of):* «IDEAS:powertypeInstance» agentReceiving - AgentReceivingType *Association (source - target):* «place2Type» agentReceiving - Receive *Attributes:* - An agentExchanging where the Agent's participation is as the receiver.
agentSending «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentSending - agentExchanging *Dependency (element - is instance of):* «IDEAS:powertypeInstance» agentSending - AgentSendingType *Association (source - target):* «place2Type» agentSending - Send *Attributes:* - An agentExchanging where the Agent's participation is as the sender.
exchangeRole «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» exchangeRole - exchangeWholePart *Generalization (element - is a subtype of):* «IDEAS:superSubtype»

exchangeRole - processWholeRoleExtentPart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
exchangeRole - ExchangeRoleType
Association (source - target): «place2Type»
exchangeRole - RoleInExchange
Attributes:
-
An exchangeWholePart where the part is a RoleInExchange.

exchangeWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exchangeWholeAndPart - exchangeWholeState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exchangeWholeAndPart - processWholeAndPart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
exchangeWholeAndPart - ExchangeWholeAndPartType
Association (source - target): «place2Type»
exchangeWholeAndPart - Exchange
Attributes:
-
An exchangeWholePart where the part is an exchange.

exchangeWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exchangeWholePart - processWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
exchangeWholePart - ExchangeWholePartType
Association (source - target): «place2Type»
exchangeWholePart - ExchangePart
Association (source - target): «place1Type»
exchangeWholePart - Exchange
Attributes:
-
A processWholePart where the whole is an Exchange and the part is an ExchangePart.

exchangeWholeState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exchangeWholeState - exchangeWholePart
Association (source - target): «place2Type»
exchangeWholeState - ExchangeState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exchangeWholeState - processWholeState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
exchangeWholeState - ExchangeWholeStateType
Attributes:
-
An exchangeWholePart where the part is a temporal state of the whole.

exchangedItemRoleInExchange «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
exchangedItemRoleInExchange - exchangeRole
Dependency (element - is instance of): «IDEAS:powertypeInstance»
exchangedItemRoleInExchange - ExchangedItemRoleInExchangeType
Association (source - target): «place2Type»
exchangedItemRoleInExchange - ExchangedItemRole
Attributes:
-
An exchangeRole where the role is that of the Individual being exchanged.

individualExchangeRole «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
individualExchangeRole - individualRole
Dependency (element - is instance of): «IDEAS:powertypeInstance»
individualExchangeRole - IndividualExchangeRoleType
Association (source - target): «place2Type»
individualExchangeRole - RoleInExchange
Attributes:
-
An individualRole where the process is an Exchange.

individualRoleAsExchangedItem «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
individualRoleAsExchangedItem - individualExchangeRole
Generalization (element - is a subtype of): «IDEAS:superSubtype»
individualRoleAsExchangedItem - individualDedicatedRole
Dependency (element - is instance of): «IDEAS:powertypeInstance»
individualRoleAsExchangedItem - IndividualRoleAsExchangedItemType
Association (source - target): «place2Type»
individualRoleAsExchangedItem - ExchangedItemRole
Attributes:
-
An individualExchangeRole the Individual is the thing being exchanged.

participationInExchange «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
participationInExchange - exchangeRole
Dependency (element - is instance of): «IDEAS:powertypeInstance»
participationInExchange - ParticipationInExchangeType
Association (source - target): «place2Type»
participationInExchange - ExchangeParticipation
Attributes:
-
An exchangeWholePart where the part is an ExchangeParticipation.

receiveInExchange «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
receiveInExchange - participationInExchange
Dependency (element - is instance of): «IDEAS:powertypeInstance»
receiveInExchange - ReceiveInExchangeType
Association (source - target): «place2Type»
receiveInExchange - Receive
Attributes:
-
A participationInExchange where the participation is a Receive.

sendInExchange «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
sendInExchange - participationInExchange
Dependency (element - is instance of): «IDEAS:powertypeInstance»
sendInExchange - SendInExchangeType
Association (source - target): «place2Type»
sendInExchange - Send
Attributes:
-
A participationInExchange where participation is a Send.

Exchange Powertypes

AgentExchangingType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentExchangingType - AgentParticipationType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentExchangingType - IndividualExchangeRoleType
Attributes:
-
The powertype of agentExchanging.

AgentReceivingType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentReceivingType - AgentExchangingType
Association (source - target): «place2Type»
AgentReceivingType - ReceiveType
Attributes:
-
The powertype of agentReceiving.

AgentSendingType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentSendingType - AgentExchangingType
Association (source - target): «place2Type»
AgentSendingType - SendType
Attributes:

-
The powertype of agentSending.

ExchangePartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangePartType - ProcessPartType
Attributes:
-
The powertype of ExchangePart.

ExchangeParticipationType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeParticipationType - RoleInExchangeType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeParticipationType - ParticipationExtentType
Attributes:
-
The powertype of ExchangeParticipation.

ExchangeRoleType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeRoleType - ProcessWholeRoleExtentPartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeRoleType - ExchangeWholePartType
Association (source - target): «place2Type»
ExchangeRoleType - RoleInExchangeType
Attributes:
-
The powertype of exchangeRole.

ExchangeStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeStateType - ExchangePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeStateType - ProcessStateType
Attributes:
-
The powertype of ExchangeState.

ExchangeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeType - ExchangeStateType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeType - ProcessType
Attributes:
-
The powertype of Exchange.

ExchangeWholeAndPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeWholeAndPartType - ExchangeWholeStateType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeWholeAndPartType - ProcessWholeAndPartType
Association (source - target): «place2Type»
ExchangeWholeAndPartType - ExchangeType
Attributes:
-
The powertype of exchangeWholeAndPart.

ExchangeWholePartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeWholePartType - ProcessWholePartType
Association (source - target): «place1Type»
ExchangeWholePartType - ExchangeType
Attributes:
-
The powertype of exchangeWholePart.

ExchangeWholeStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeWholeStateType - ExchangeWholePartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangeWholeStateType - ProcessWholeStateType
Association (source - target): «place2Type»
ExchangeWholeStateType - ExchangeStateType
Attributes:
-
The powertype of exchangeWholeState.

ExchangedItemRoleInExchangeType «IDEAS:Powertype»
Connectors:
Association (source - target): «place2Type»
ExchangedItemRoleInExchangeType - ExchangedItemRoleType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangedItemRoleInExchangeType - ExchangeRoleType
Attributes:
-
The powertype of exchangedItemRoleInExchange.

ExchangedItemRoleType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ExchangedItemRoleType - RoleInExchangeType
Attributes:
-
The powertype of ExchangedItemRole.

IndividualExchangeRoleType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
IndividualExchangeRoleType - IndividualRoleType
Association (source - target): «place2Type»
IndividualExchangeRoleType - RoleInExchangeType
Attributes:
-
The powertype of individualExchangeRole.

IndividualRoleAsExchangedItemType «IDEAS:Powertype»
Connectors:
Association (source - target): «place2Type»
IndividualRoleAsExchangedItemType - ExchangedItemRoleType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
IndividualRoleAsExchangedItemType - IndividualExchangeRoleType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
IndividualRoleAsExchangedItemType - IndividualDedicatedRoleType
Attributes:
-
The powertype of individualRoleAsExchangedItem.

ParticipationInExchangeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ParticipationInExchangeType - ExchangeRoleType
Association (source - target): «place2Type»
ParticipationInExchangeType - ExchangeParticipationType
Attributes:
-
The powertype of participationInExchange.

ReceiveInExchangeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ReceiveInExchangeType - ParticipationInExchangeType
Association (source - target): «place2Type»
ReceiveInExchangeType - ReceiveType
Attributes:
-
The powertype of receiveInExchange.

ReceiveType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ReceiveType - ExchangeParticipationType
Attributes:
-
The powertype of Receive.

RoleInExchangeType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» RoleInExchangeType - RoleExtentType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» RoleInExchangeType - ExchangePartType *Attributes:* - The powertype of RoleInExchange.
SendInExchangeType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SendInExchangeType - ParticipationInExchangeType *Association (source - target):* «place2Type» SendInExchangeType - SendType *Attributes:* - The powertype of sendInExchange.
SendType «IDEAS:Powertype» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» SendType - ExchangeParticipationType *Attributes:* - The powertype of Send.

3.4.9 Agent diagrams

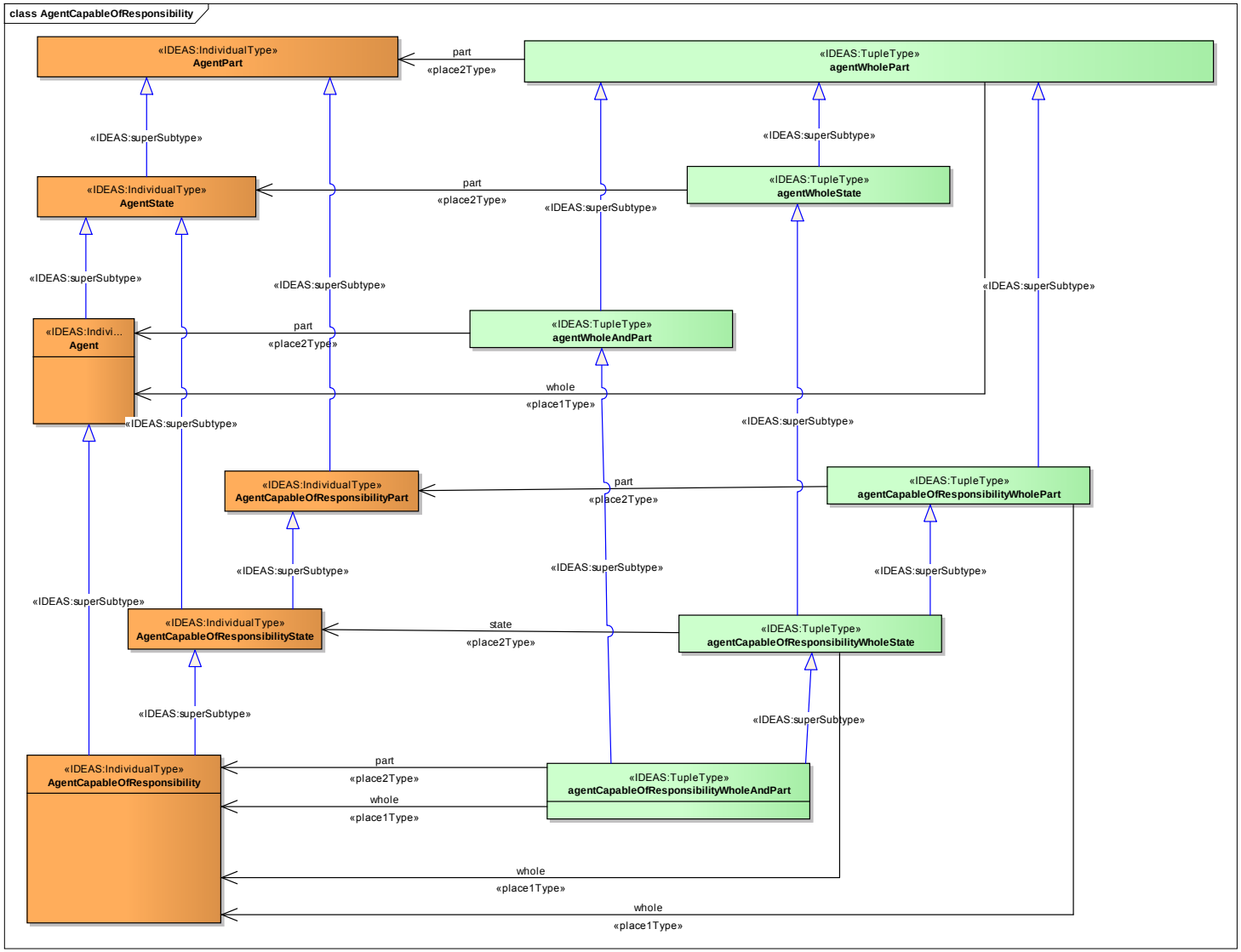


Figure 133 : AgentCapableOfResponsibility

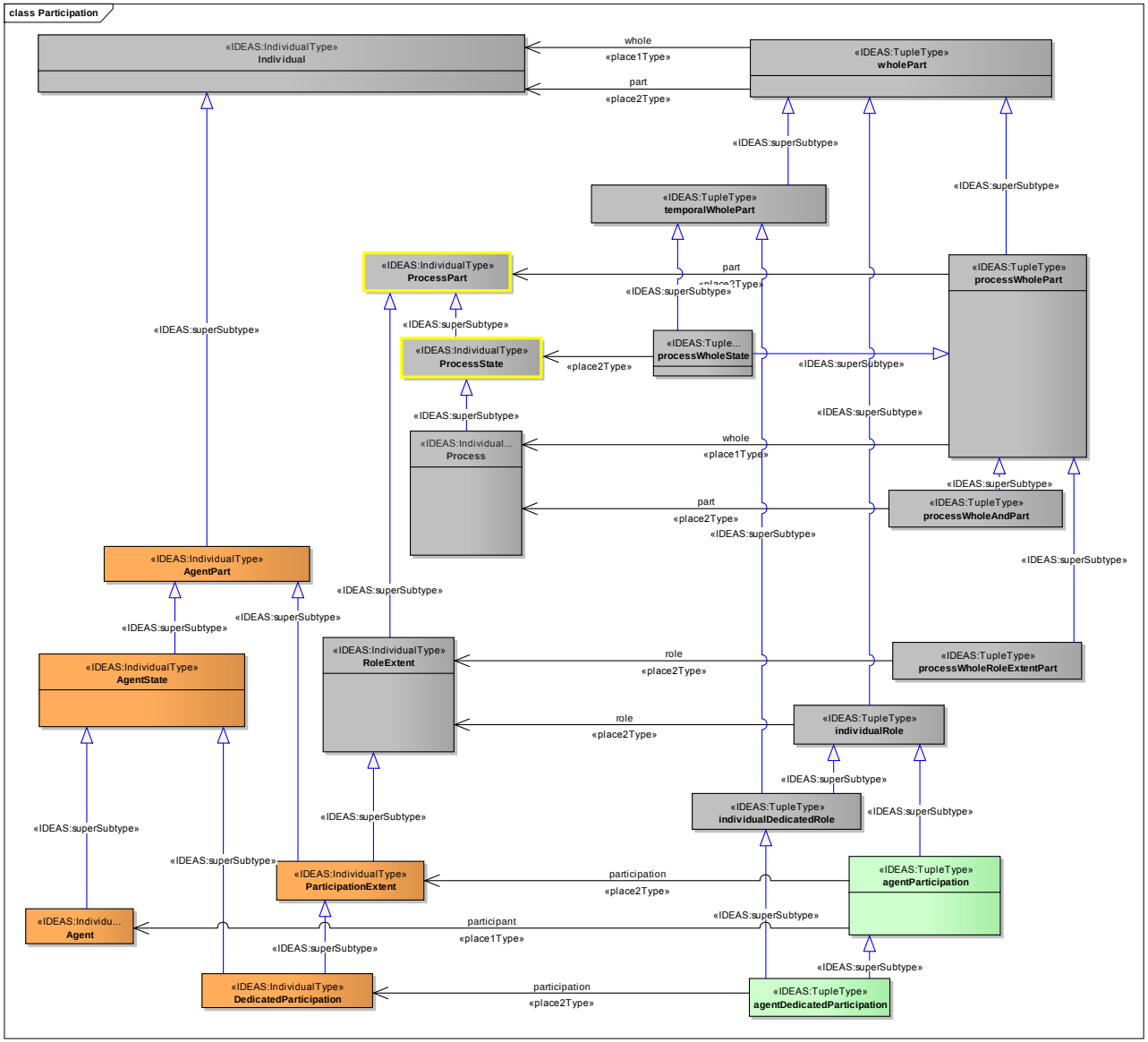


Figure 134 : Participation

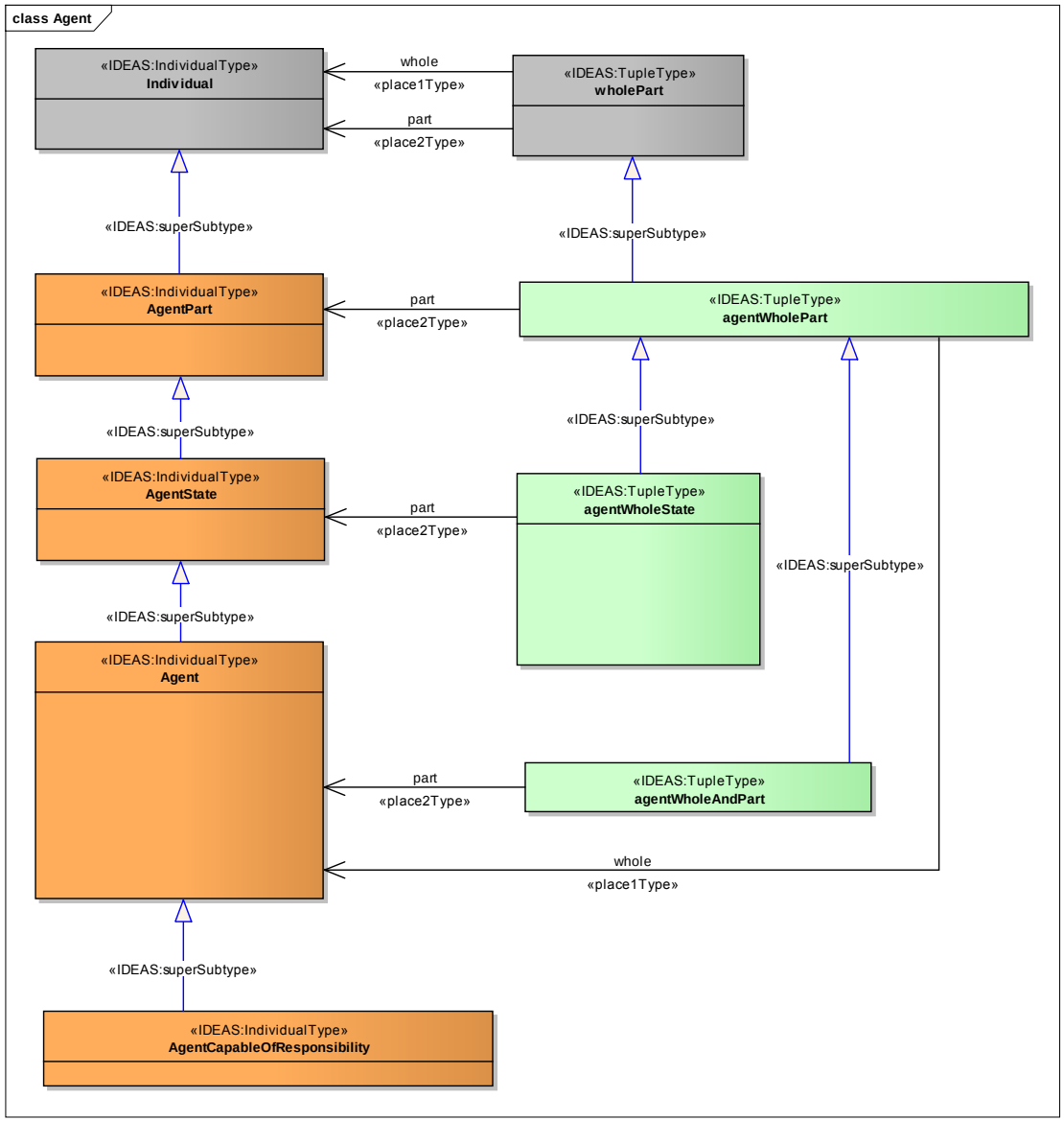


Figure 135 : Agent

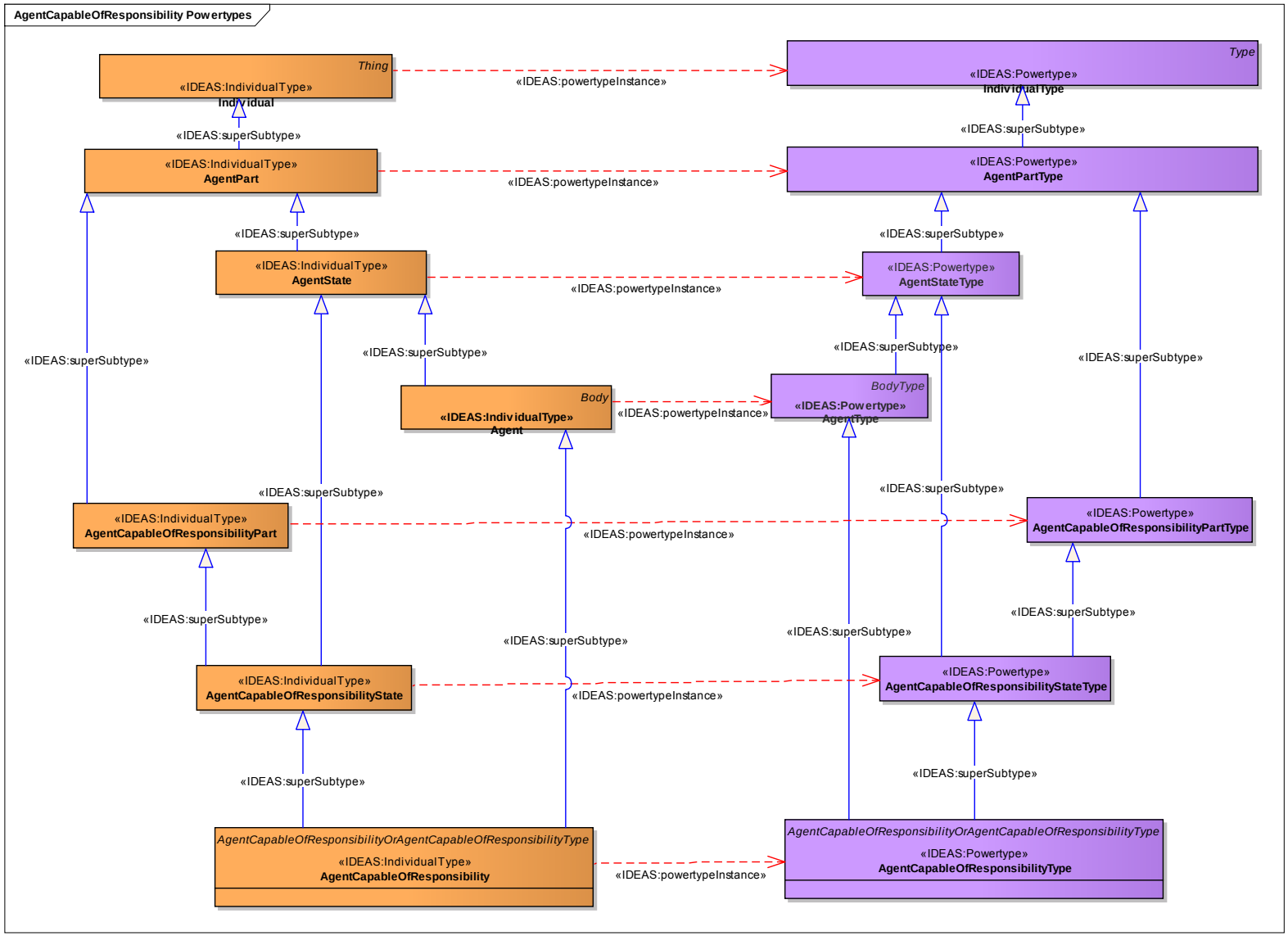


Figure 136 : AgentCapableOfResponsibility Powertypes

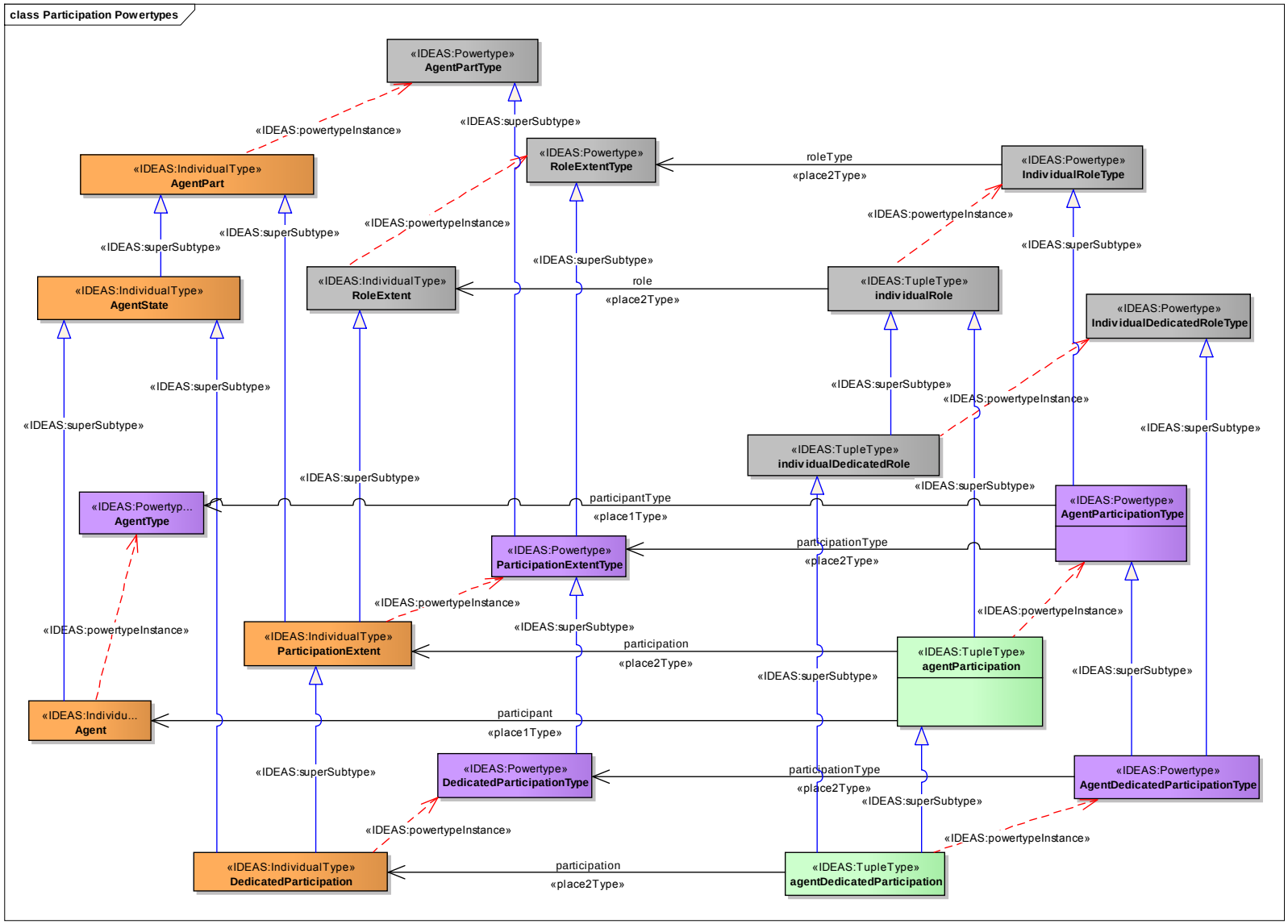


Figure 137 : Participation Powertypes

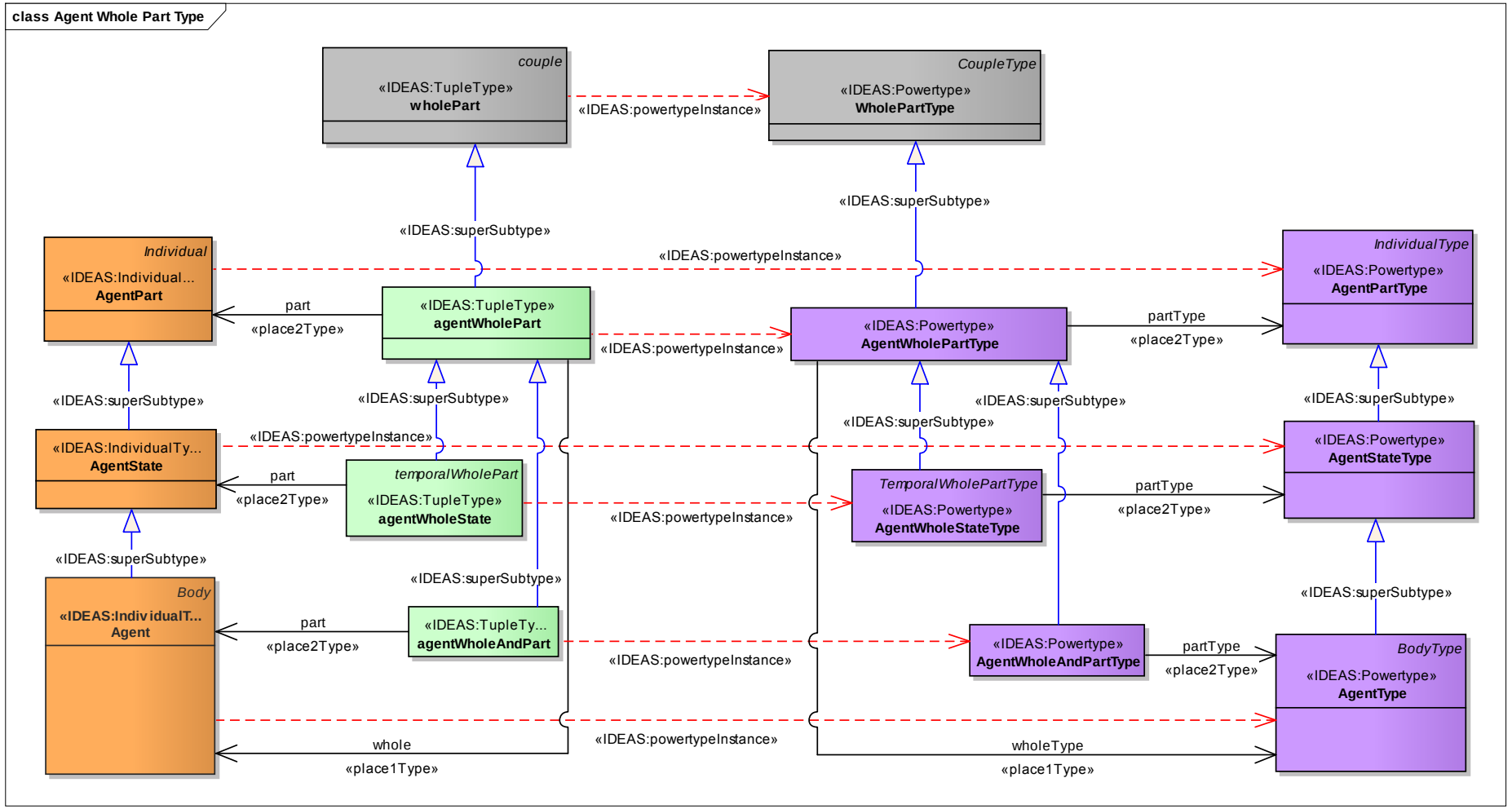


Figure 138 : Agent Whole Part Type

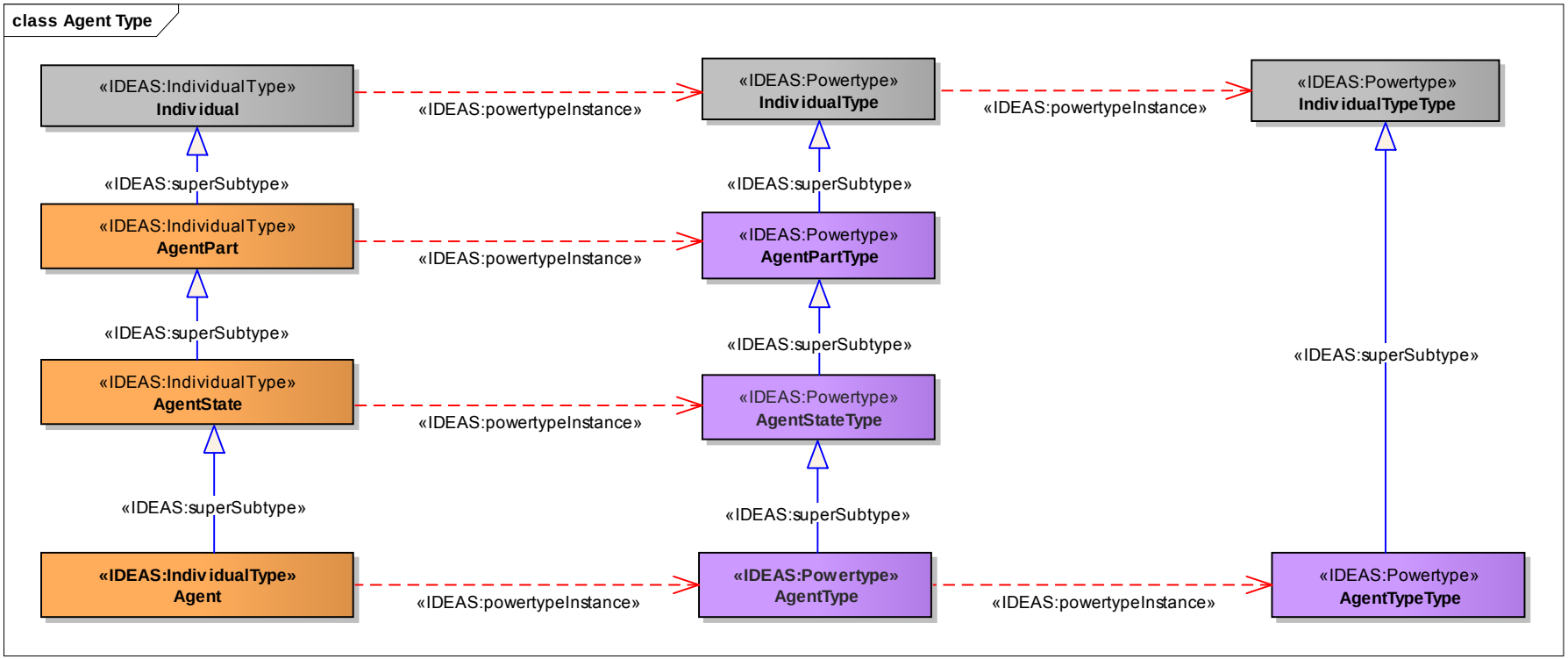


Figure 139 : Agent Type

3.4.10 Agent elements list

Agent
AgentCapableOfResponsibility «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentCapableOfResponsibility - AgentCapableOfResponsibilityOrAgentCapableOfResponsibilityType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentCapableOfResponsibility - Agent *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentCapableOfResponsibility - AgentCapableOfResponsibilityState *Dependency (element - is instance of):* «IDEAS:powertypeInstance» AgentCapableOfResponsibility - AgentCapableOfResponsibilityType *Attributes:* - An Agent that, from a legal perspective, has responsibility for its actions.
AgentCapableOfResponsibilityPart «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentCapableOfResponsibilityPart - AgentPart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» AgentCapableOfResponsibilityPart - AgentCapableOfResponsibilityPartType *Attributes:* - An AgentPart that is part of an AgentCapableOfResponsibility.
AgentCapableOfResponsibilityState «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentCapableOfResponsibilityState - AgentState *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentCapableOfResponsibilityState - AgentCapableOfResponsibilityPart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» AgentCapableOfResponsibilityState - AgentCapableOfResponsibilityStateType *Attributes:* - An AgentState that is a temporal part of an AgentCapableOfResponsibility.
AgentPart «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» AgentPart - Individual *Dependency (element - is instance of):* «IDEAS:powertypeInstance» AgentPart - AgentPartType *Attributes:* - An Individual that is part of an Agent.

AgentState «IDEAS:IndividualType»
Connectors:
Dependency (element - is instance of): «IDEAS:powertypeInstance»
AgentState - AgentStateType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentState - AgentPart
Attributes:
-
An AgentPart that is a temporal part of an Agent.

DedicatedParticipation «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
DedicatedParticipation - AgentState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
DedicatedParticipation - DedicatedParticipationType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
DedicatedParticipation - ParticipationExtent
Attributes:
-
A ParticipationExtent which is also an AgentState - i.e. a temporal part of an Agent. Note: A DedicatedParticipation may be temporally scattered - i.e. the fusion of the all the participations of an Agent.

ParticipationExtent «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ParticipationExtent - RoleExtent
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ParticipationExtent - AgentPart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
ParticipationExtent - ParticipationExtentType
Attributes:
-
A RoleExtent where the involved Individual is an Agent that participates actively in the Process.

agentCapableOfResponsibilityWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
agentCapableOfResponsibilityWholeAndPart - agentWholeAndPart
Generalization (element - is a subtype of): «IDEAS:superSubtype»
agentCapableOfResponsibilityWholeAndPart - agentCapableOfResponsibilityWholeState
Association (source - target): «place2Type»
agentCapableOfResponsibilityWholeAndPart - AgentCapableOfResponsibility
Association (source - target): «place1Type»
agentCapableOfResponsibilityWholeAndPart - AgentCapableOfResponsibility
Attributes:
-
An AgentCapableOfResponsibility where both the whole and part are AgentsCapableOfResponsibility.

agentCapableOfResponsibilityWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
agentCapableOfResponsibilityWholePart - agentWholePart
Association (source - target): «place2Type»
agentCapableOfResponsibilityWholePart - AgentCapableOfResponsibilityPart
Association (source - target): «place1Type»
agentCapableOfResponsibilityWholePart - AgentCapableOfResponsibility
Attributes:
-
An agentWholePart where the whole is an AgentCapableOfResponsibility.

agentCapableOfResponsibilityWholeState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of):«IDEAS:superSubtype»
agentCapableOfResponsibilityWholeState - agentWholeState
Generalization (element - is a subtype of):«IDEAS:superSubtype»
agentCapableOfResponsibilityWholeState - agentCapableOfResponsibilityWholePart
Association (source - target):«place2Type»
agentCapableOfResponsibilityWholeState - AgentCapableOfResponsibilityState
Association (source - target):«place1Type»
agentCapableOfResponsibilityWholeState - AgentCapableOfResponsibility
Attributes:
-
A temporalWholePart and an agentCapableOfResponsibilityWholePart where an AgentCapableOfResponsibilityState is a temporal part of an Agent.

agentDedicatedParticipation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
agentDedicatedParticipation - agentParticipation
Generalization (element - is a subtype of): «IDEAS:superSubtype»
agentDedicatedParticipation - individualDedicatedRole
Dependency (element - is instance of): «IDEAS:powertypeInstance»
agentDedicatedParticipation - AgentDedicatedParticipationType
Association (source - target): «place2Type»
agentDedicatedParticipation - DedicatedParticipation
Attributes:
-
An agentParticipation which is also an individualDedicatedRole and the participation is a DedicatedParticipation.

agentParticipation «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
agentParticipation - individualRole
Dependency (element - is instance of): «IDEAS:powertypeInstance»
agentParticipation - AgentParticipationType
Association (source - target): «place1Type»
agentParticipation - Agent
Association (source - target): «place2Type»
agentParticipation - ParticipationExtent
Attributes:
-

An individualRole where the role is a ParticipationExtent.
agentWholeAndPart «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentWholeAndPart - agentWholePart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» agentWholeAndPart - AgentWholeAndPartType *Association (source - target):* «place2Type» agentWholeAndPart - Agent *Attributes:* - An agentWholePart where both the whole and part are Agents.
agentWholePart «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentWholePart - wholePart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» agentWholePart - AgentWholePartType *Association (source - target):* «place2Type» agentWholePart - AgentPart *Association (source - target):* «place1Type» agentWholePart - Agent *Attributes:* - A wholePart where the whole is an Agent.
agentWholeState «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentWholeState - temporalWholePart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» agentWholeState - AgentWholeStateType *Generalization (element - is a subtype of):* «IDEAS:superSubtype» agentWholeState - agentWholePart *Association (source - target):* «place2Type» agentWholeState - AgentState *Attributes:* - A temporalWholePart and an agentWholePart where an AgentState is a temporal part of an Agent.
overlapTypeIndividualInstance «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» overlapTypeIndividualInstance - typeInstance *Association (source - target):* «place1Type» overlapTypeIndividualInstance - SetOfOverlappingIndividuals *Association (source - target):* «place2Type» overlapTypeIndividualInstance - Individual *Attributes:* -

A typeInstance where an Individual is an instance of a SetOfOverlappingIndividuals.
Agent «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Agent - AgentState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Agent - Body
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Agent - AgentType

Attributes:
-
The powertype of agentDedicatedParticipation.

AgentPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentPartType - IndividualType
Attributes:
-
An IndividualType that is the Powertype of AgentPart.

AgentParticipationType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentParticipationType - IndividualRoleType
Association (source - target): «place1Type»
AgentParticipationType - AgentType
Association (source - target): «place2Type»
AgentParticipationType - ParticipationExtentType
Attributes:
-
The powertype of agentParticipation.

AgentStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentStateType - AgentPartType
Attributes:
-
The powertype of AgentState.

AgentType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentType - AgentStateType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentType - BodyType
Dependency (element - is instance of): «IDEAS:powertypeInstance»
AgentType - AgentTypeType
Attributes:
-
The powertype of Agent.

AgentTypeType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
AgentTypeType - IndividualTypeType
Attributes:
-
The powertype of AgentType.

AgentWholeAndPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of):

ParticipationExtentType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ParticipationExtentType - RoleExtentType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ParticipationExtentType - AgentPartType
Attributes:
-
The powertype of ParticipationExtent.

3.4.11 Process diagrams

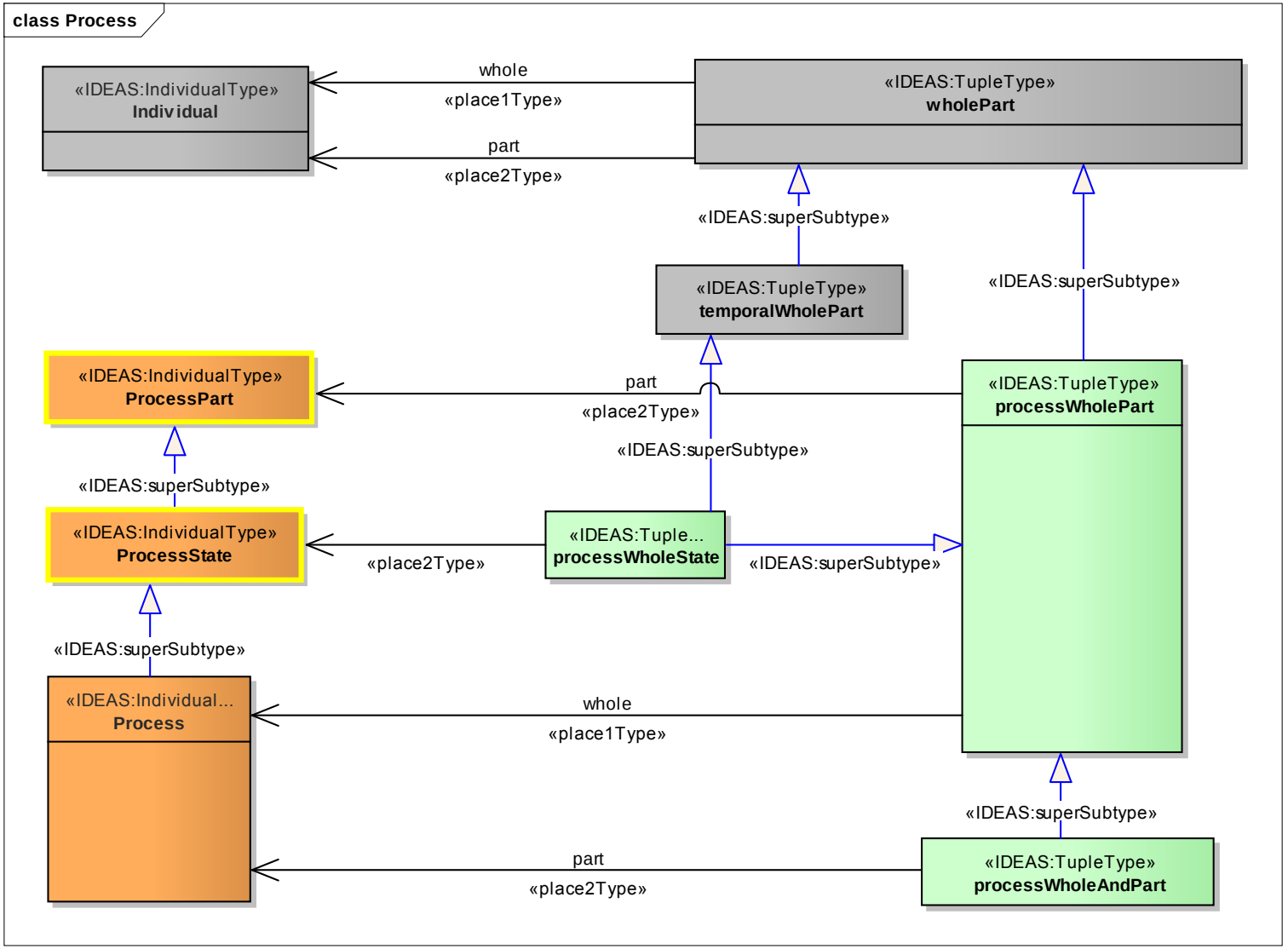


Figure 140 : Process

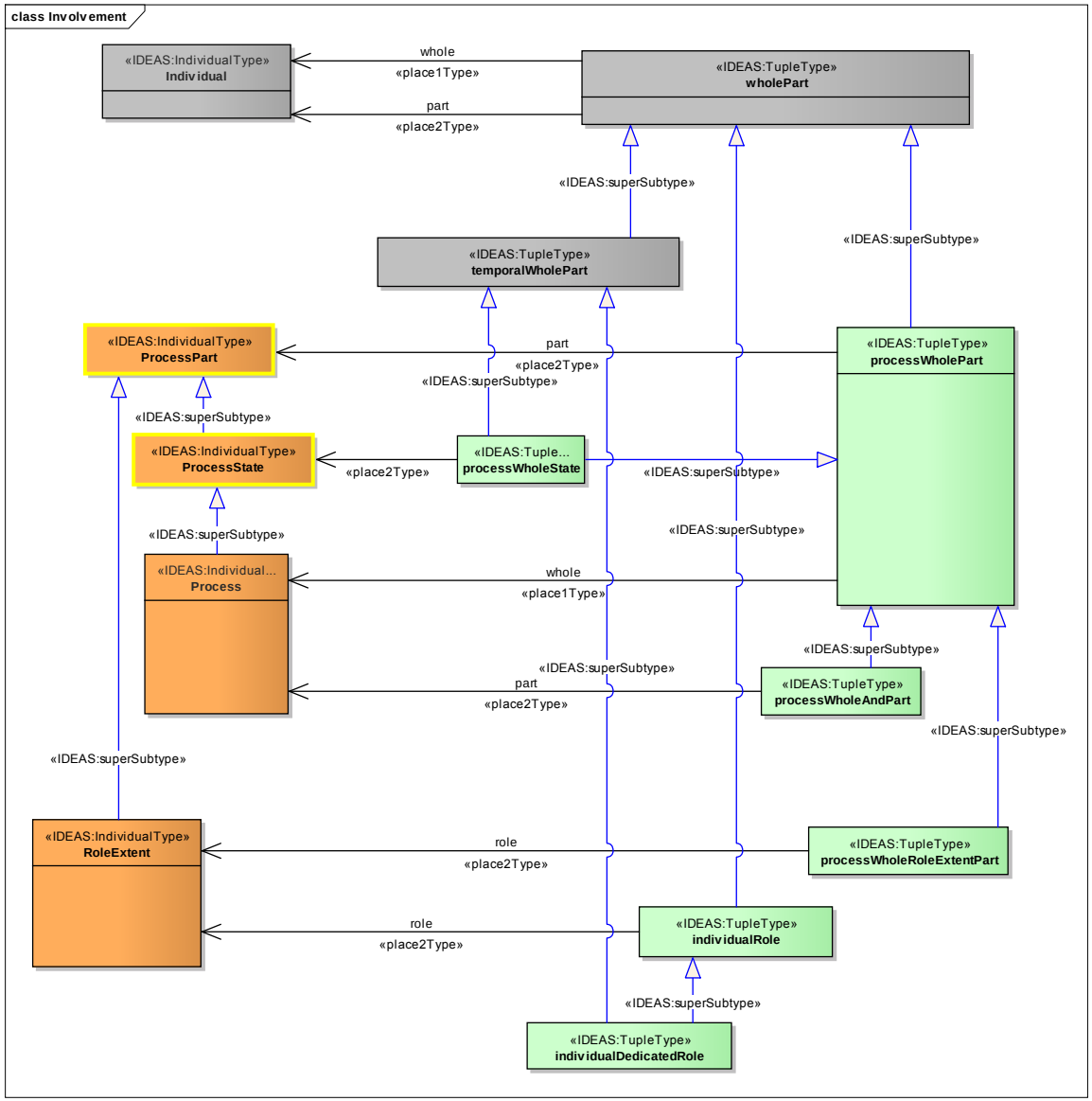


Figure 141 : Involvement

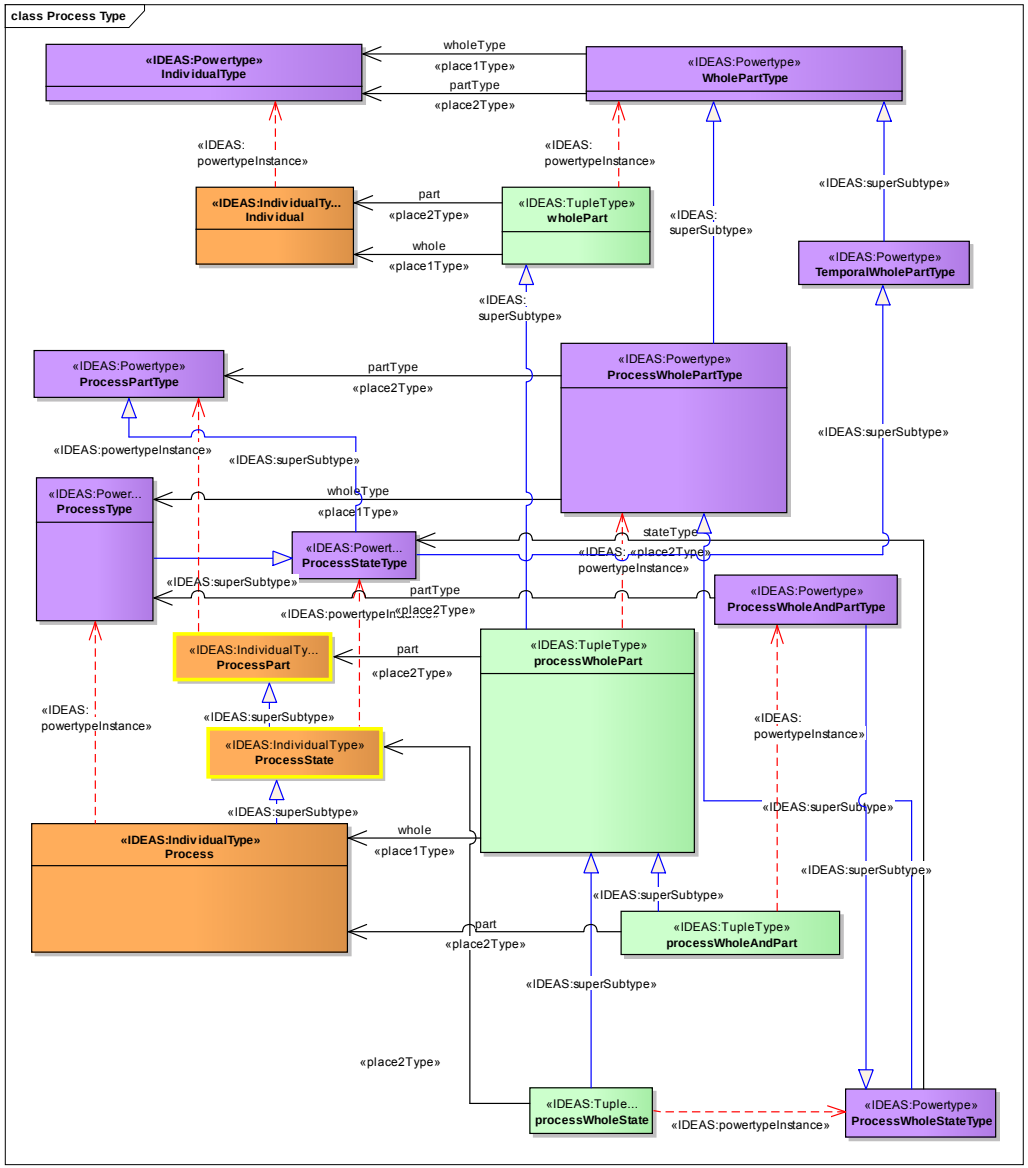


Figure 142 : ProcessType

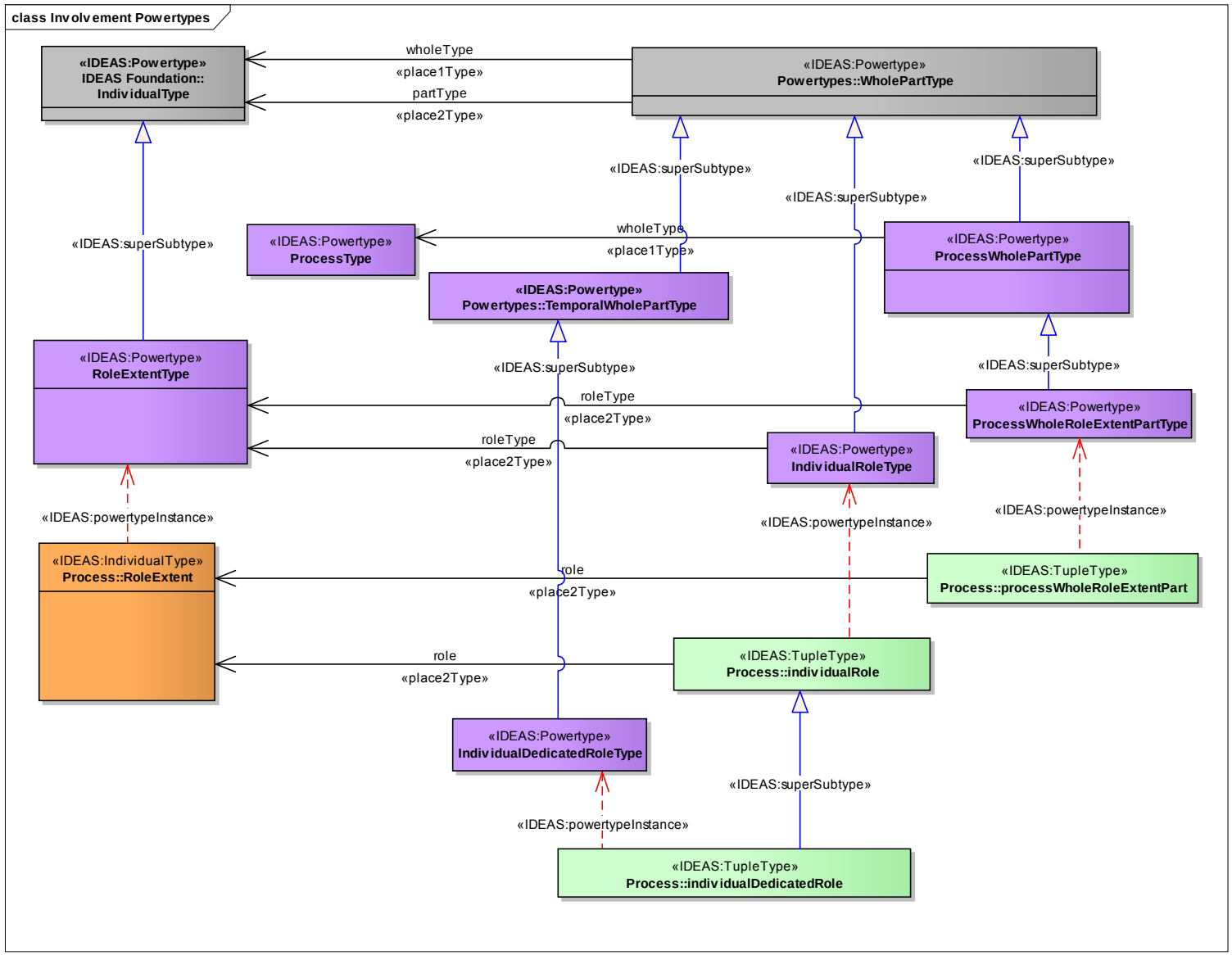


Figure 143 : Involvement Powertypes

3.4.12 Process elements list

Process
ProcessPart «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ProcessPart - BodyPart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ProcessPart - ProcessPartType *Attributes:* - An Individual that is part of a Process.
ProcessState «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» ProcessState - ProcessPart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» ProcessState - ProcessStateType *Attributes:* - A ProcessPart that is a temporal part of a Process.
RoleExtent «IDEAS:IndividualType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» RoleExtent - ProcessPart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» RoleExtent - RoleExtentType *Attributes:* - A ProcessPart that is the extent of an Individual's involvement in a Process.
individualDedicatedRole «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» individualDedicatedRole - temporalWholePart *Generalization (element - is a subtype of):* «IDEAS:superSubtype» individualDedicatedRole - individualRole *Dependency (element - is instance of):* «IDEAS:powertypeInstance» individualDedicatedRole - IndividualDedicatedRoleType *Attributes:* - An individualRole where the RoleExtent is a temporal part of the Individual involved in the Process - i.e. all the Individual for a period of time. Note: The RoleExtent may be temporally scattered - i.e. the fusion of all occasions the Individual was involved.
individualRole «IDEAS:TupleType» *Connectors:* *Generalization (element - is a subtype of):* «IDEAS:superSubtype» individualRole - wholePart *Dependency (element - is instance of):* «IDEAS:powertypeInstance» individualRole - IndividualRoleType *Association (source - target):* «place2Type»

individualRole - RoleExtent
Attributes:
-
A wholePart where the part is the extent of the Role played by the Individual in a particular Process.

processWholeRoleExtentPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
processWholeRoleExtentPart - processWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
processWholeRoleExtentPart - ProcessWholeRoleExtentPartType
Association (source - target): «place2Type»
processWholeRoleExtentPart - RoleExtent
Attributes:
-
A processWholePart where the part is a RoleExtent - i.e. the extent of an Individual's role in the Process.

processWholeState «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
processWholeState - processWholePart
Association (source - target): «place2Type»
processWholeState - ProcessState
Generalization (element - is a subtype of): «IDEAS:superSubtype»
processWholeState - temporalWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
processWholeState - ProcessWholeStateType
Attributes:
-
A processWholeState where the part is a ProcessState - i.e. all of the spatial extent of the process for a period of time.

Process «IDEAS:IndividualType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
Process - ProcessState
Dependency (element - is instance of): «IDEAS:powertypeInstance»
Process - ProcessType
Attributes:
-
A ProcessPart that is an Individual whose extent is defined by its involvements.

processWholeAndPart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
processWholeAndPart - processWholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
processWholeAndPart - ProcessWholeAndPartType
Association (source - target): «place2Type»
processWholeAndPart - Process
Attributes:
-
A processWholePart that asserts a Process is part of a Process.

processWholePart «IDEAS:TupleType»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
processWholePart - wholePart
Dependency (element - is instance of): «IDEAS:powertypeInstance»
processWholePart - ProcessWholePartType
Association (source - target): «place2Type»
processWholePart - ProcessPart
Association (source - target): «place1Type»
processWholePart - Process
Attributes:
-
A wholePart that asserts an Individual is part of a Process.

Process Powertypes

IndividualDedicatedRoleType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
IndividualDedicatedRoleType - TemporalWholePartType
Attributes:
-
The powertype of individualDedicatedRole.

IndividualRoleType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
IndividualRoleType - WholePartType
Association (source - target): «place1Type»
IndividualRoleType - RoleExtentType
Attributes:
-
The powertype of individualRole.

ProcessPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessPartType - BodyPartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessPartType - ModemIndividualElementType
Attributes:
-
The powertype of ProcessPart.

ProcessStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessStateType - ProcessPartType
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessStateType - TemporalWholePartType
Attributes:
-
The powertype of ProcessState.

ProcessWholeAndPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessWholeAndPartType - ProcessWholeStateType
Association (source - target): «place2Type»
ProcessWholeAndPartType - ProcessType
Attributes:
-
The ProcessWholePartType that is the Powertype of processWholeAndPart.

ProcessWholeRoleExtentPartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessWholeRoleExtentPartType - ProcessWholePartType
Association (source - target): «place2Type»
ProcessWholeRoleExtentPartType - RoleExtentType
Attributes:
-
The powertype of processWholeRoleExtentPart.

ProcessWholeStateType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessWholeStateType - ProcessWholePartType
Association (source - target): «place2Type»
ProcessWholeStateType - ProcessStateType
Attributes:
-
The powertype of processWholeState.

RoleExtentType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
RoleExtentType - IndividualType
Attributes:
-
The powertype of RoleExtent.

ProcessType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessType - ProcessStateType
Attributes:
-
The powertype of Process.

ProcessWholePartType «IDEAS:Powertype»
Connectors:
Generalization (element - is a subtype of): «IDEAS:superSubtype»
ProcessWholePartType - WholePartType
Association (source - target): «place2Type»
ProcessWholePartType - ProcessPartType
Association (source - target): «place1Type»
ProcessWholePartType - ProcessType
Attributes:
-
The powertype of processWholePart.

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